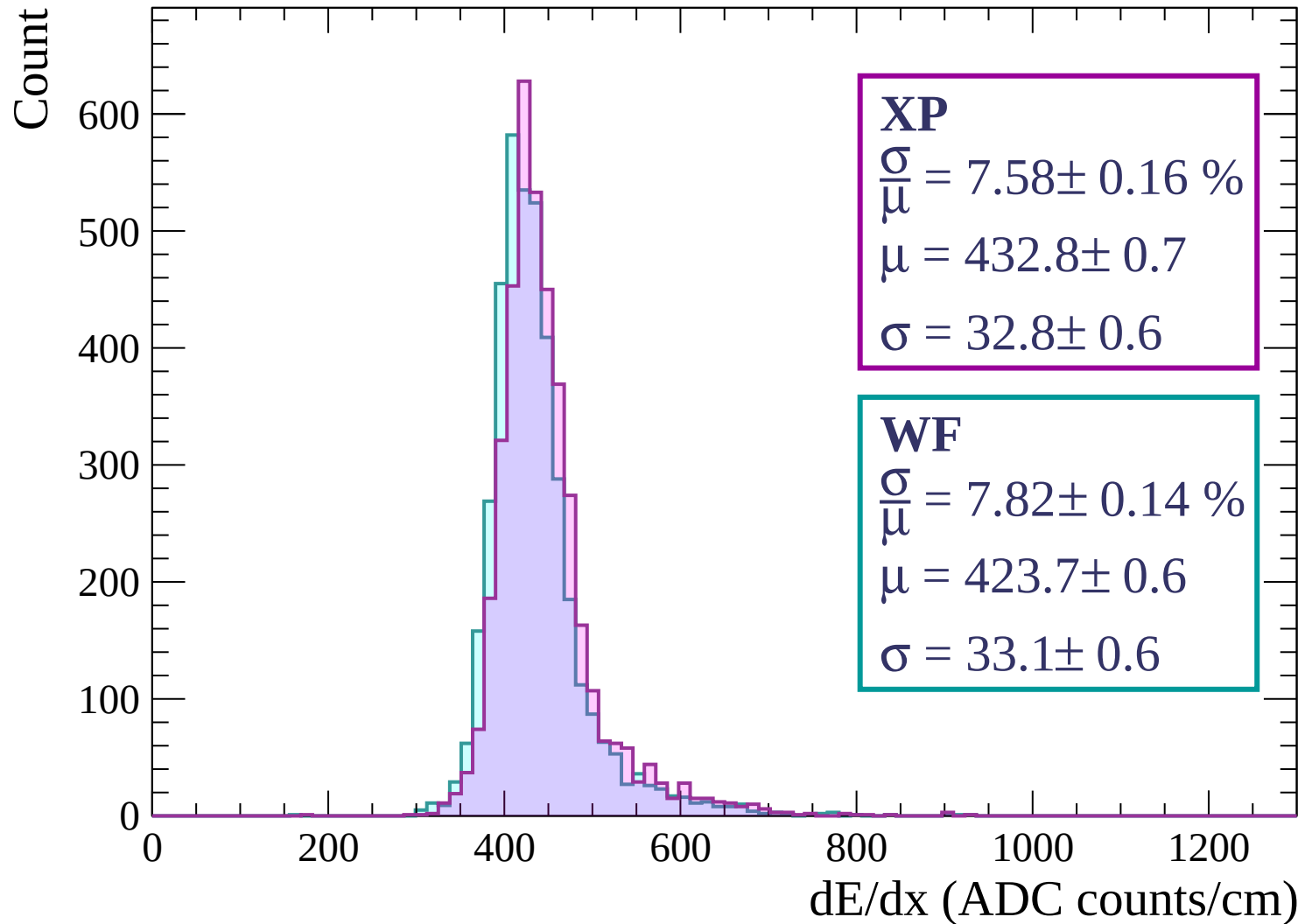
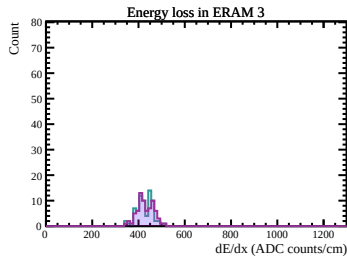
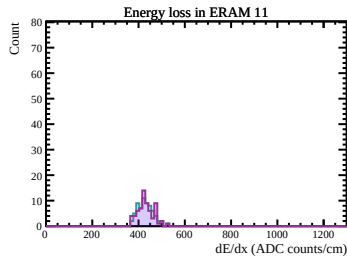
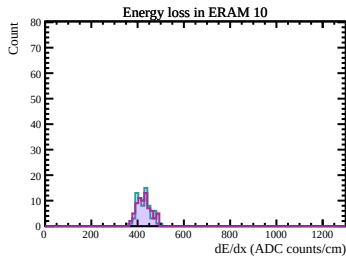
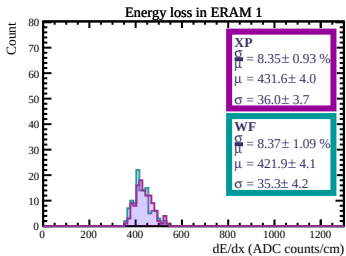
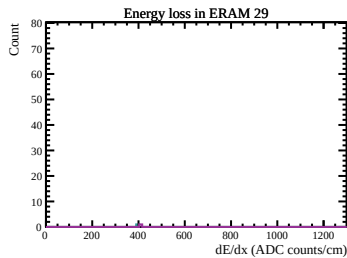
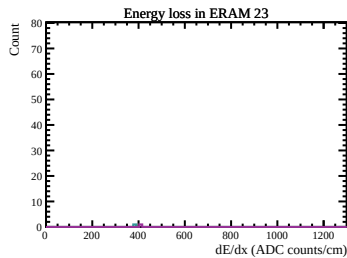
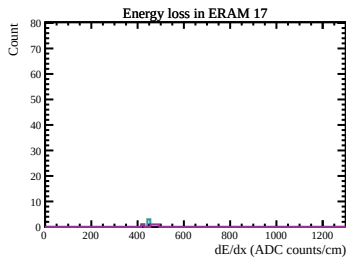
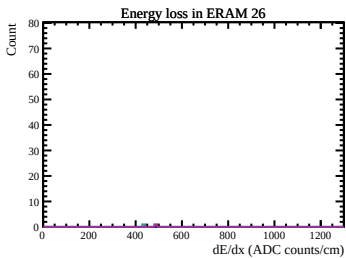
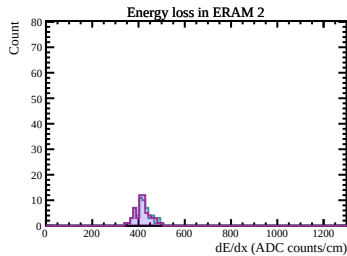
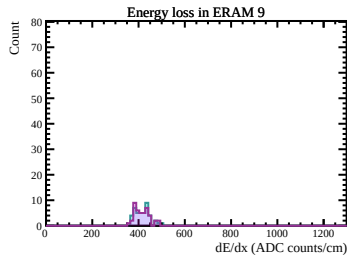
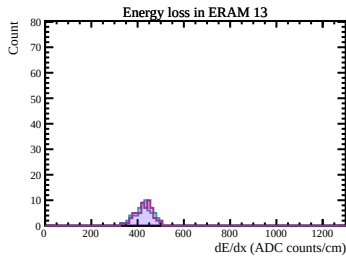
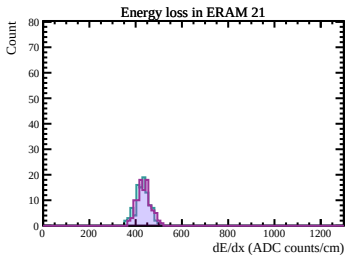
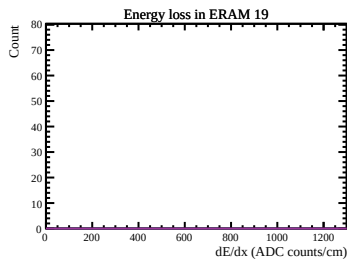
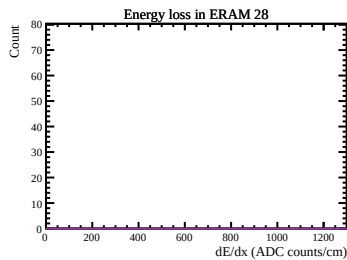
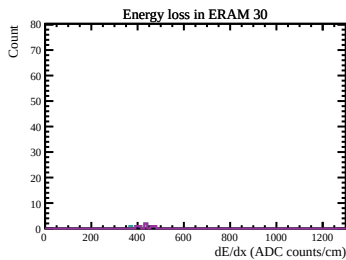
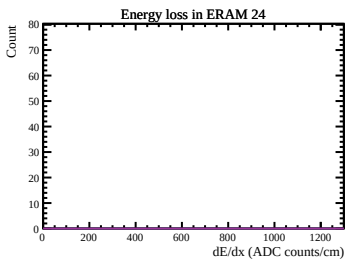
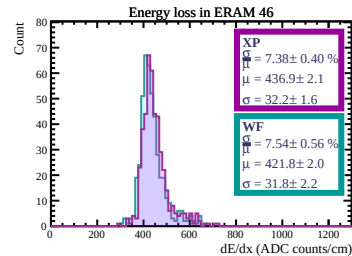
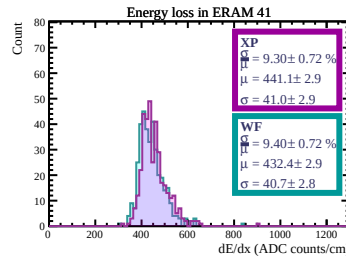
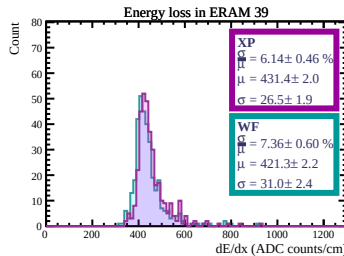
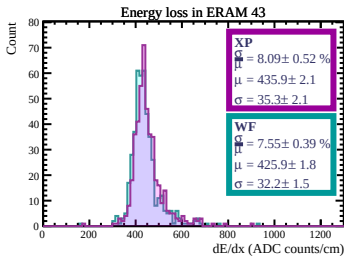
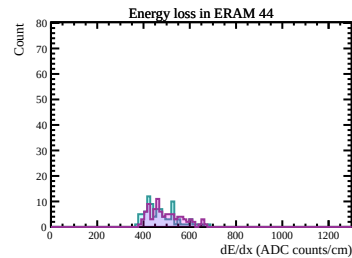
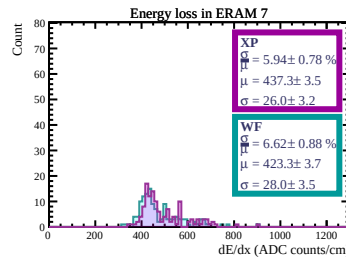
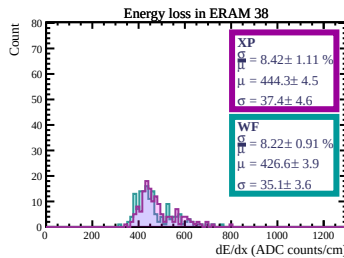
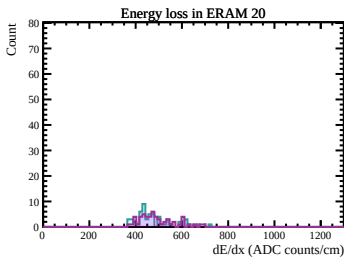
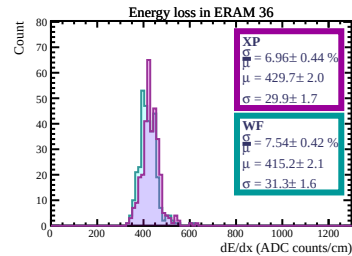
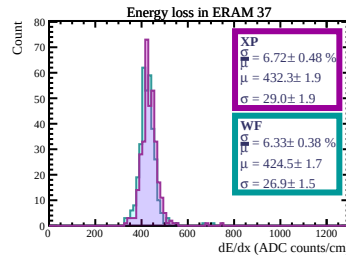
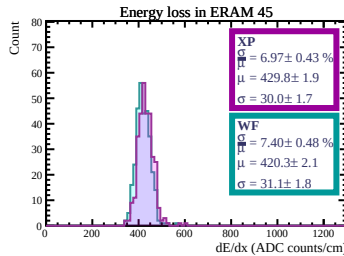
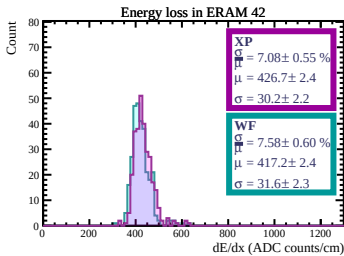
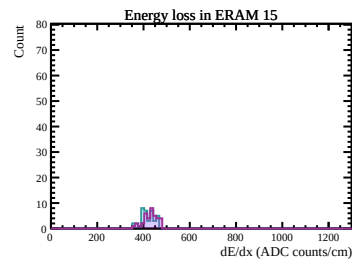
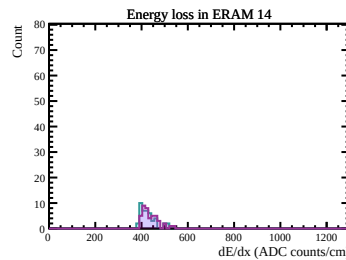
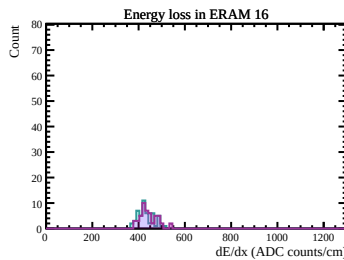
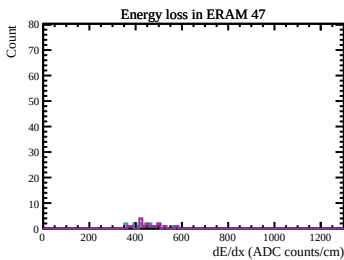
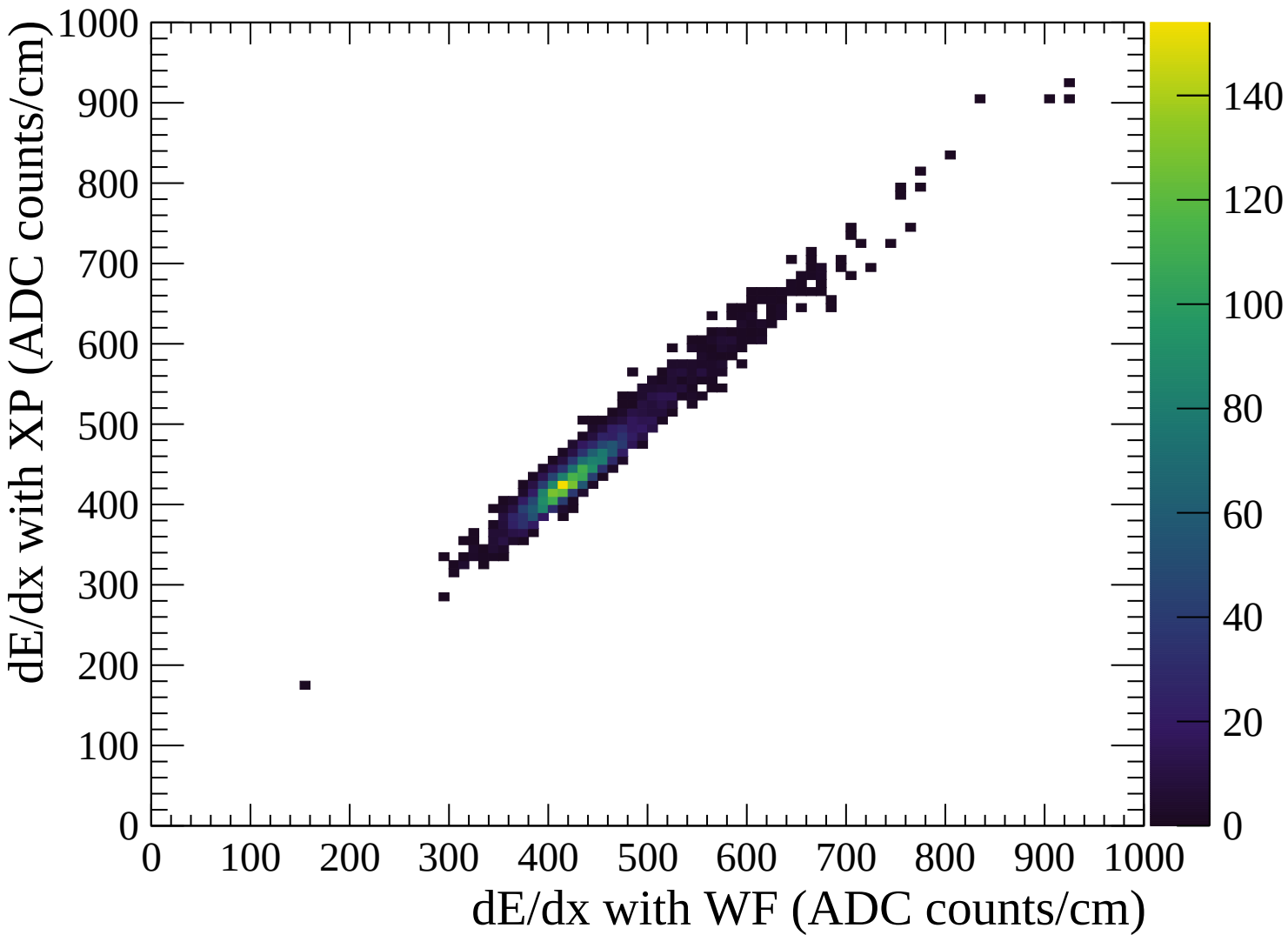
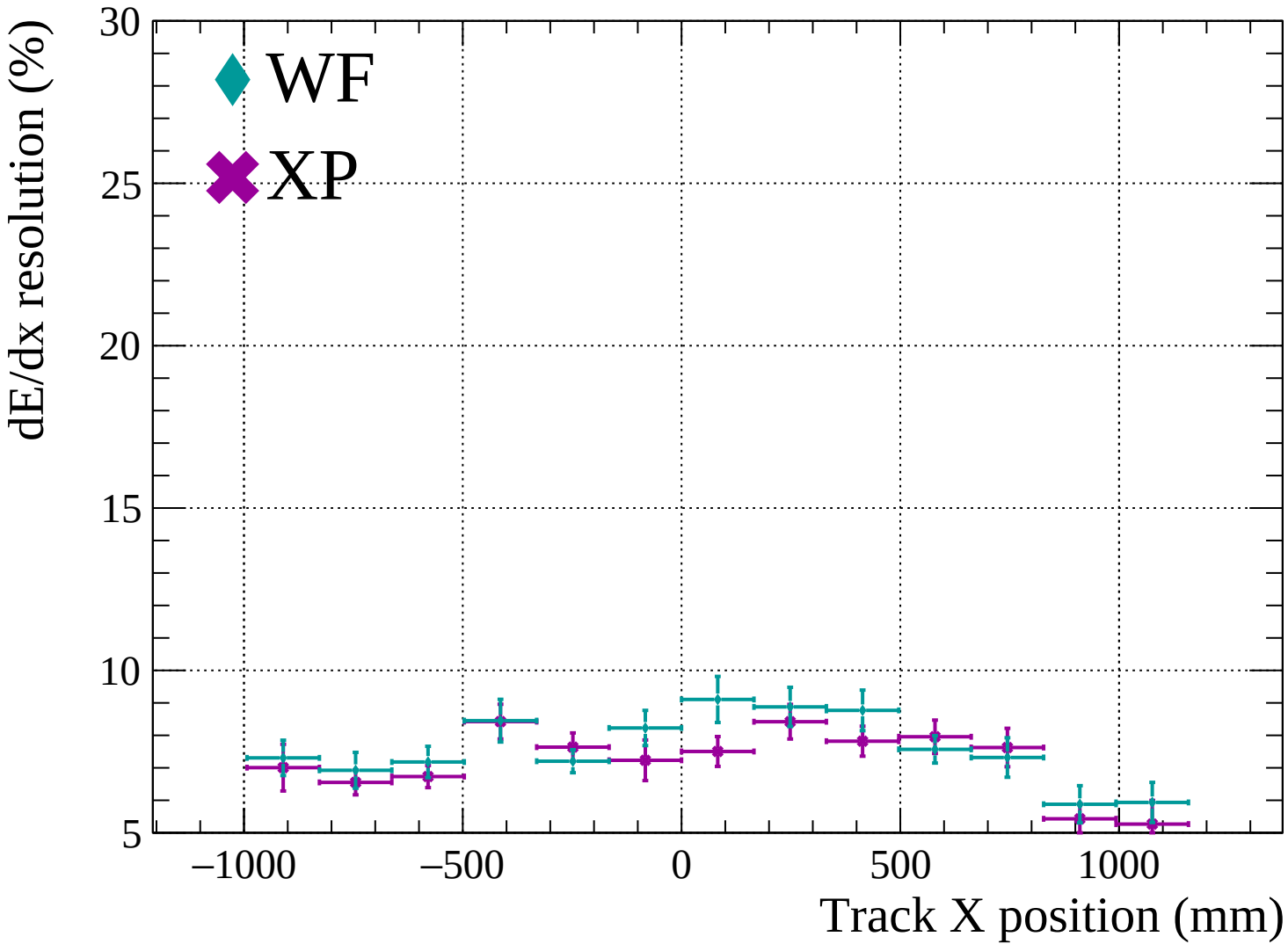


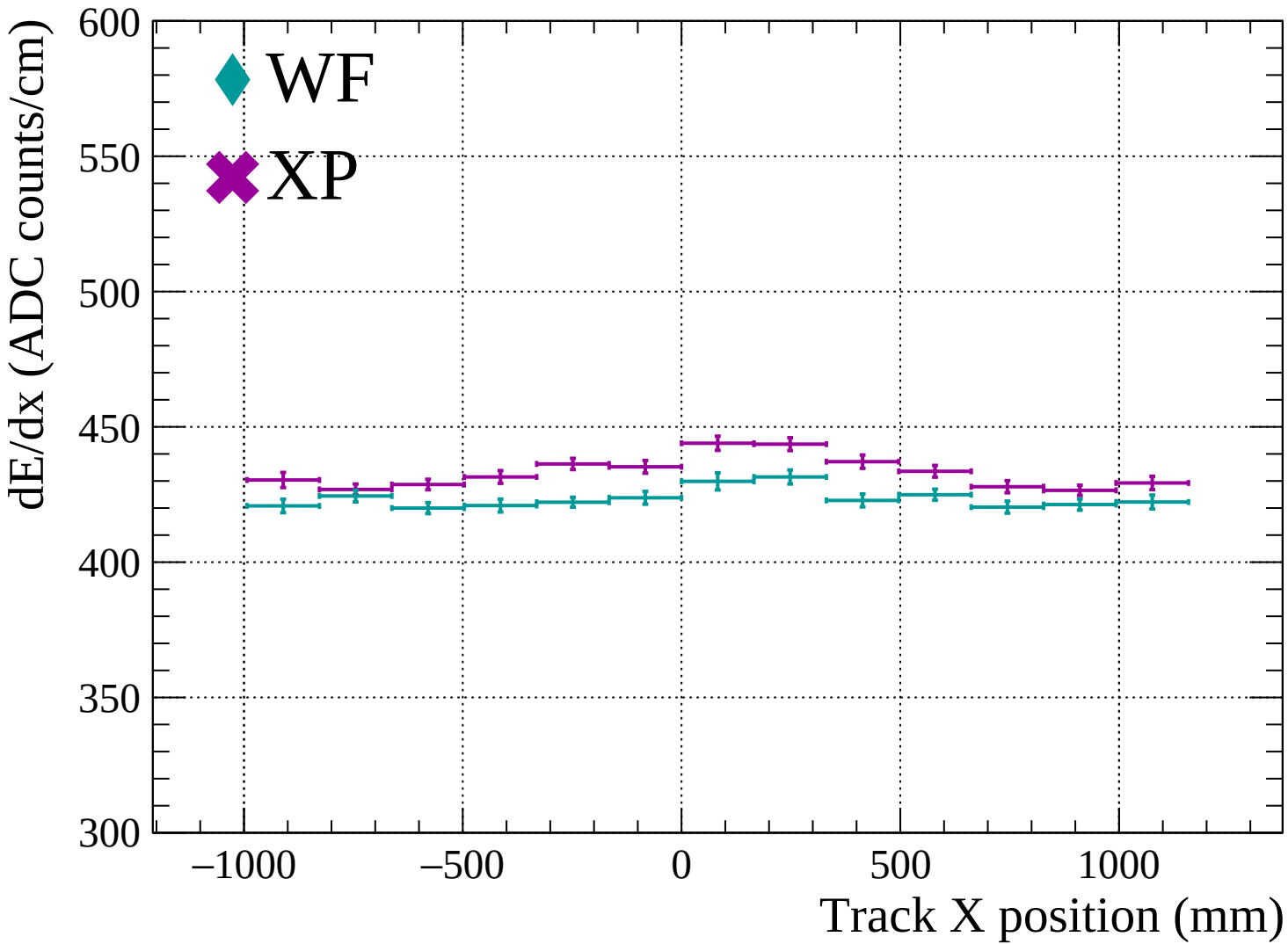


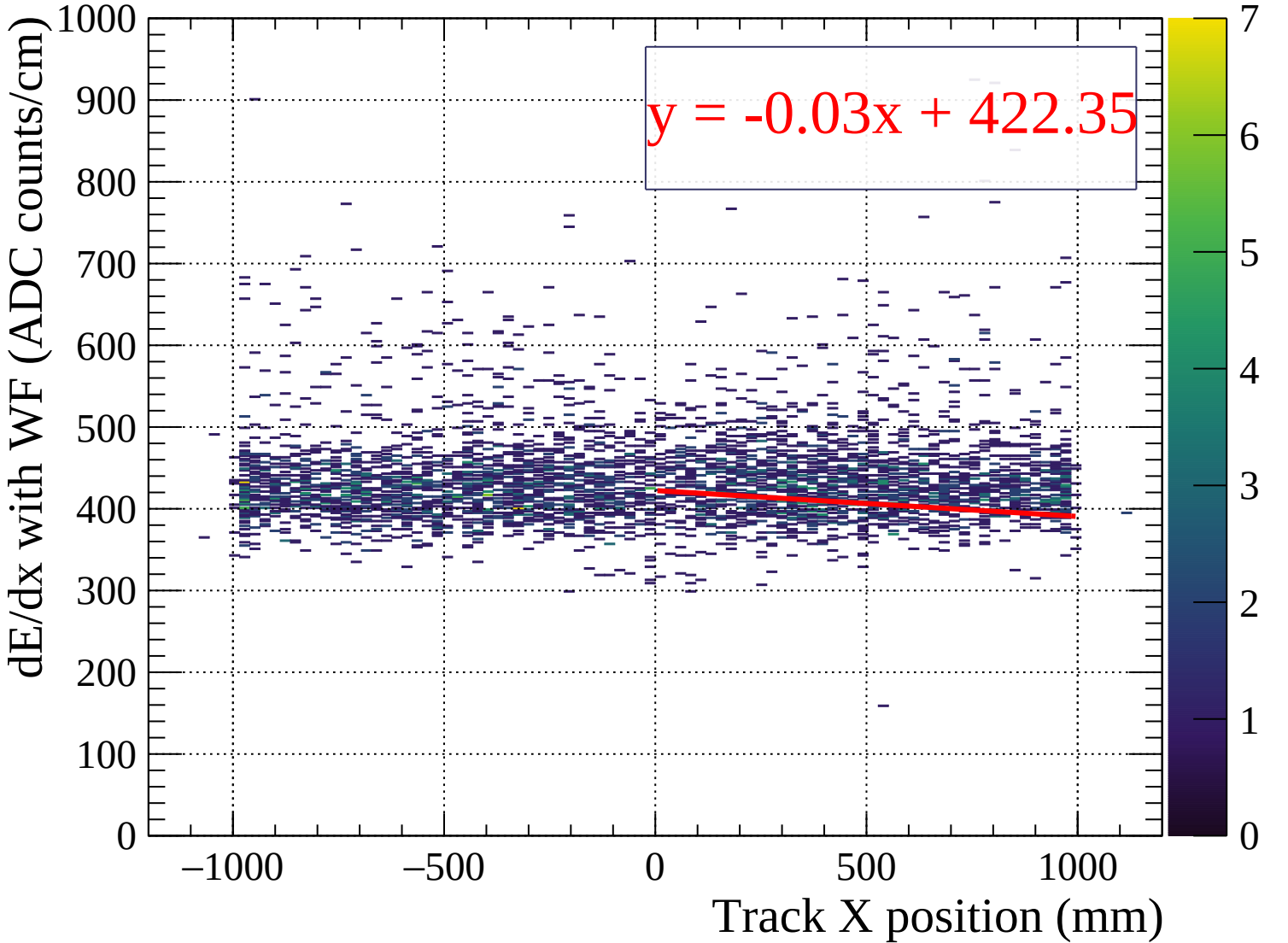


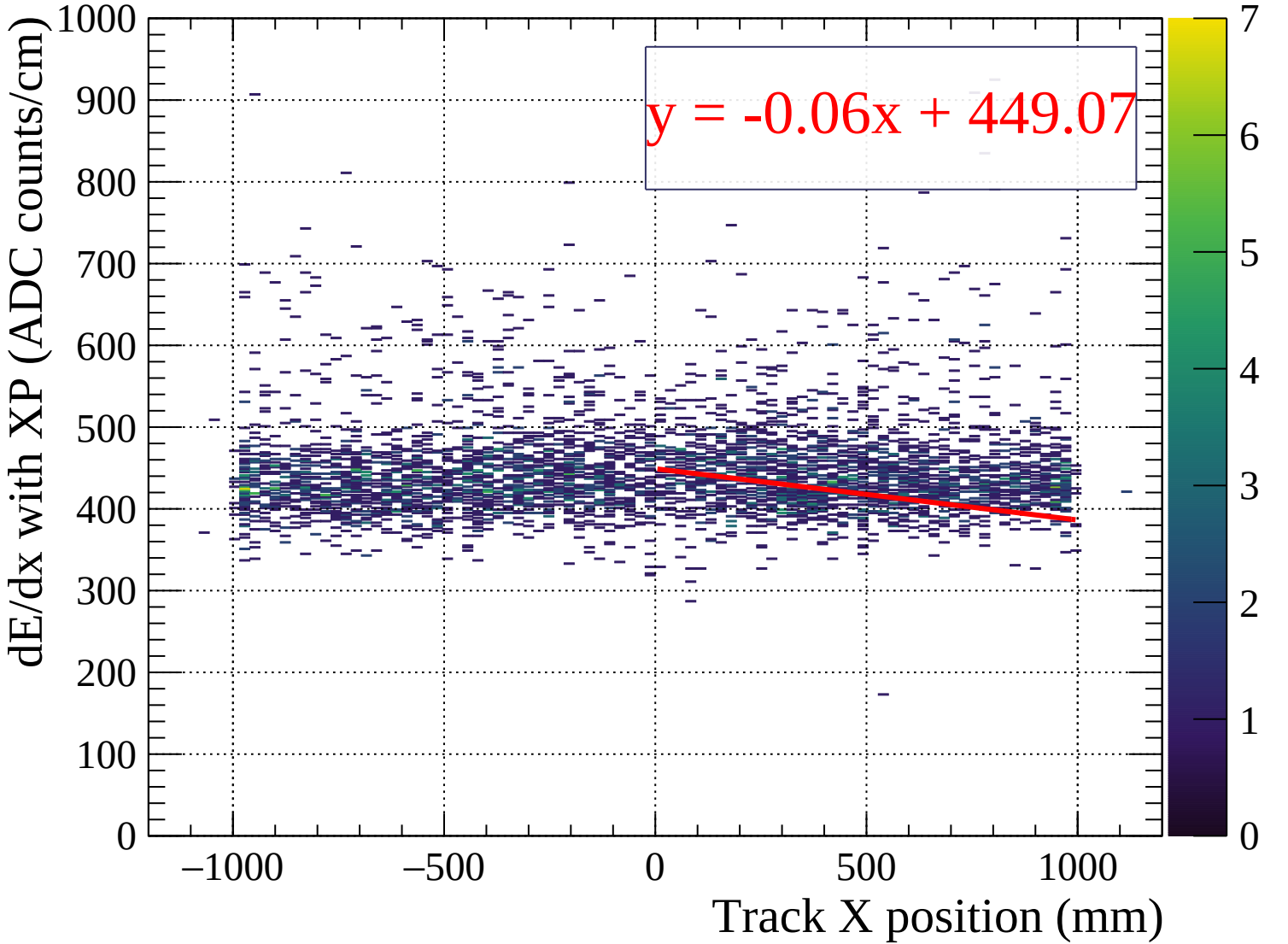


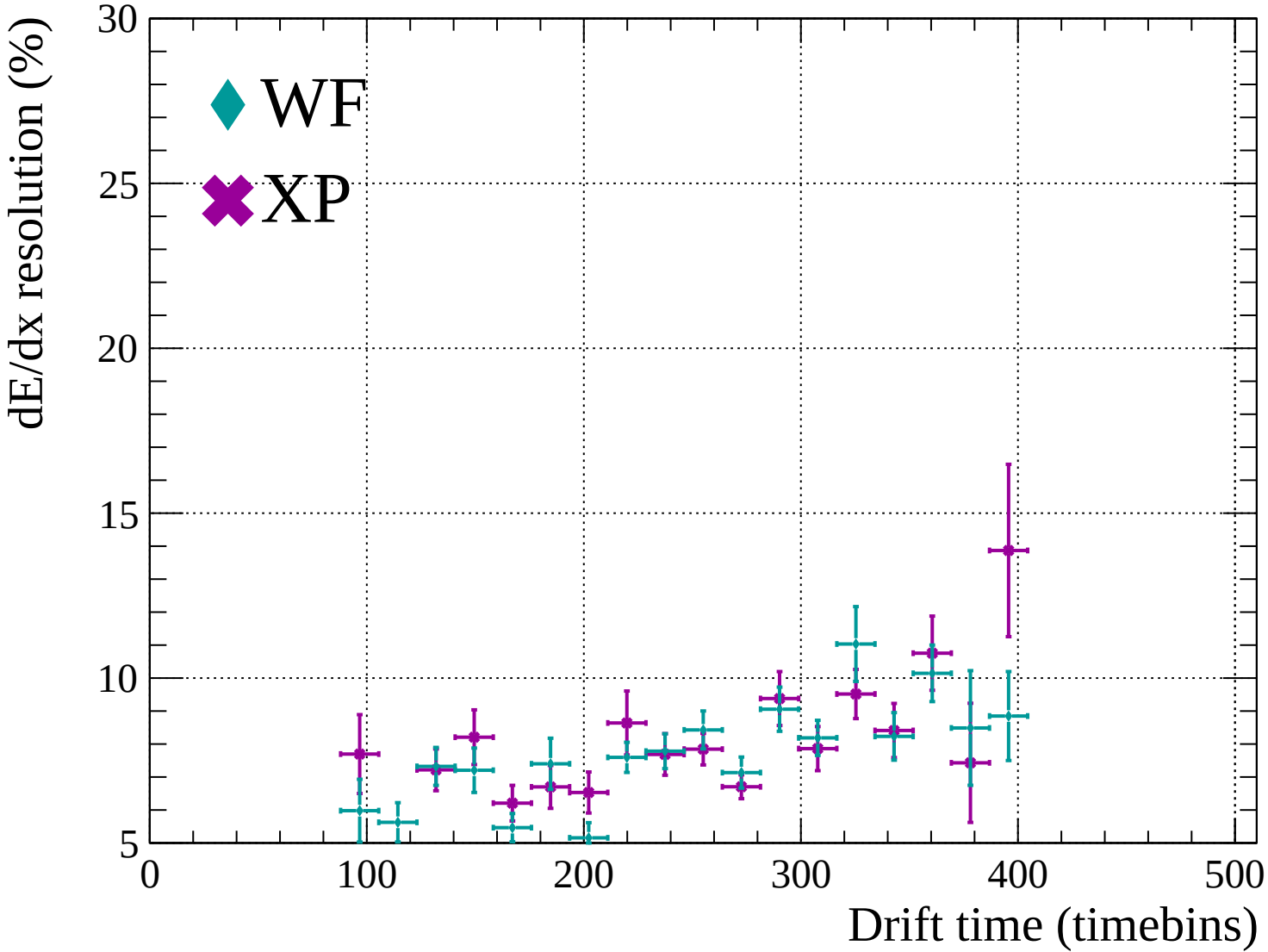


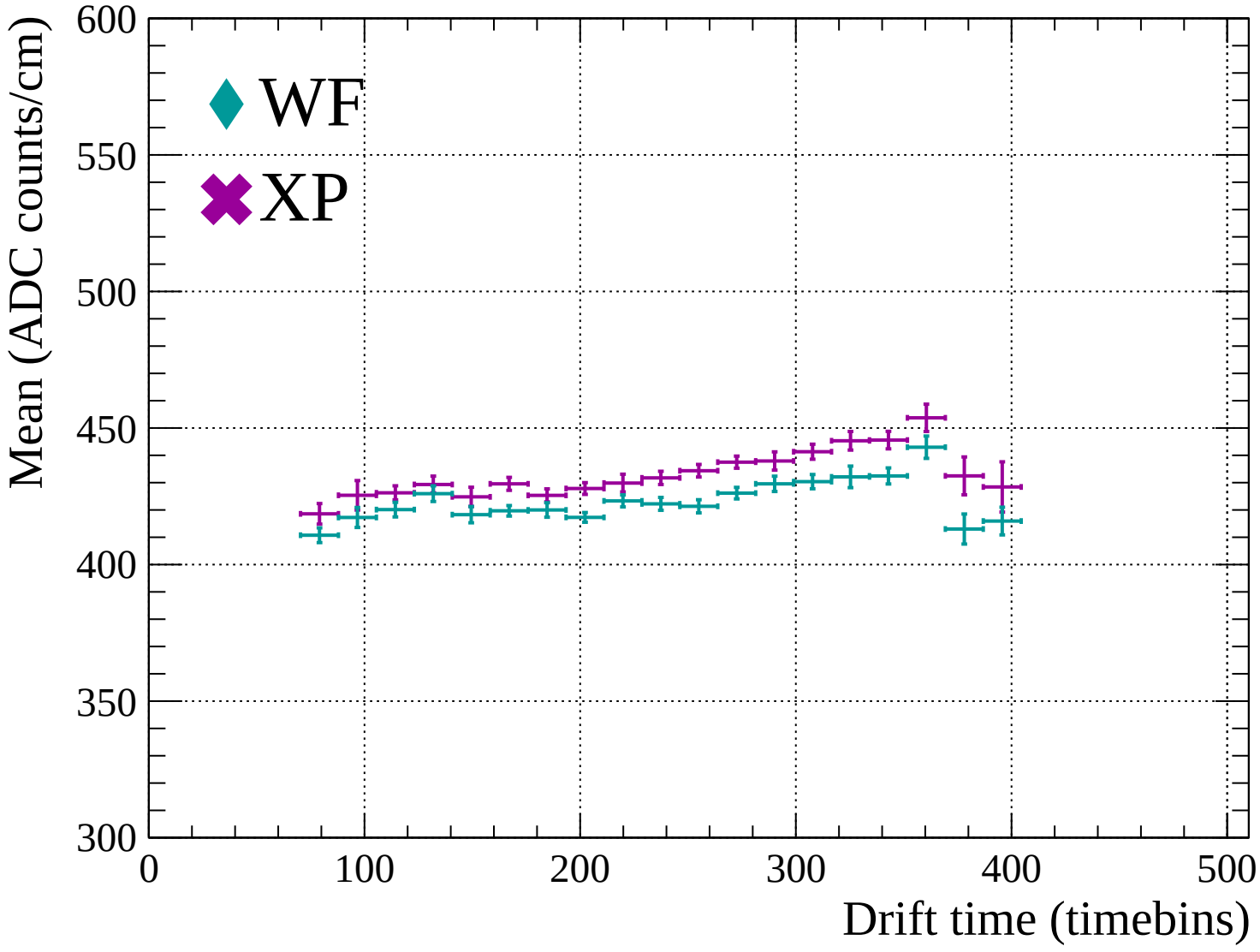


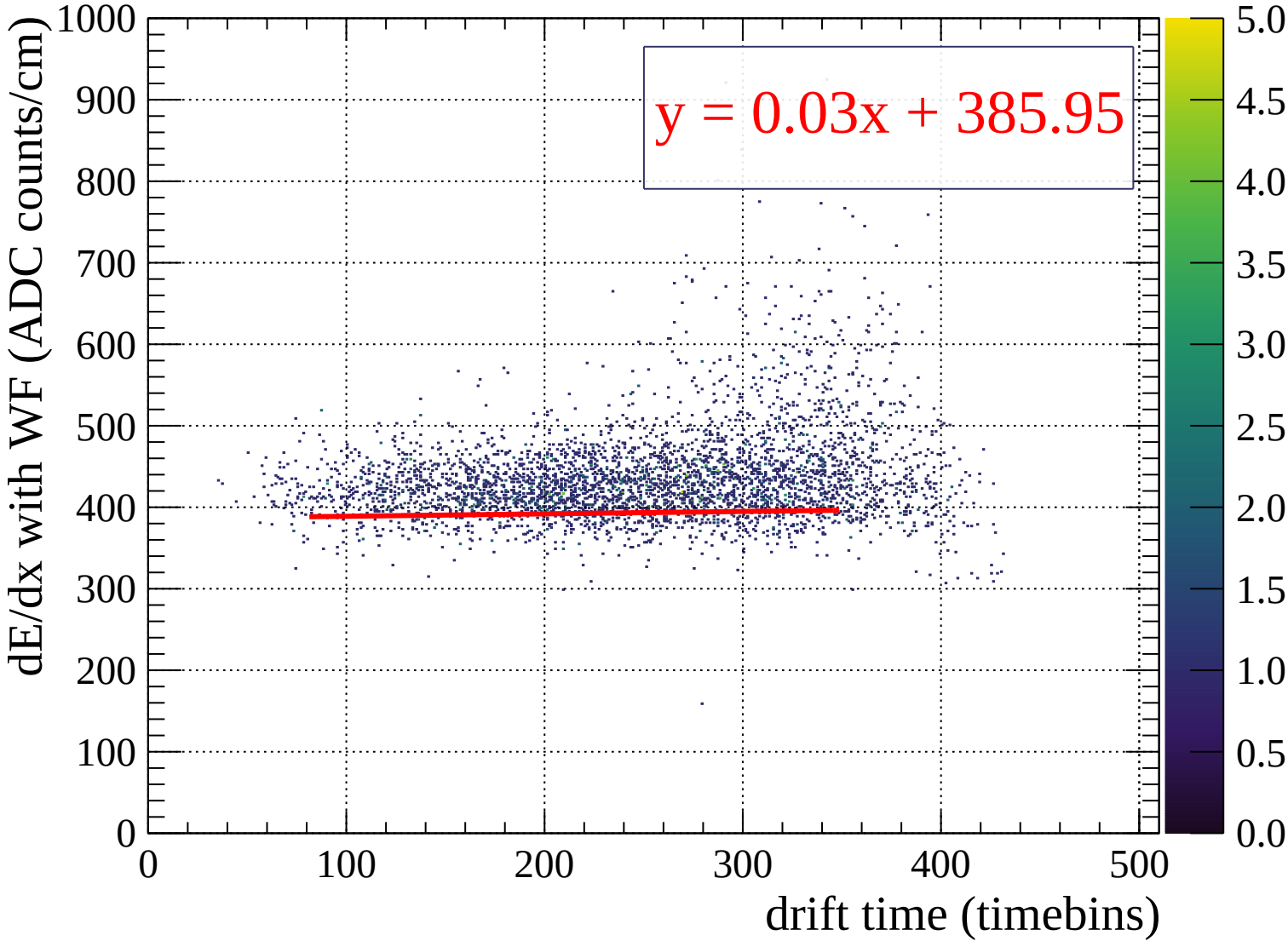


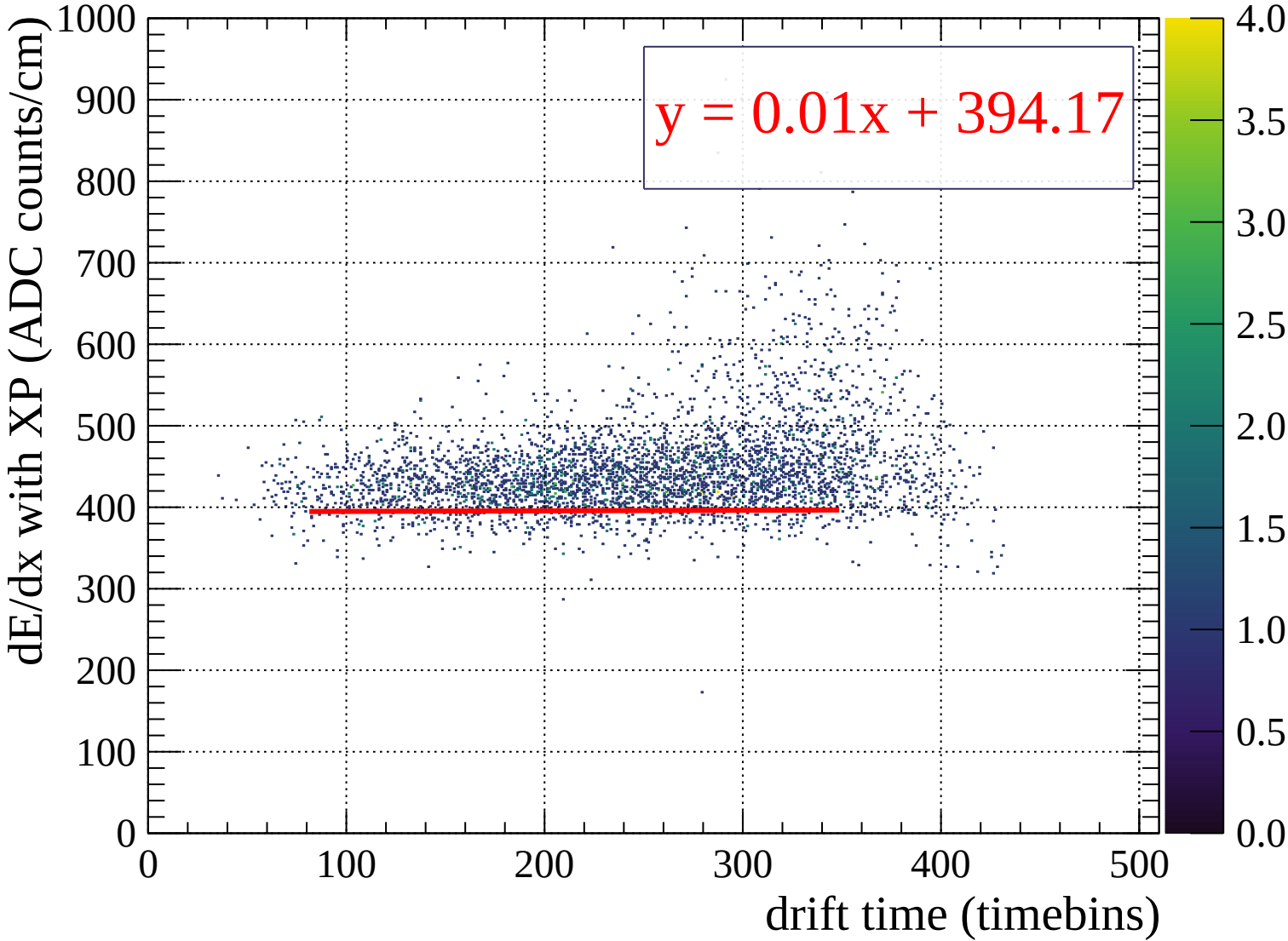


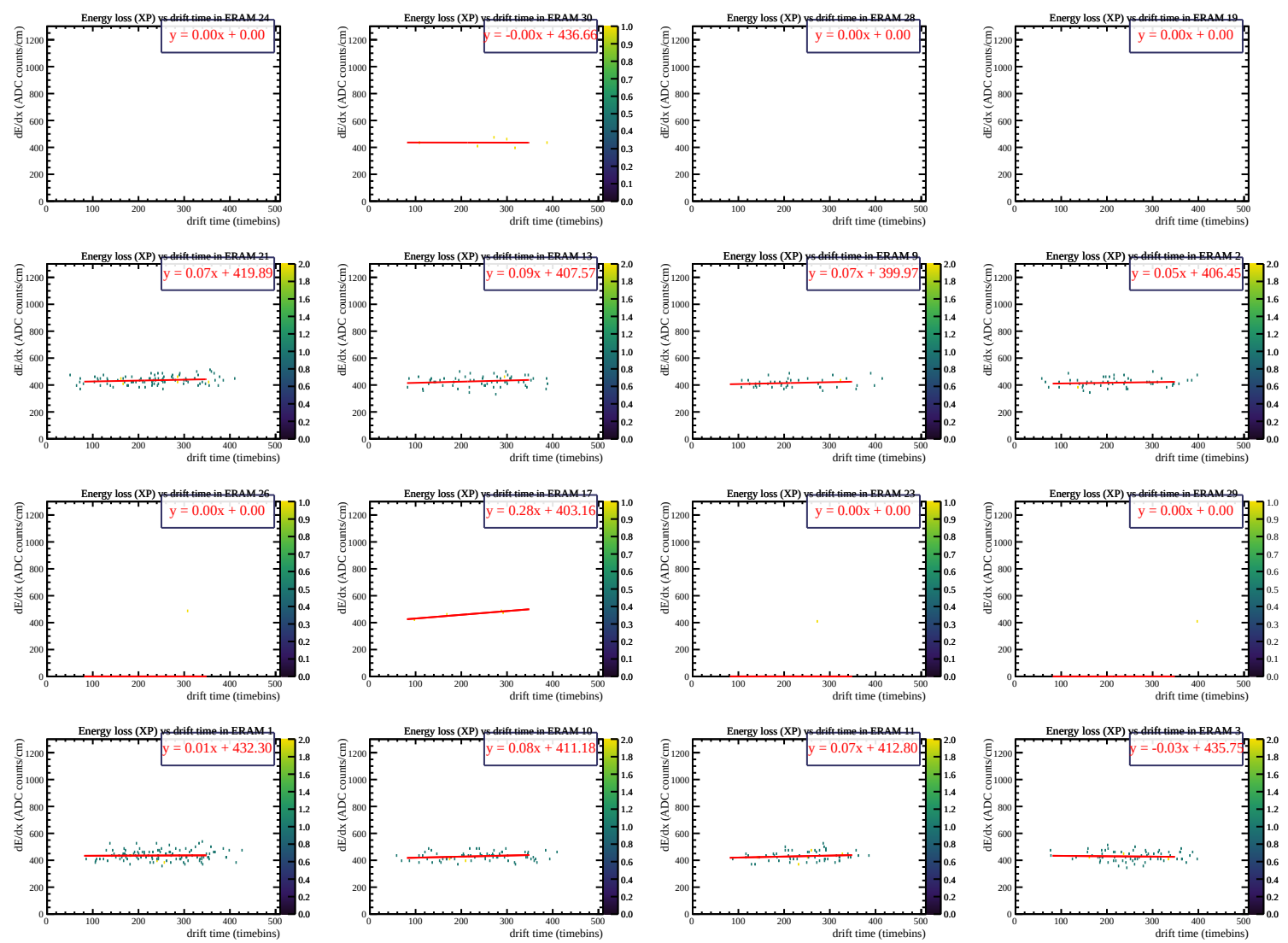


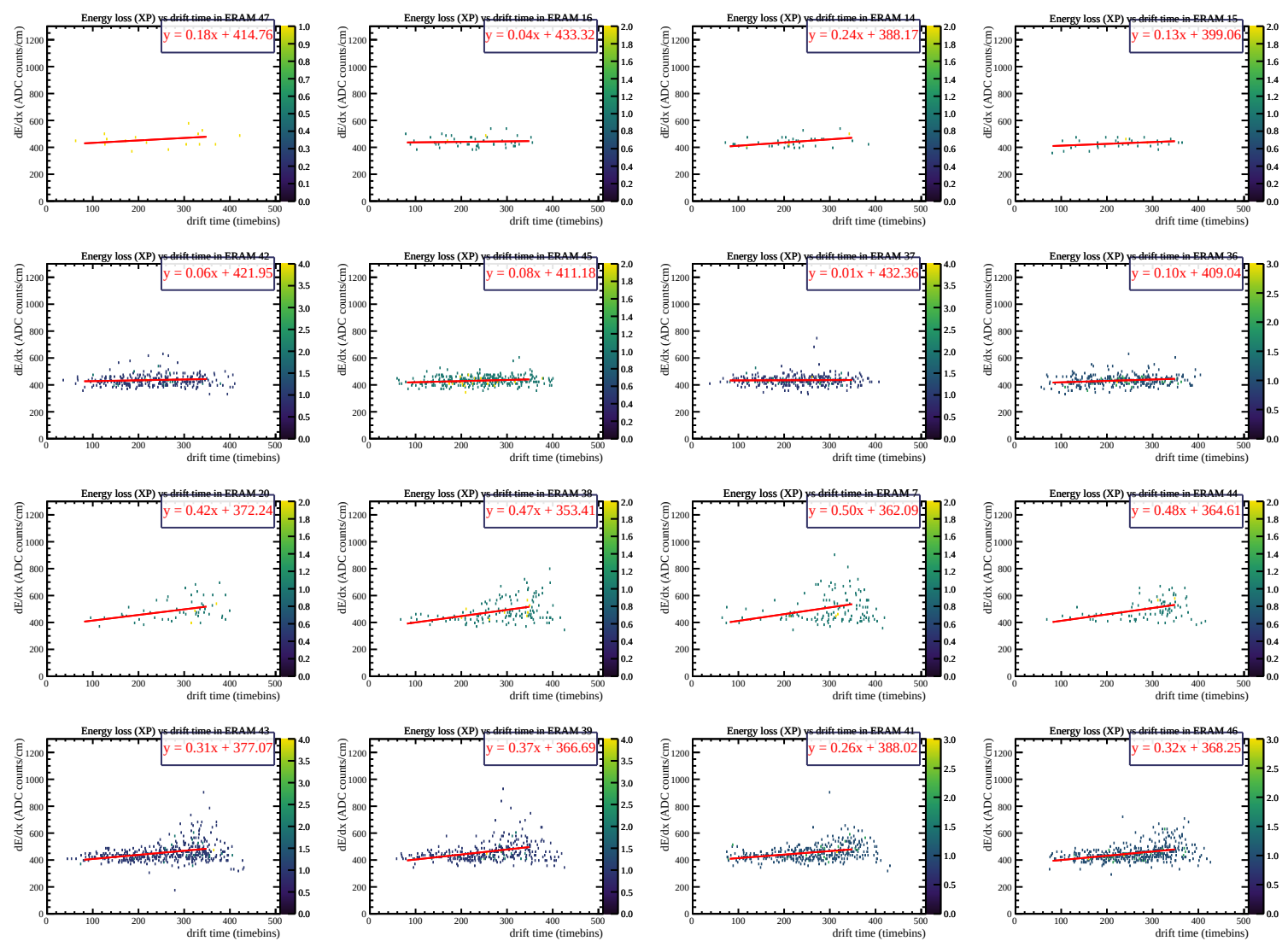


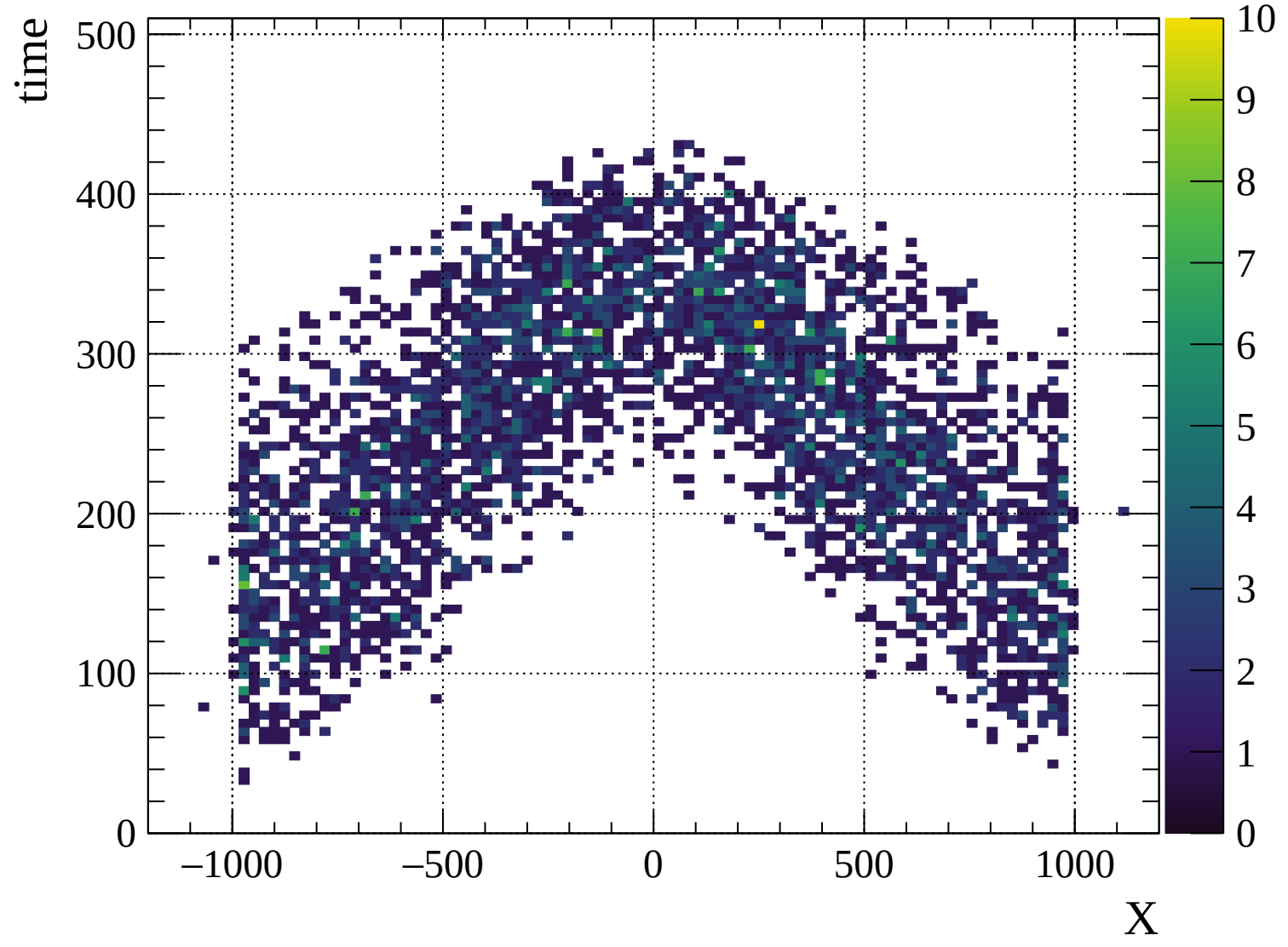


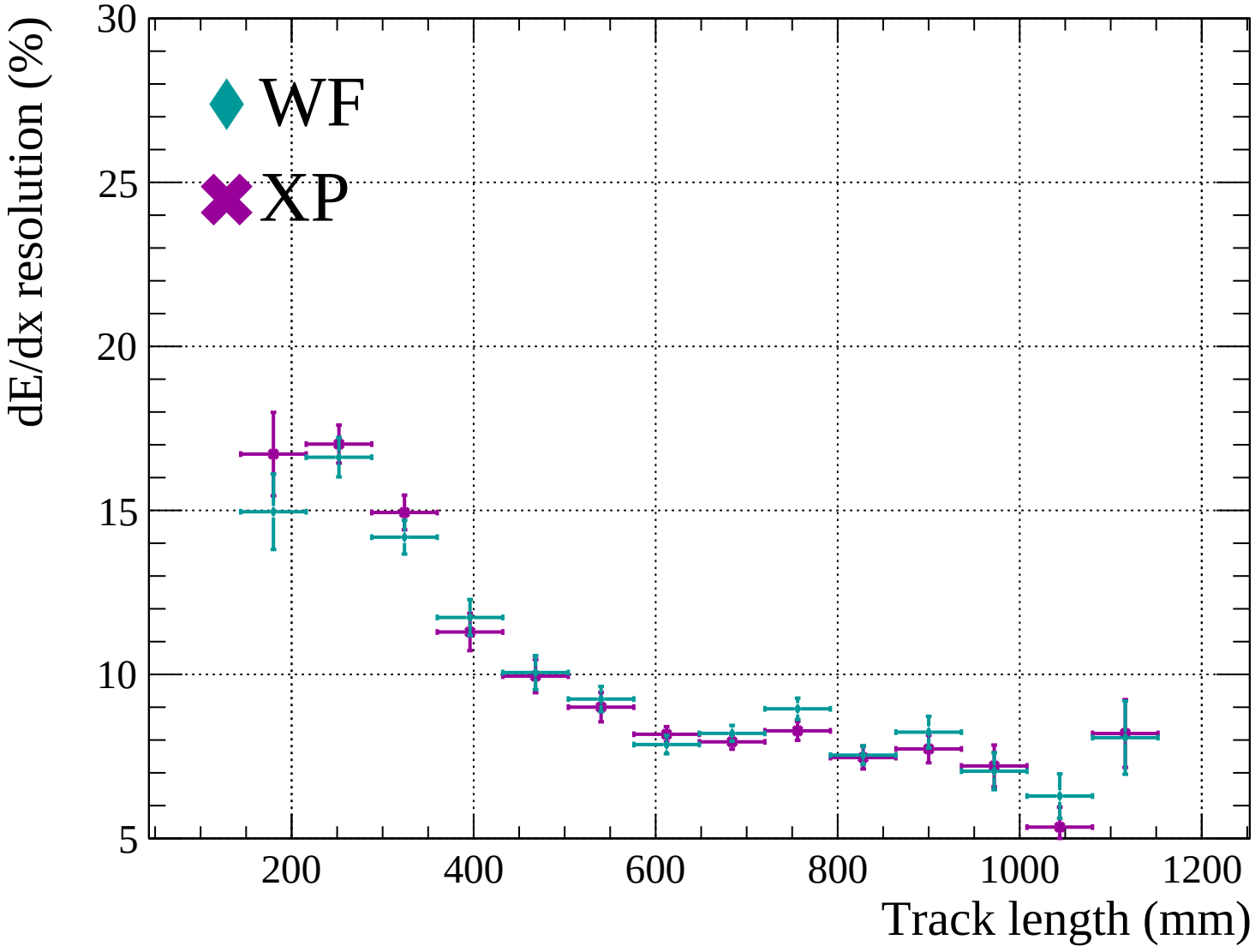


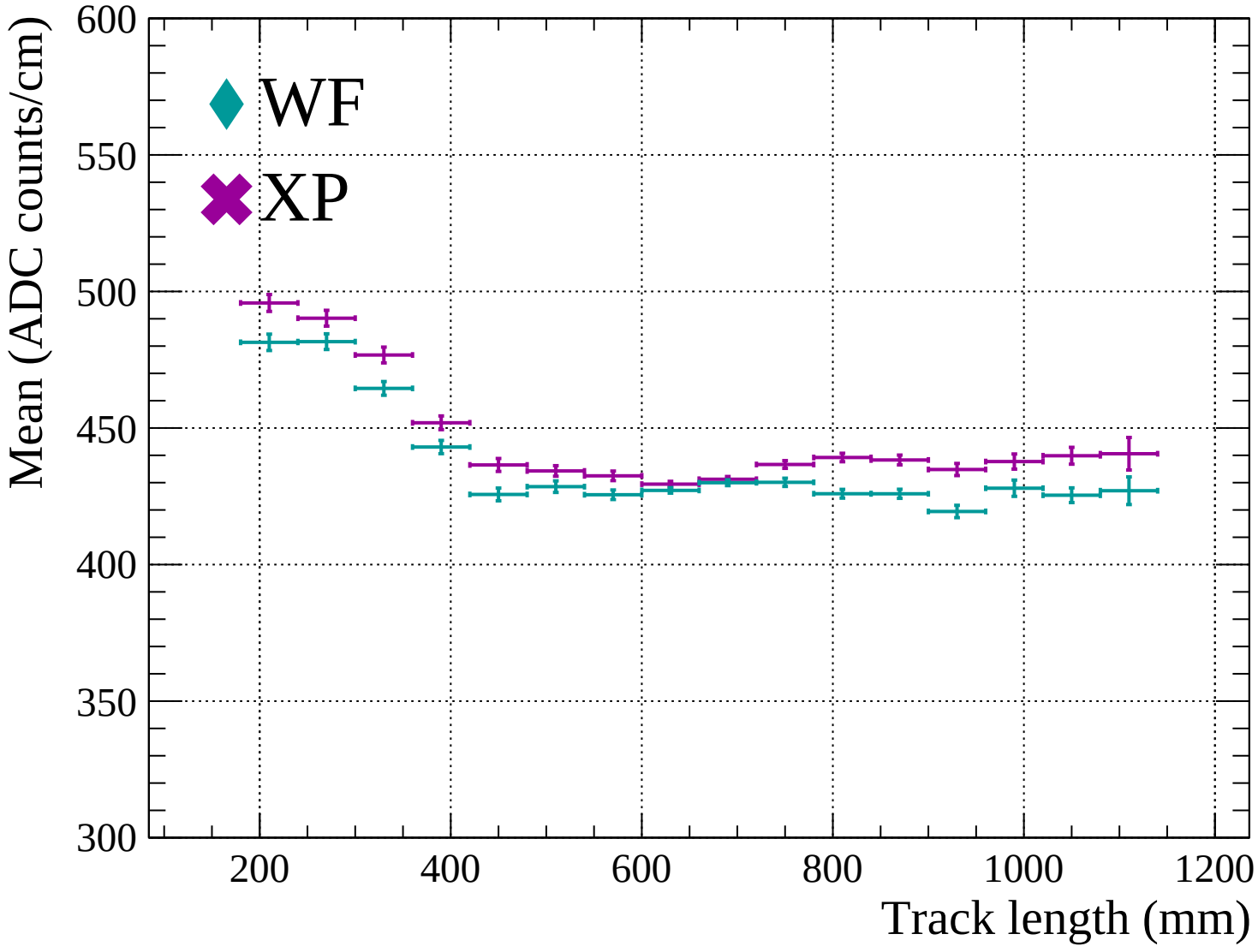


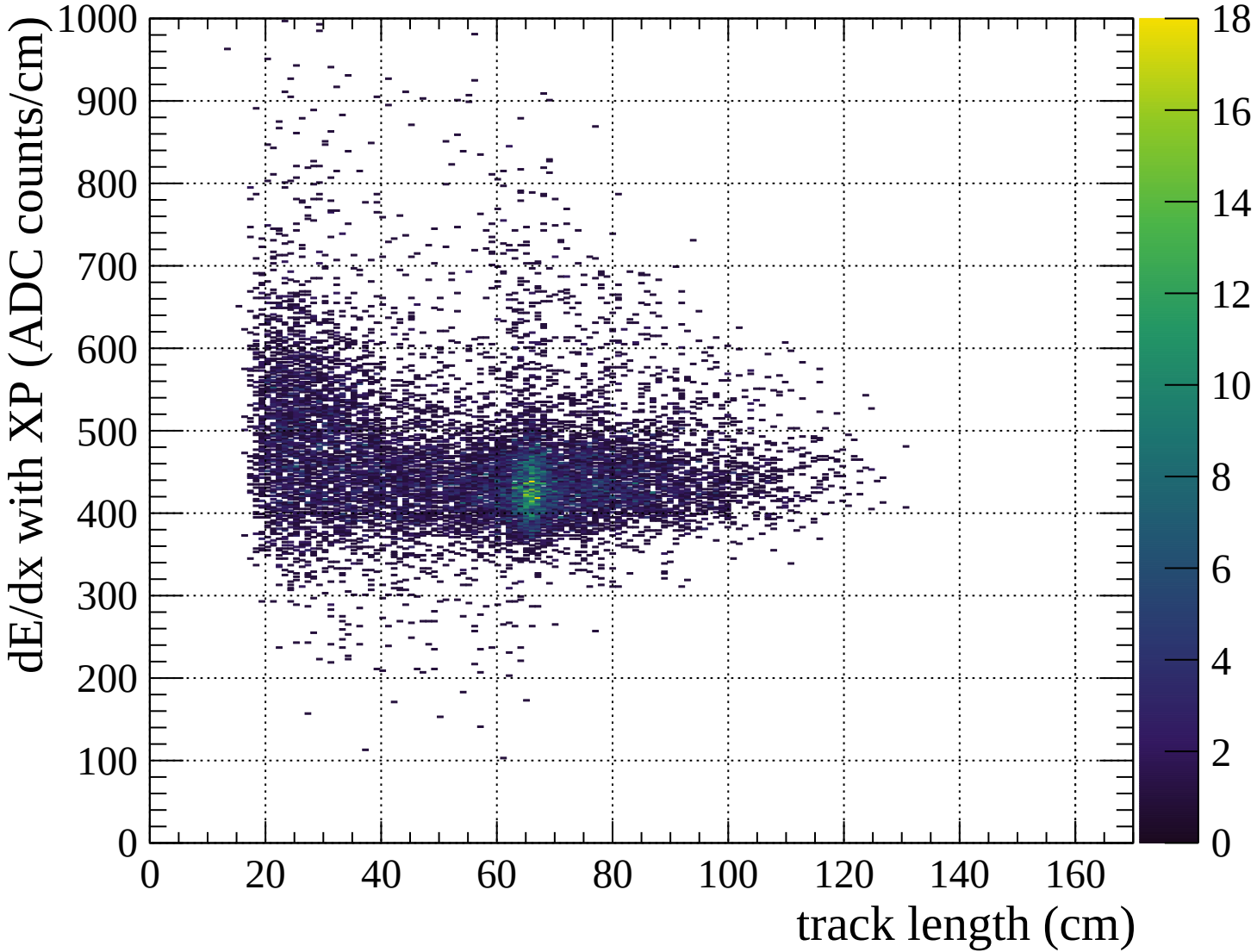


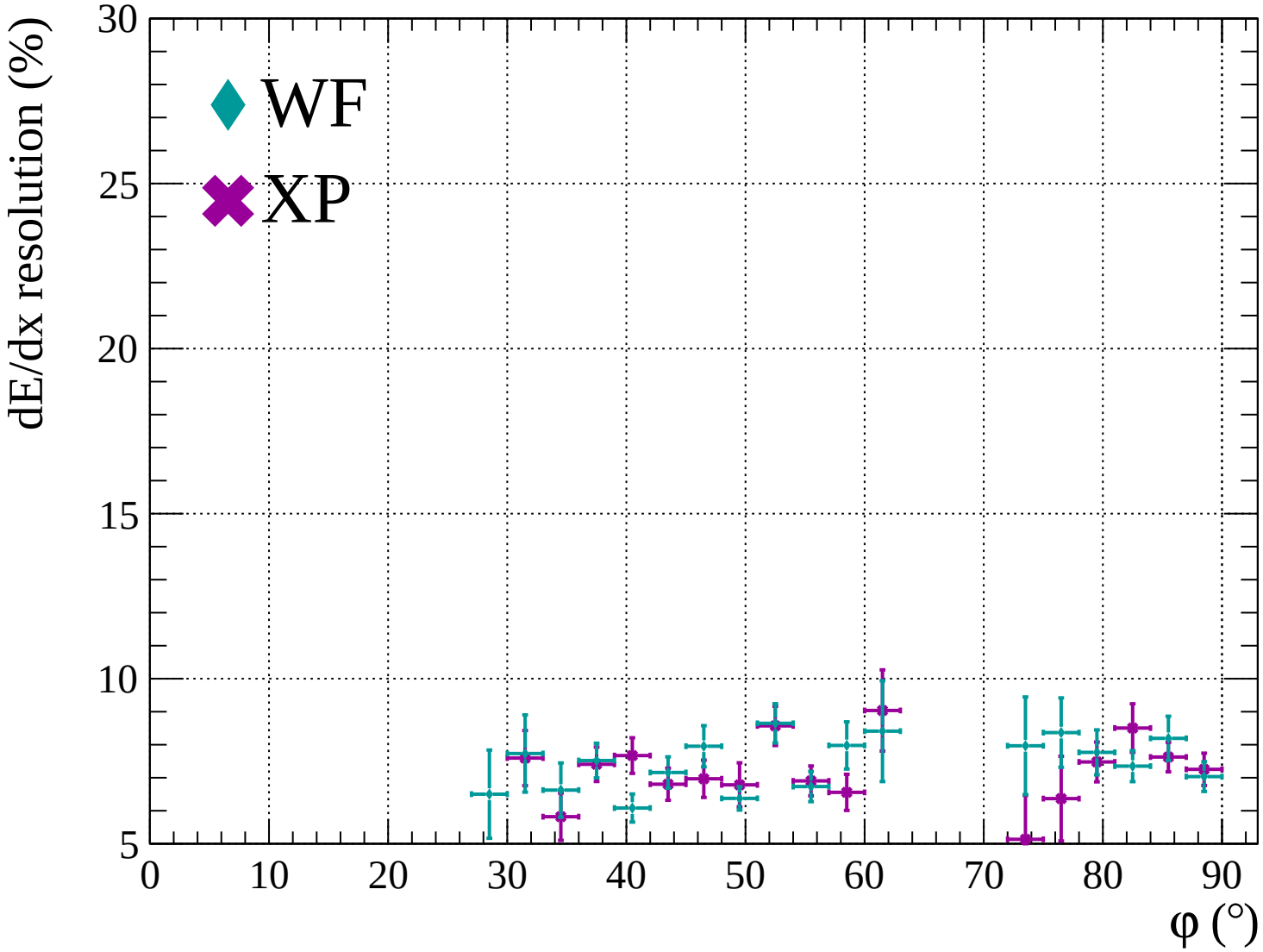


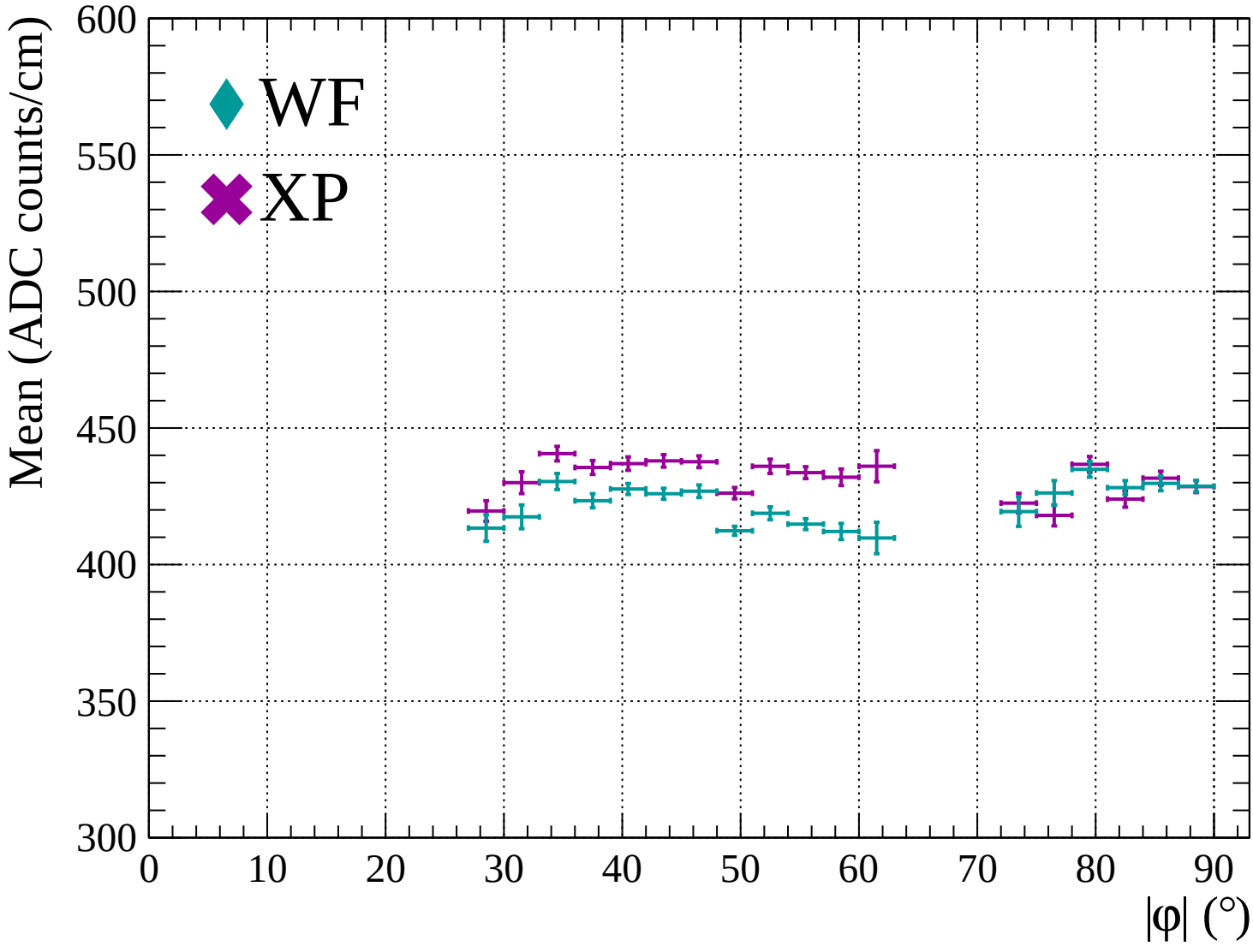


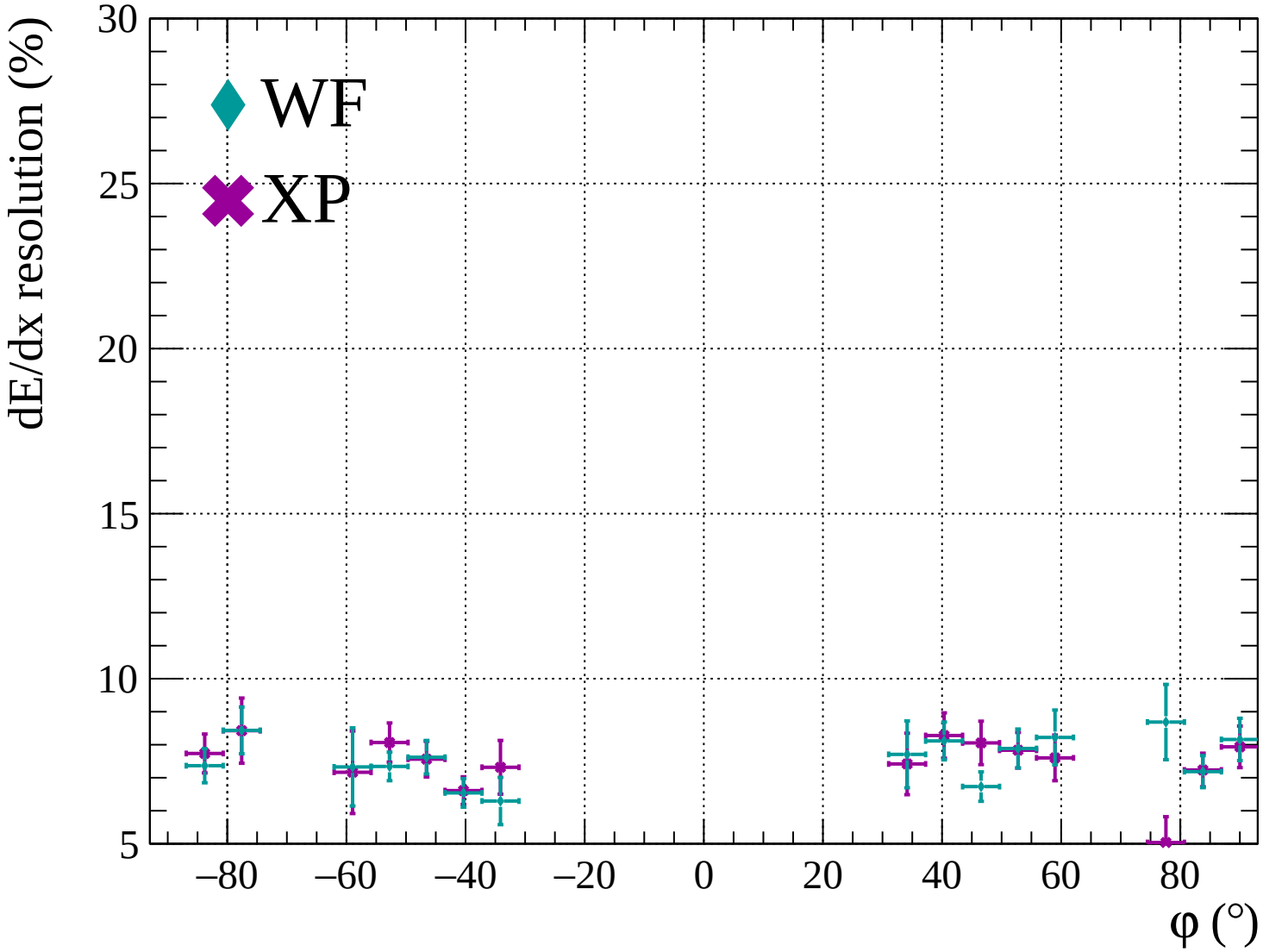


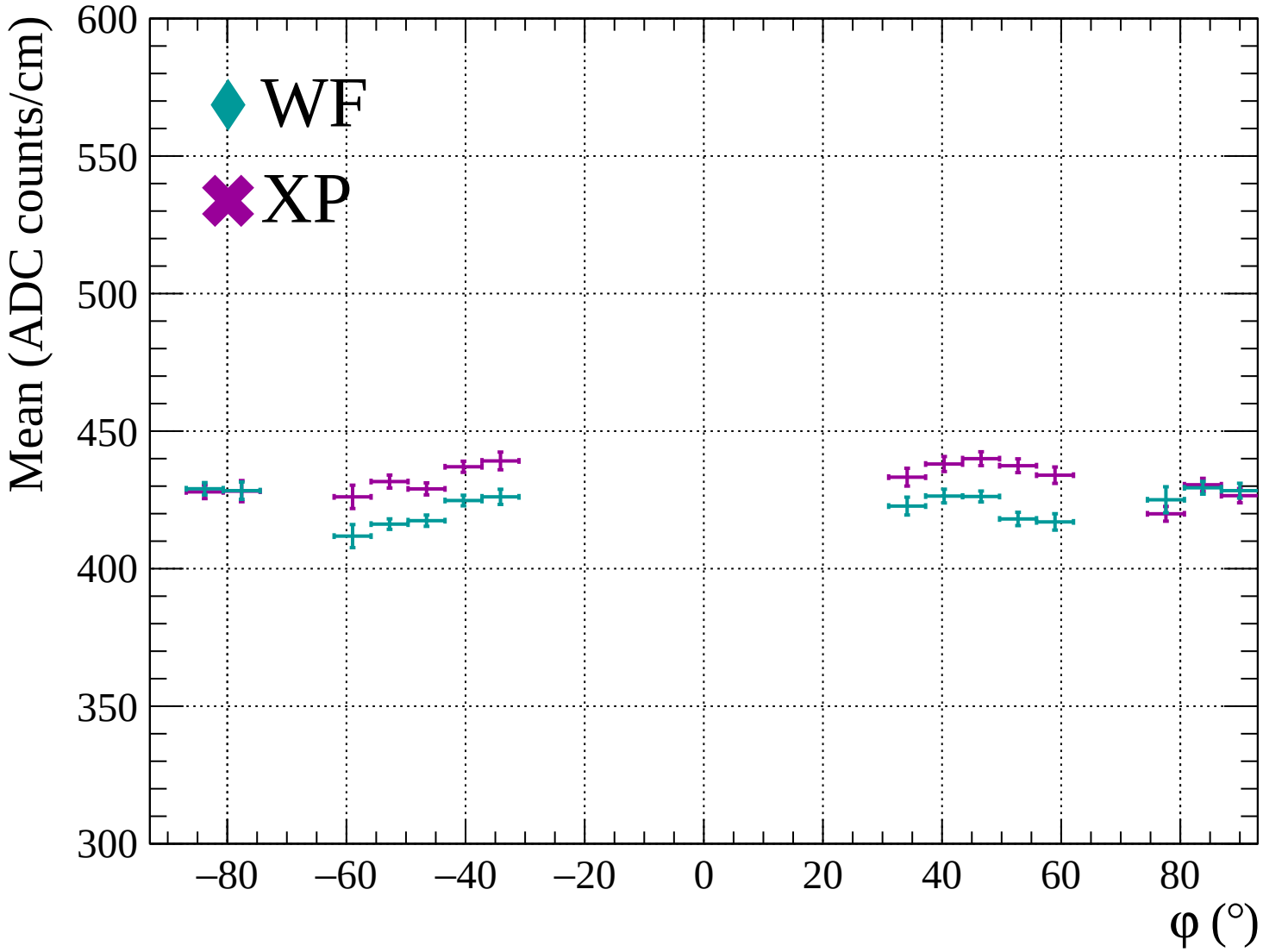


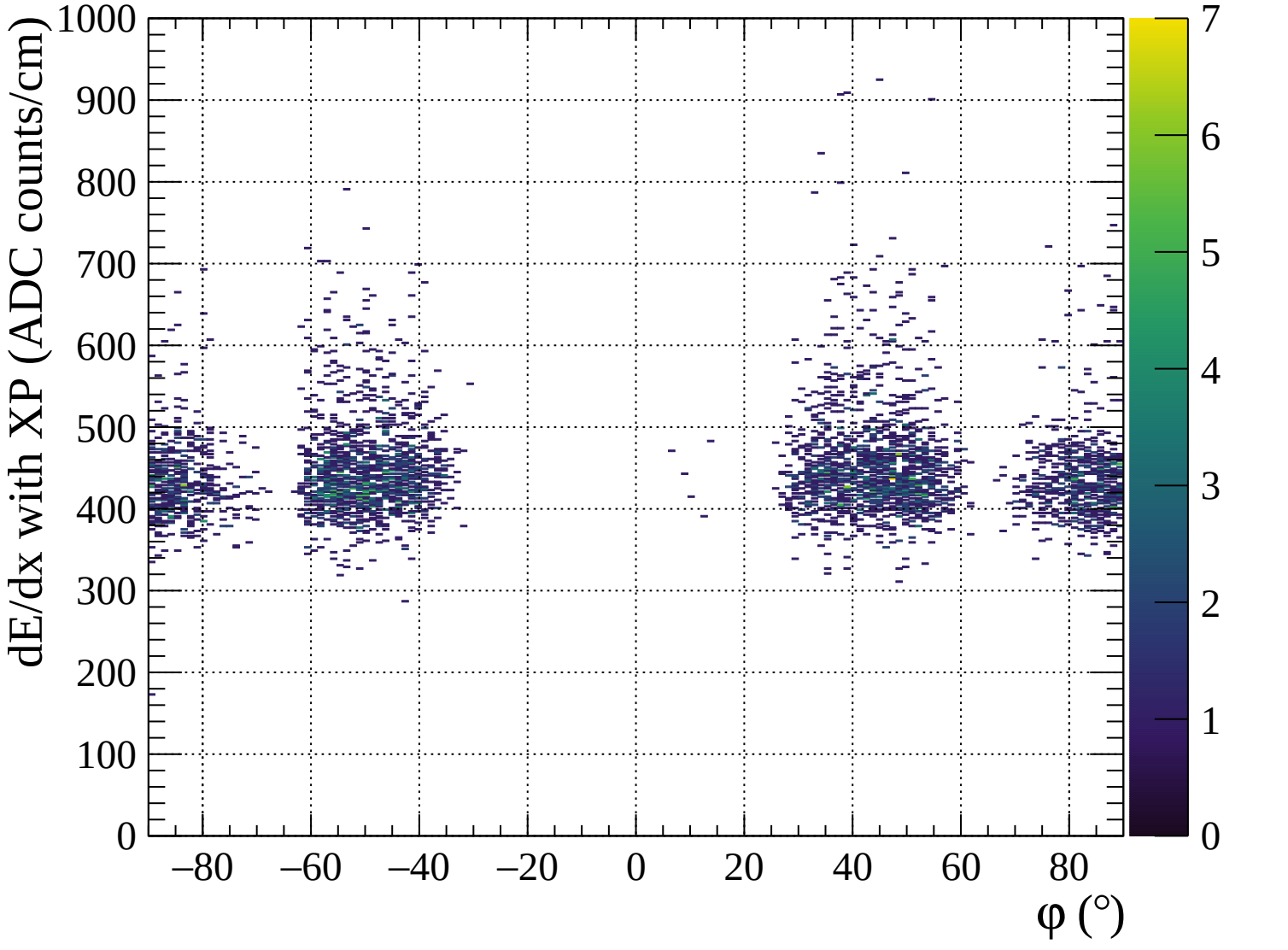


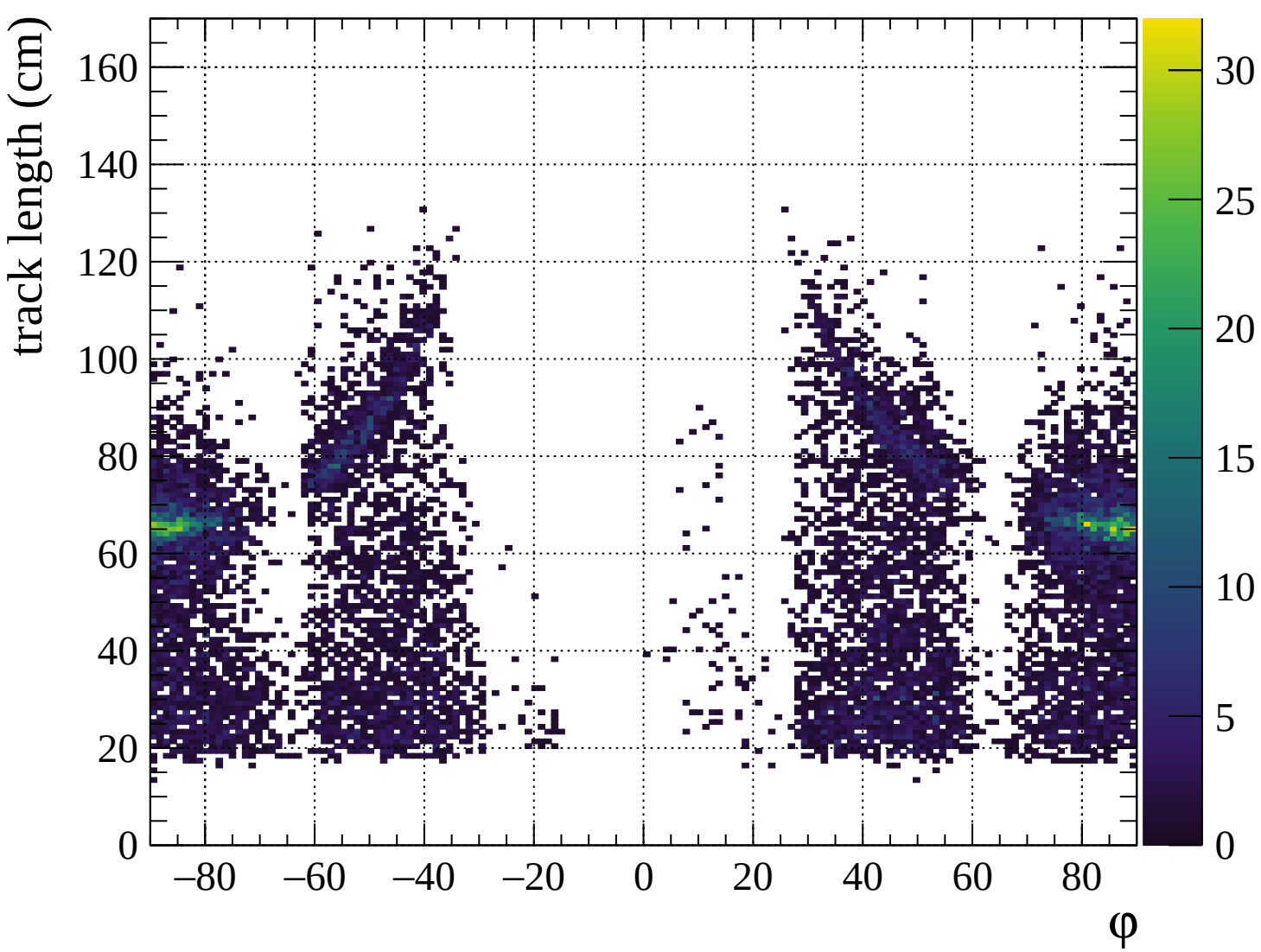


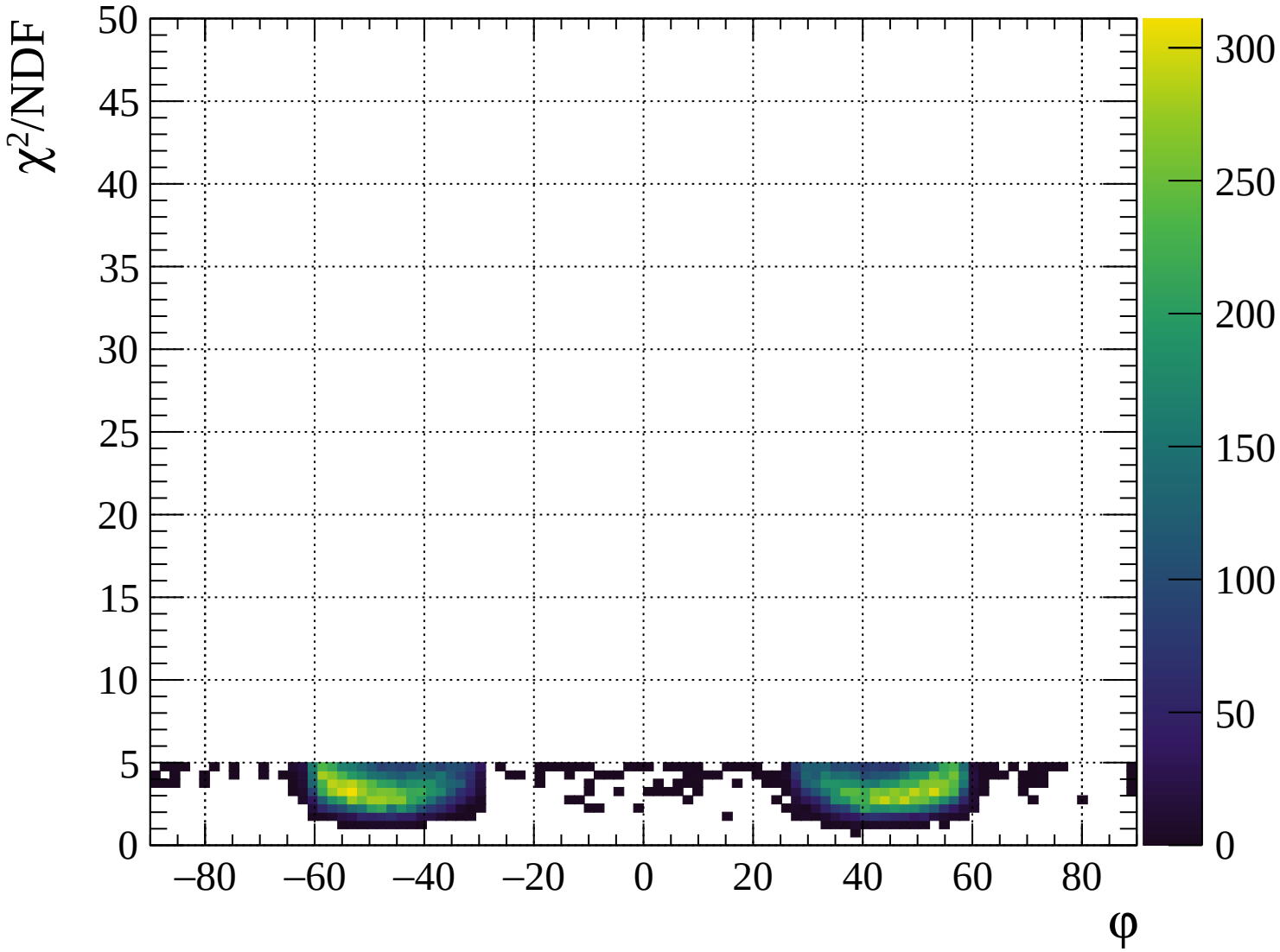


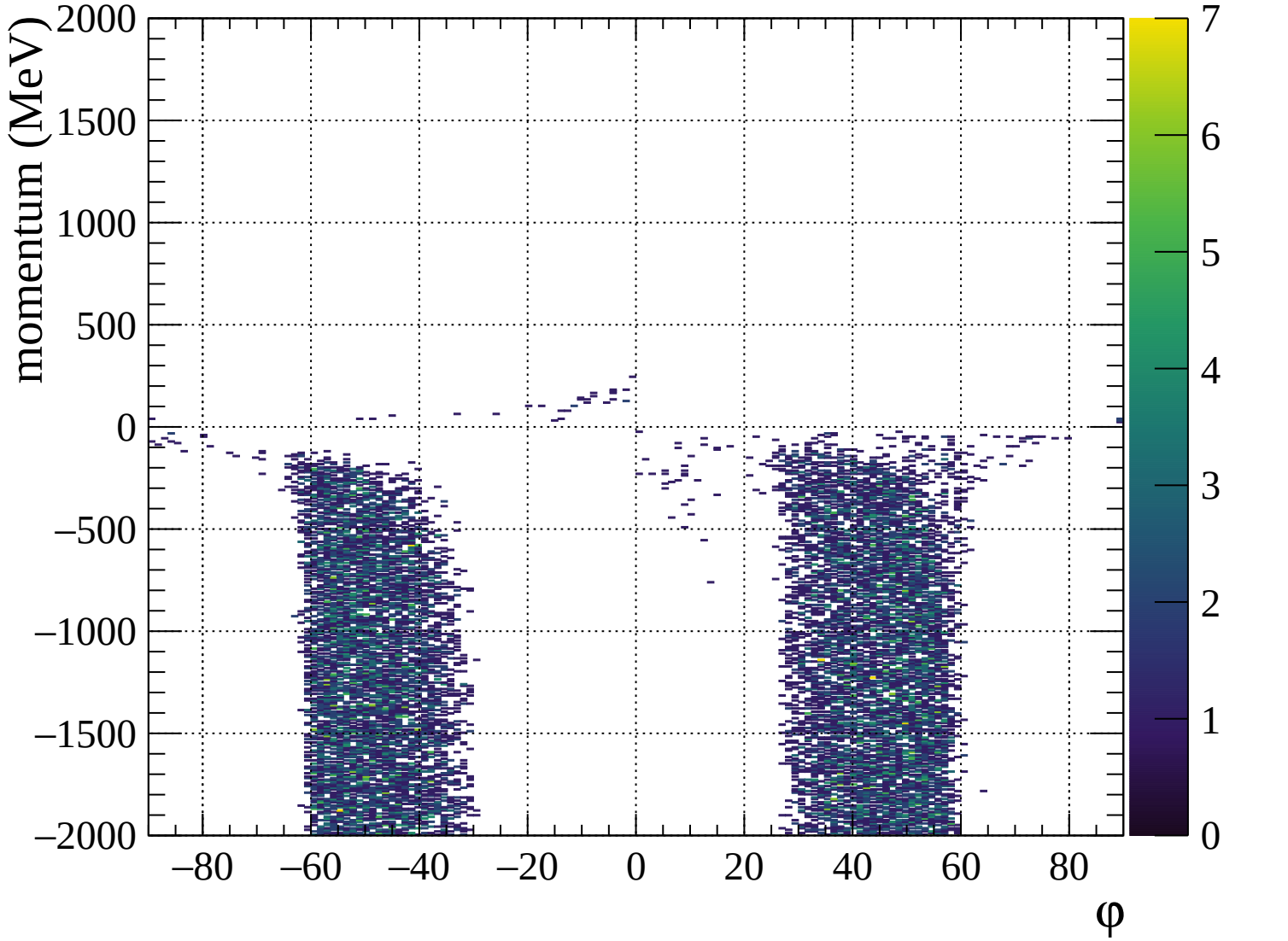


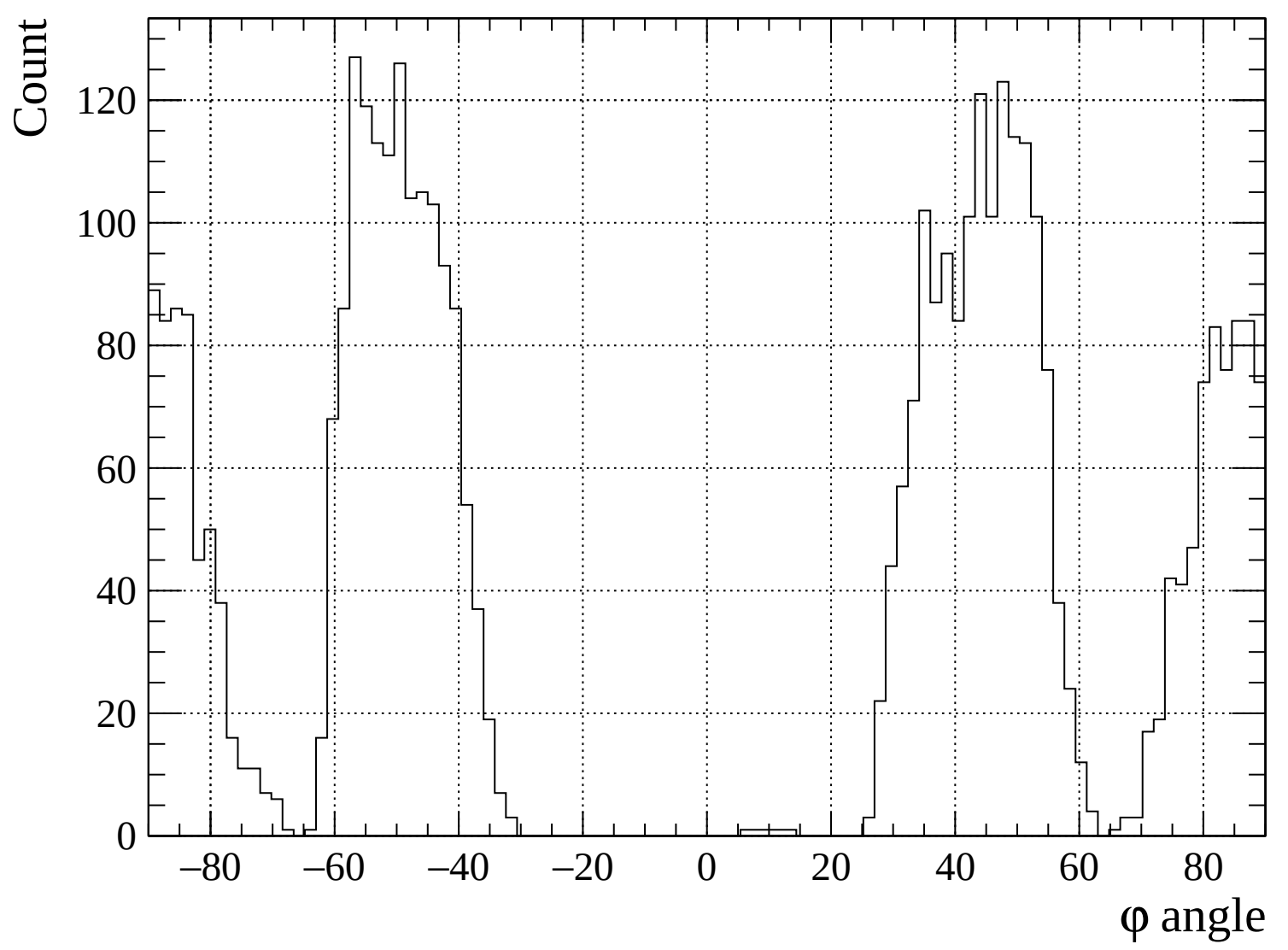


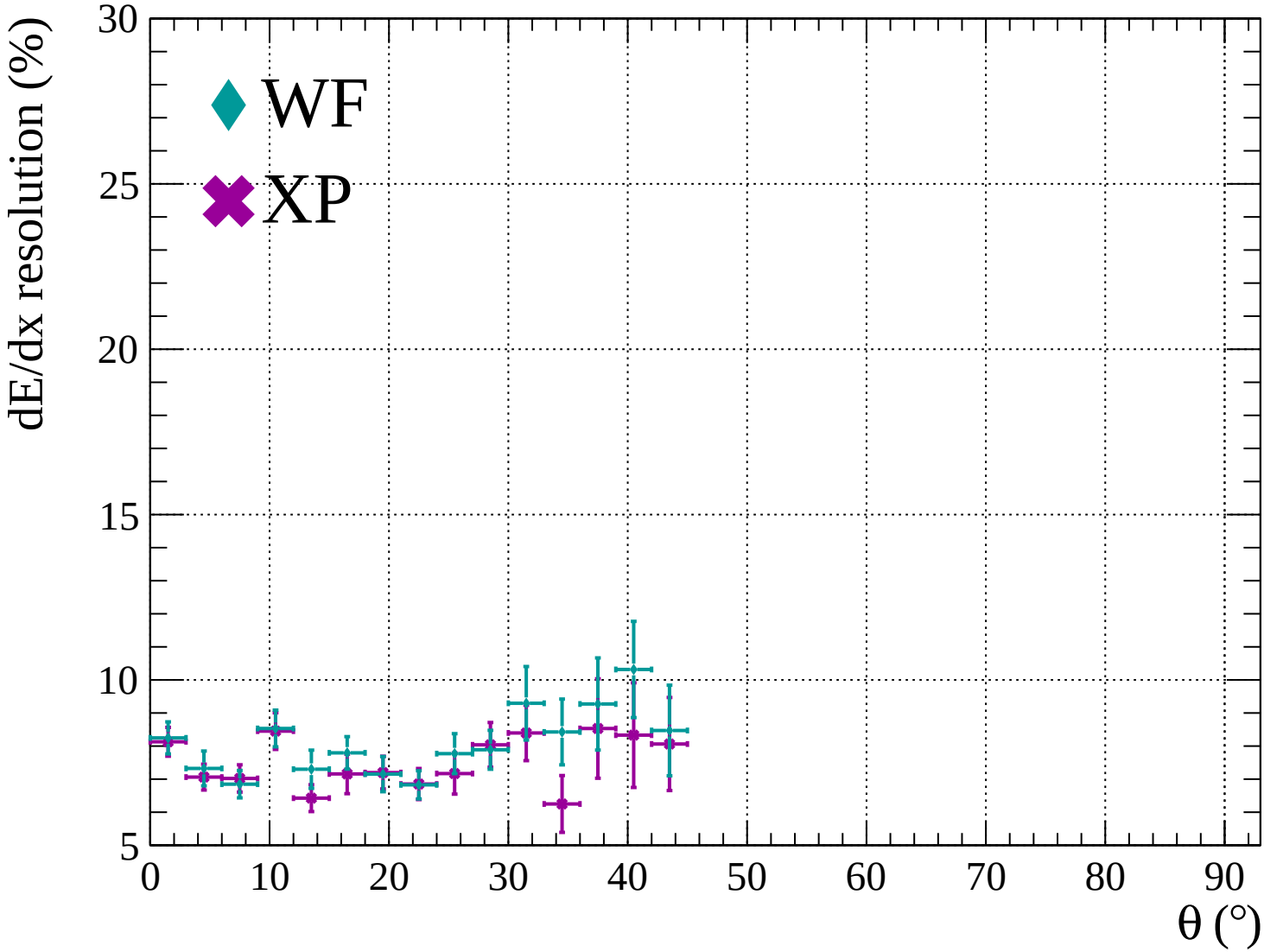


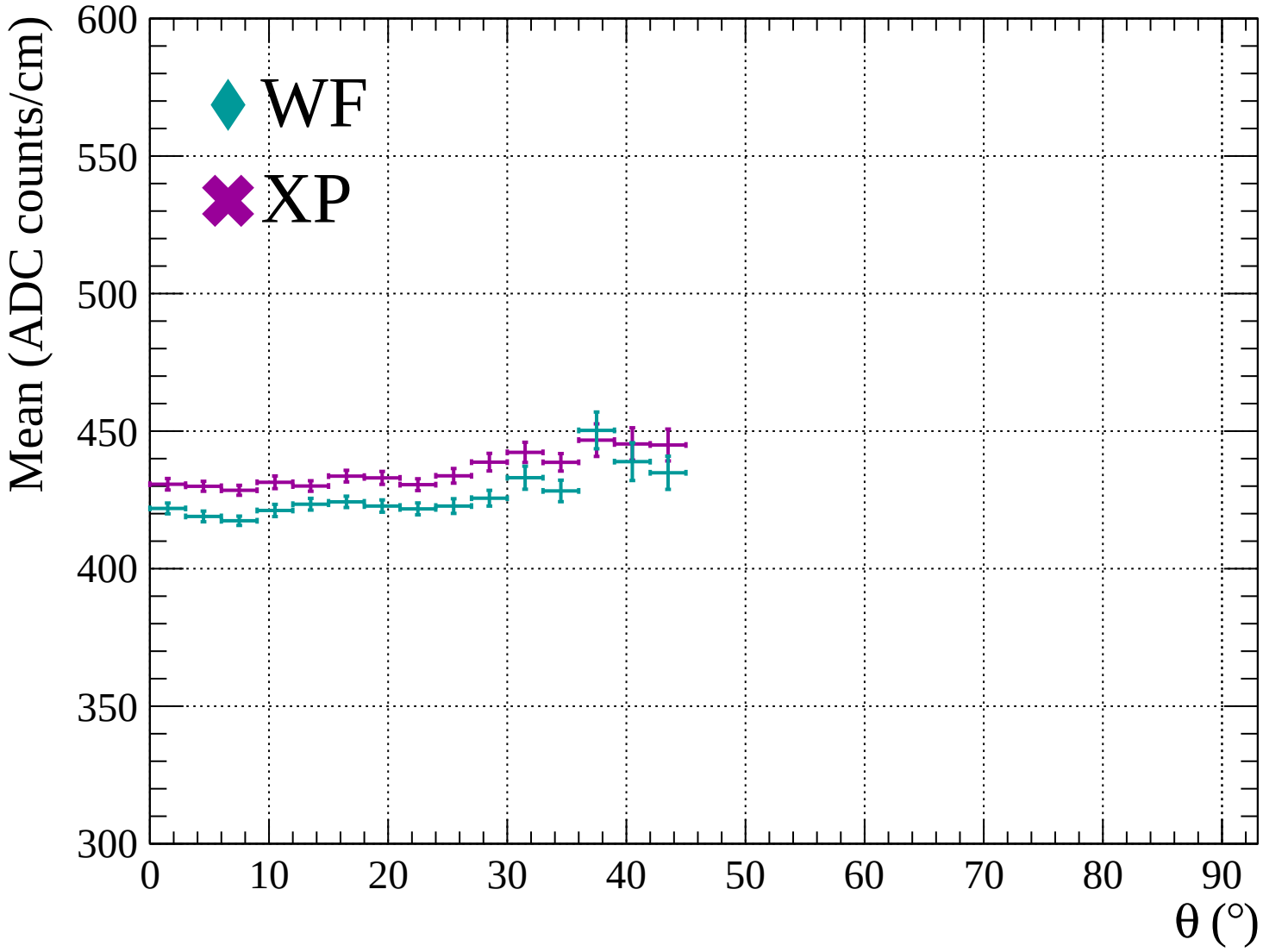


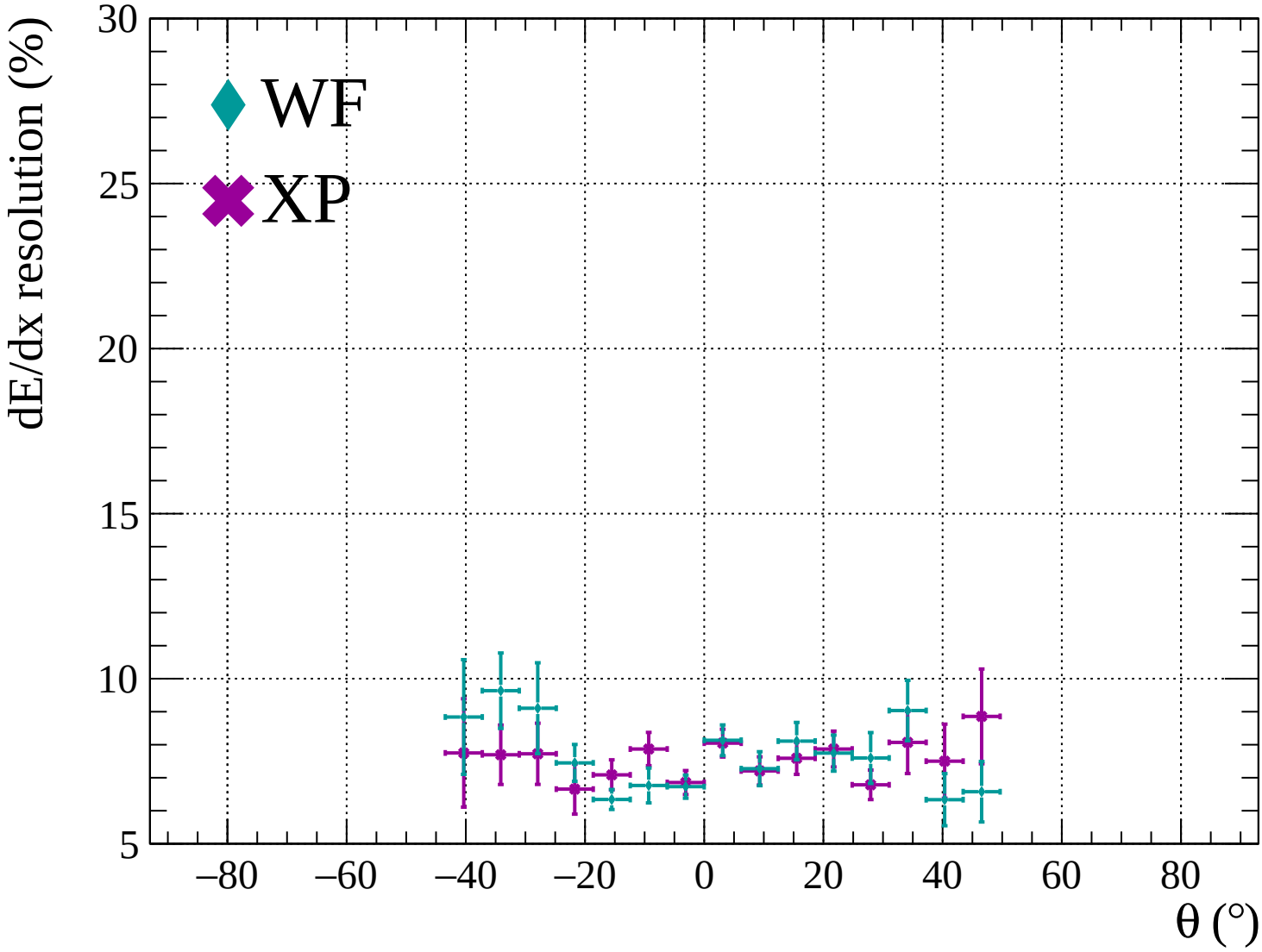


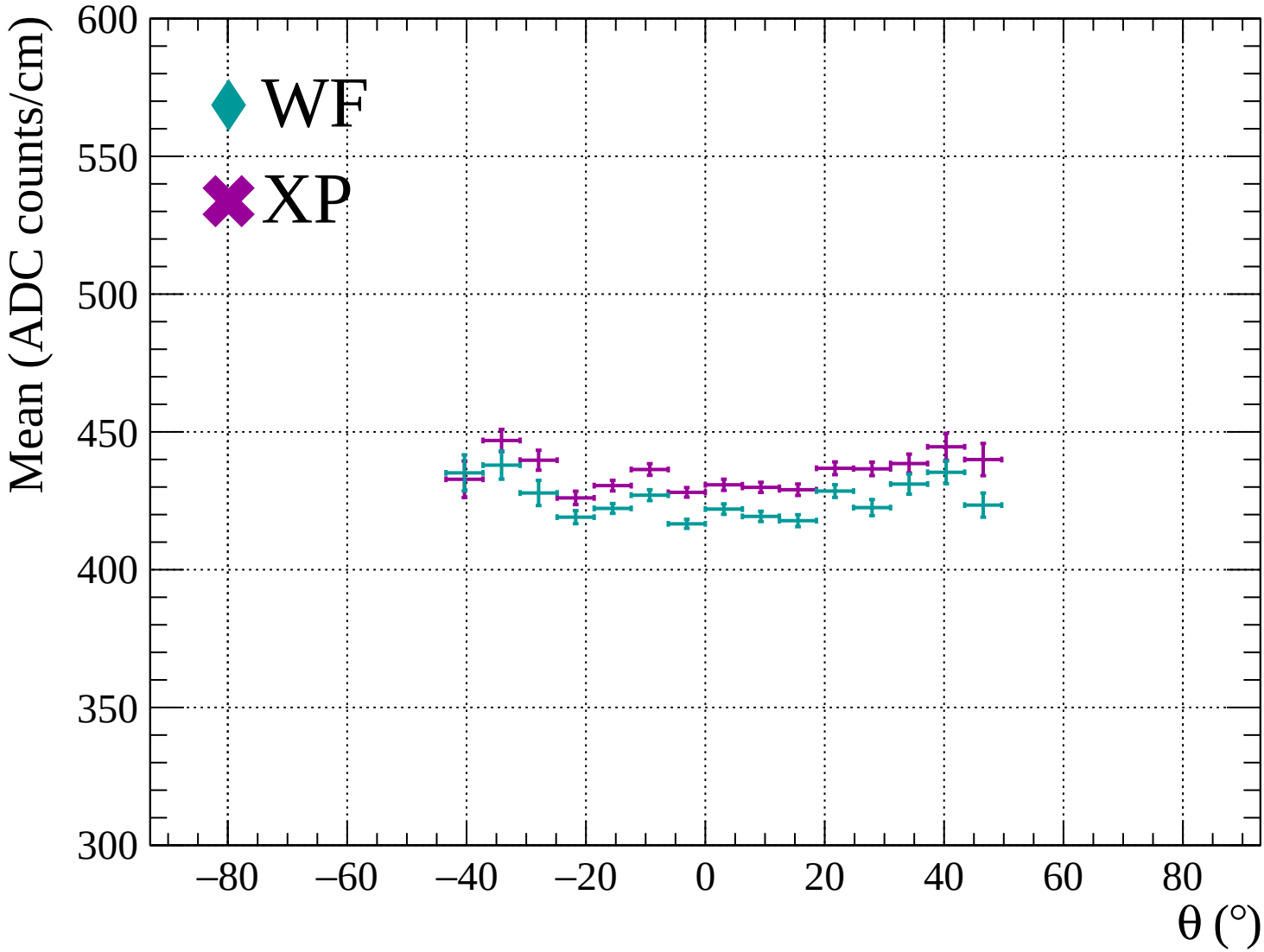


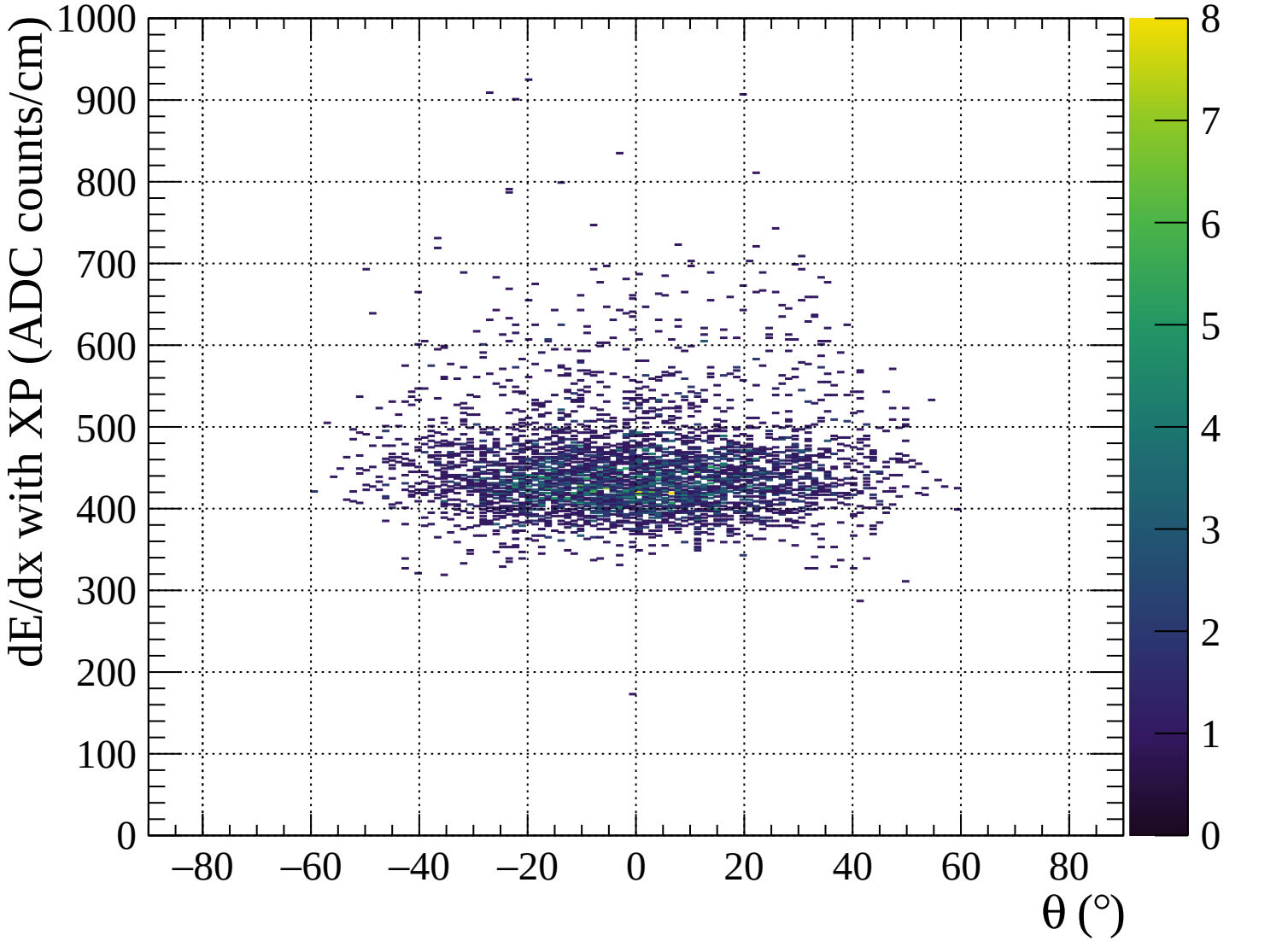


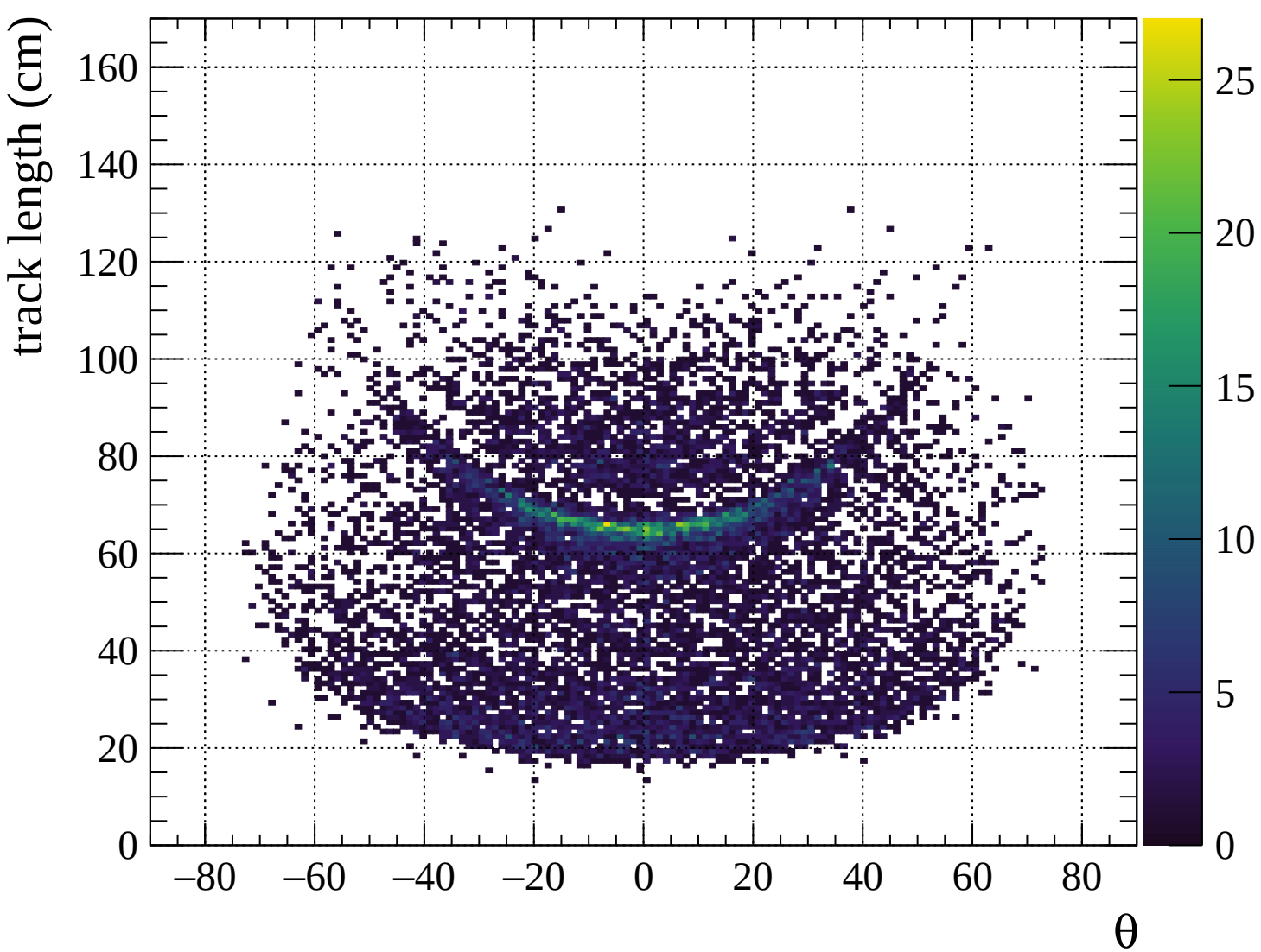


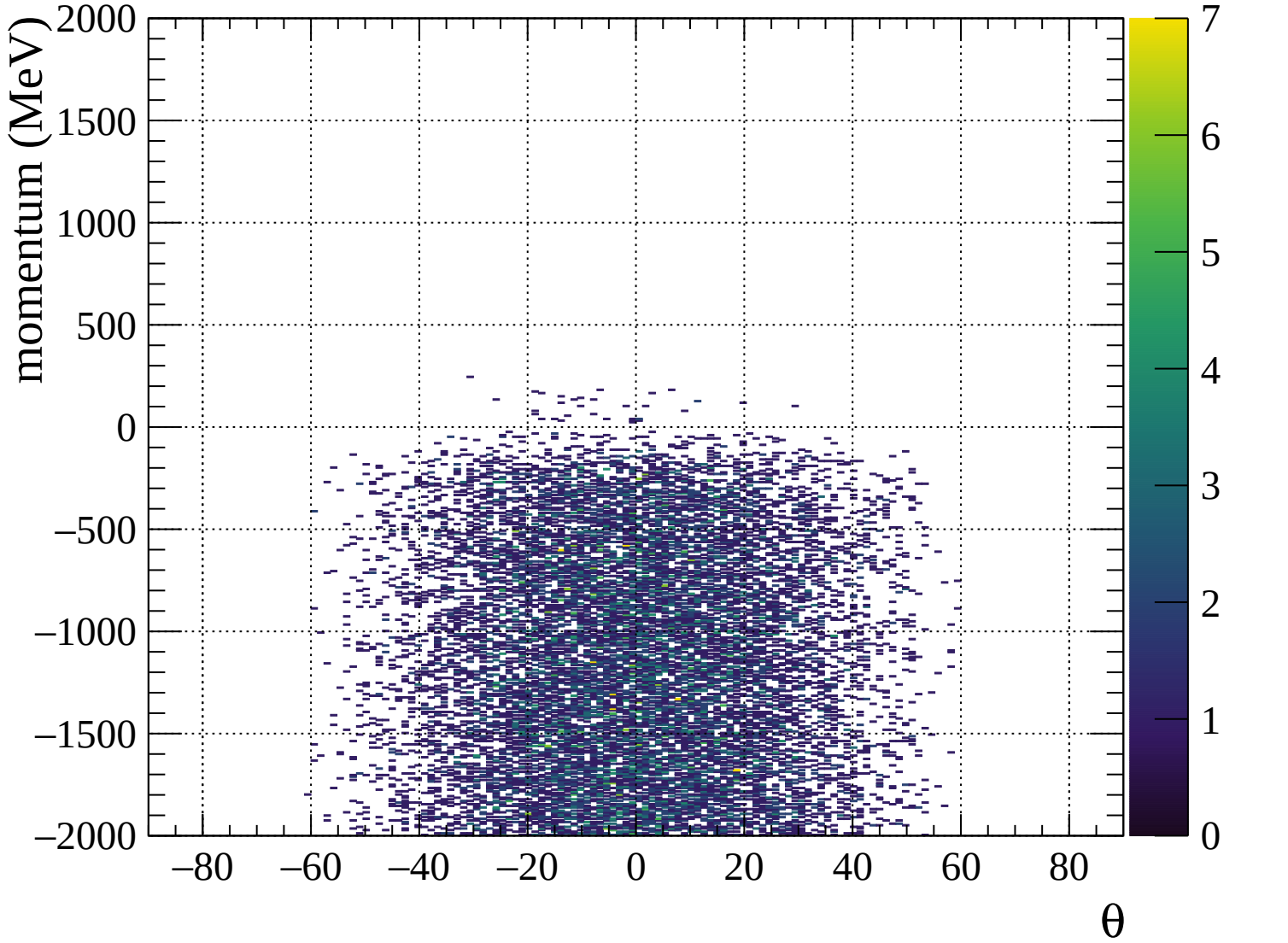


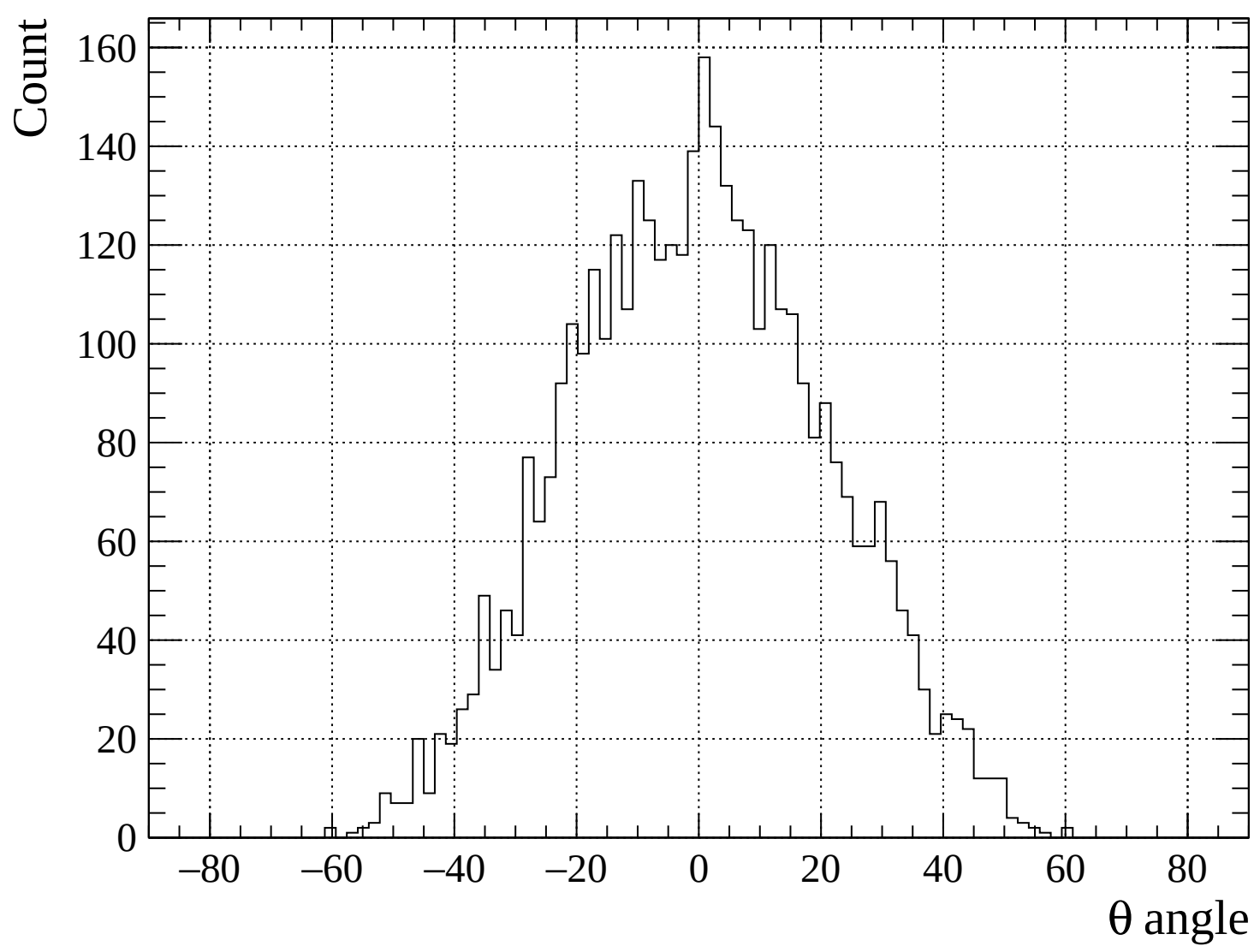


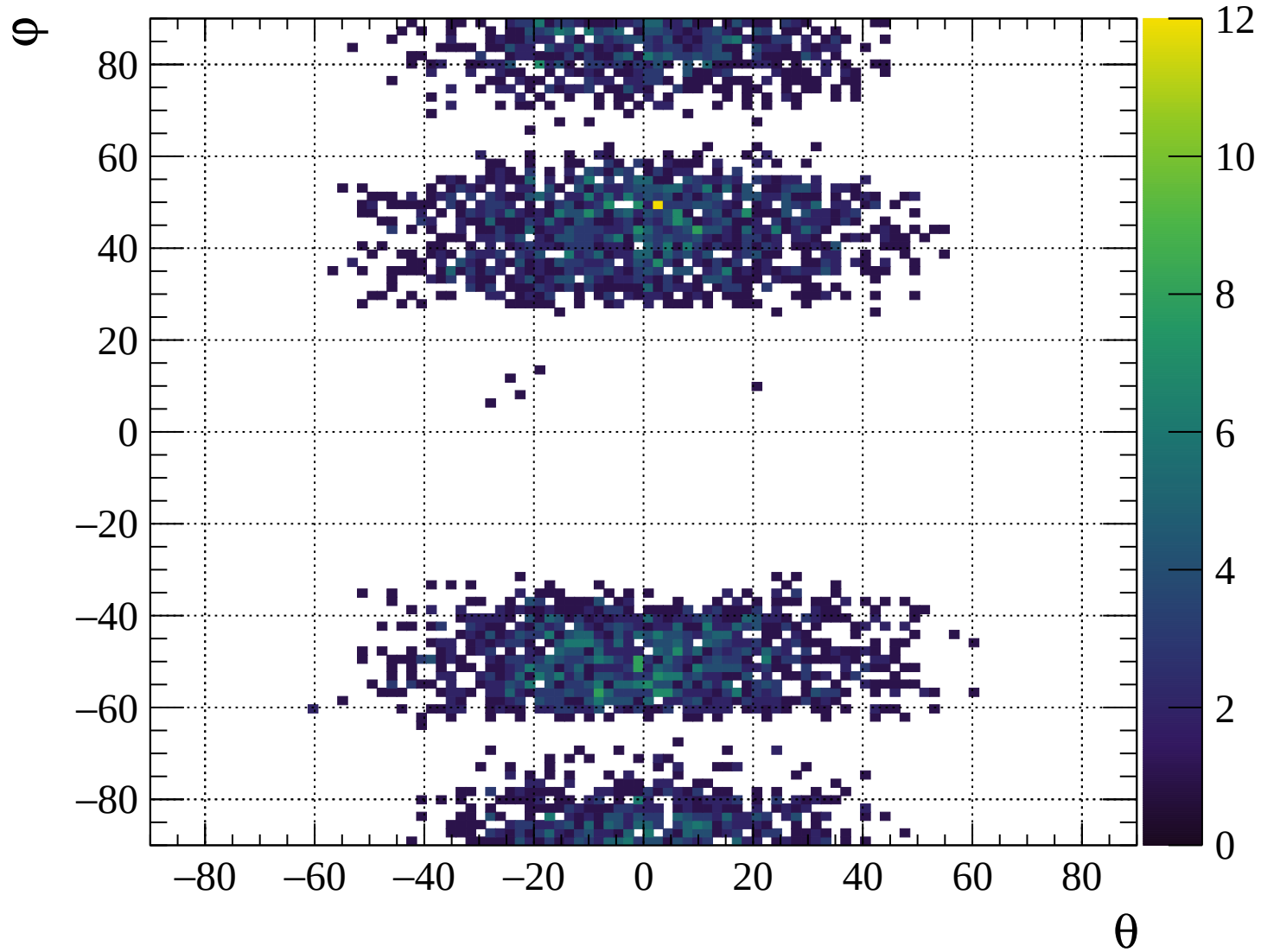


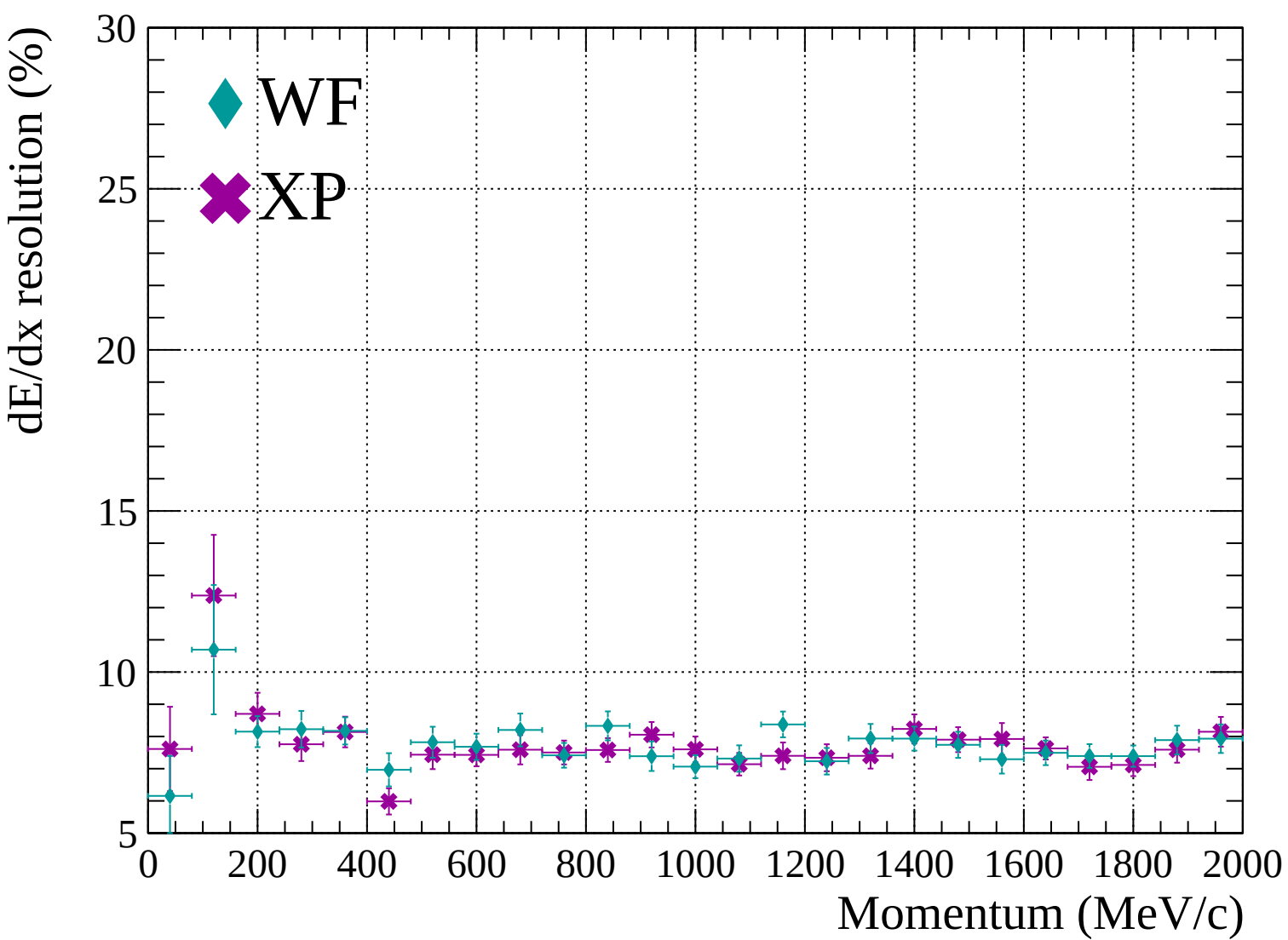


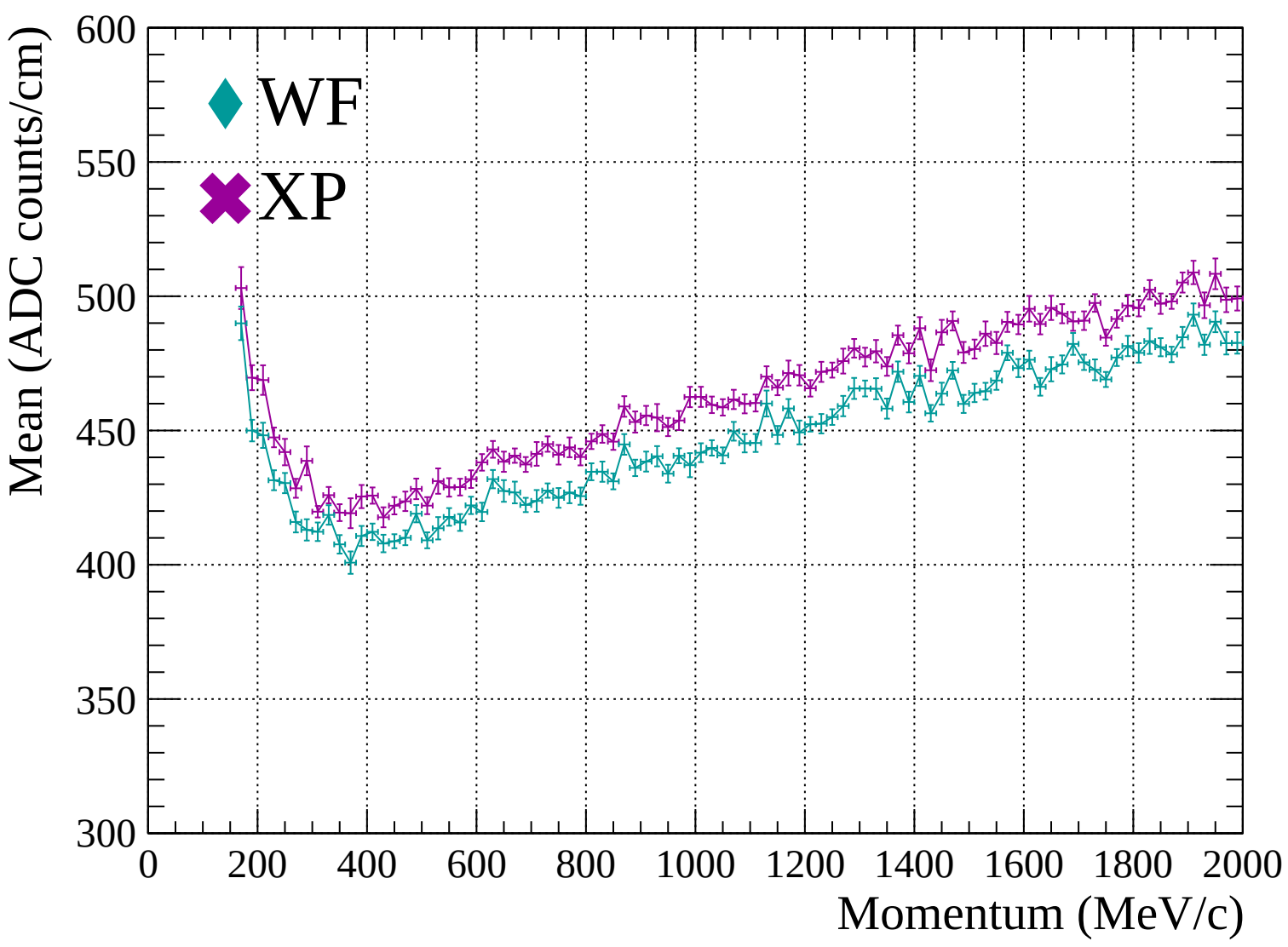


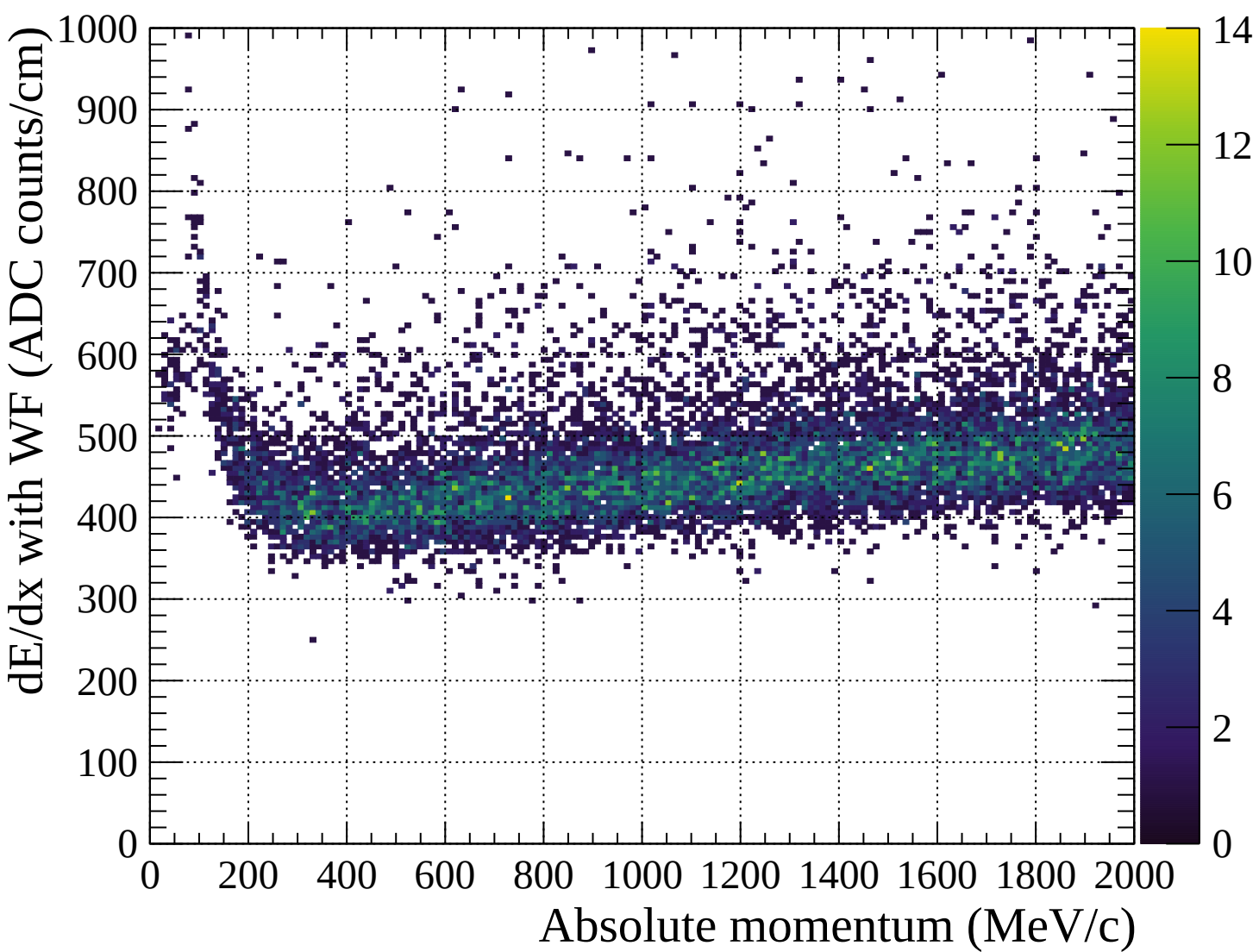


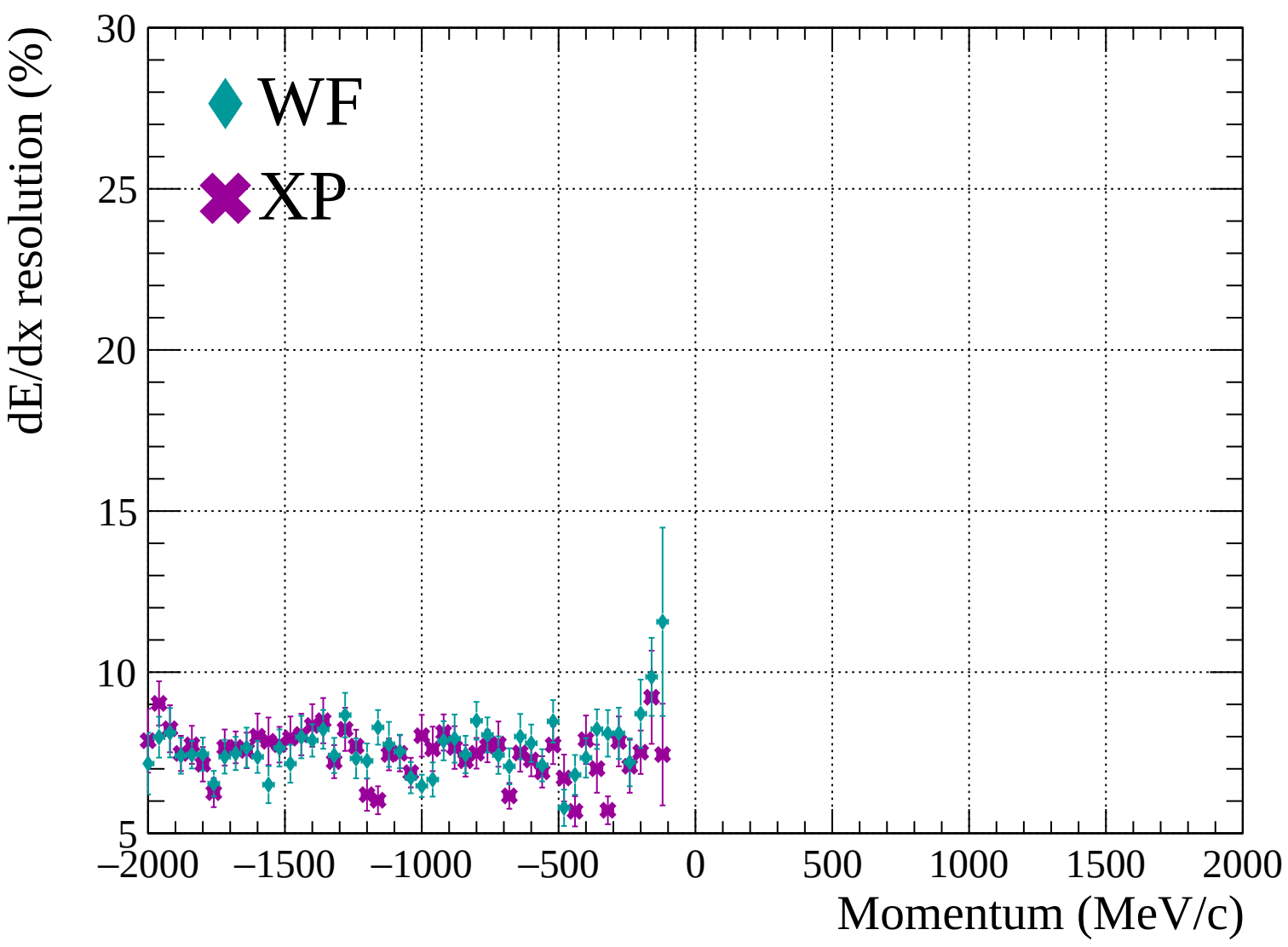


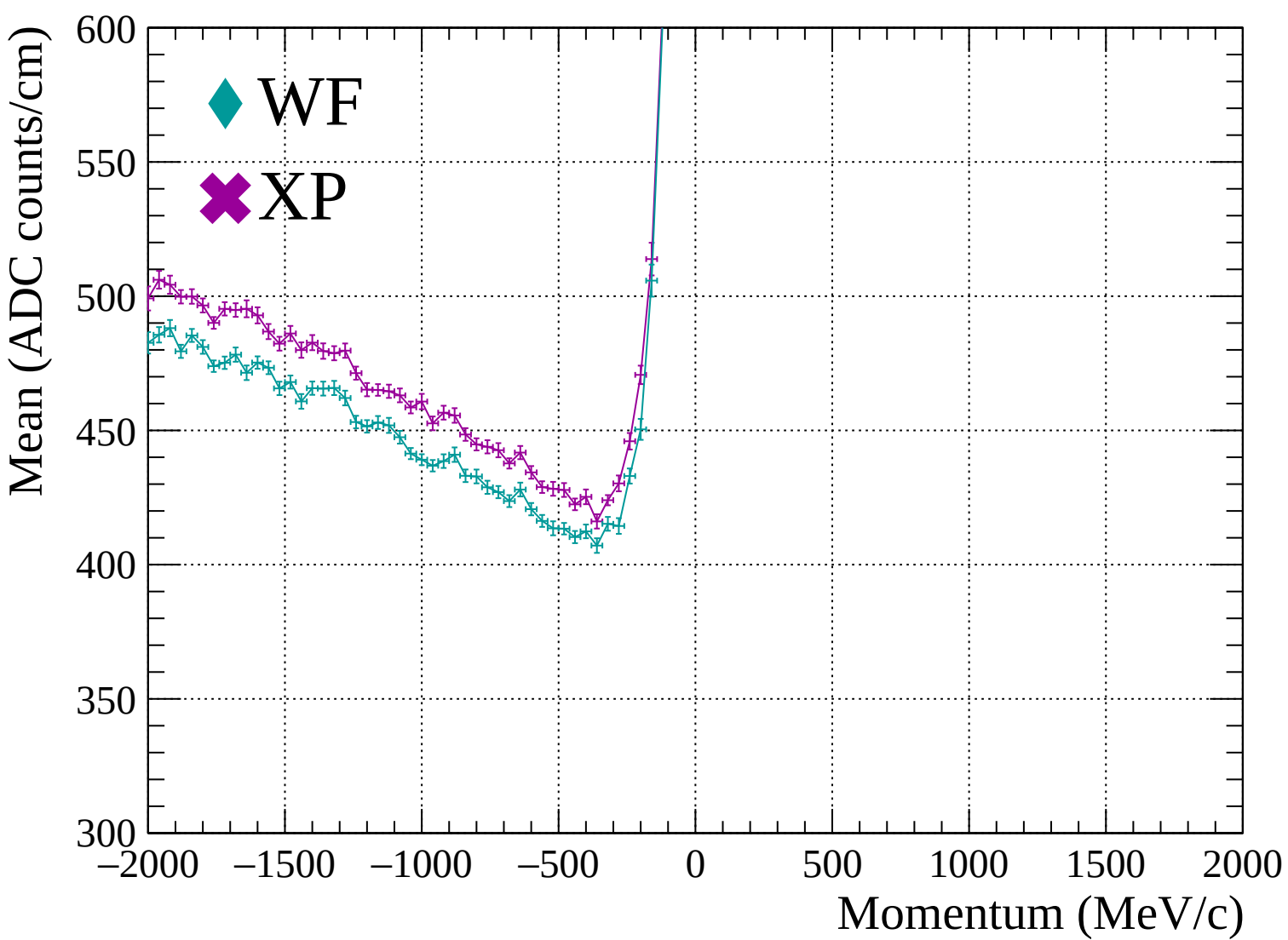


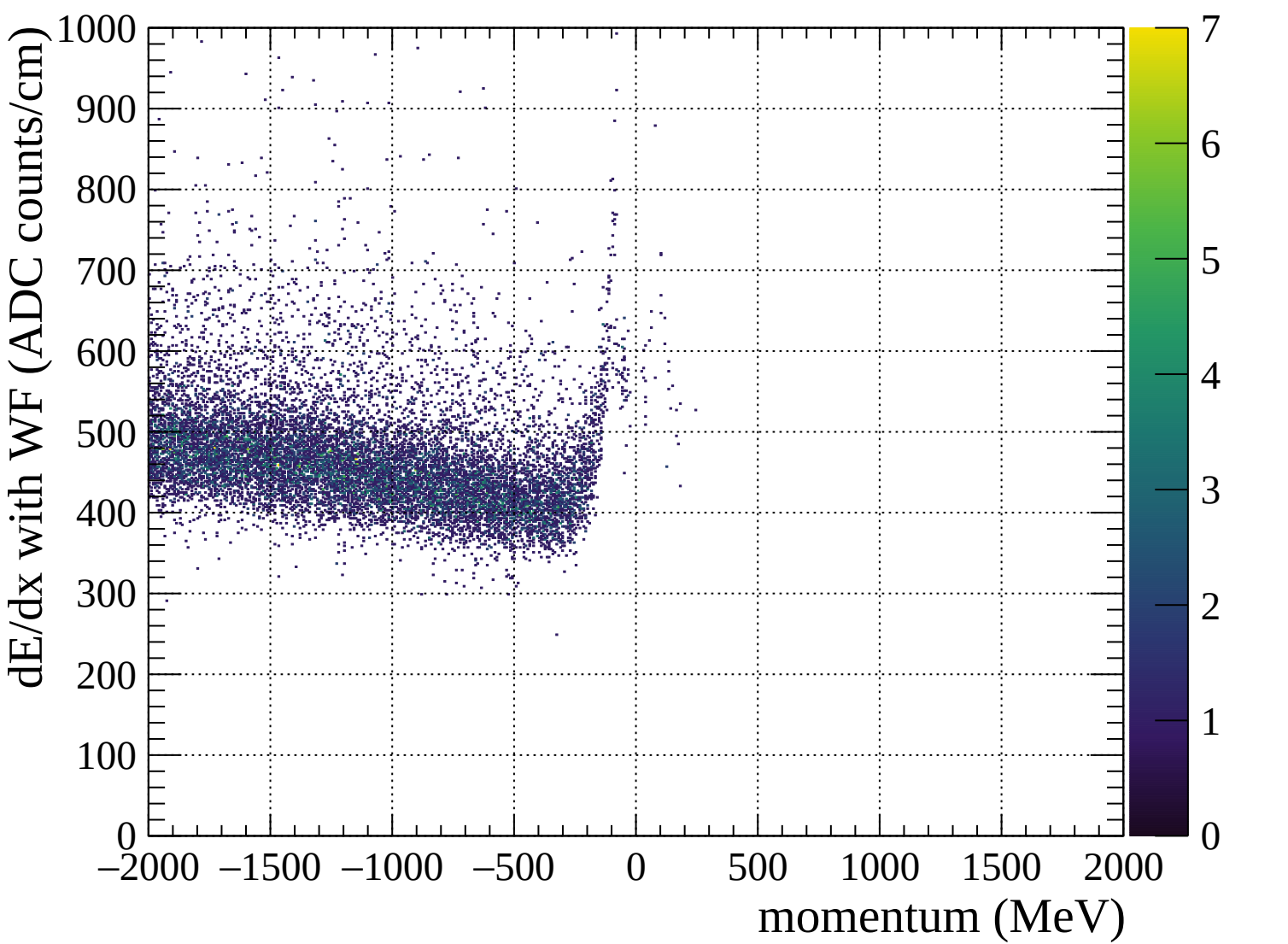


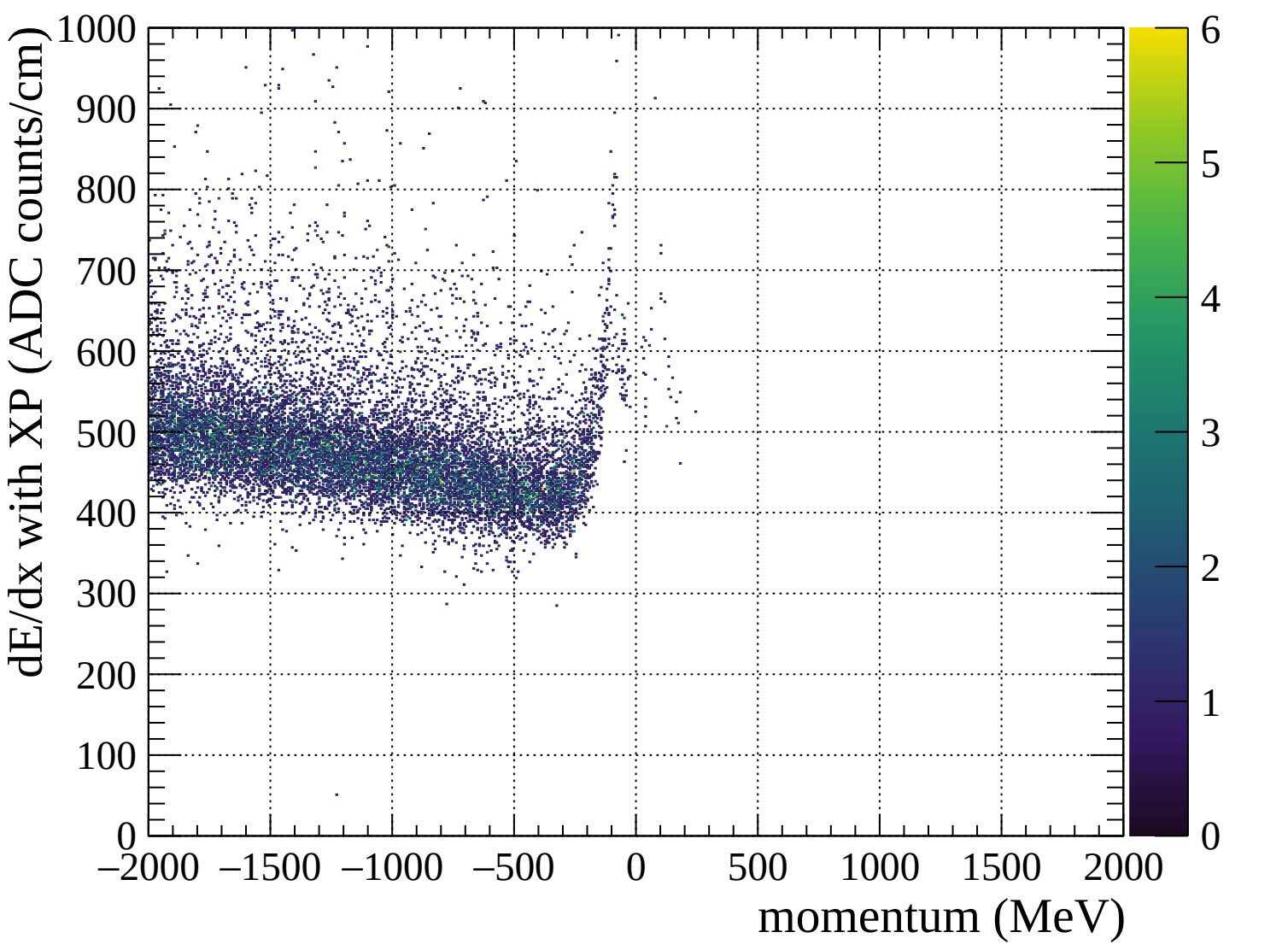


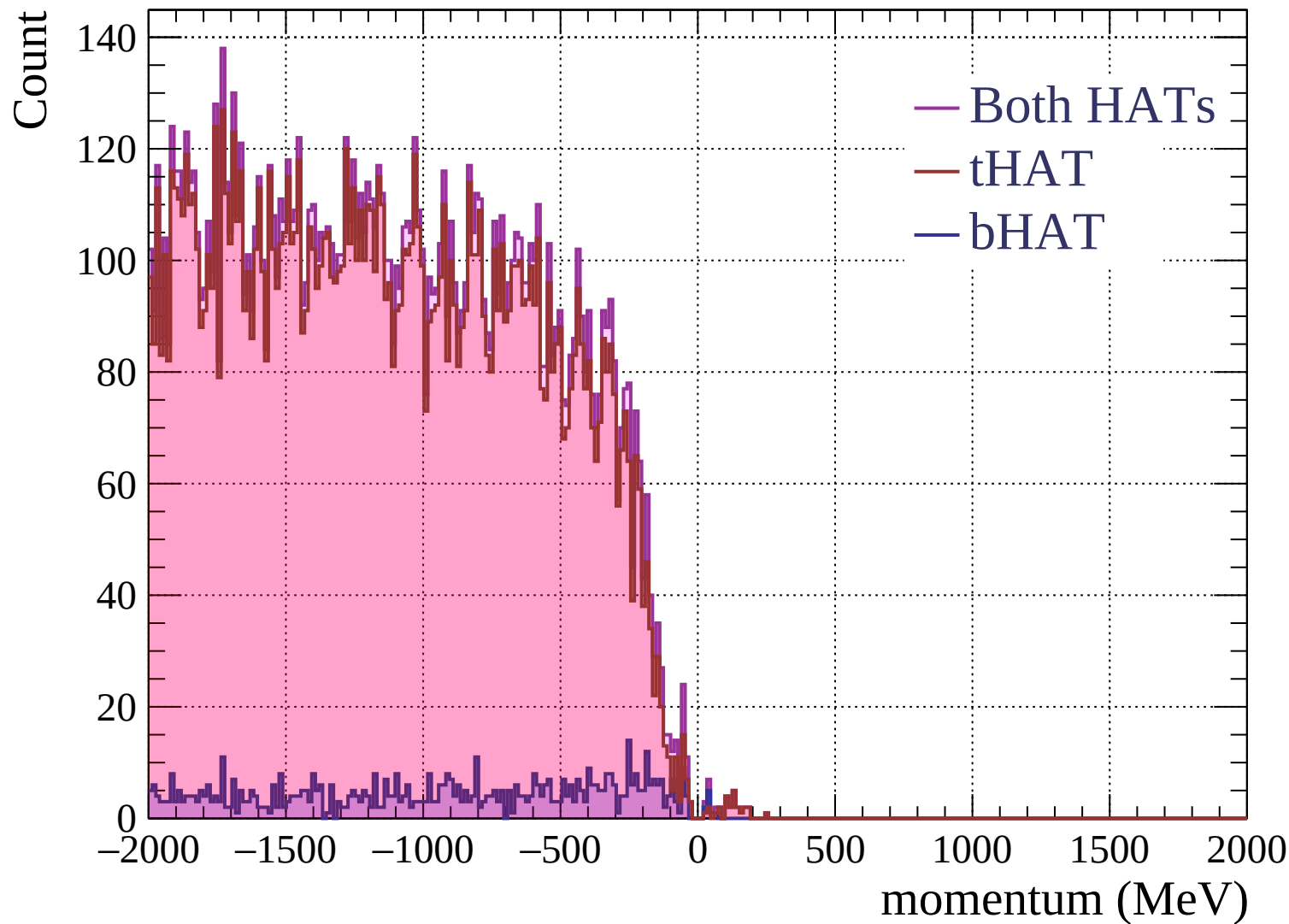


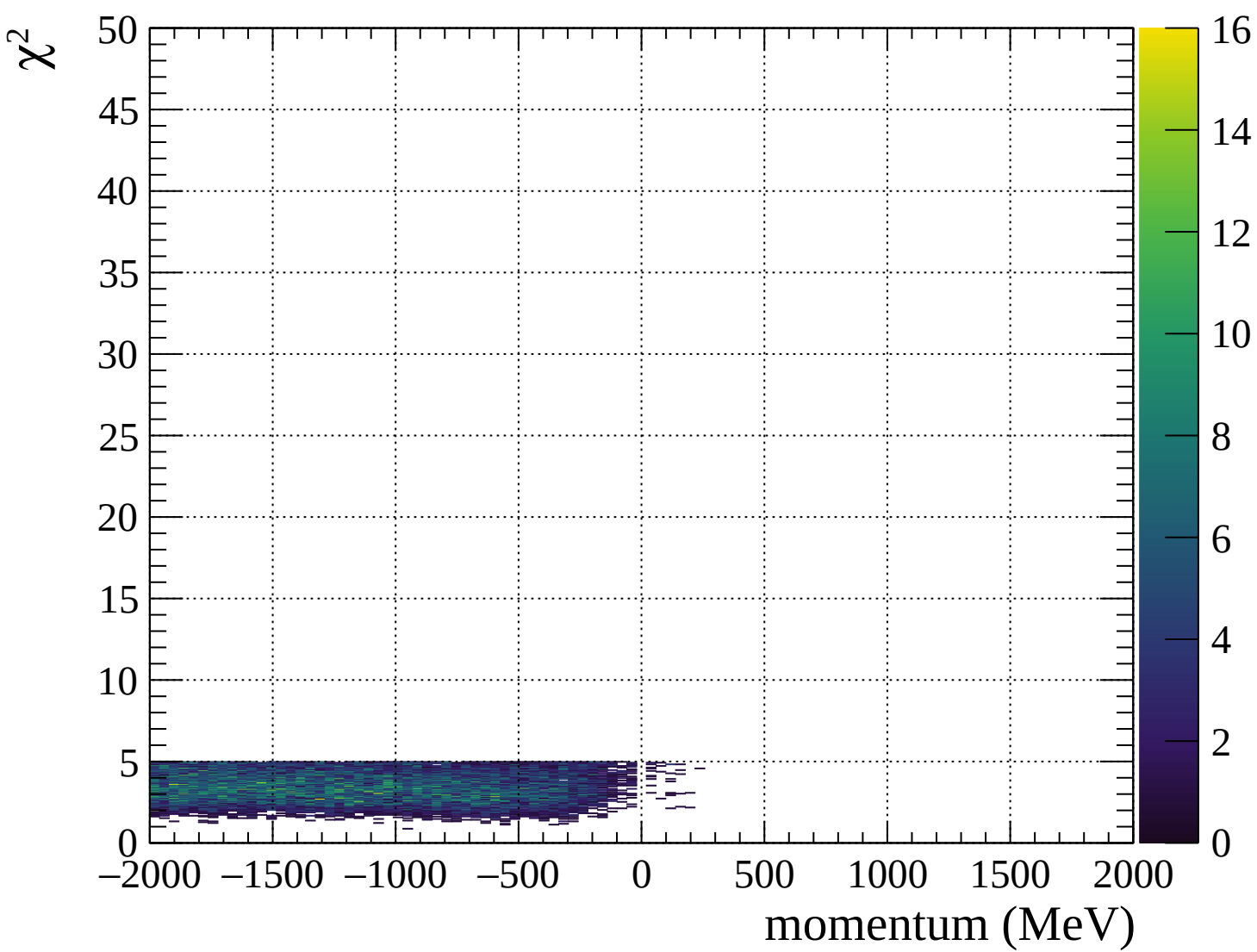


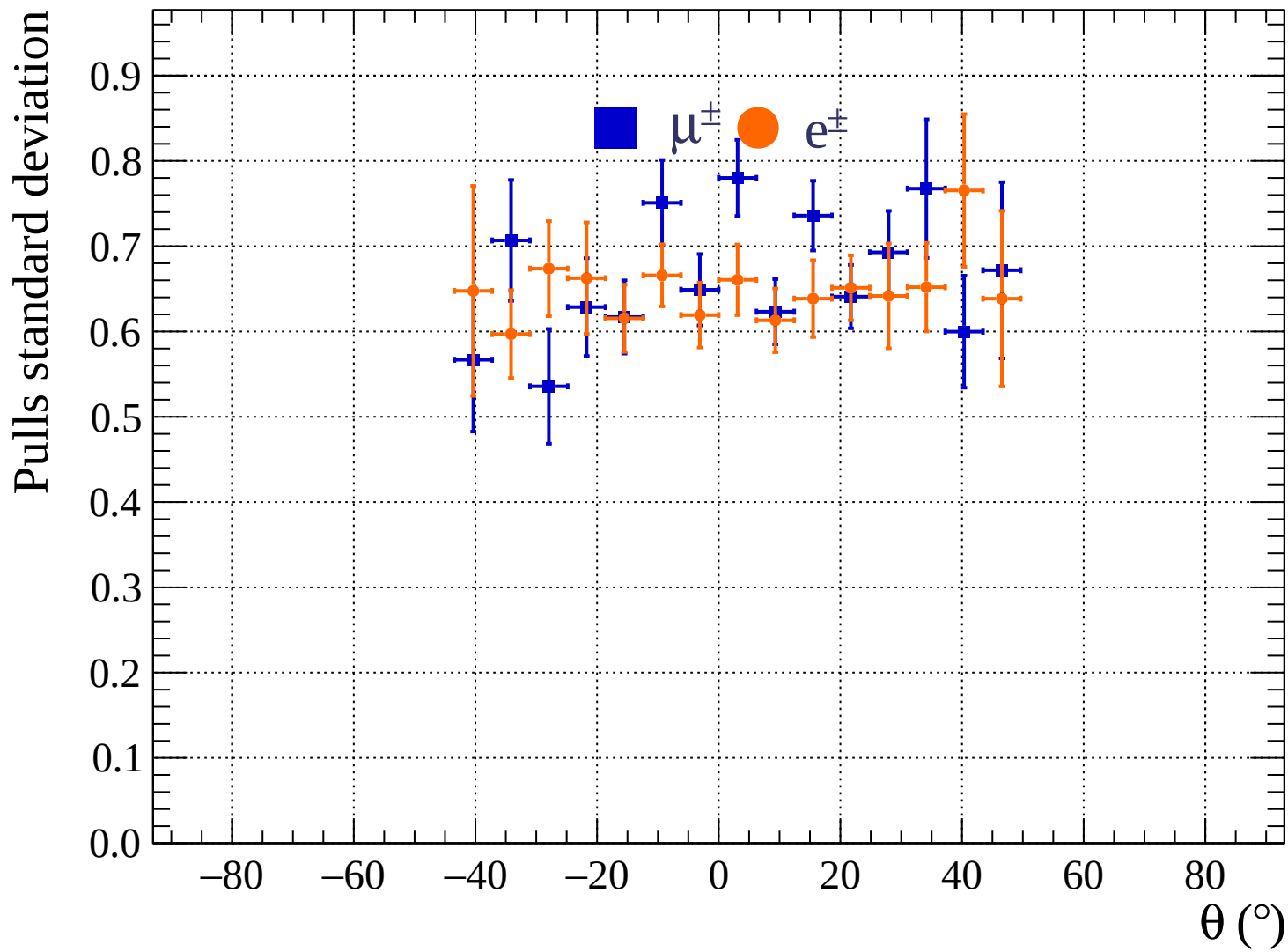




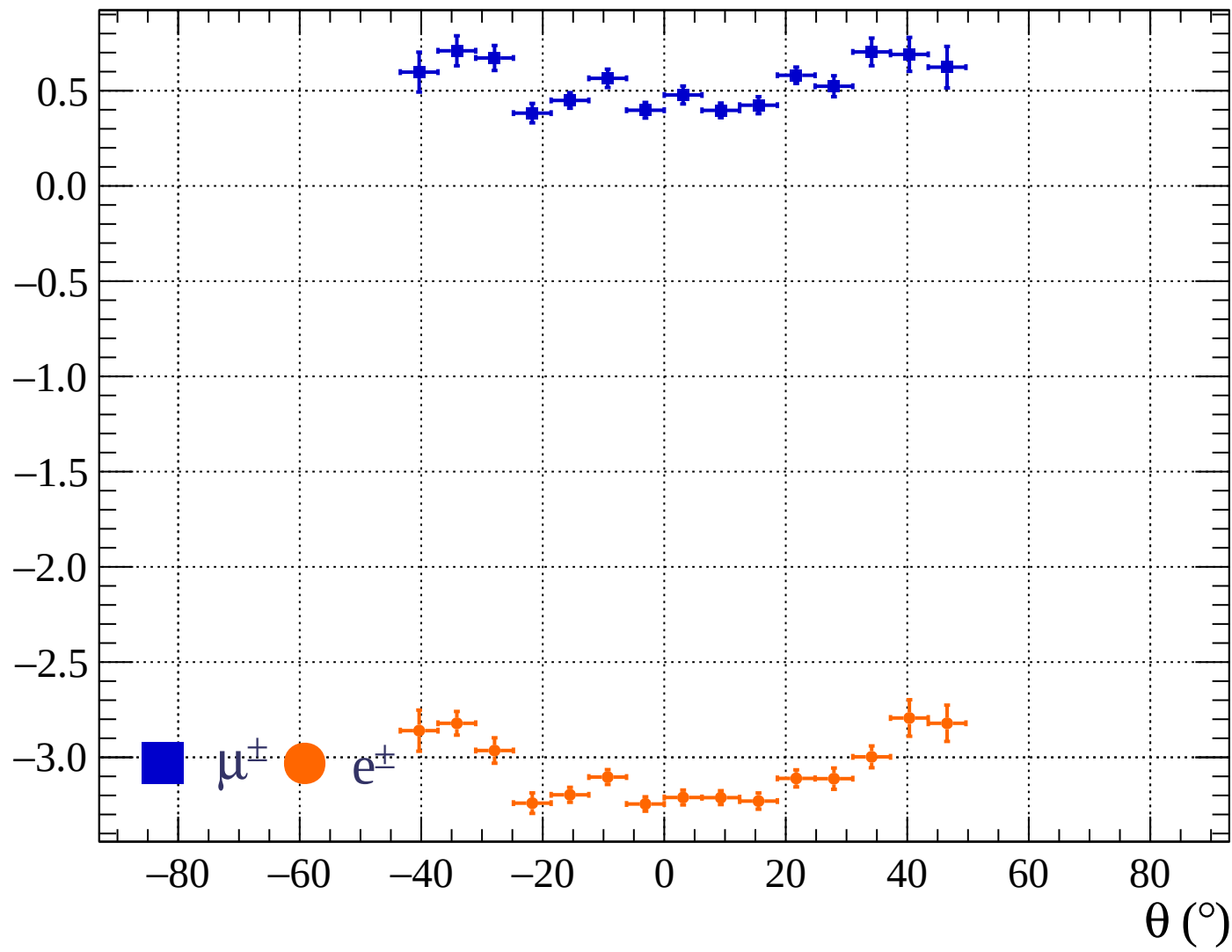




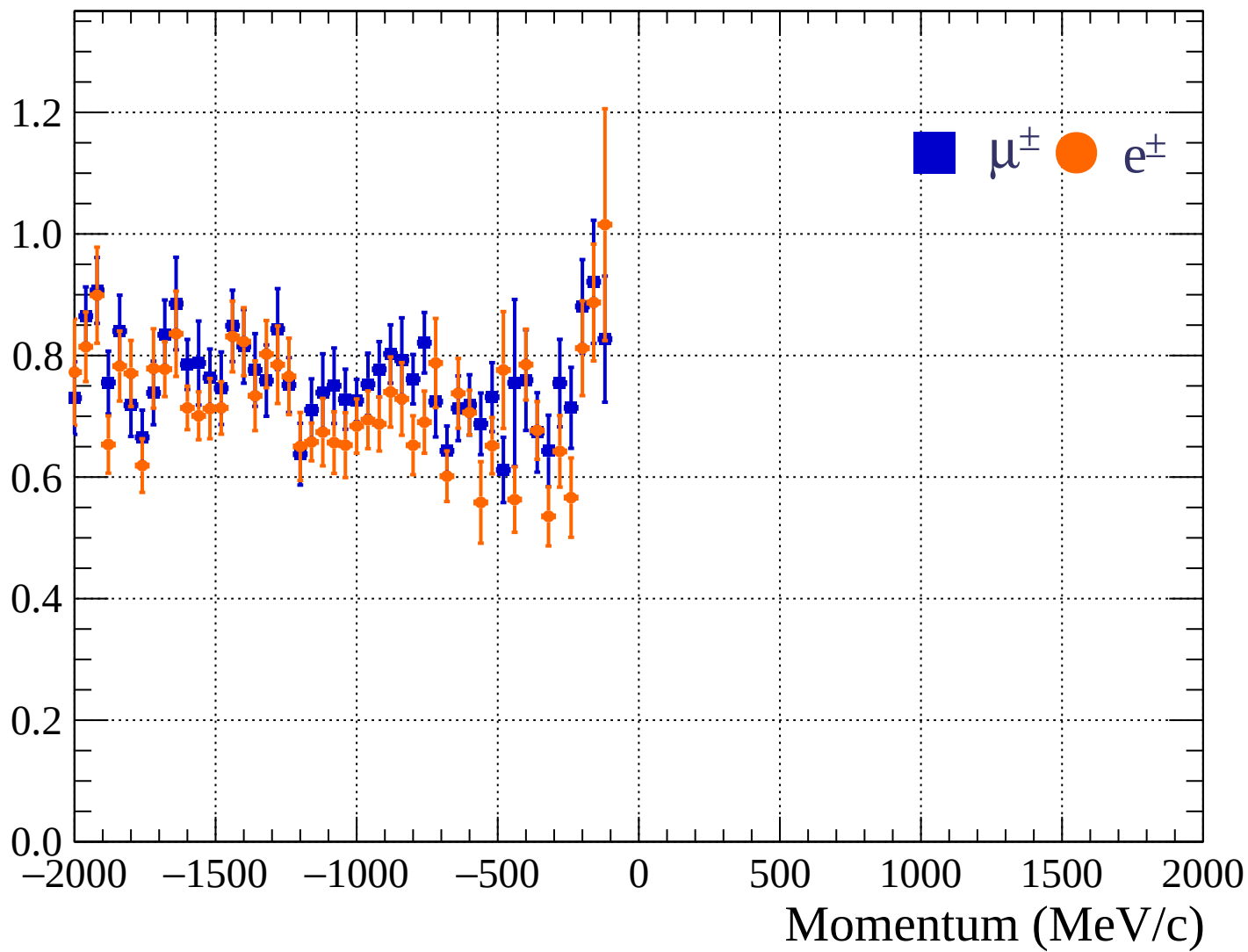




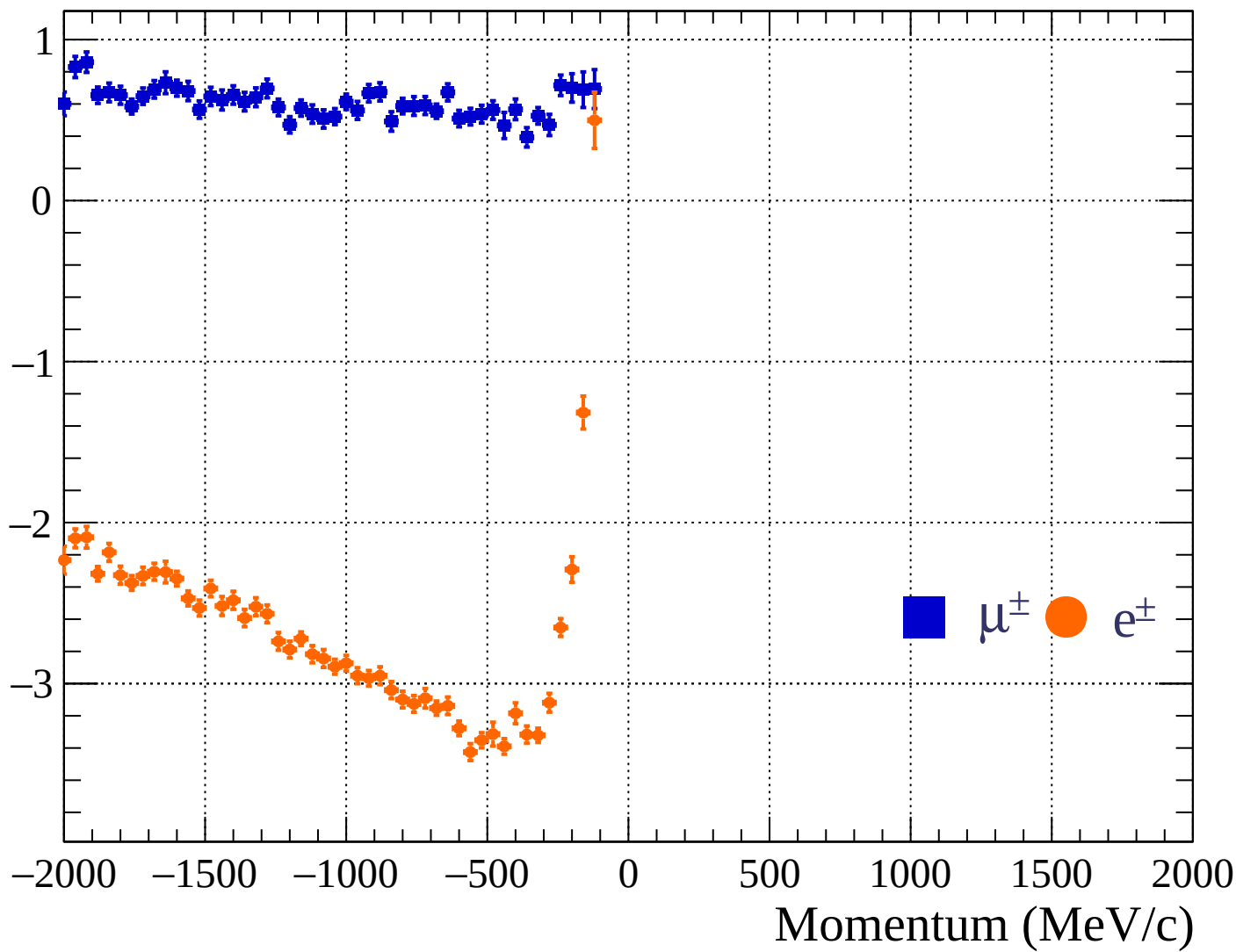
Pulls mean

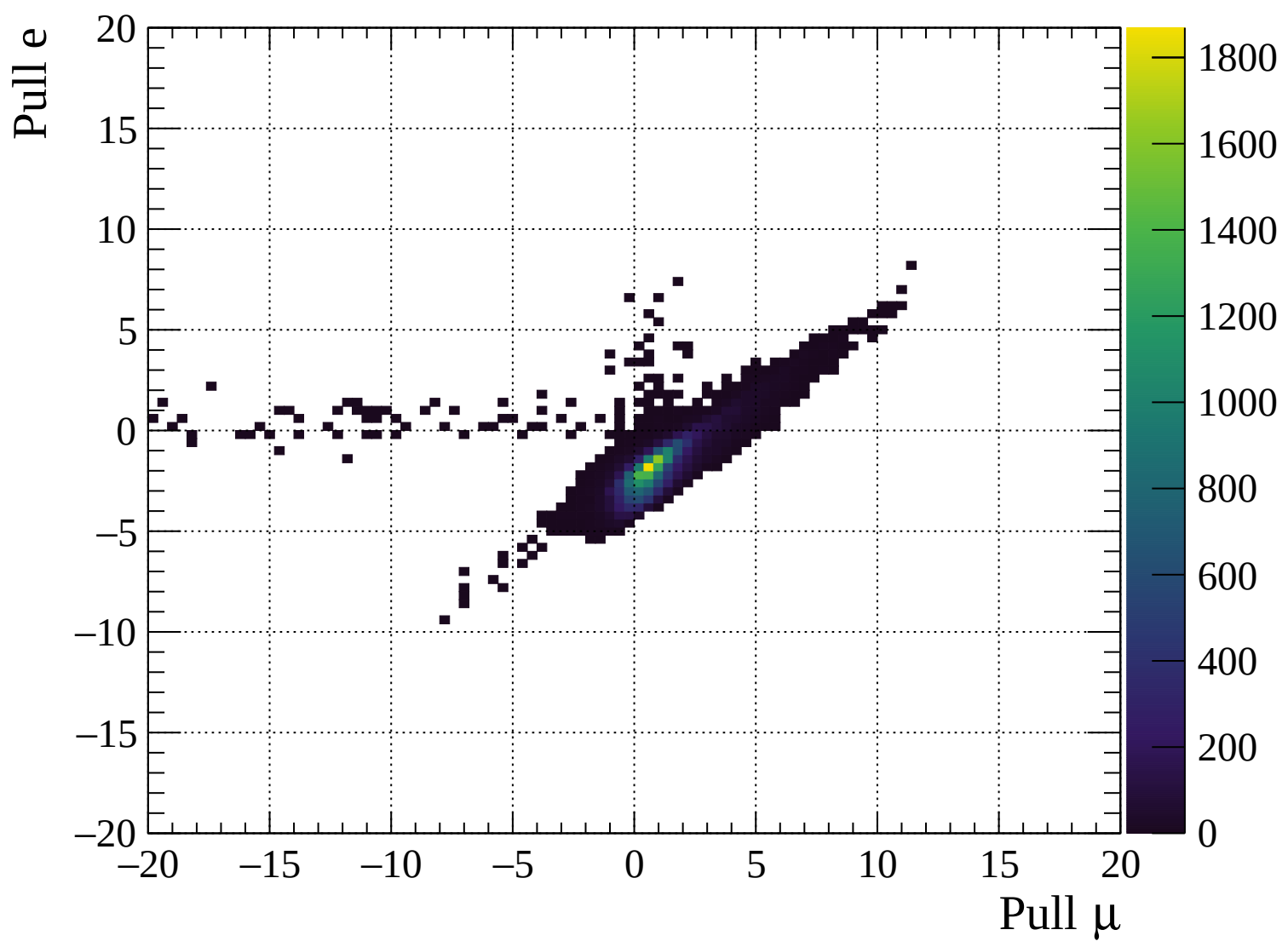


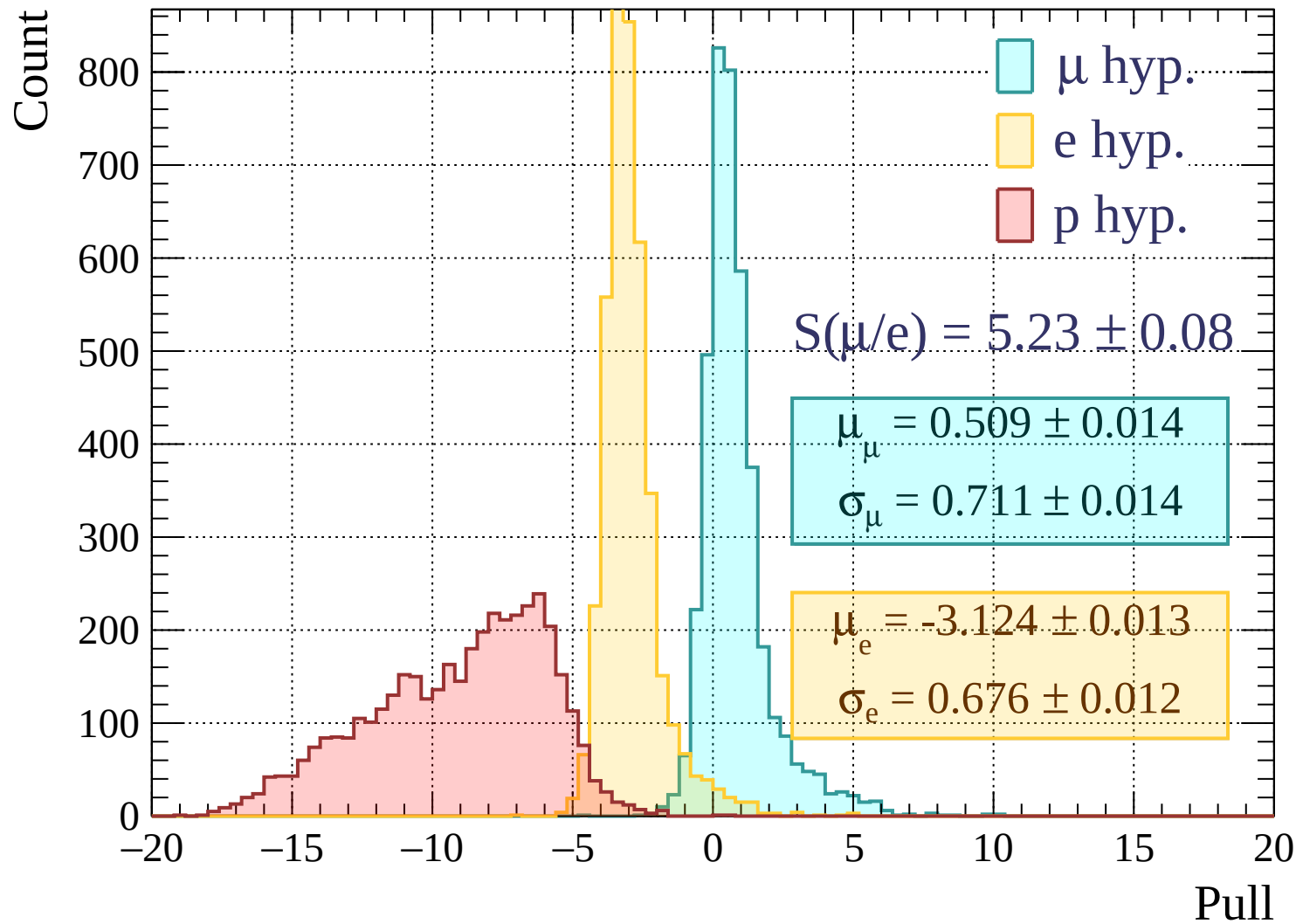
Pulls standard deviation

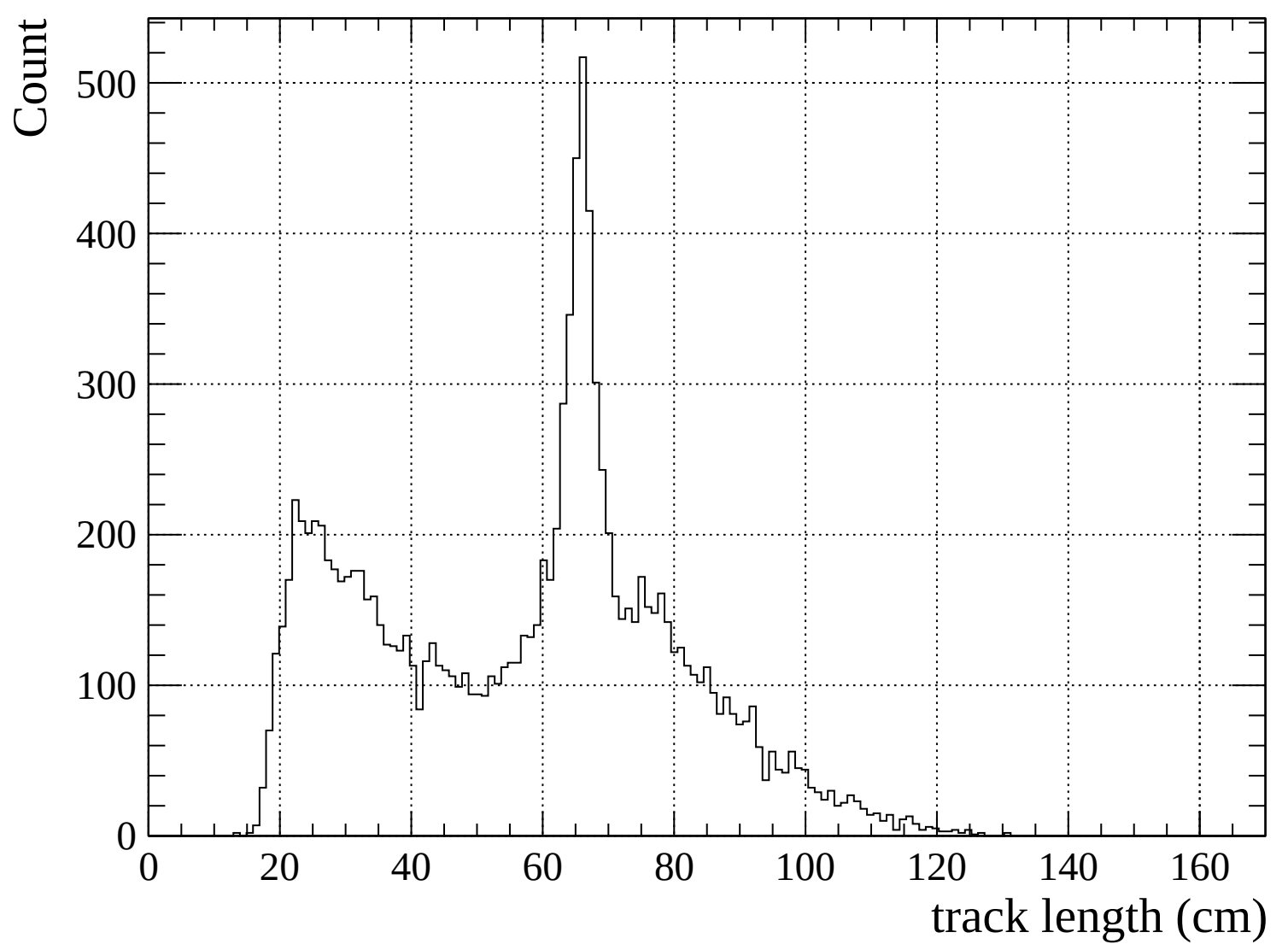


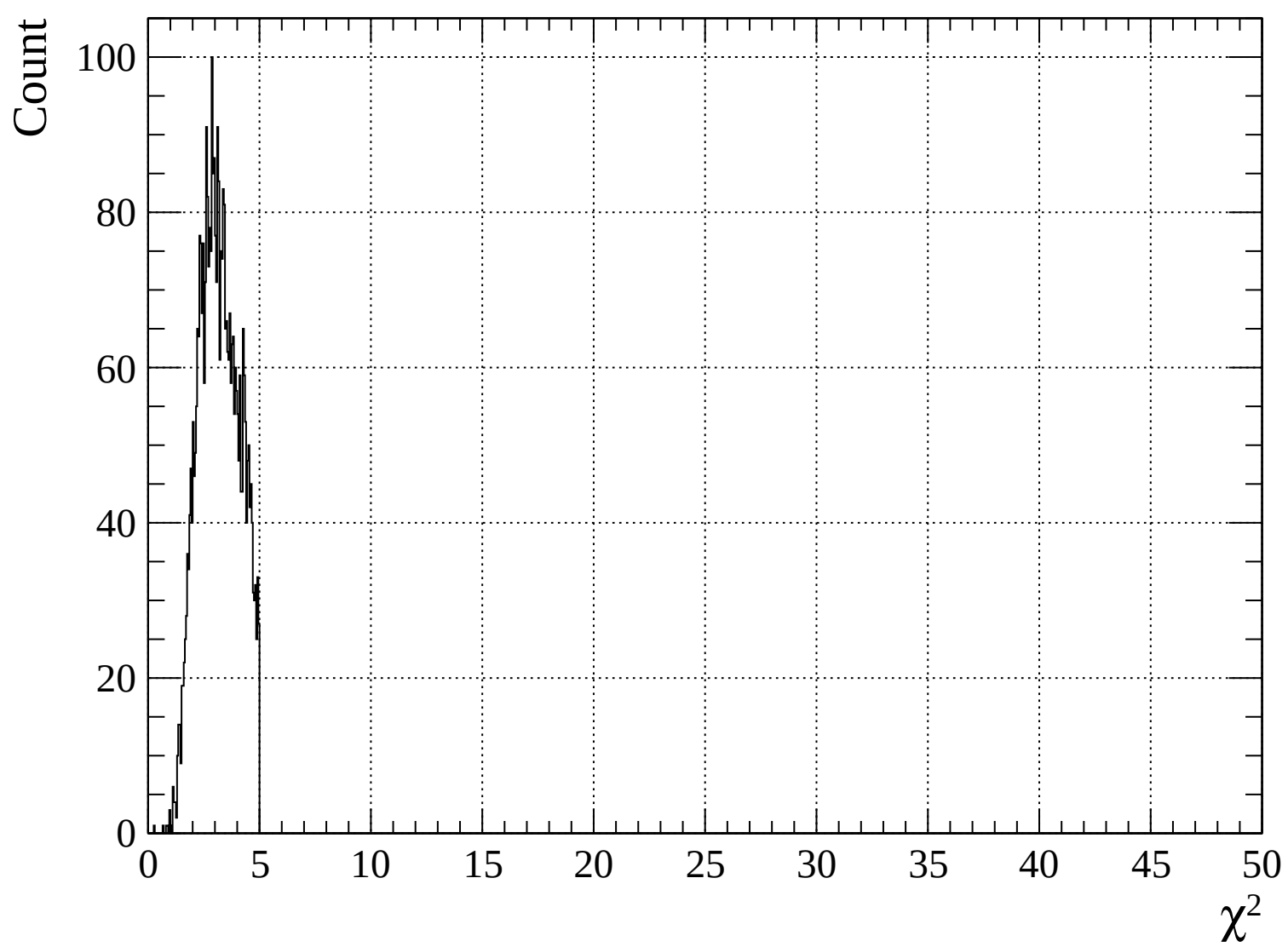
Pulls mean

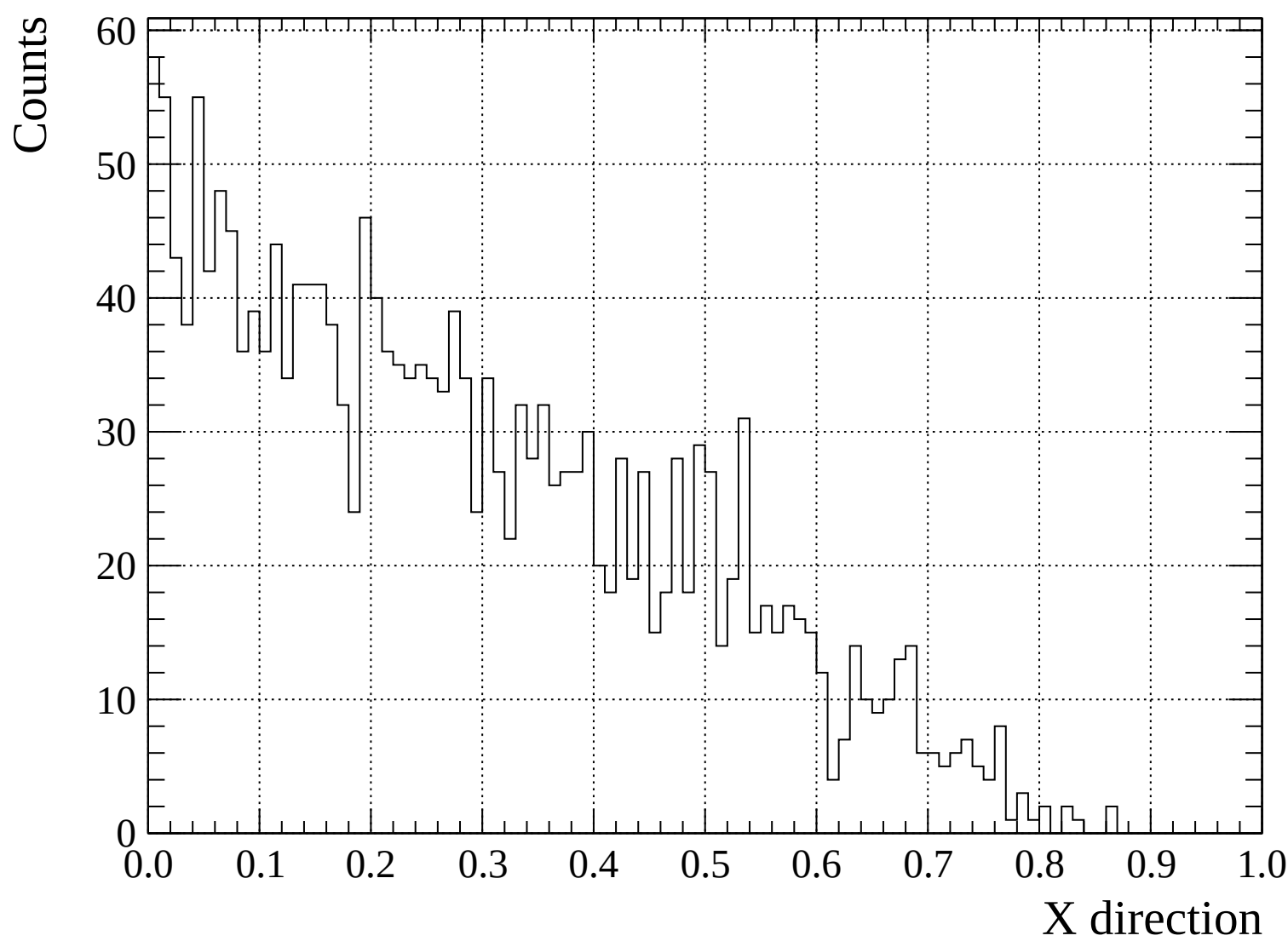


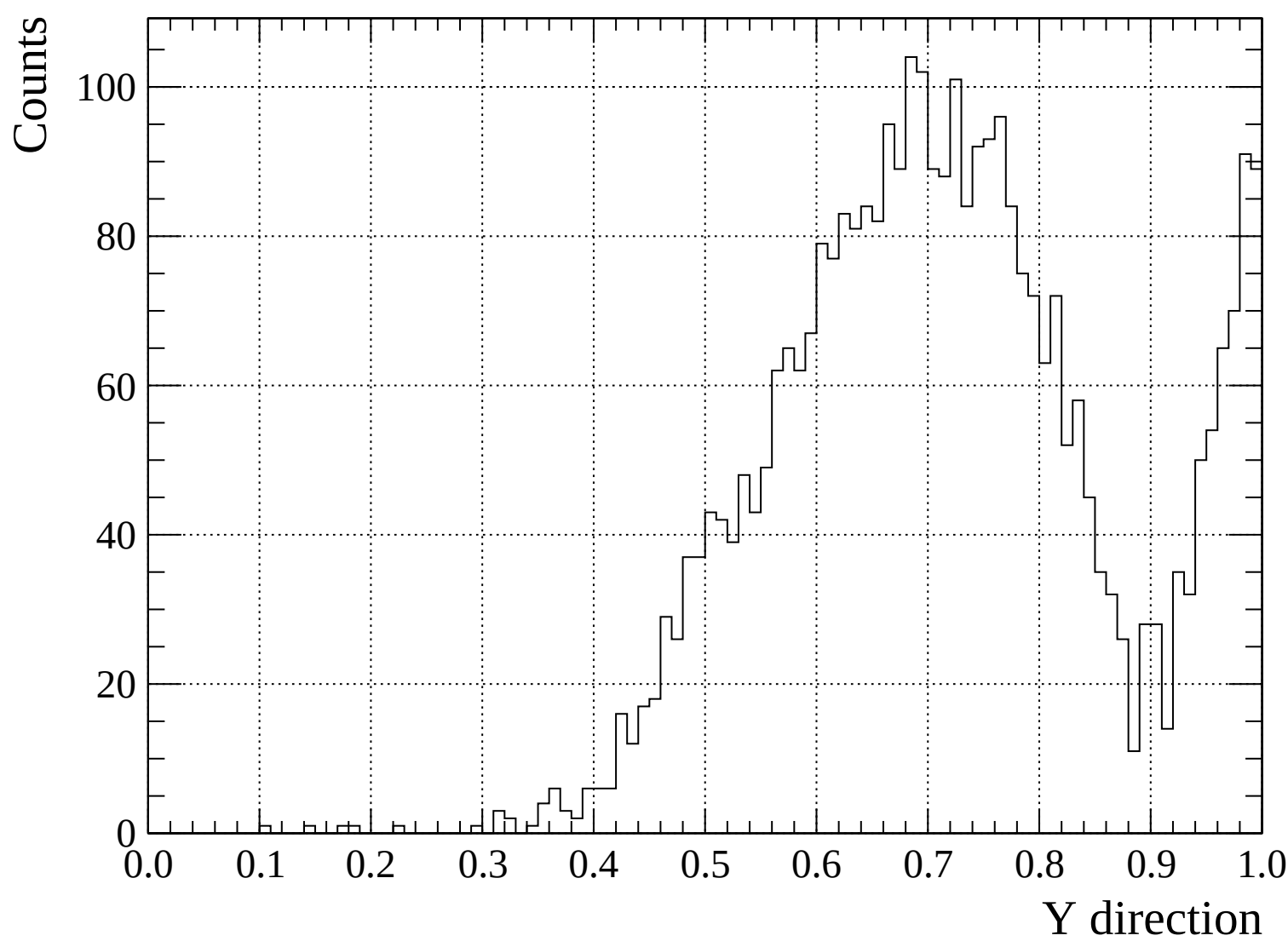


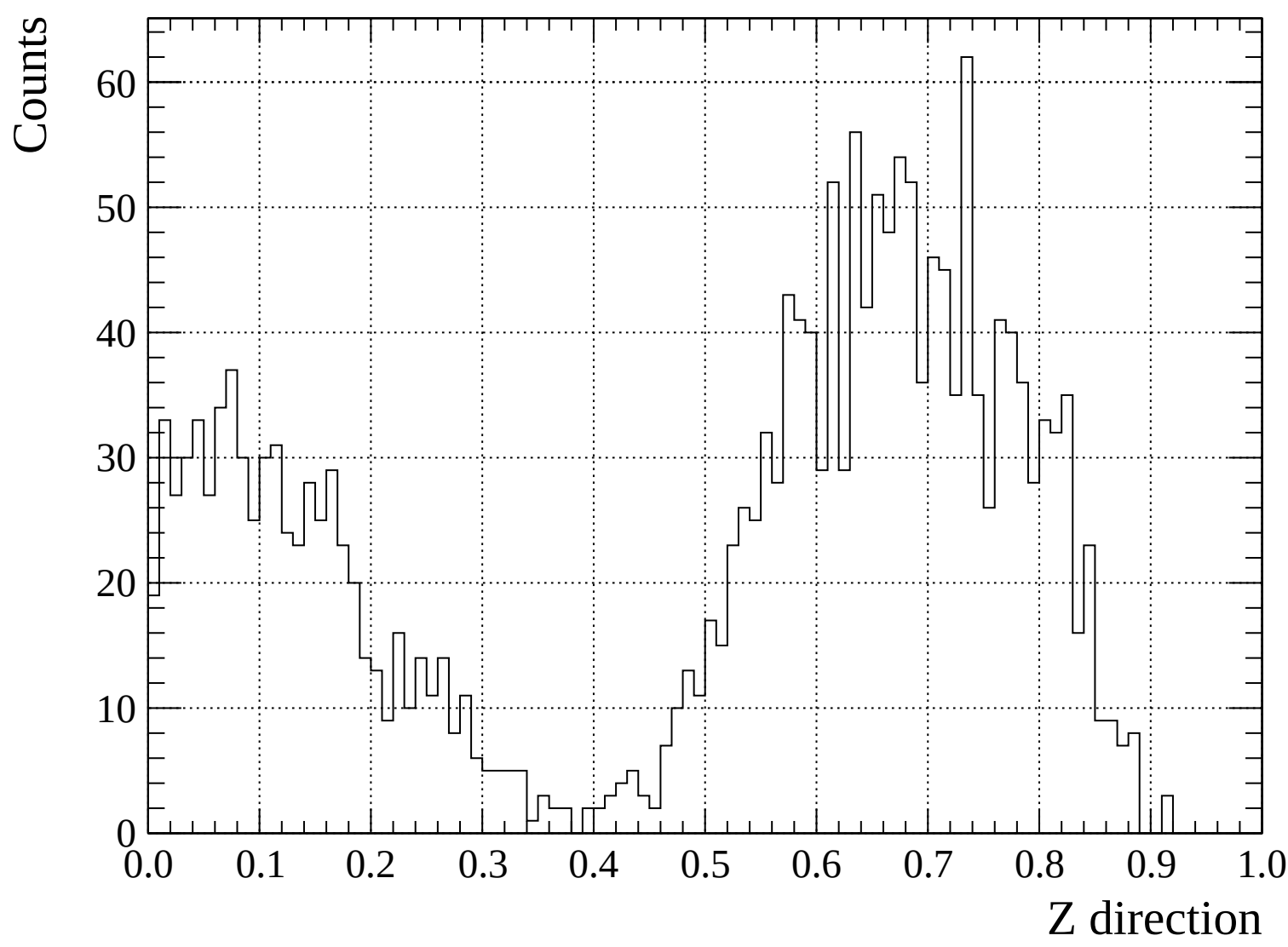


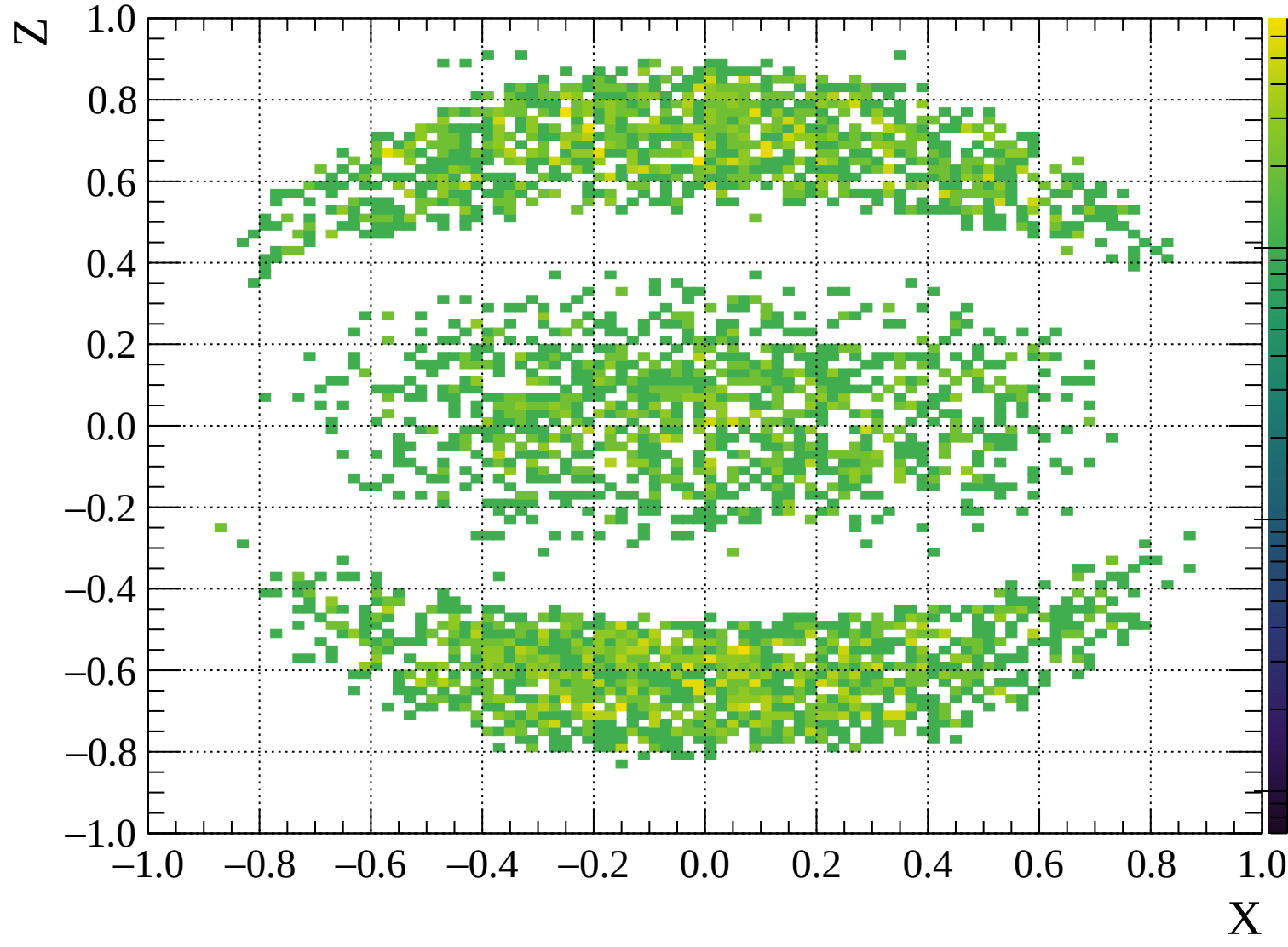


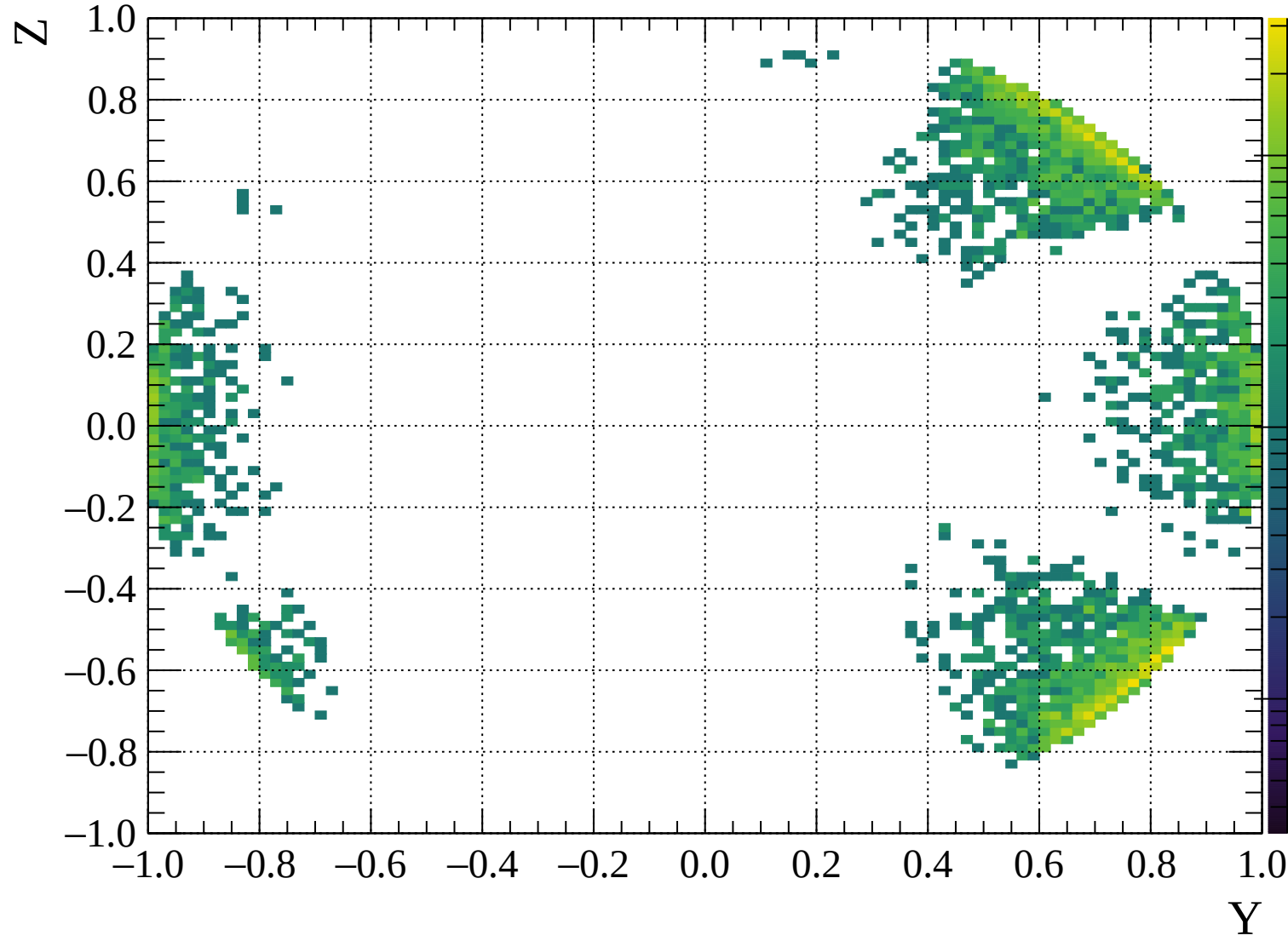


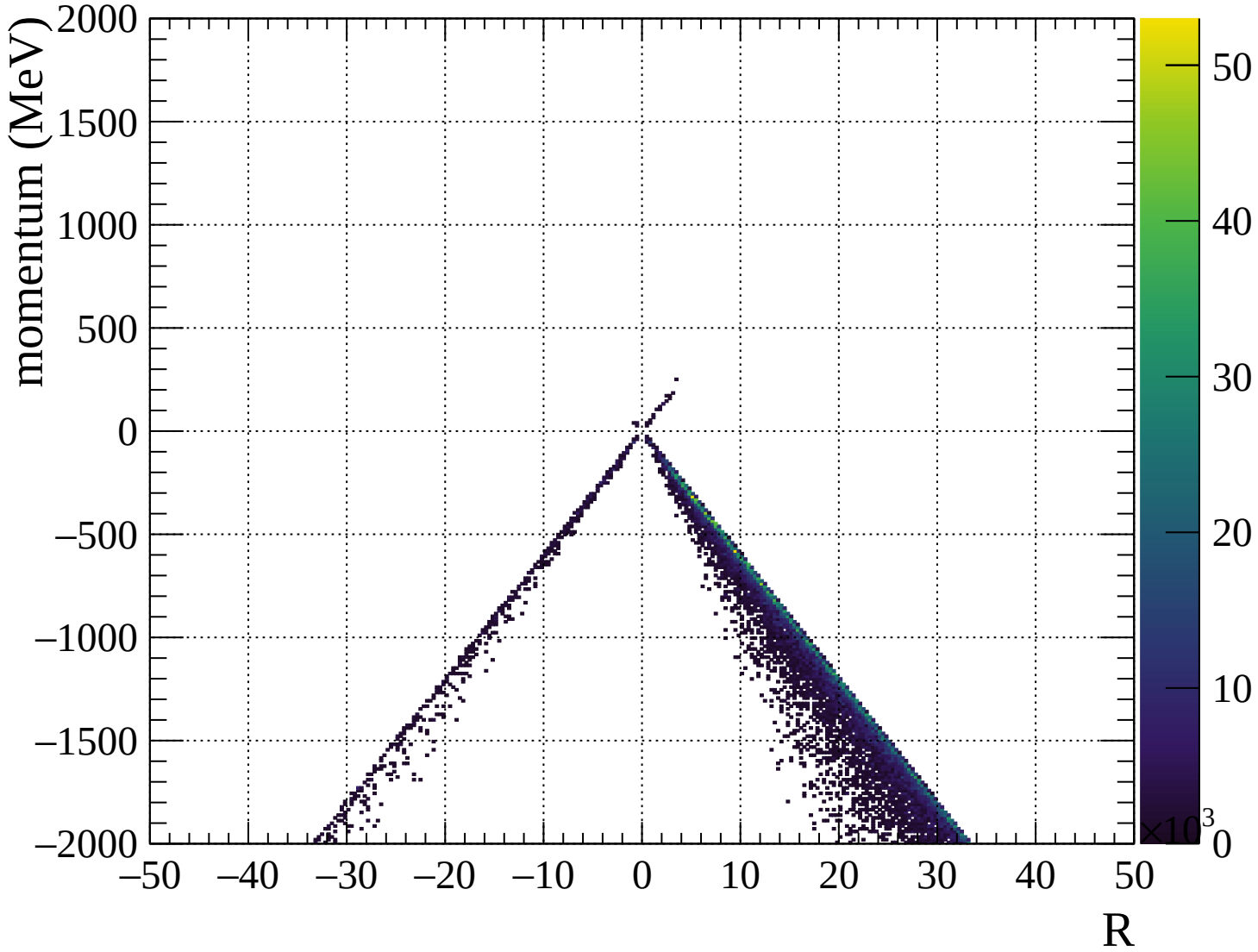


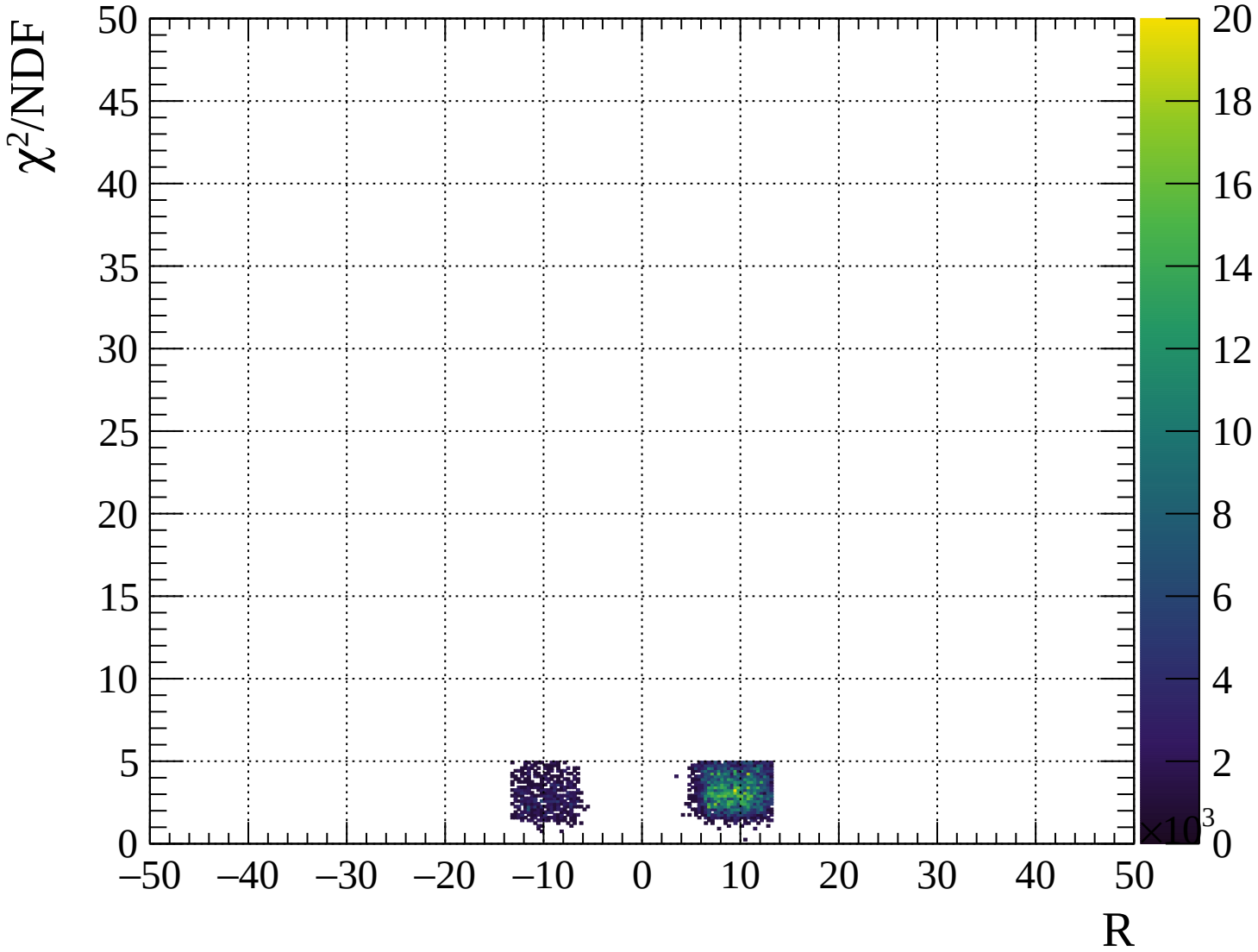






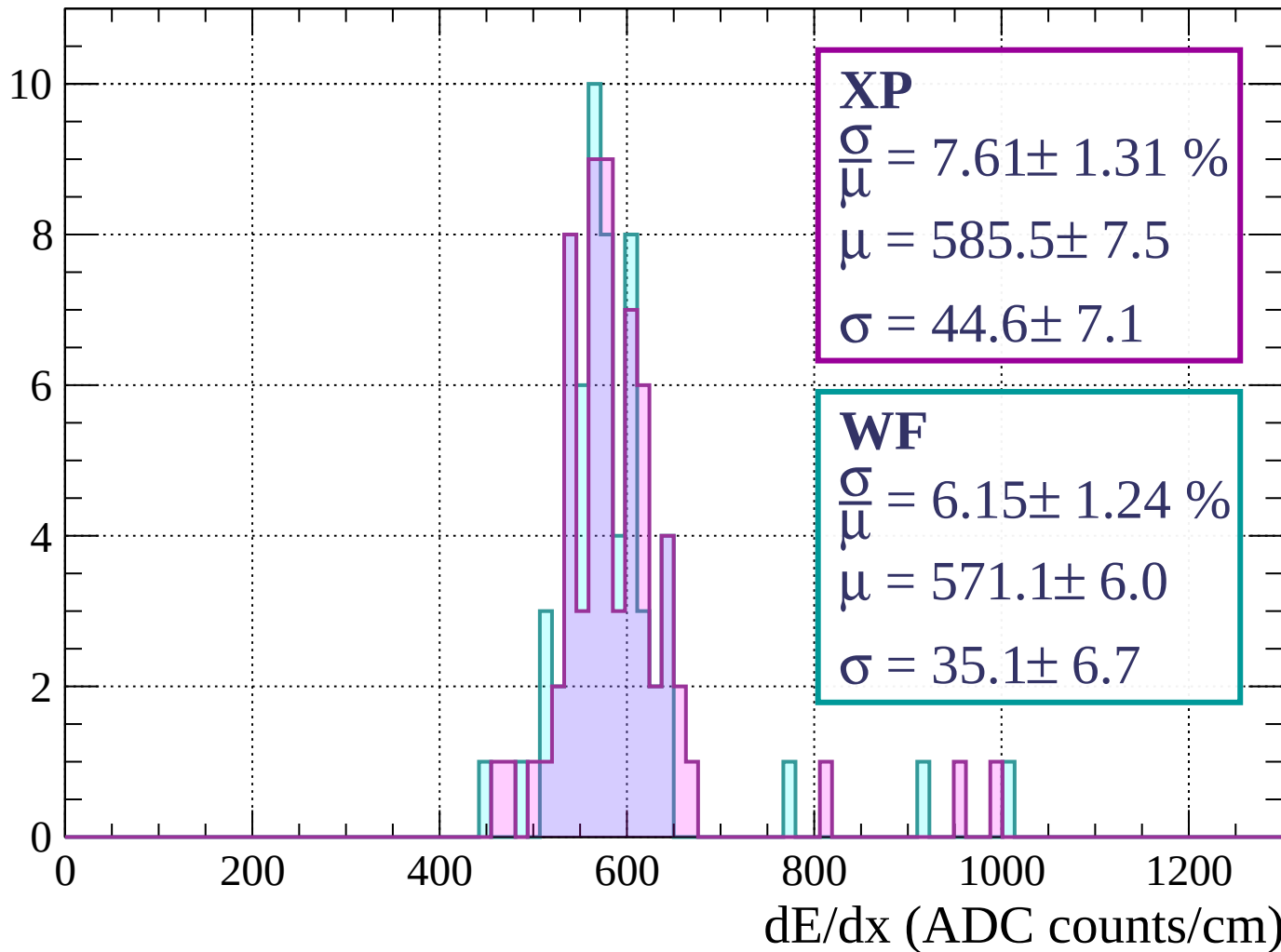






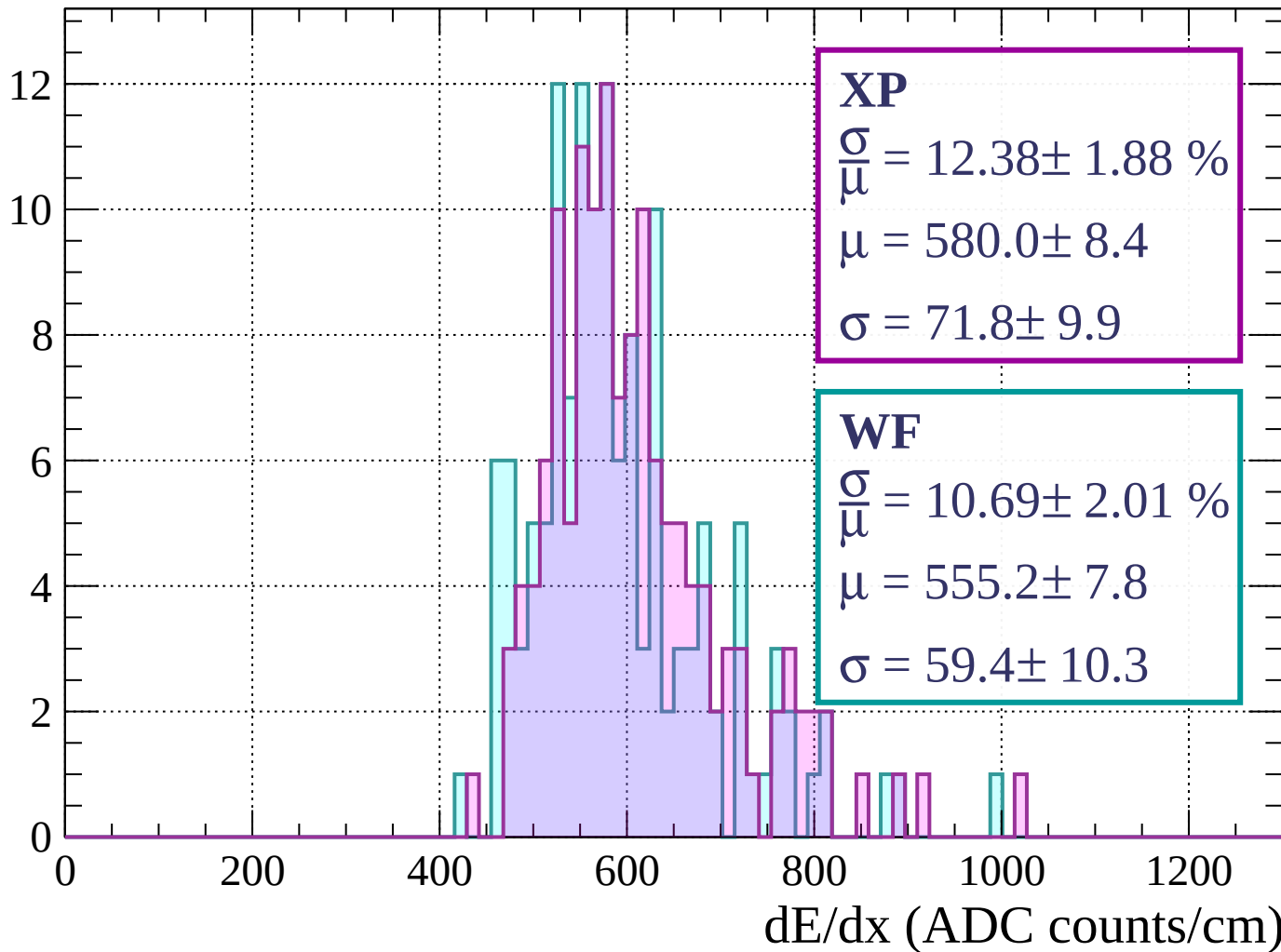
Energy loss | $0 < p < 80$

Count



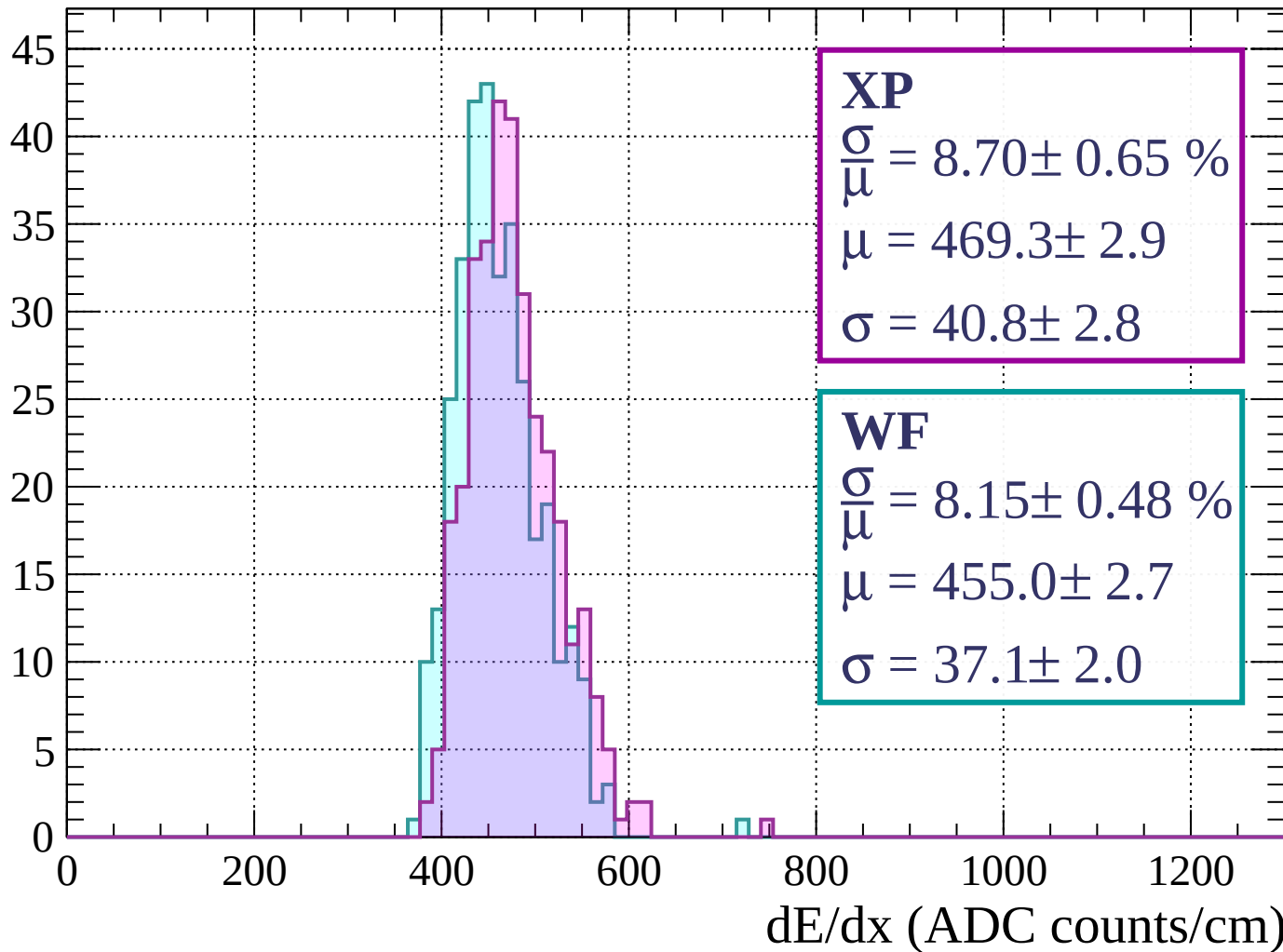
Energy loss | $80 < p < 160$

Count



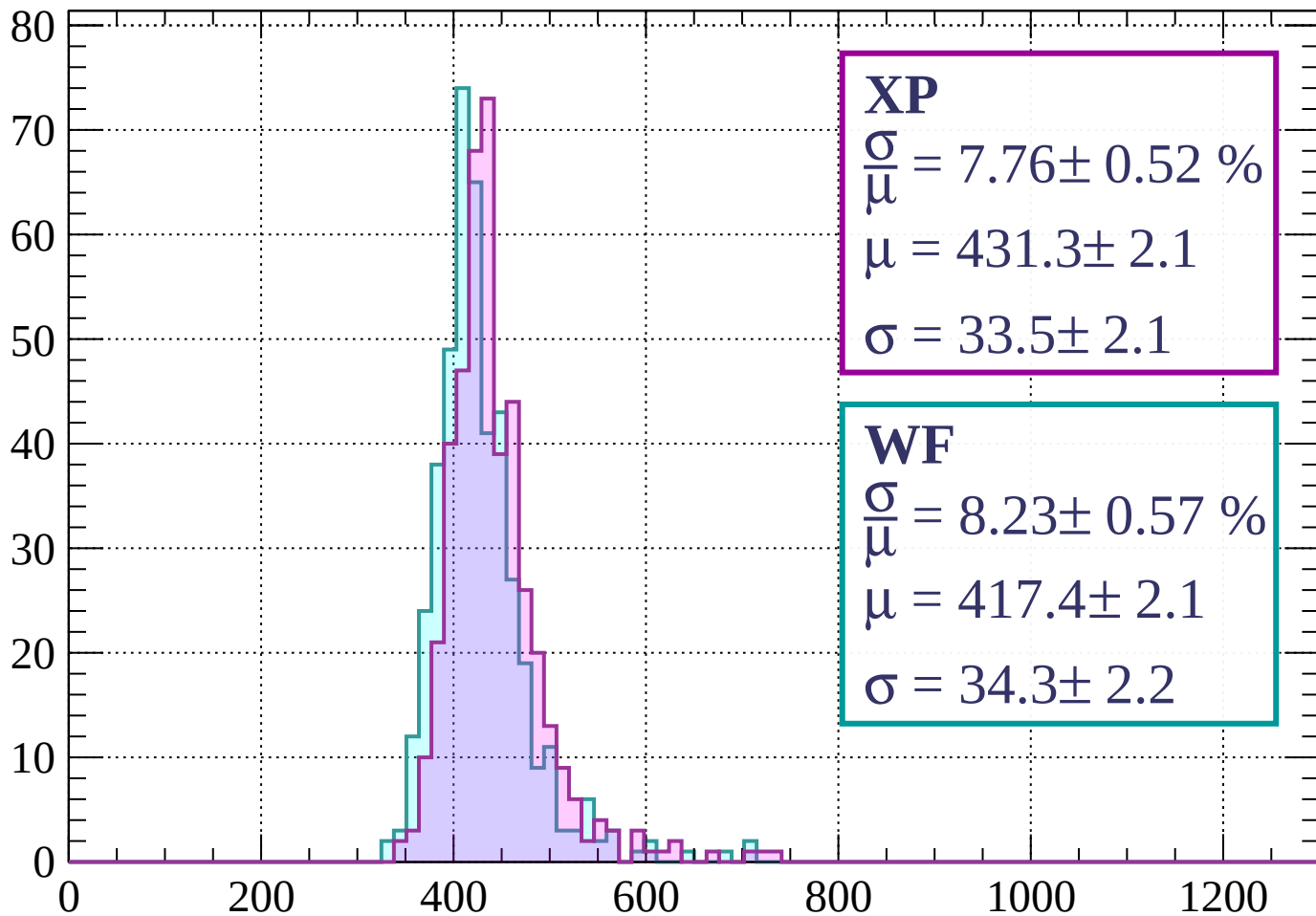
Energy loss | $160 < p < 240$

Count



Energy loss | $240 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 7.76 \pm 0.52 \%$$

$$\mu = 431.3 \pm 2.1$$

$$\sigma = 33.5 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 8.23 \pm 0.57 \%$$

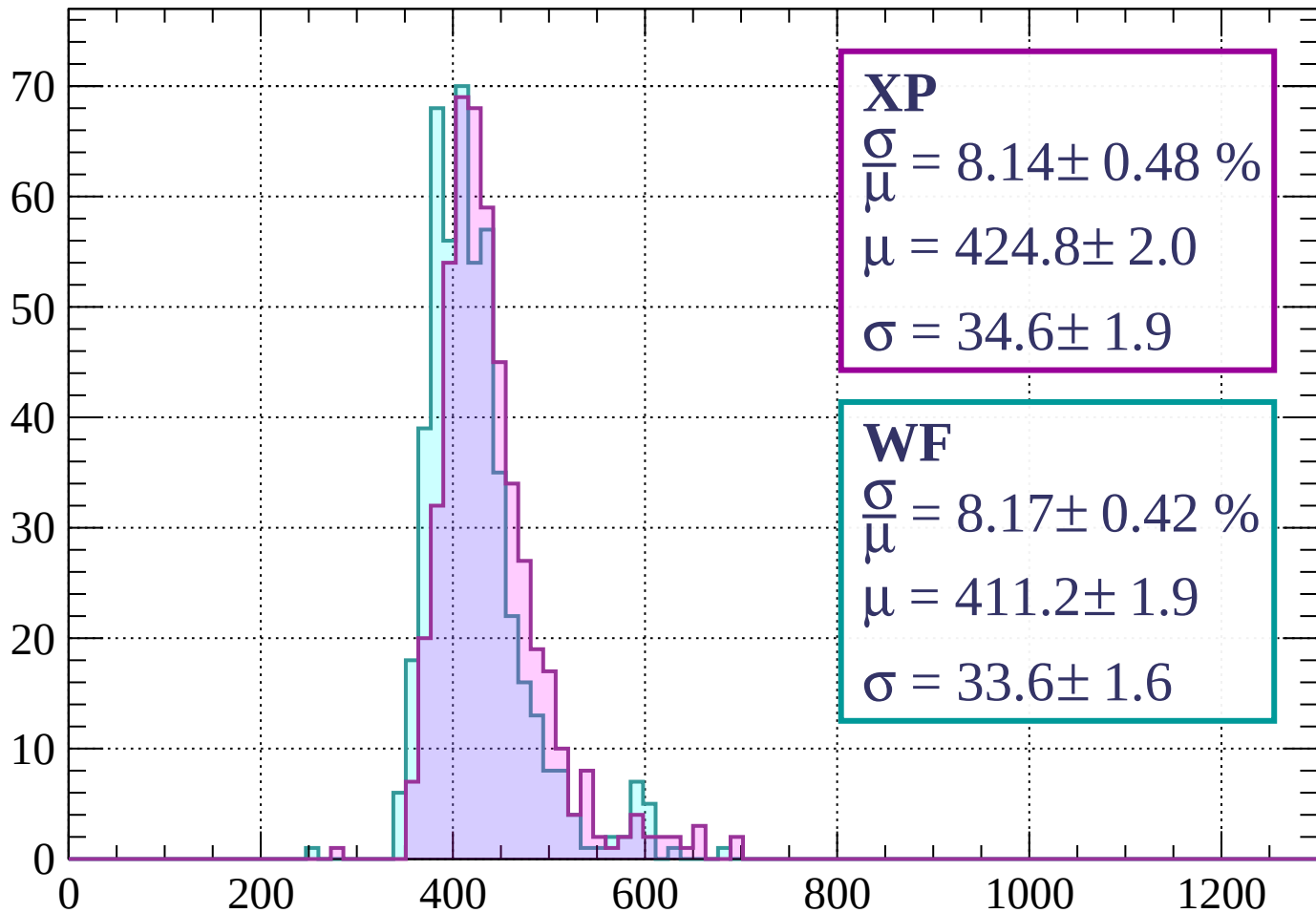
$$\mu = 417.4 \pm 2.1$$

$$\sigma = 34.3 \pm 2.2$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 400$

Count



XP

$$\frac{\sigma}{\mu} = 8.14 \pm 0.48 \%$$

$$\mu = 424.8 \pm 2.0$$

$$\sigma = 34.6 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 8.17 \pm 0.42 \%$$

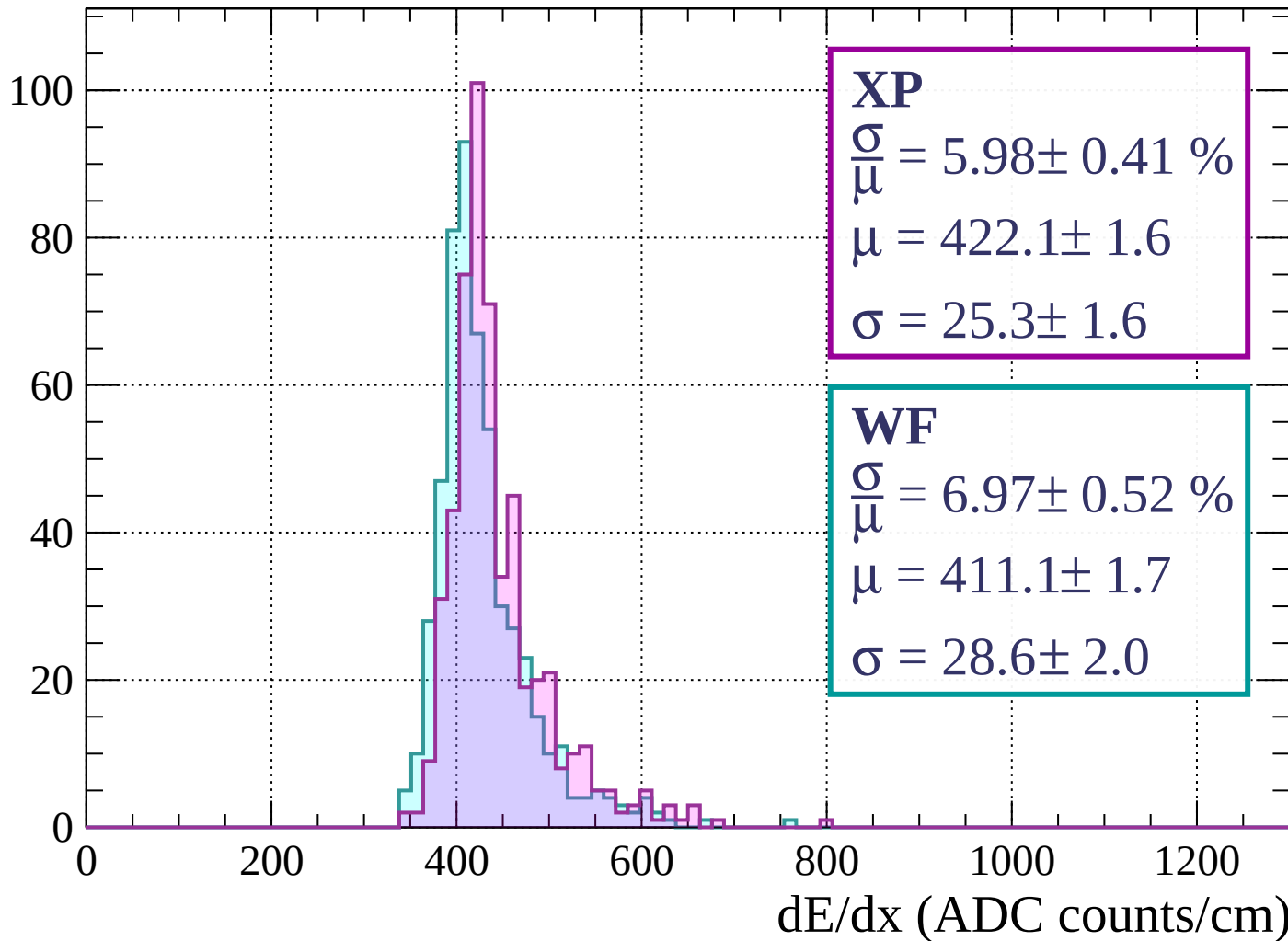
$$\mu = 411.2 \pm 1.9$$

$$\sigma = 33.6 \pm 1.6$$

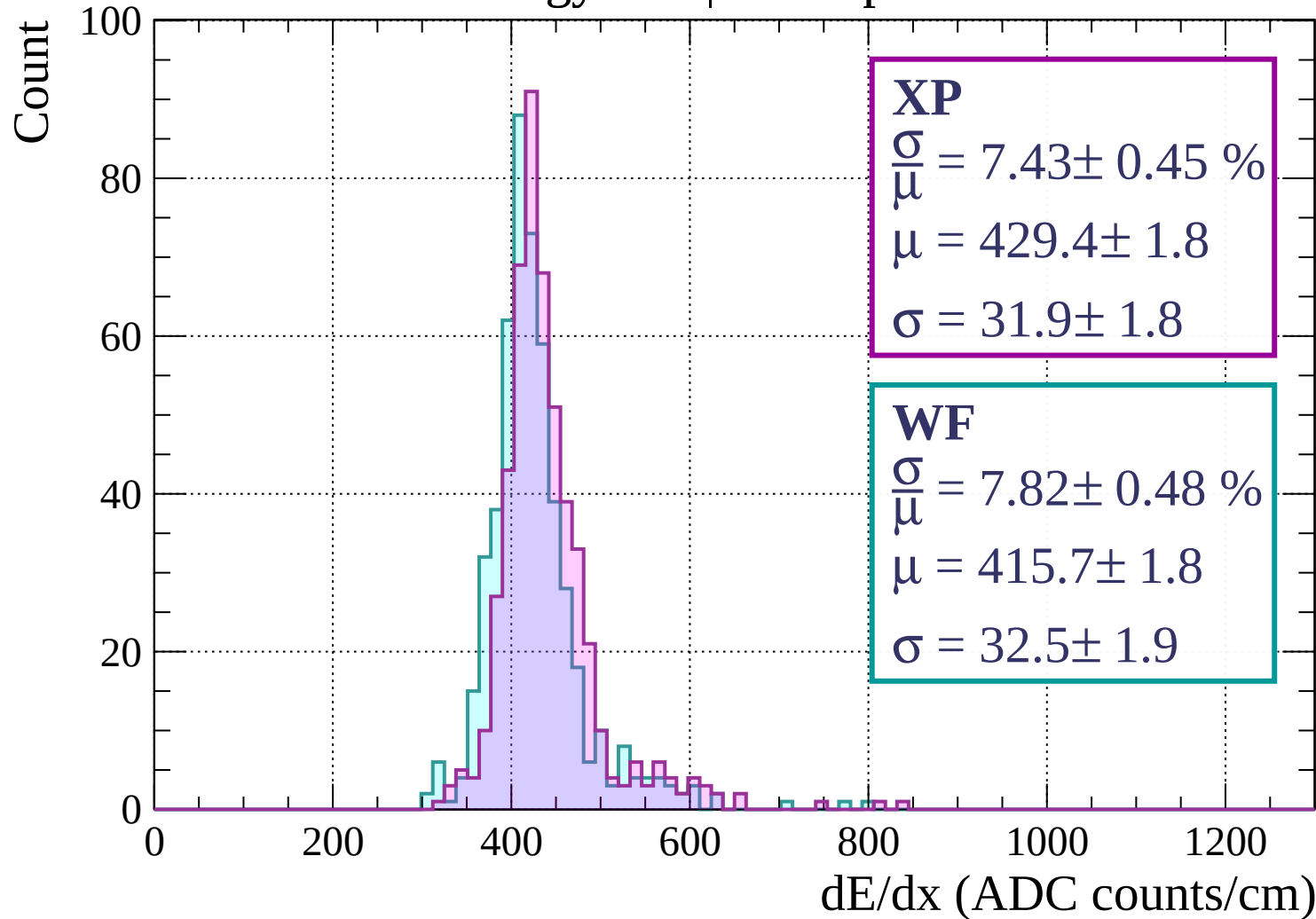
dE/dx (ADC counts/cm)

Energy loss | $400 < p < 480$

Count

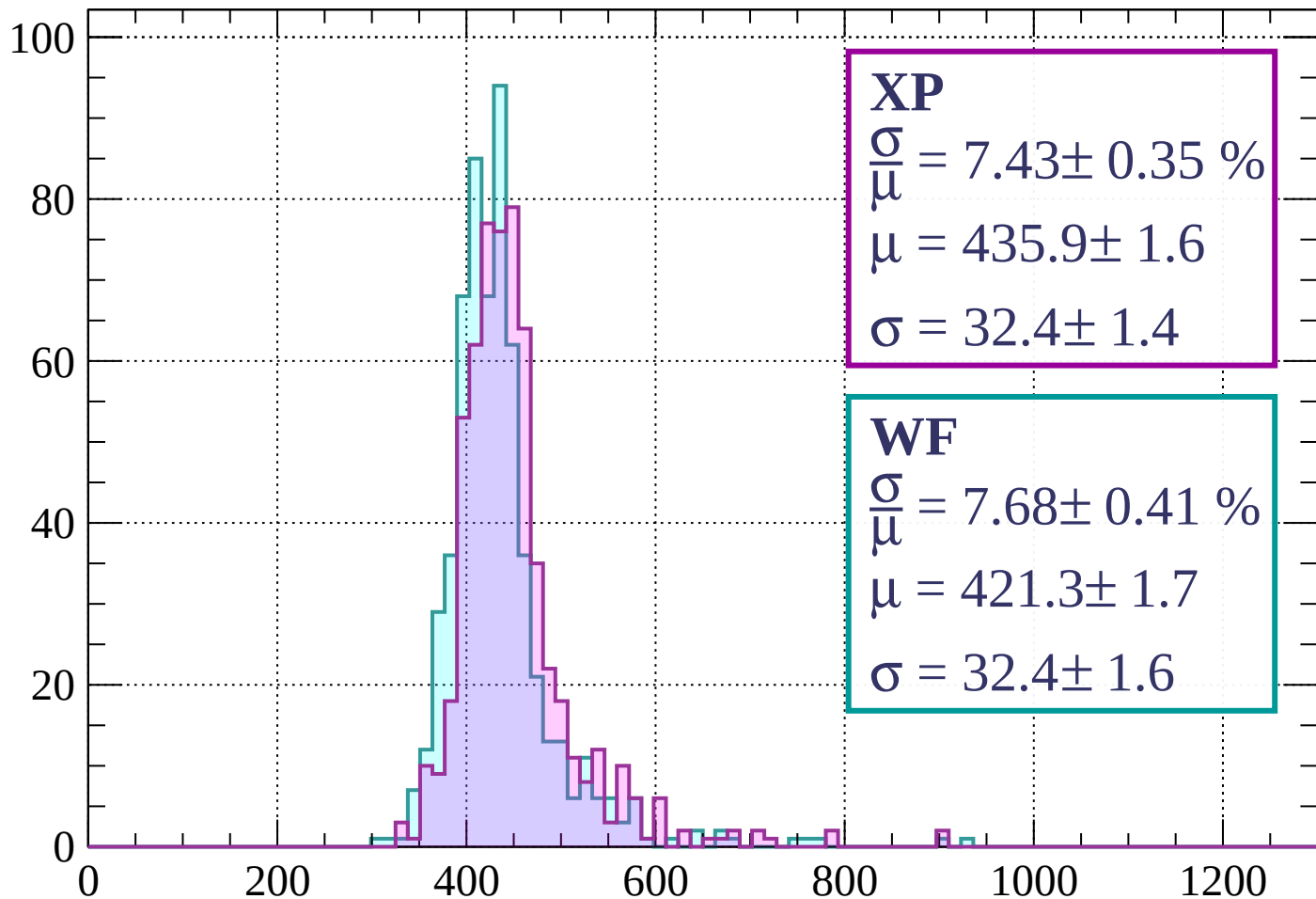


Energy loss | $480 < p < 560$



Energy loss | $560 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 7.43 \pm 0.35 \%$$

$$\mu = 435.9 \pm 1.6$$

$$\sigma = 32.4 \pm 1.4$$

WF

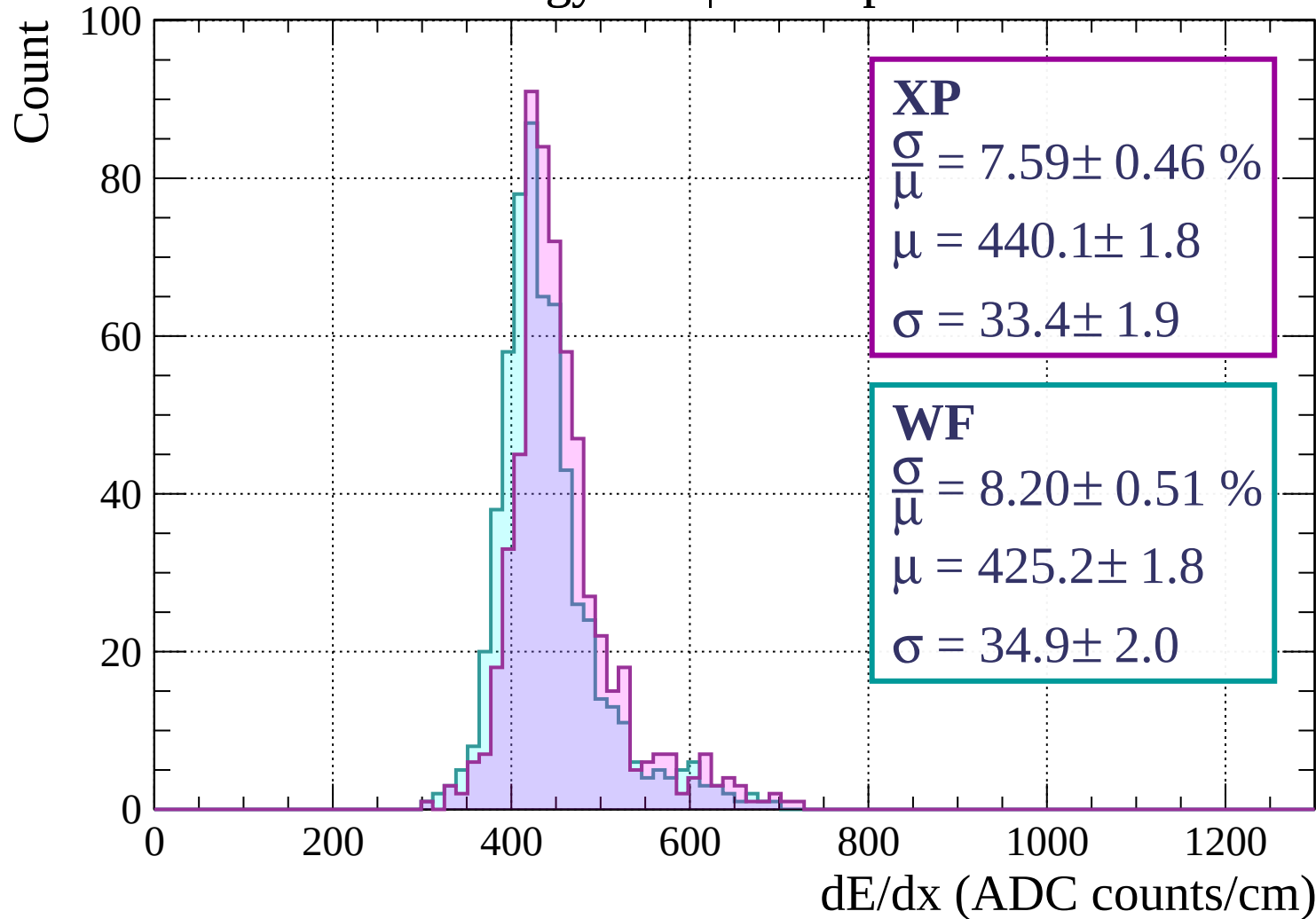
$$\frac{\sigma}{\mu} = 7.68 \pm 0.41 \%$$

$$\mu = 421.3 \pm 1.7$$

$$\sigma = 32.4 \pm 1.6$$

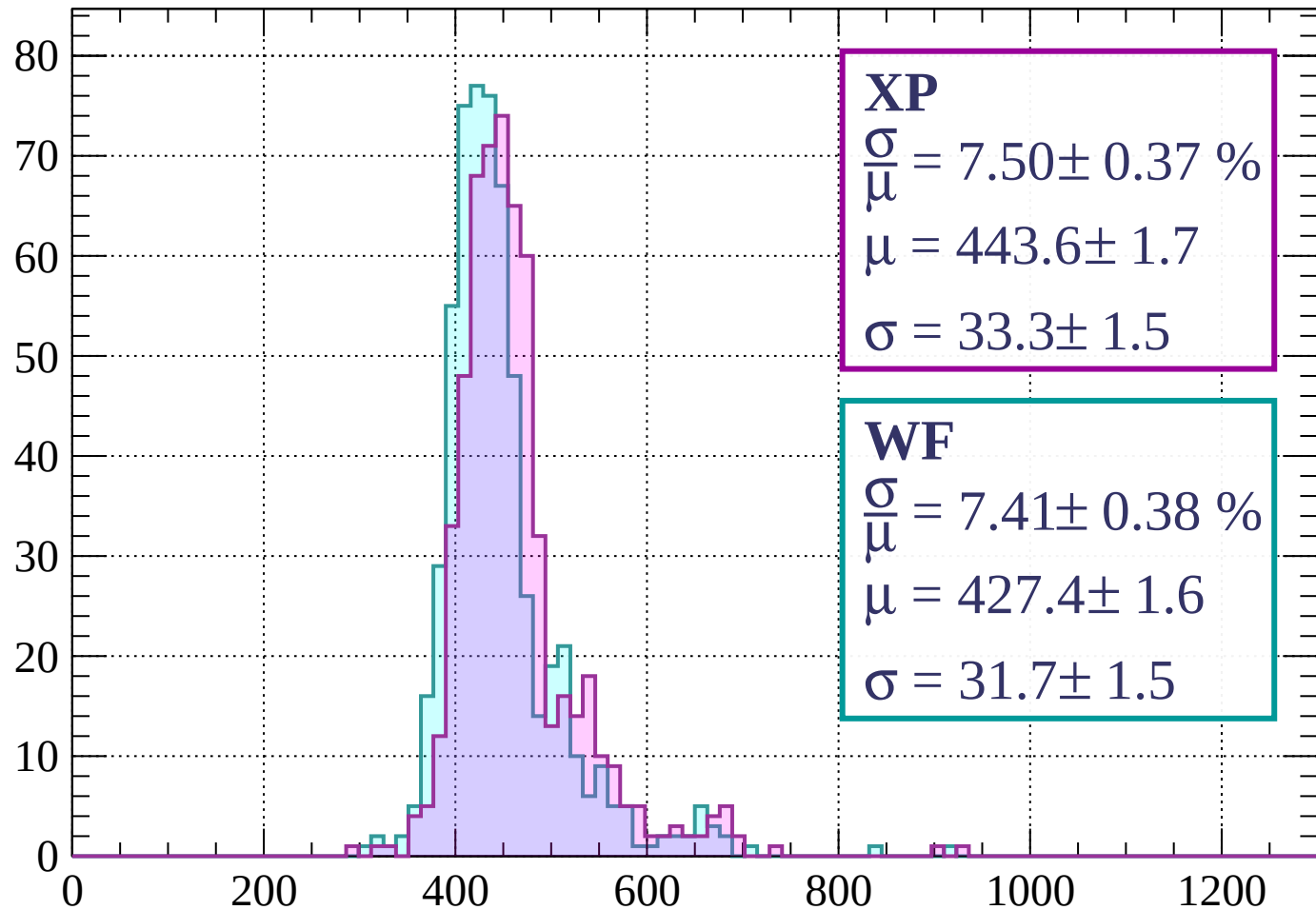
dE/dx (ADC counts/cm)

Energy loss | $640 < p < 720$



Energy loss | $720 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 7.50 \pm 0.37 \%$$

$$\mu = 443.6 \pm 1.7$$

$$\sigma = 33.3 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 7.41 \pm 0.38 \%$$

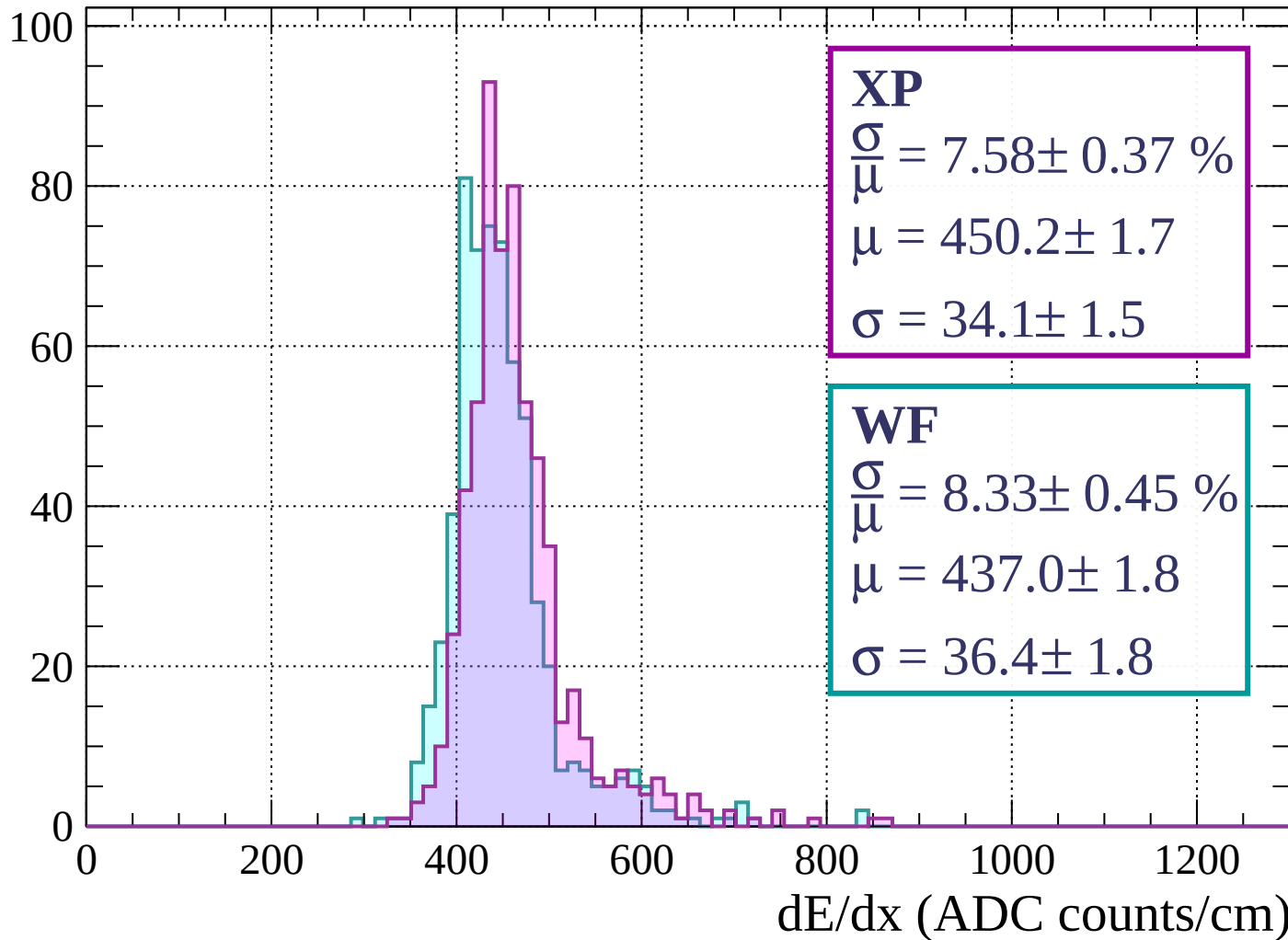
$$\mu = 427.4 \pm 1.6$$

$$\sigma = 31.7 \pm 1.5$$

dE/dx (ADC counts/cm)

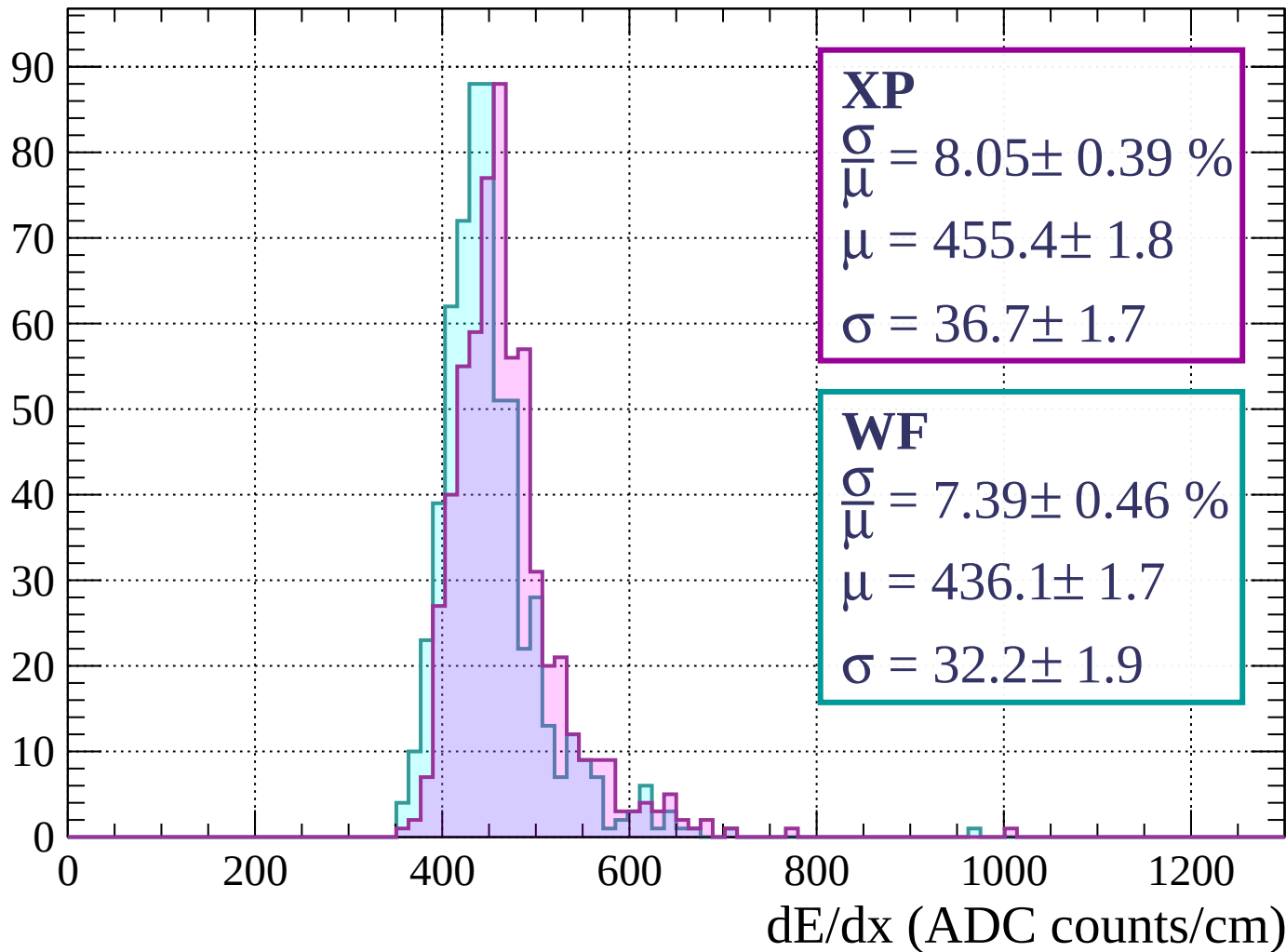
Energy loss | $800 < p < 880$

Count



Energy loss | $880 < p < 960$

Count



Energy loss | $960 < p < 1040$

Count

100

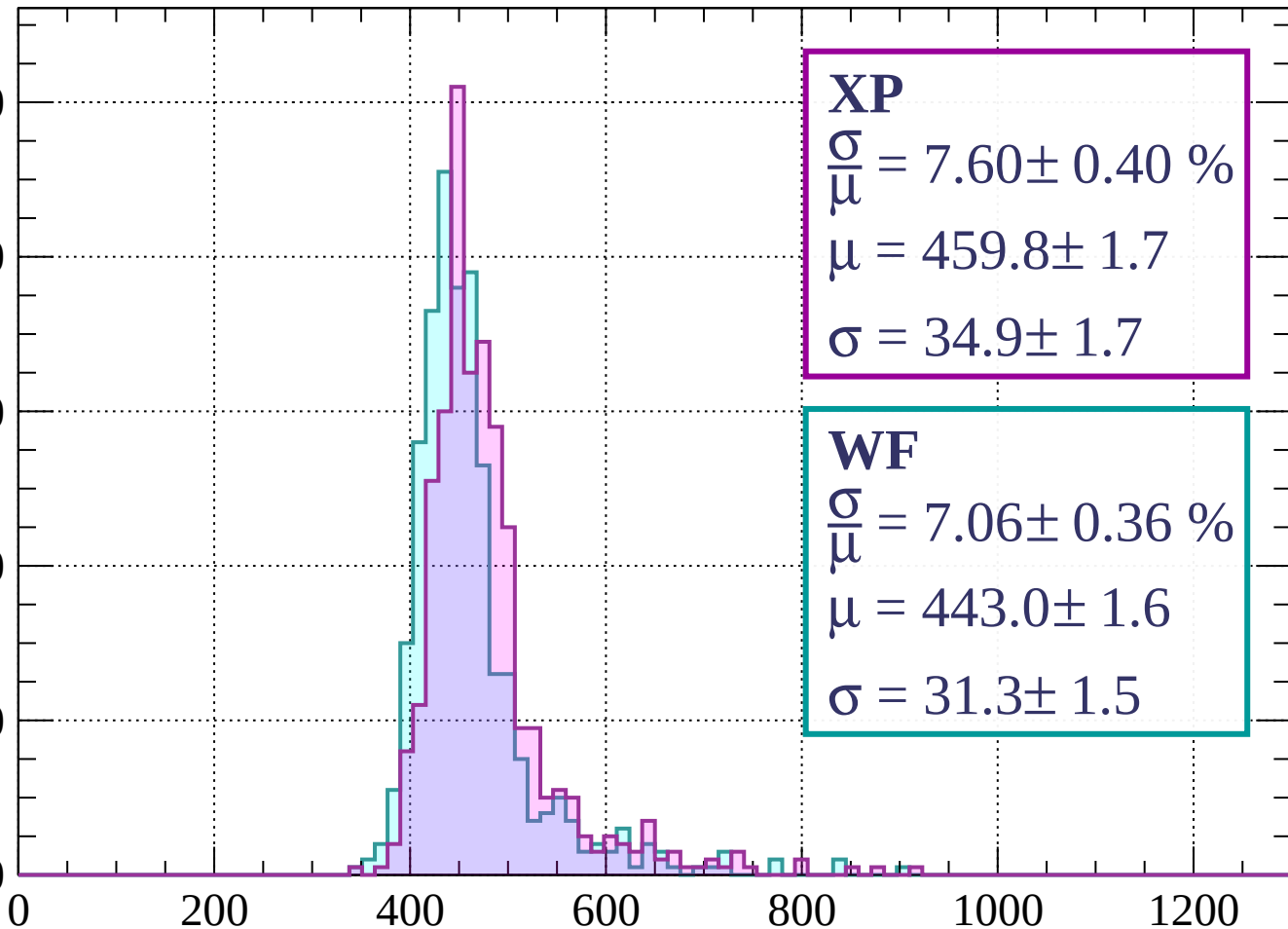
80

60

40

20

0



XP

$$\frac{\sigma}{\mu} = 7.60 \pm 0.40 \%$$

$$\mu = 459.8 \pm 1.7$$

$$\sigma = 34.9 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 7.06 \pm 0.36 \%$$

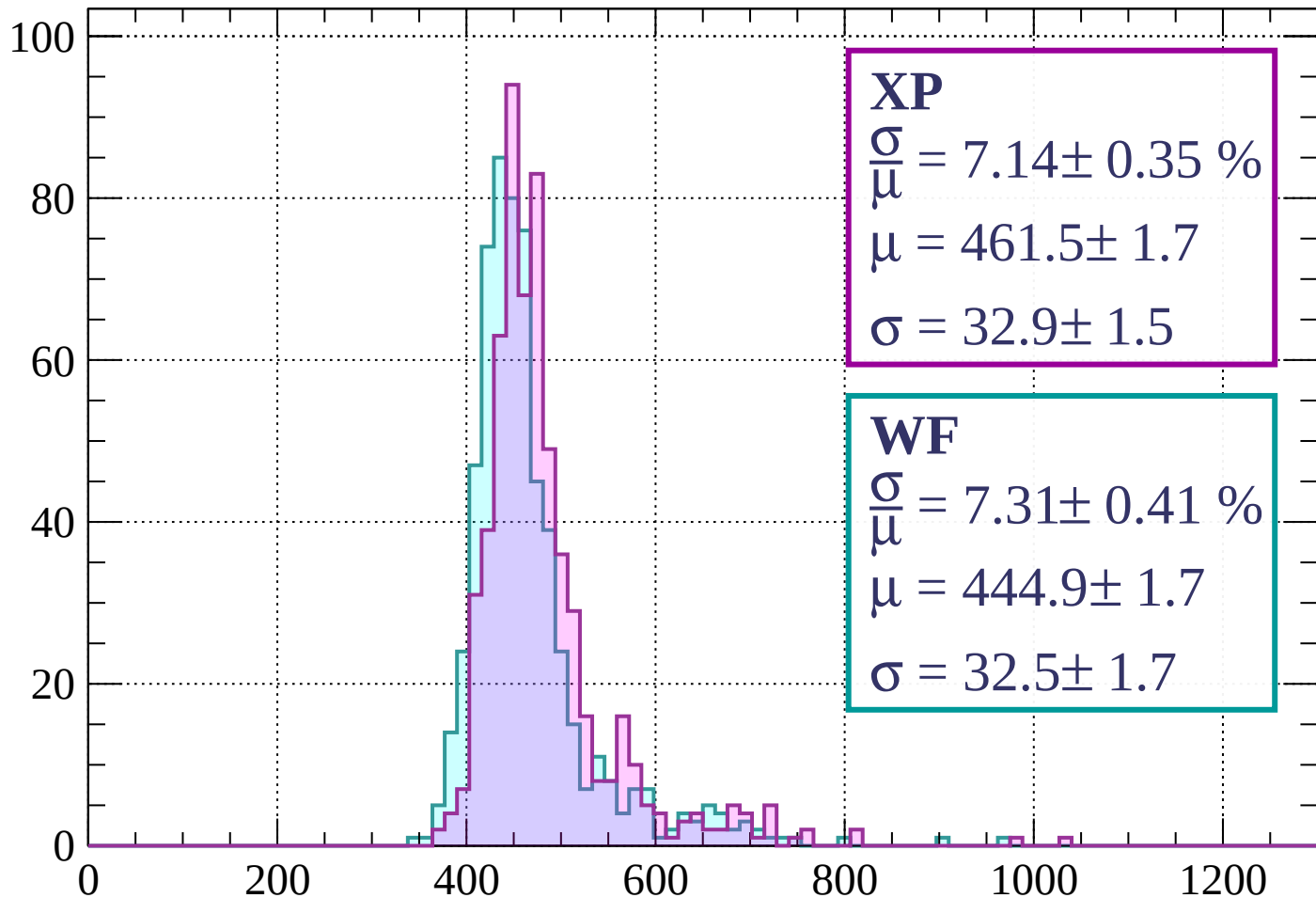
$$\mu = 443.0 \pm 1.6$$

$$\sigma = 31.3 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $1040 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 7.14 \pm 0.35 \%$$

$$\mu = 461.5 \pm 1.7$$

$$\sigma = 32.9 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 7.31 \pm 0.41 \%$$

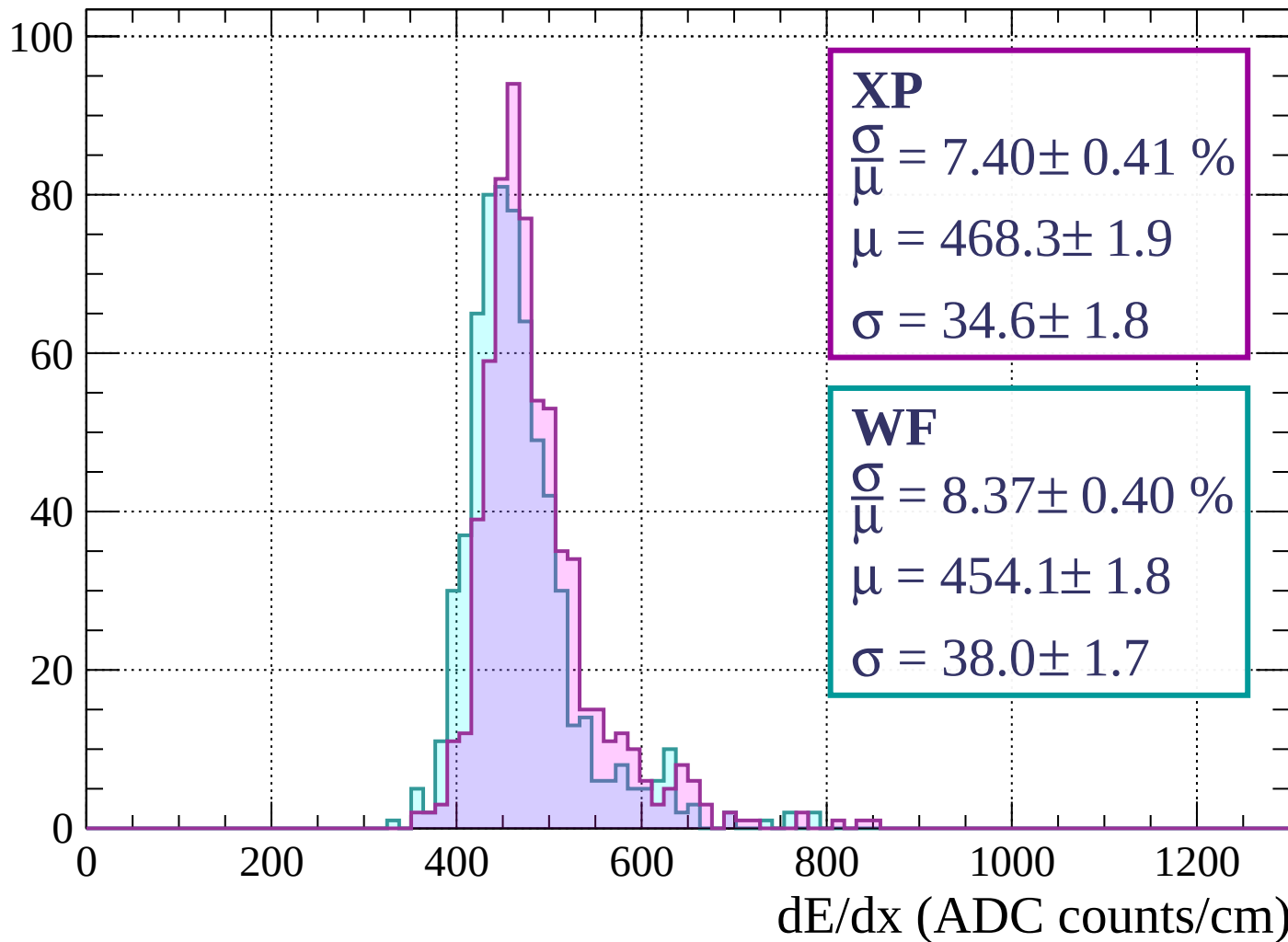
$$\mu = 444.9 \pm 1.7$$

$$\sigma = 32.5 \pm 1.7$$

dE/dx (ADC counts/cm)

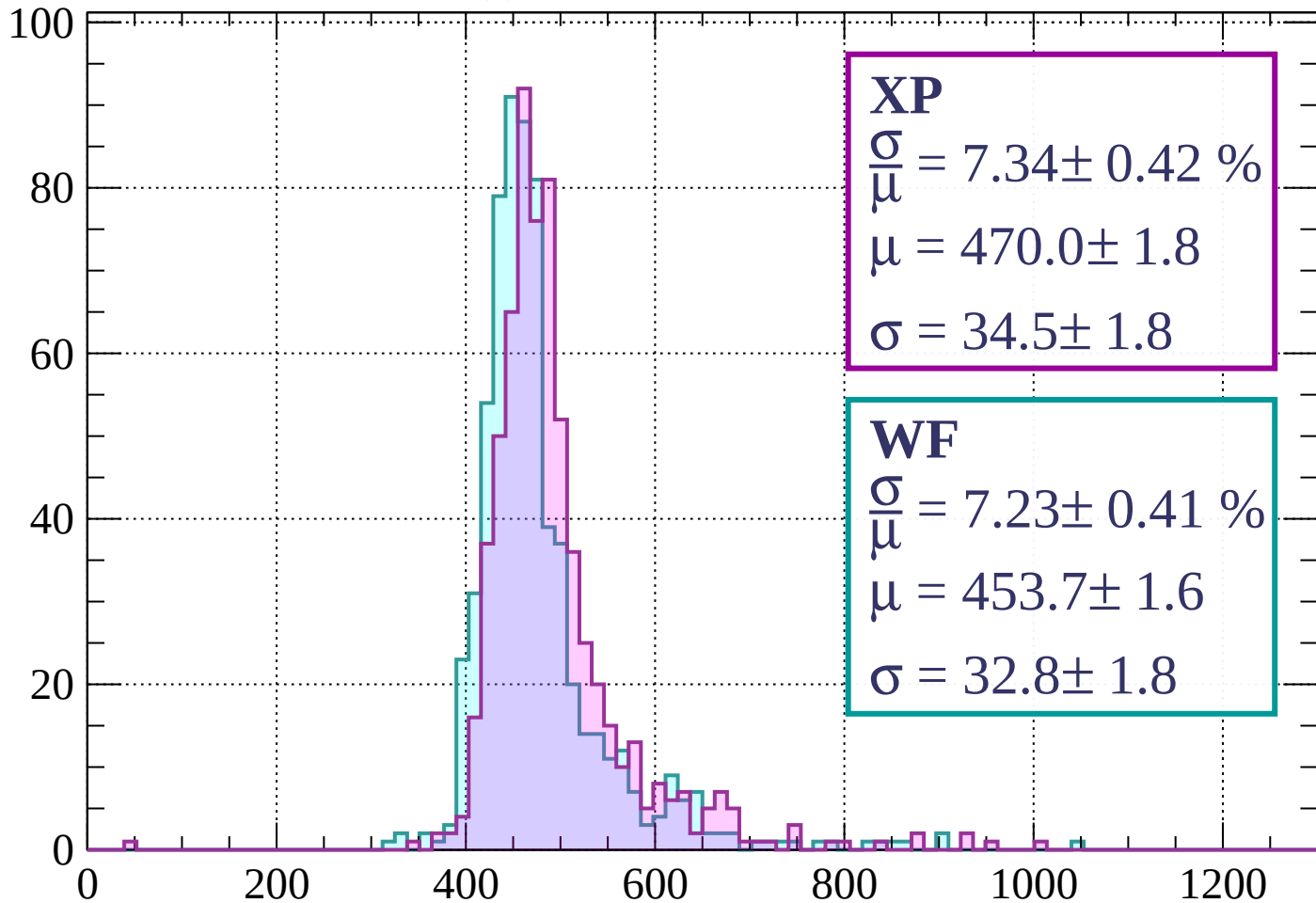
Energy loss | $1120 < p < 1200$

Count



Energy loss | $1200 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 7.34 \pm 0.42 \%$$

$$\mu = 470.0 \pm 1.8$$

$$\sigma = 34.5 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 7.23 \pm 0.41 \%$$

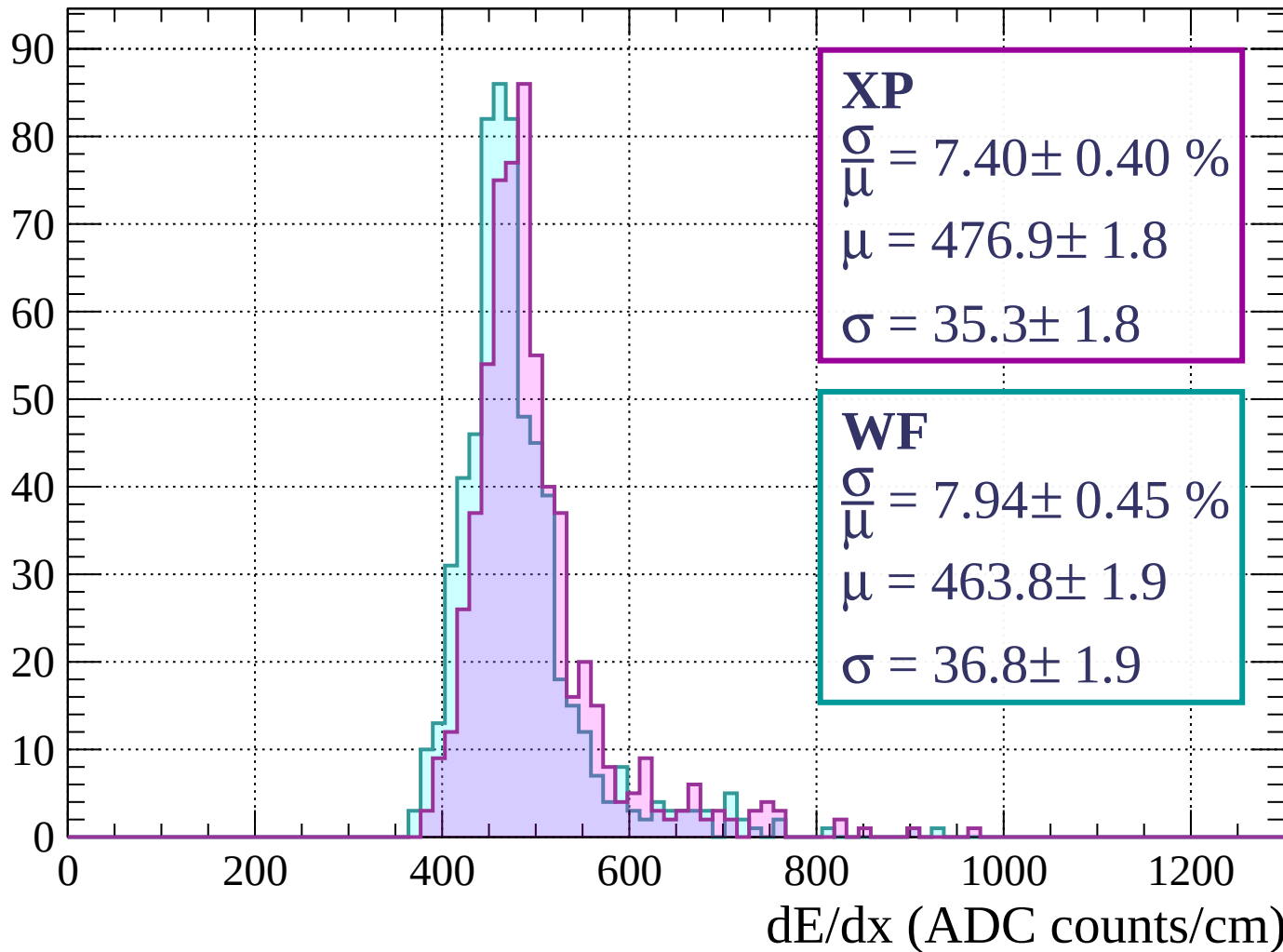
$$\mu = 453.7 \pm 1.6$$

$$\sigma = 32.8 \pm 1.8$$

dE/dx (ADC counts/cm)

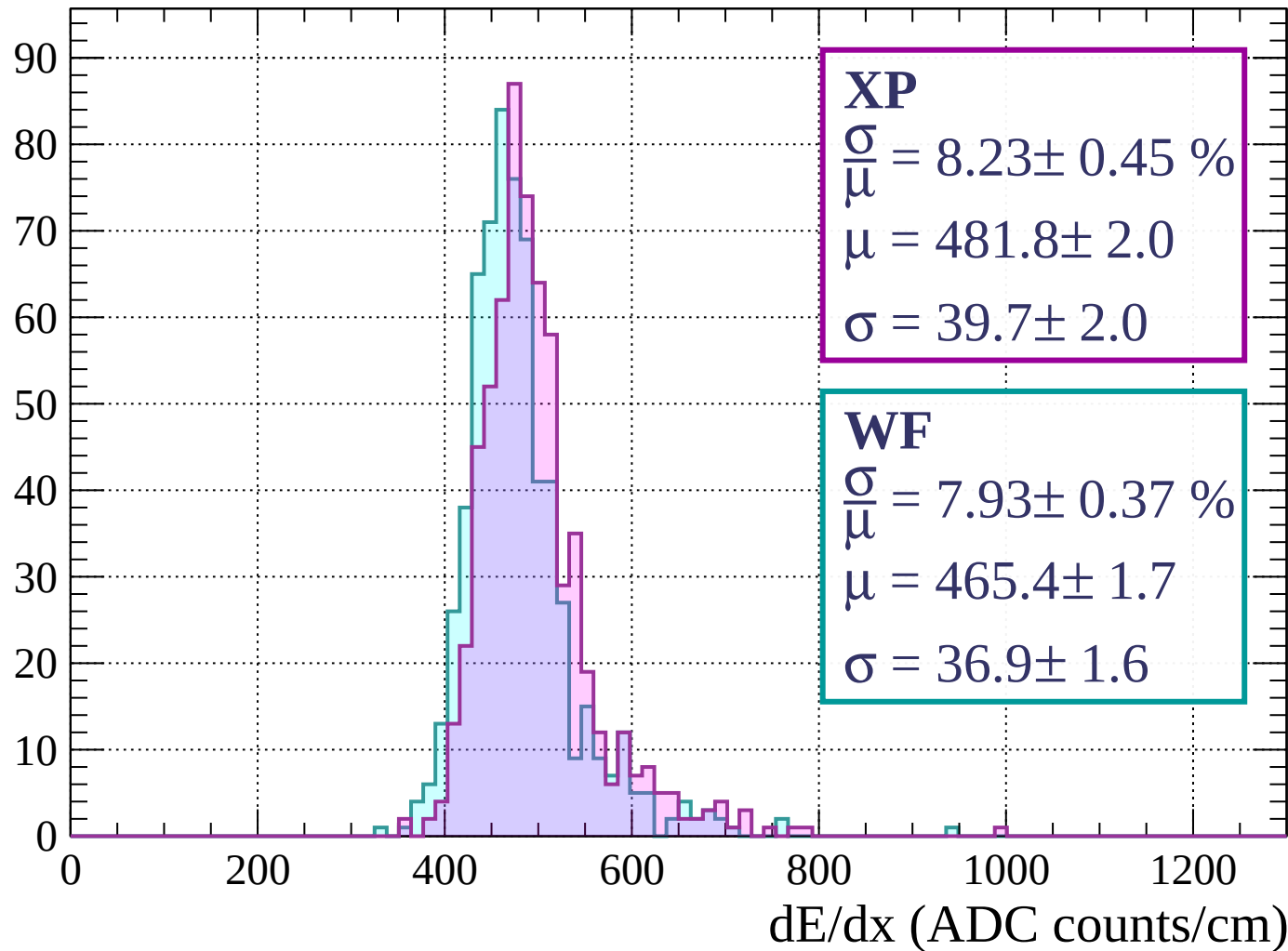
Energy loss | $1280 < p < 1360$

Count



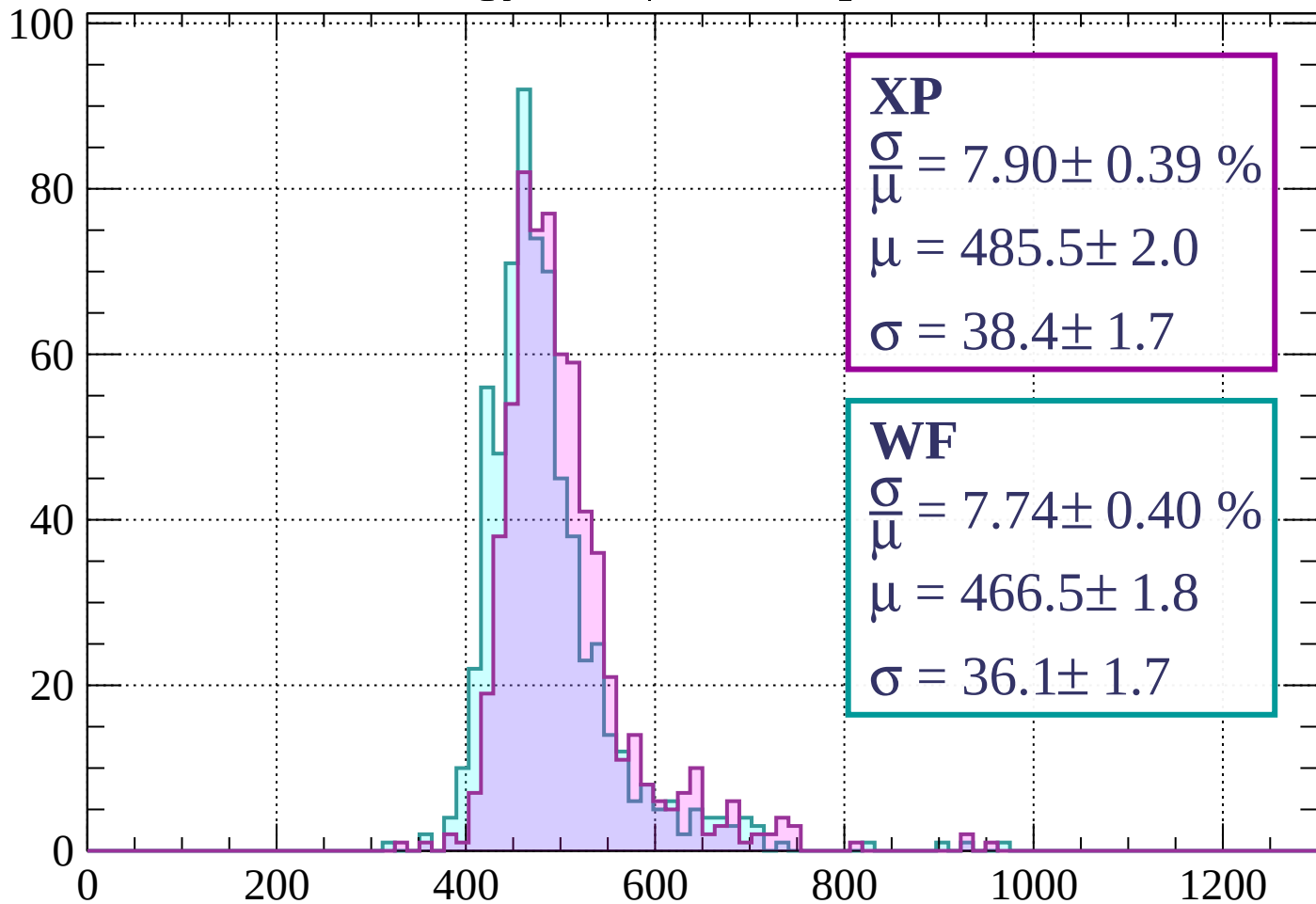
Energy loss | $1360 < p < 1440$

Count



Energy loss | $1440 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 7.90 \pm 0.39 \%$$

$$\mu = 485.5 \pm 2.0$$

$$\sigma = 38.4 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 7.74 \pm 0.40 \%$$

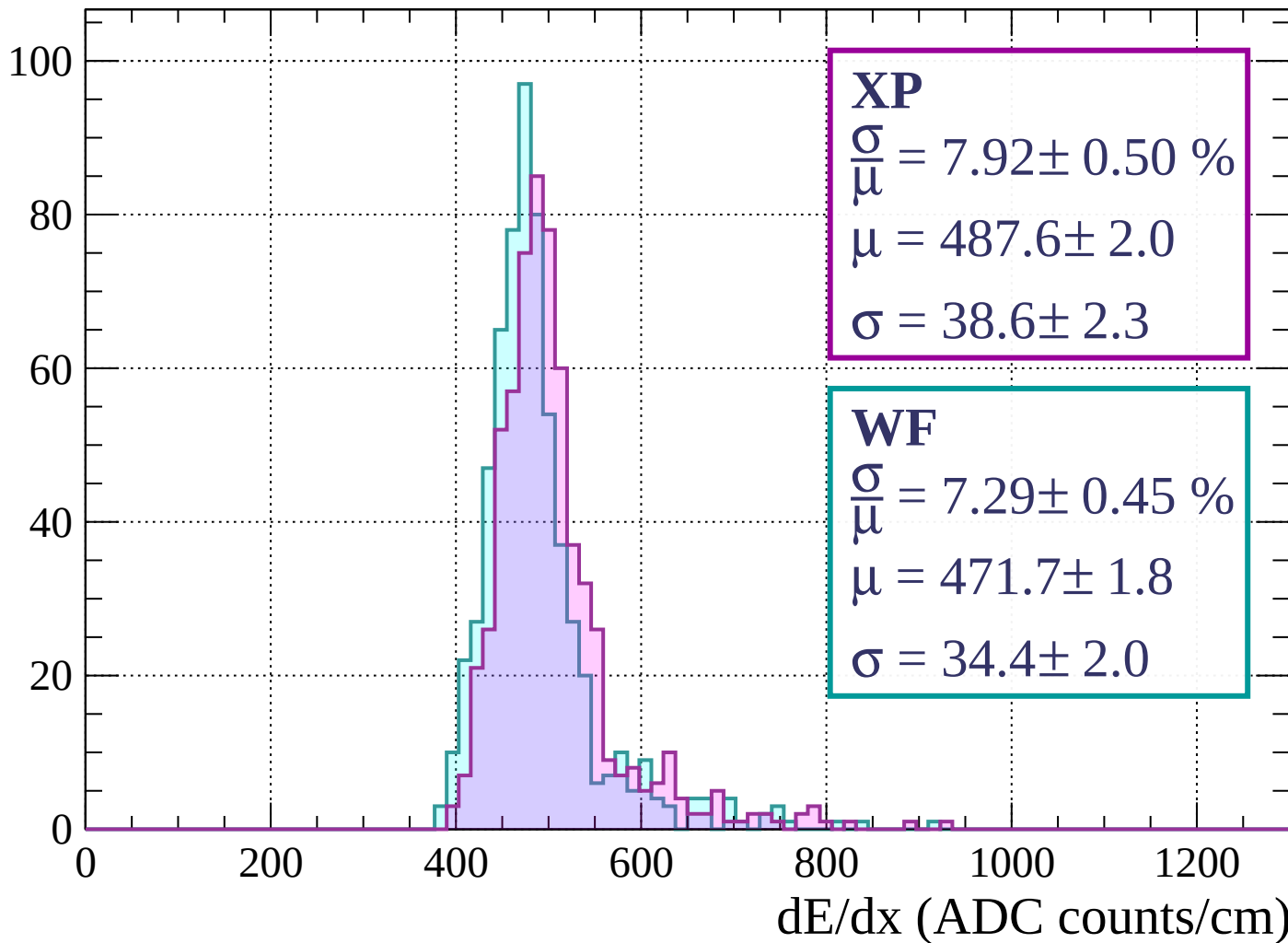
$$\mu = 466.5 \pm 1.8$$

$$\sigma = 36.1 \pm 1.7$$

dE/dx (ADC counts/cm)

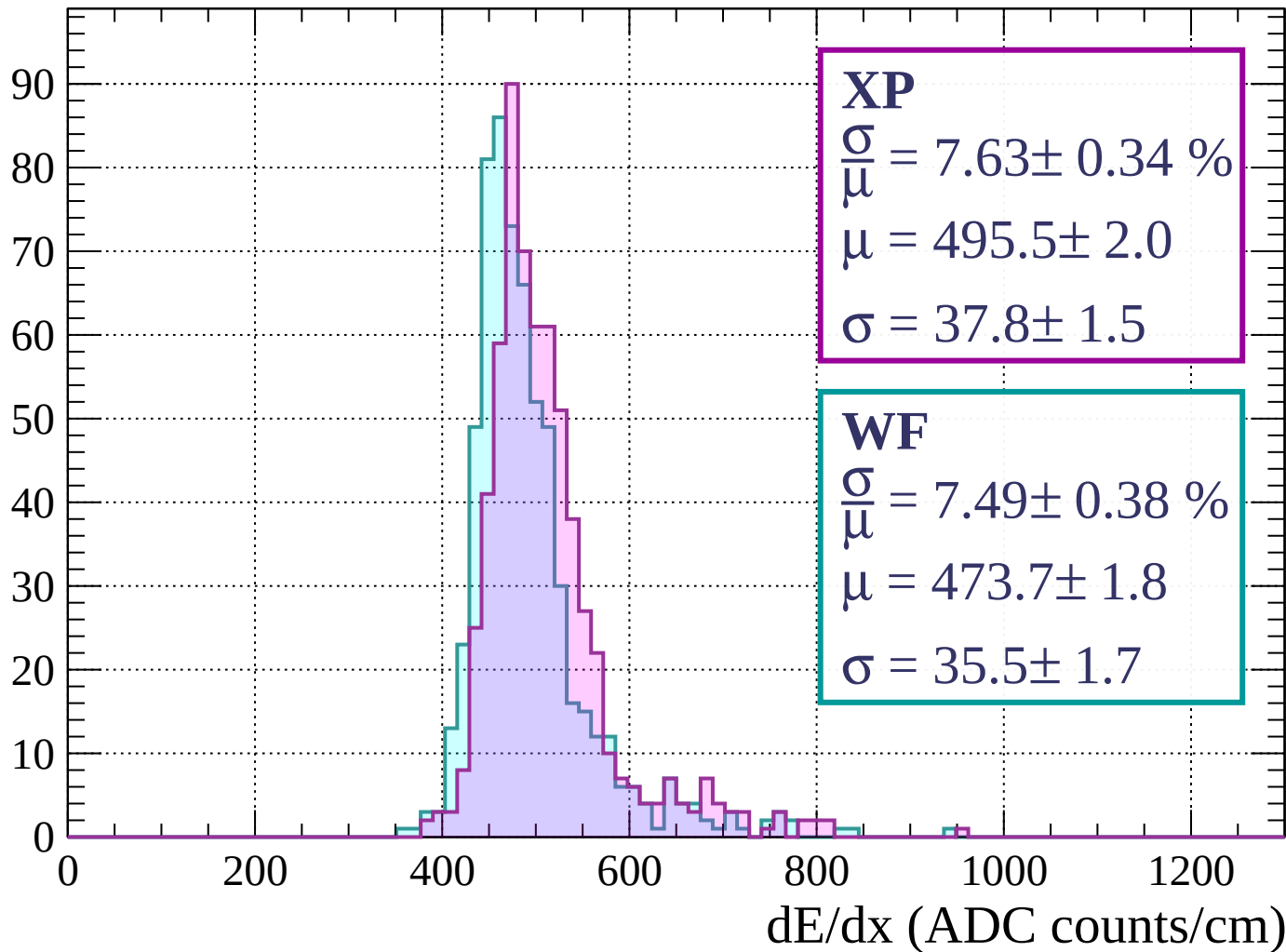
Energy loss | $1520 < p < 1600$

Count



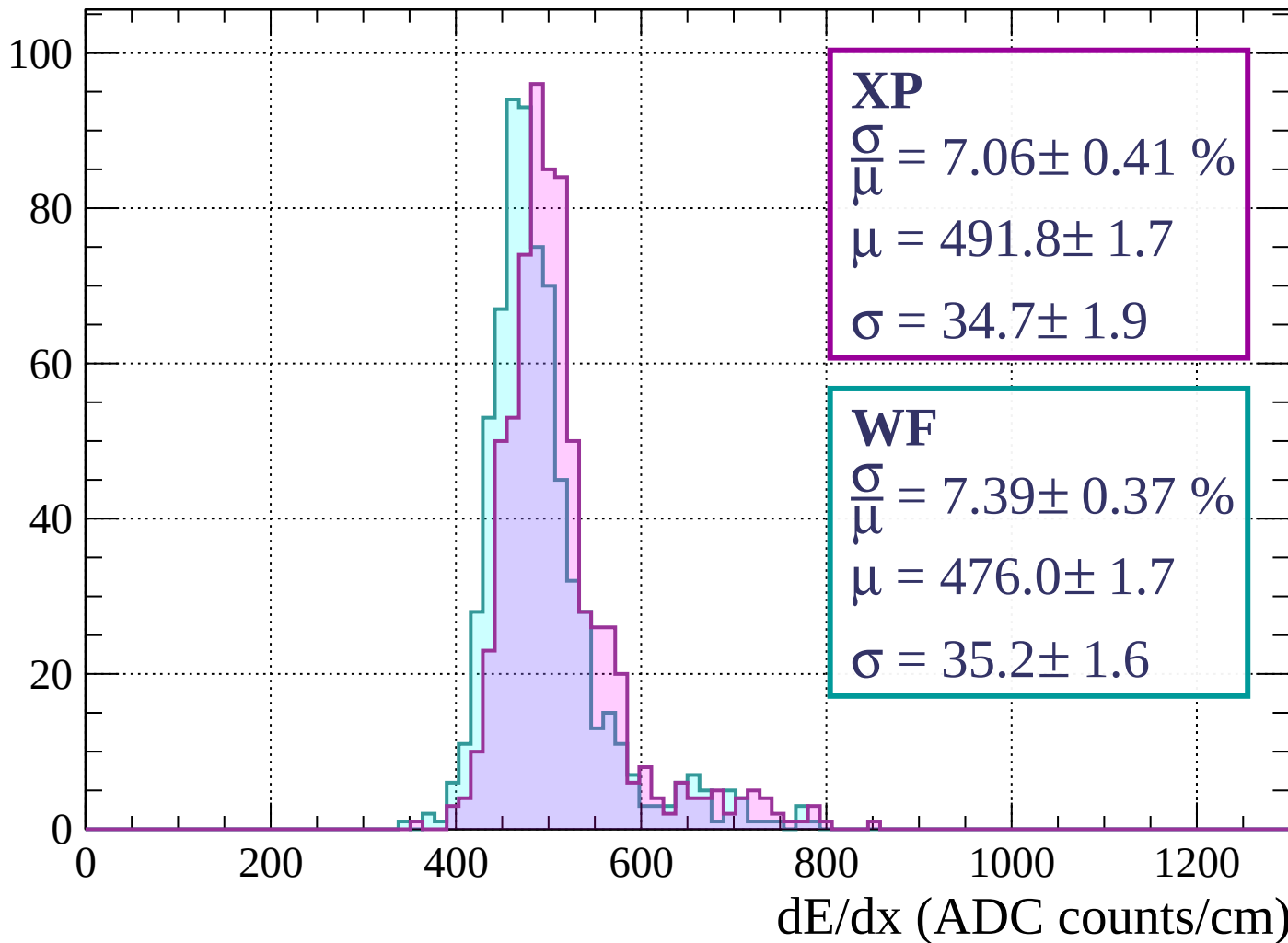
Energy loss | $1600 < p < 1680$

Count



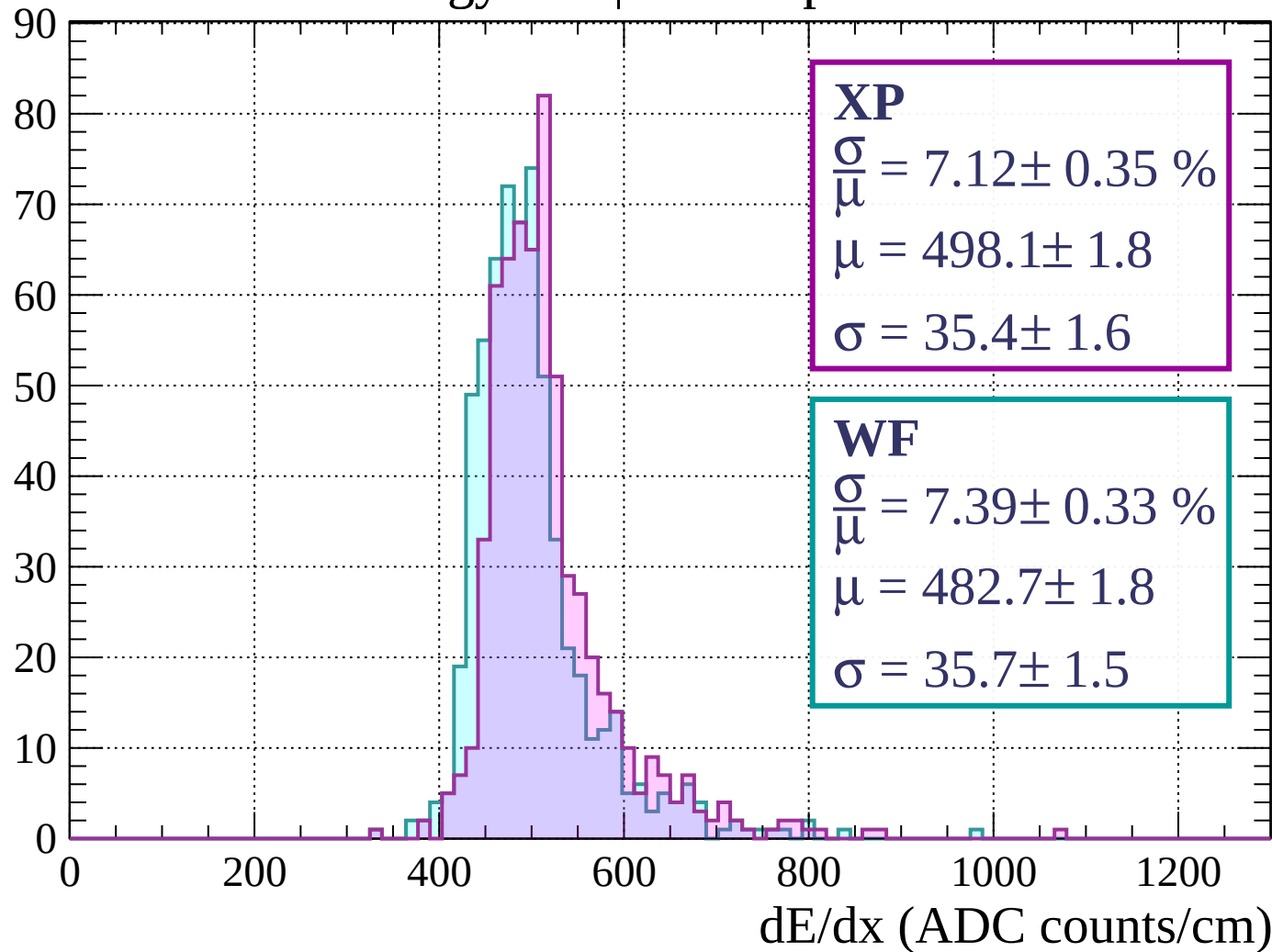
Energy loss | $1680 < p < 1760$

Count



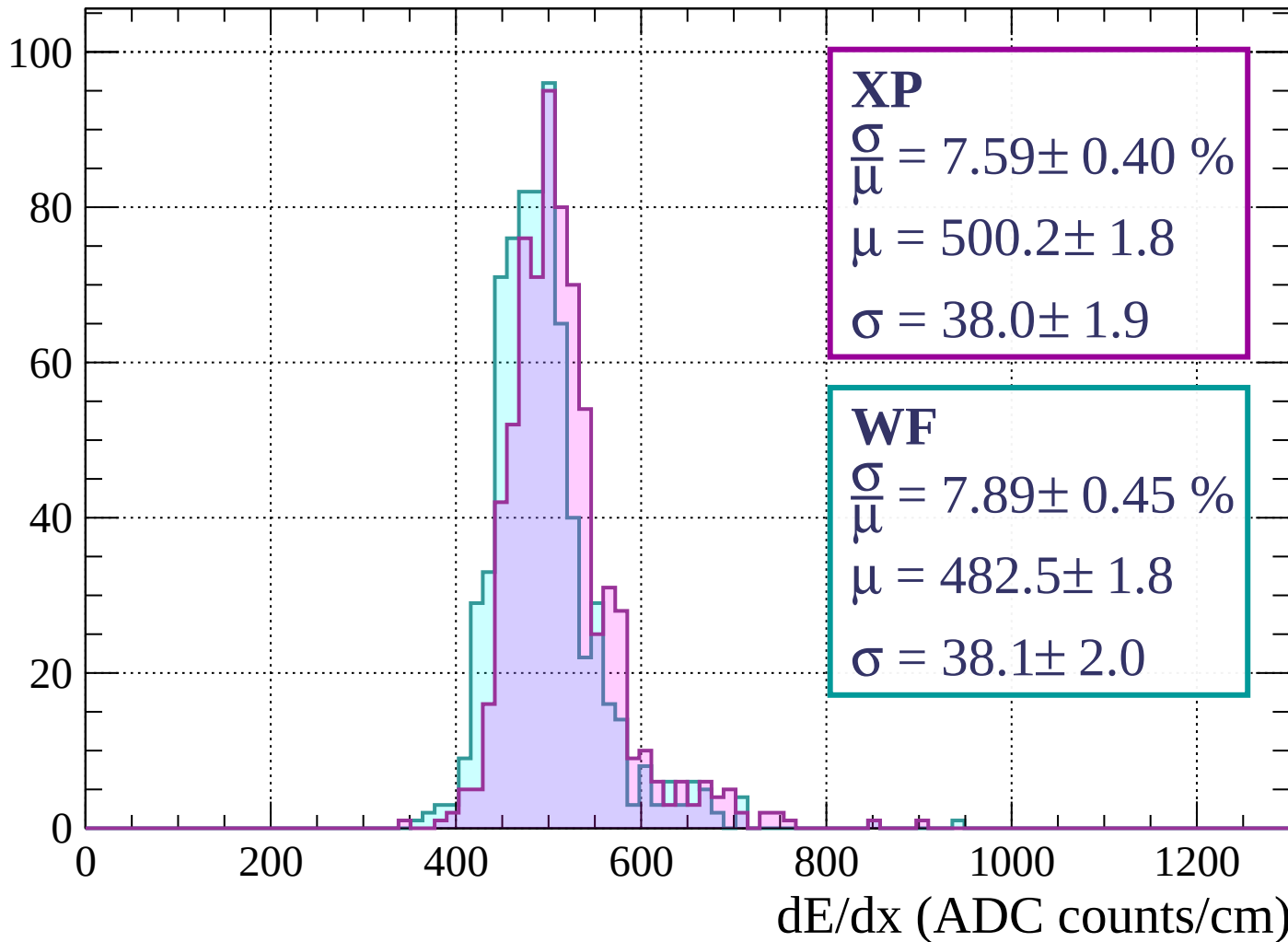
Energy loss | $1760 < p < 1840$

Count



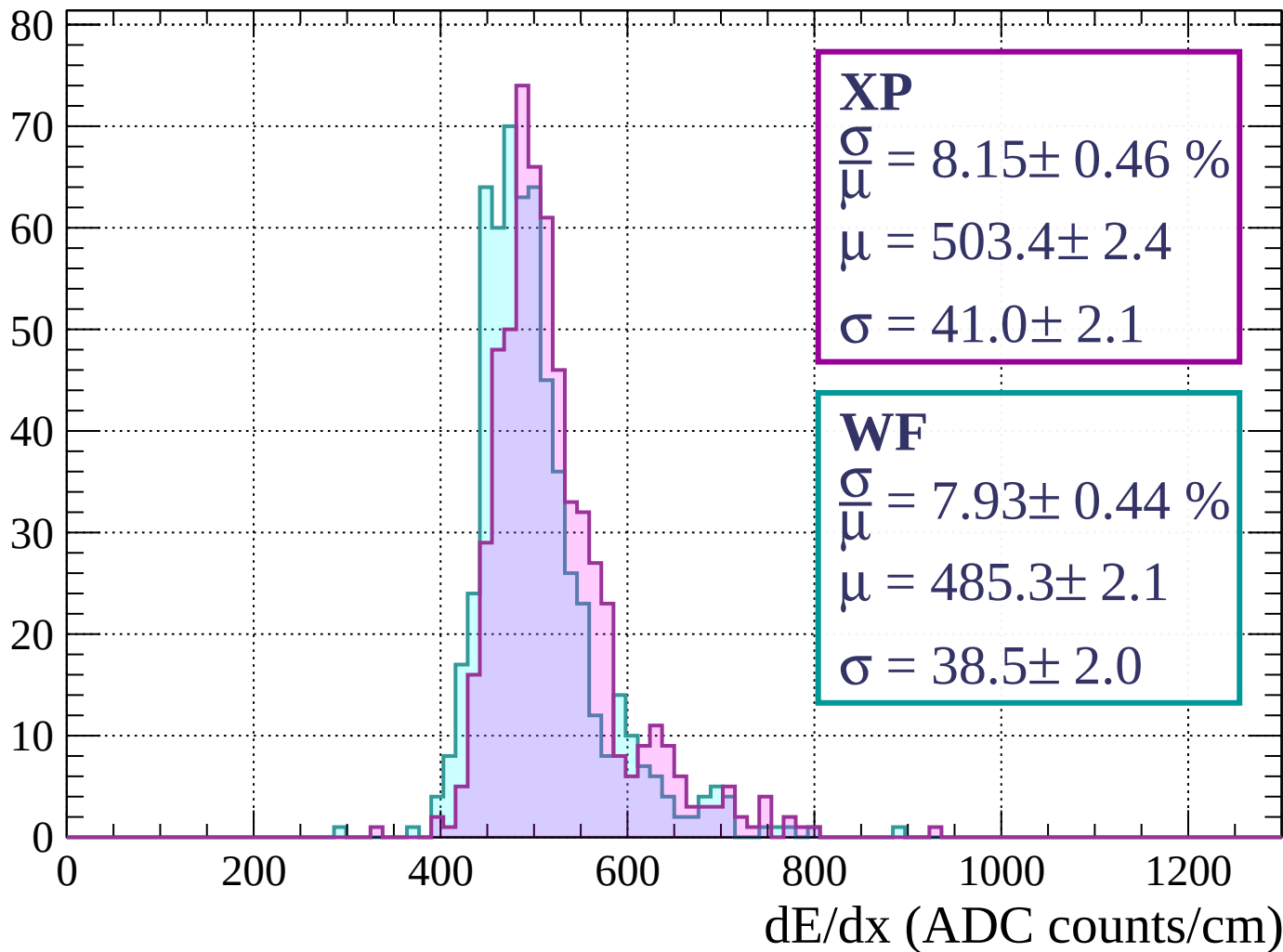
Energy loss | $1840 < p < 1920$

Count



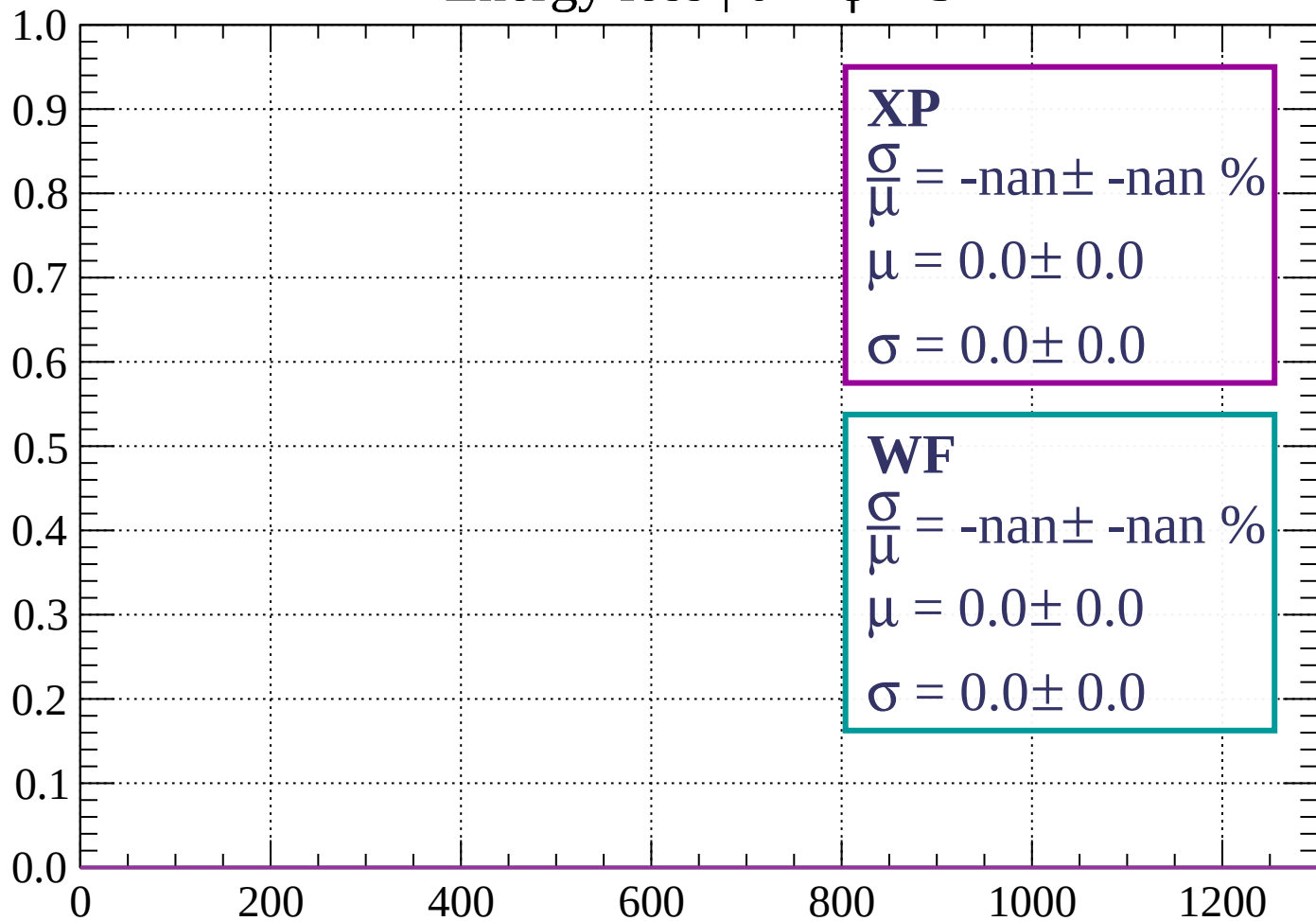
Energy loss | $1920 < p < 2000$

Count



Energy loss | $0 < \phi < 3$

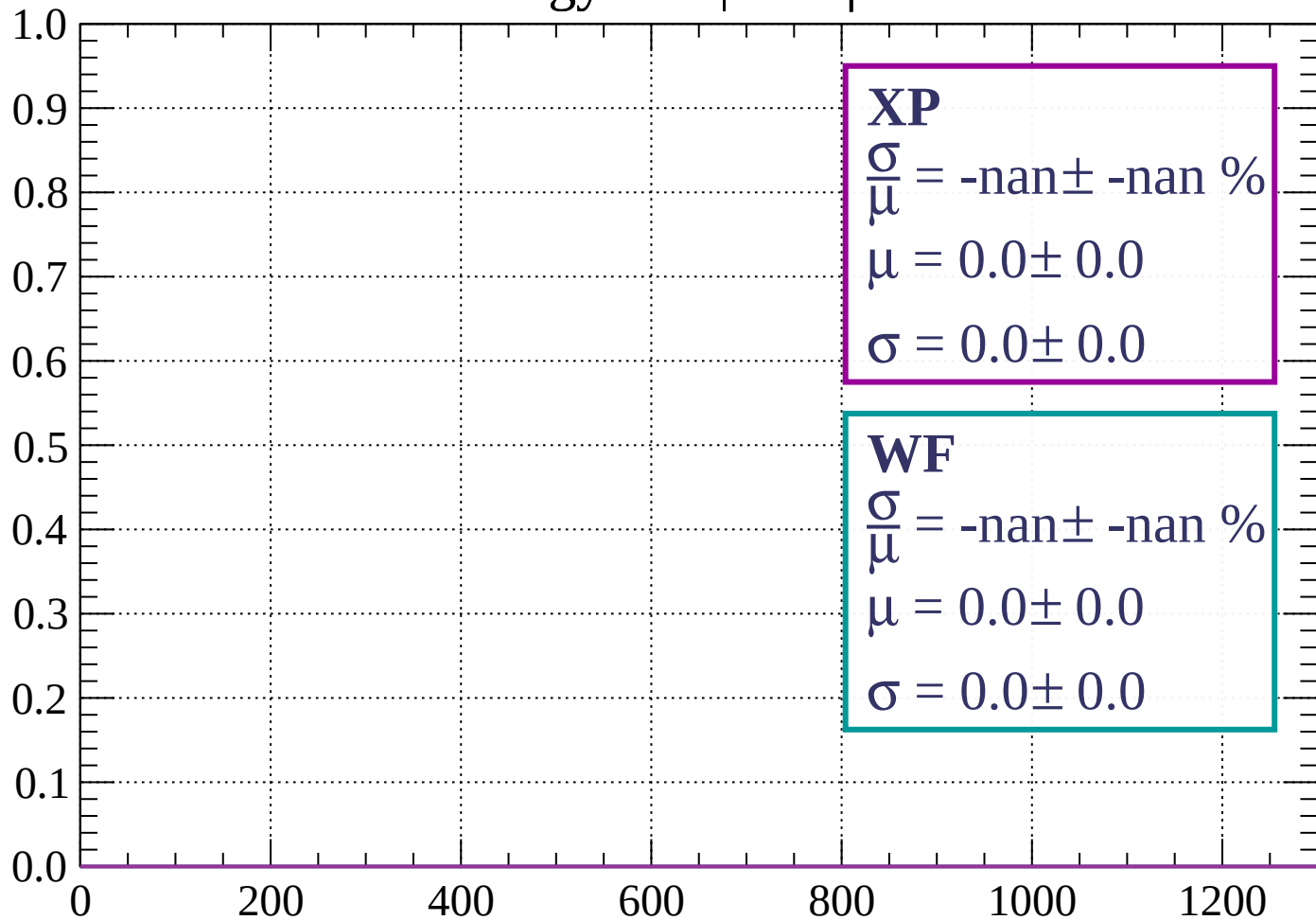
Count



dE/dx (ADC counts/cm)

Energy loss | $3 < \phi < 6$

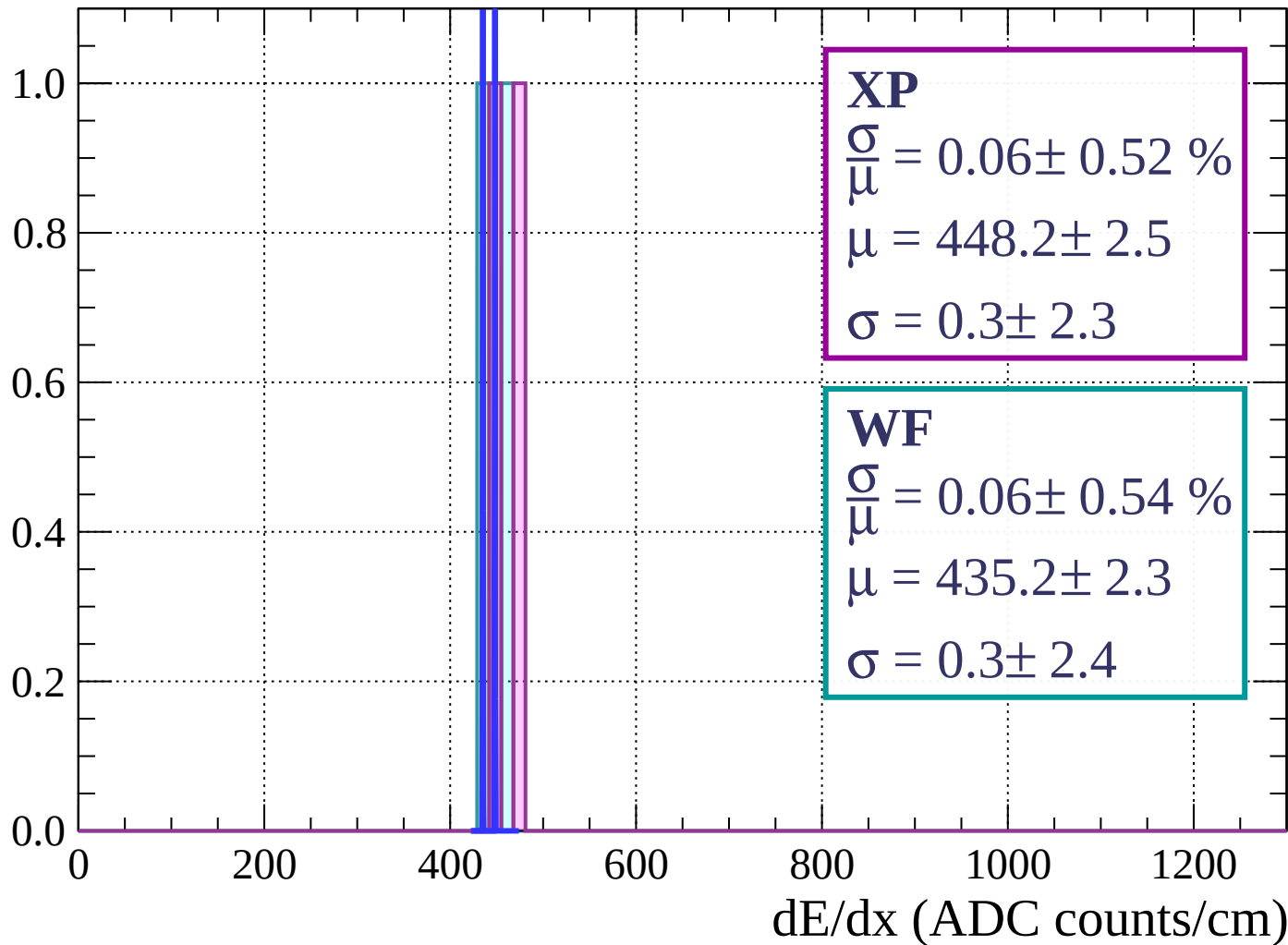
Count



dE/dx (ADC counts/cm)

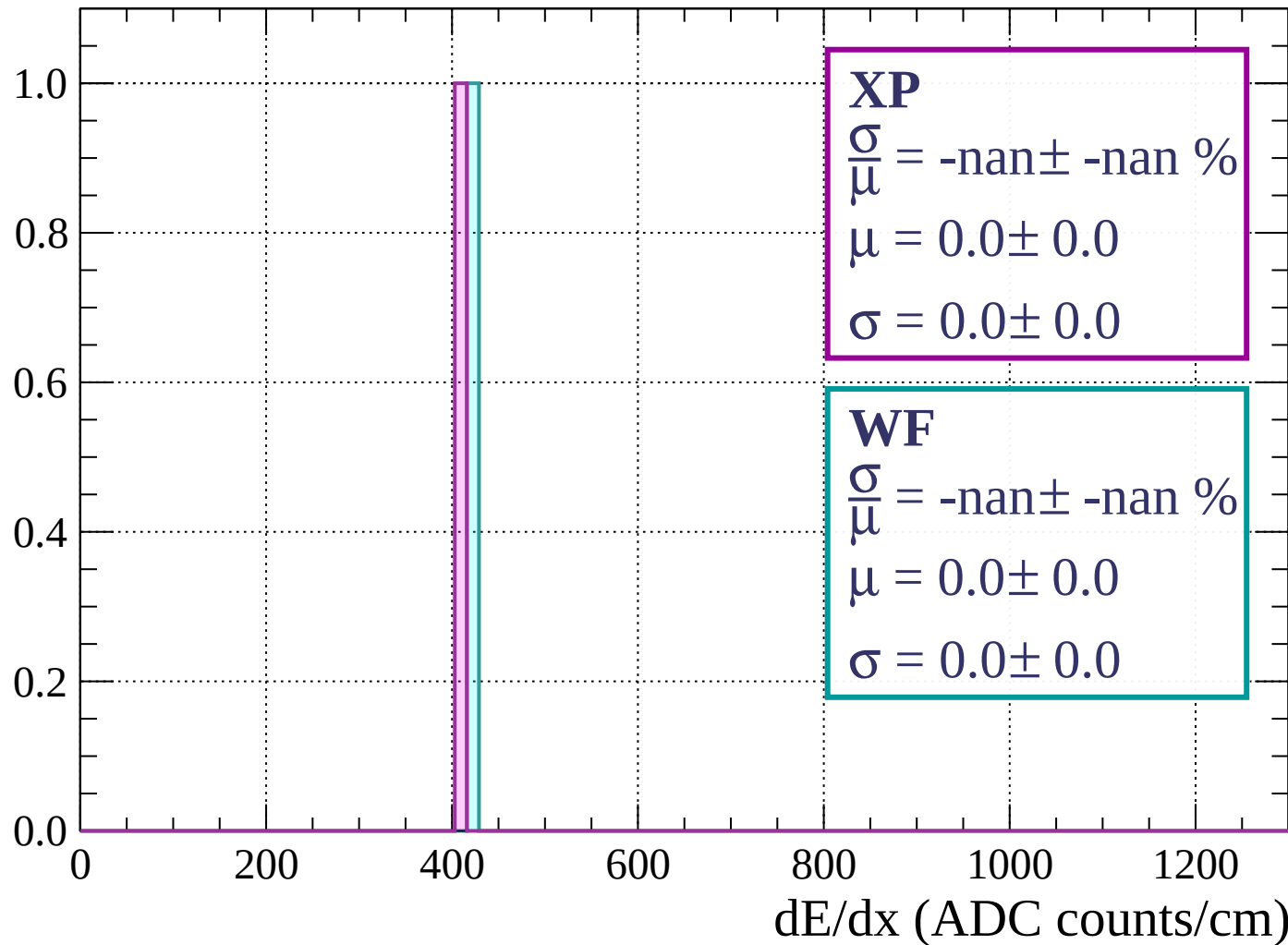
Energy loss | $6 < \phi < 9$

Count



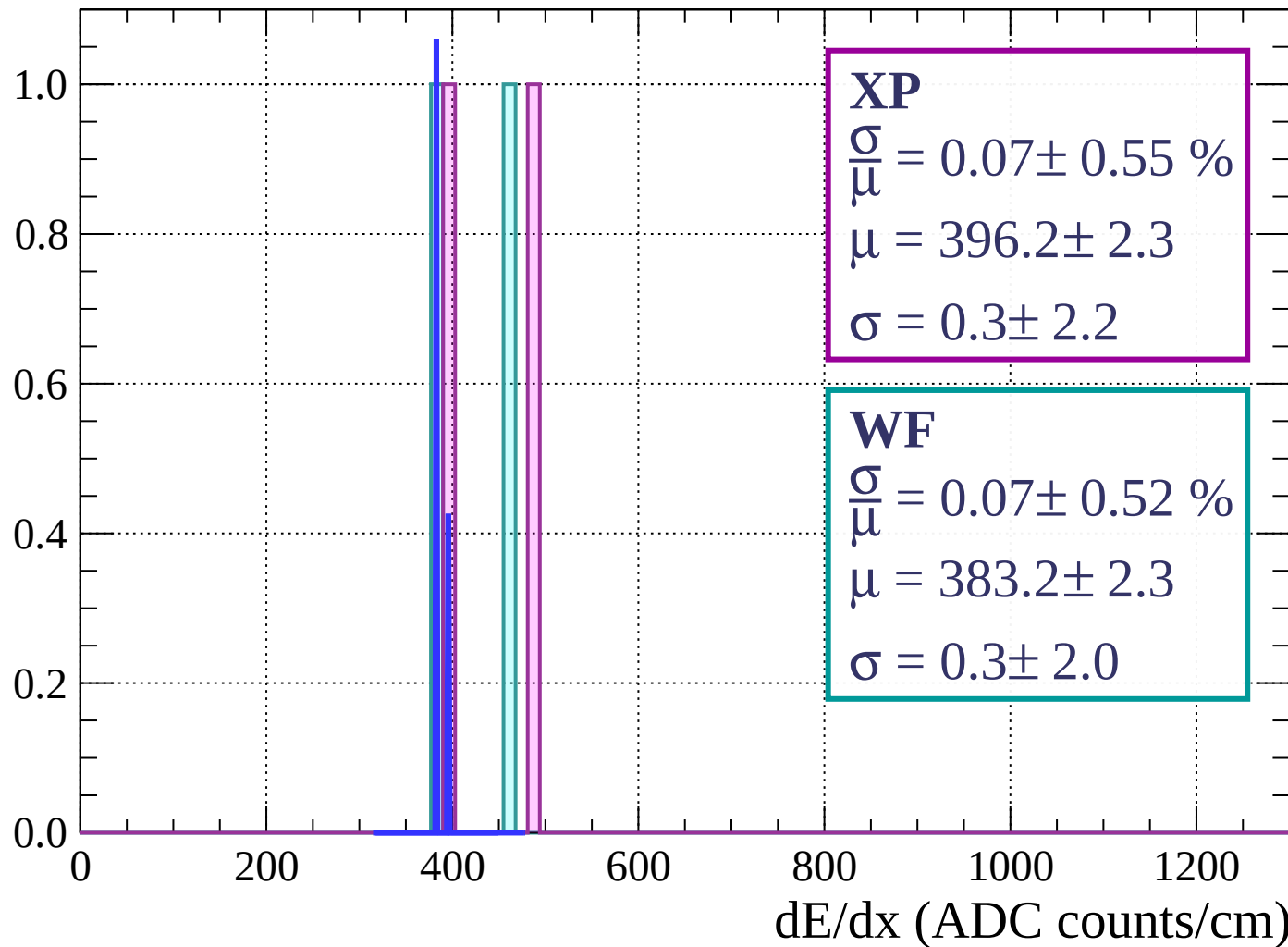
Energy loss | $9 < \phi < 12$

Count



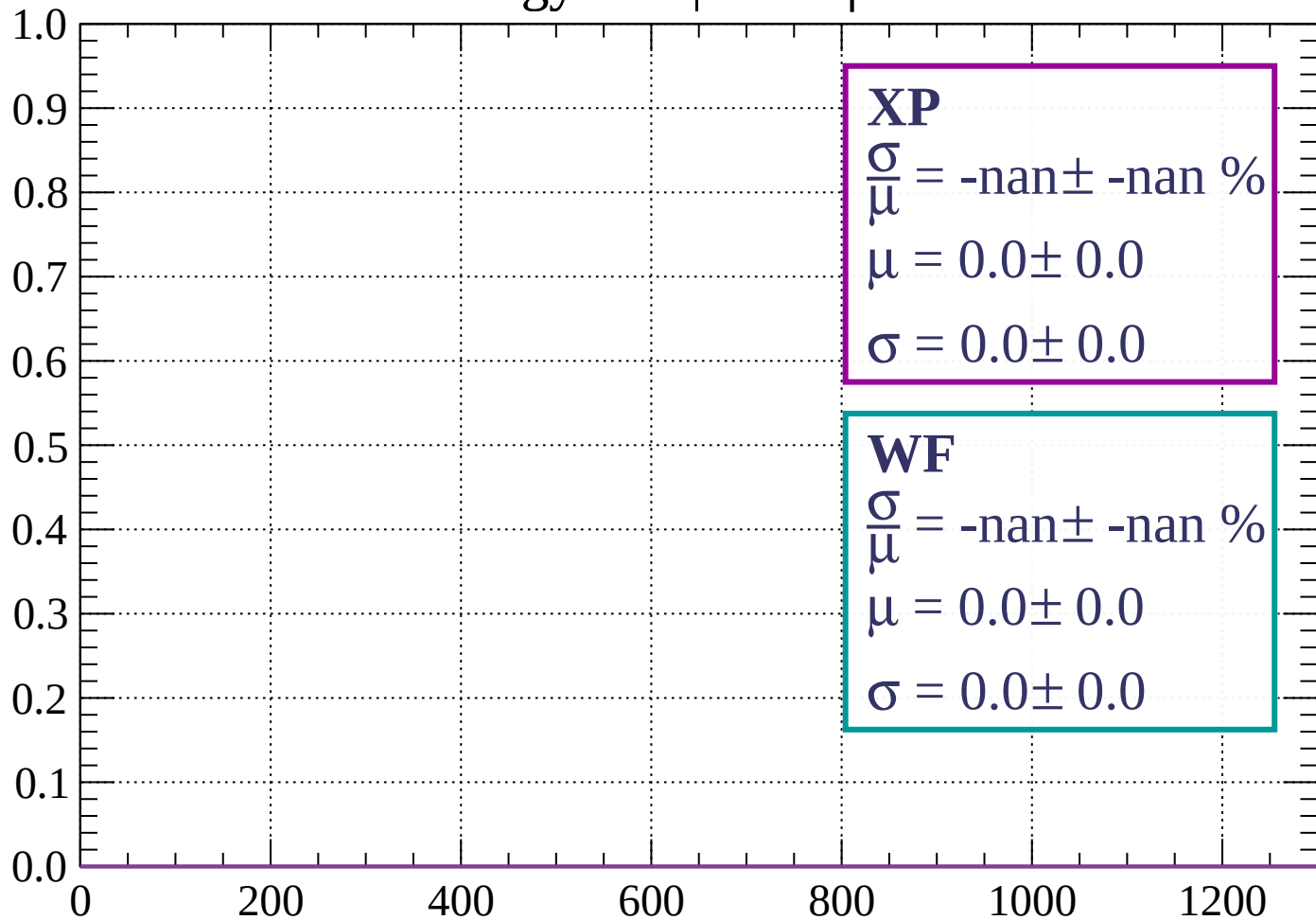
Energy loss | $12 < \phi < 15$

Count



Energy loss | $15 < \phi < 18$

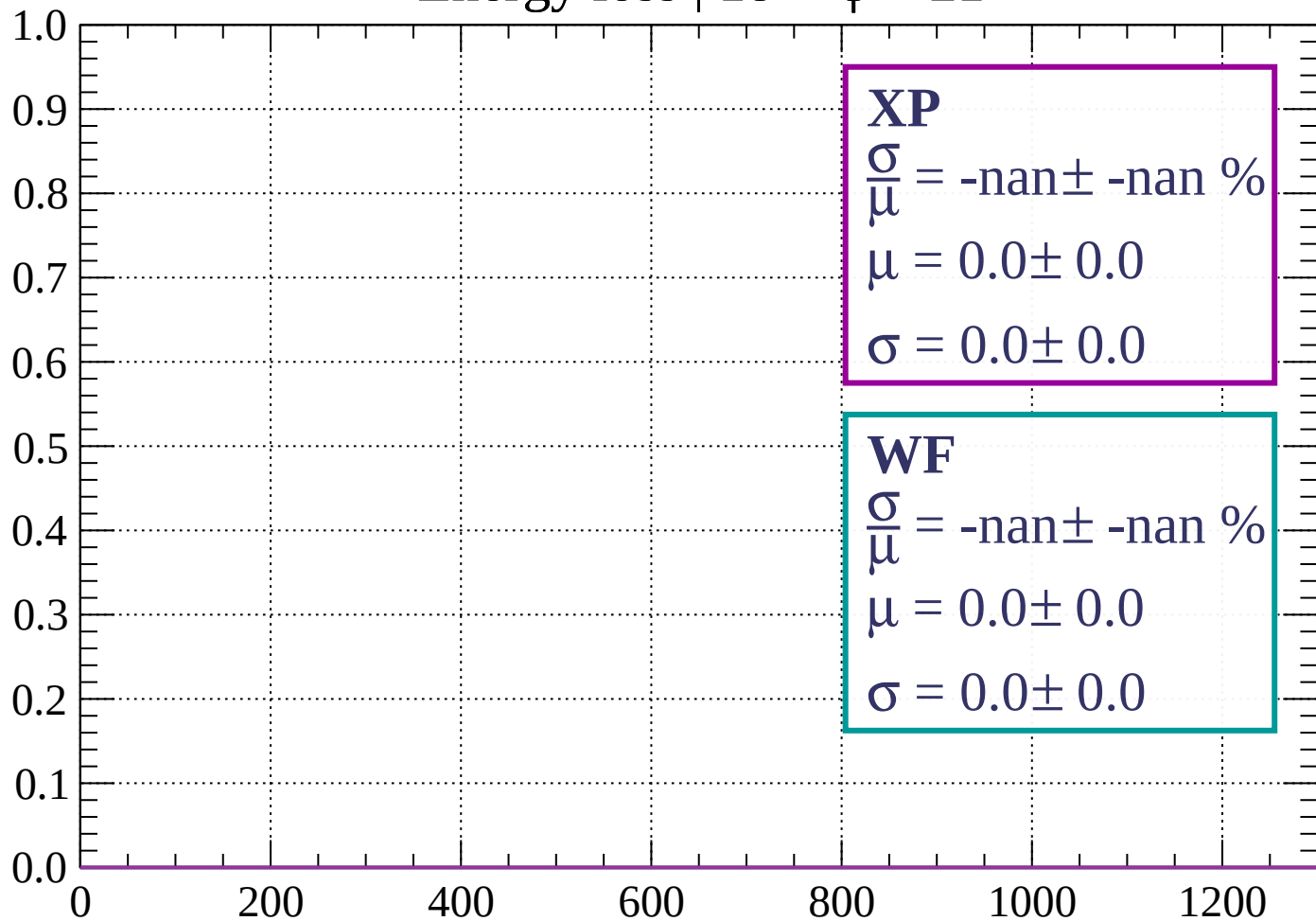
Count



dE/dx (ADC counts/cm)

Energy loss | $18 < \phi < 21$

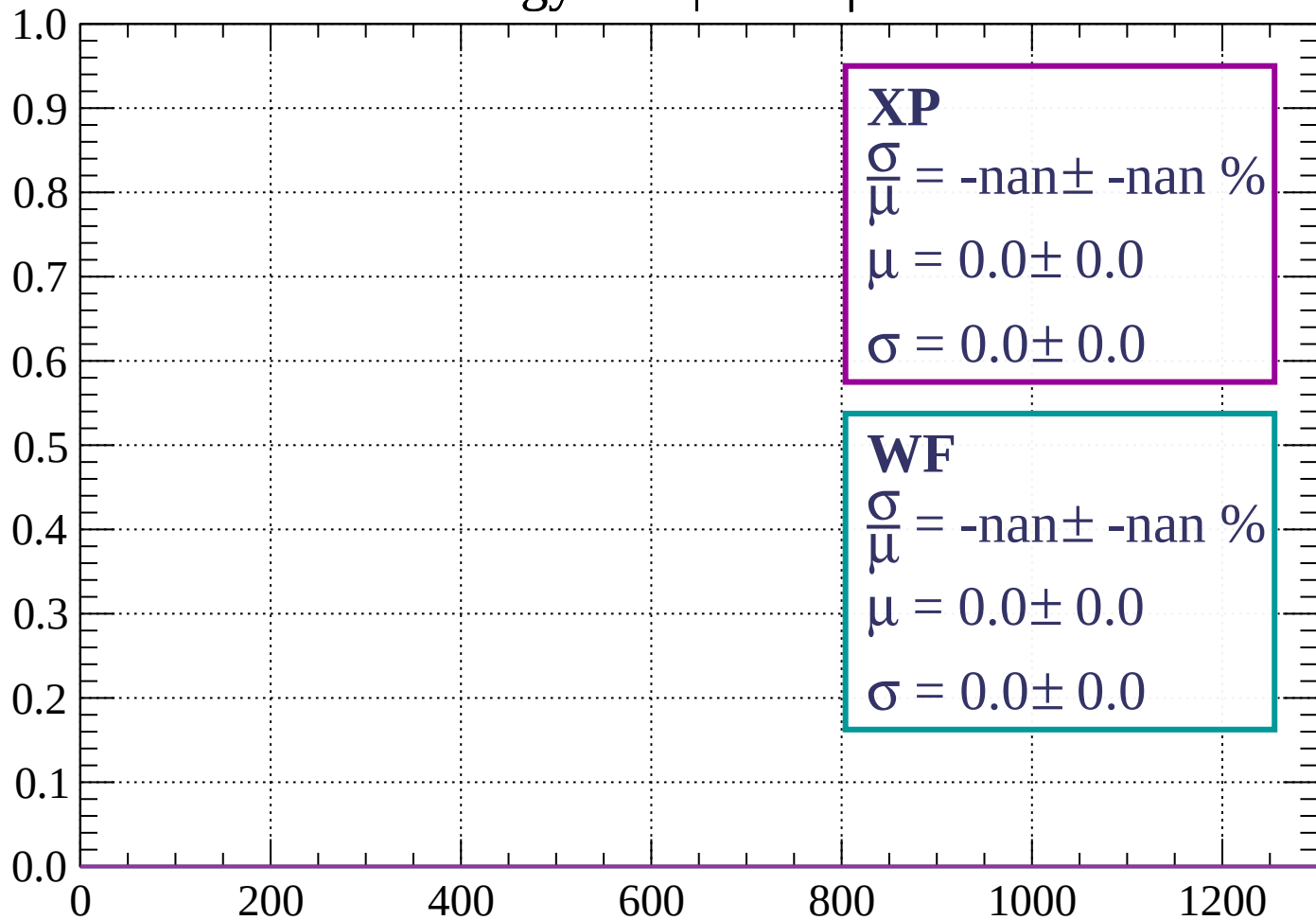
Count



dE/dx (ADC counts/cm)

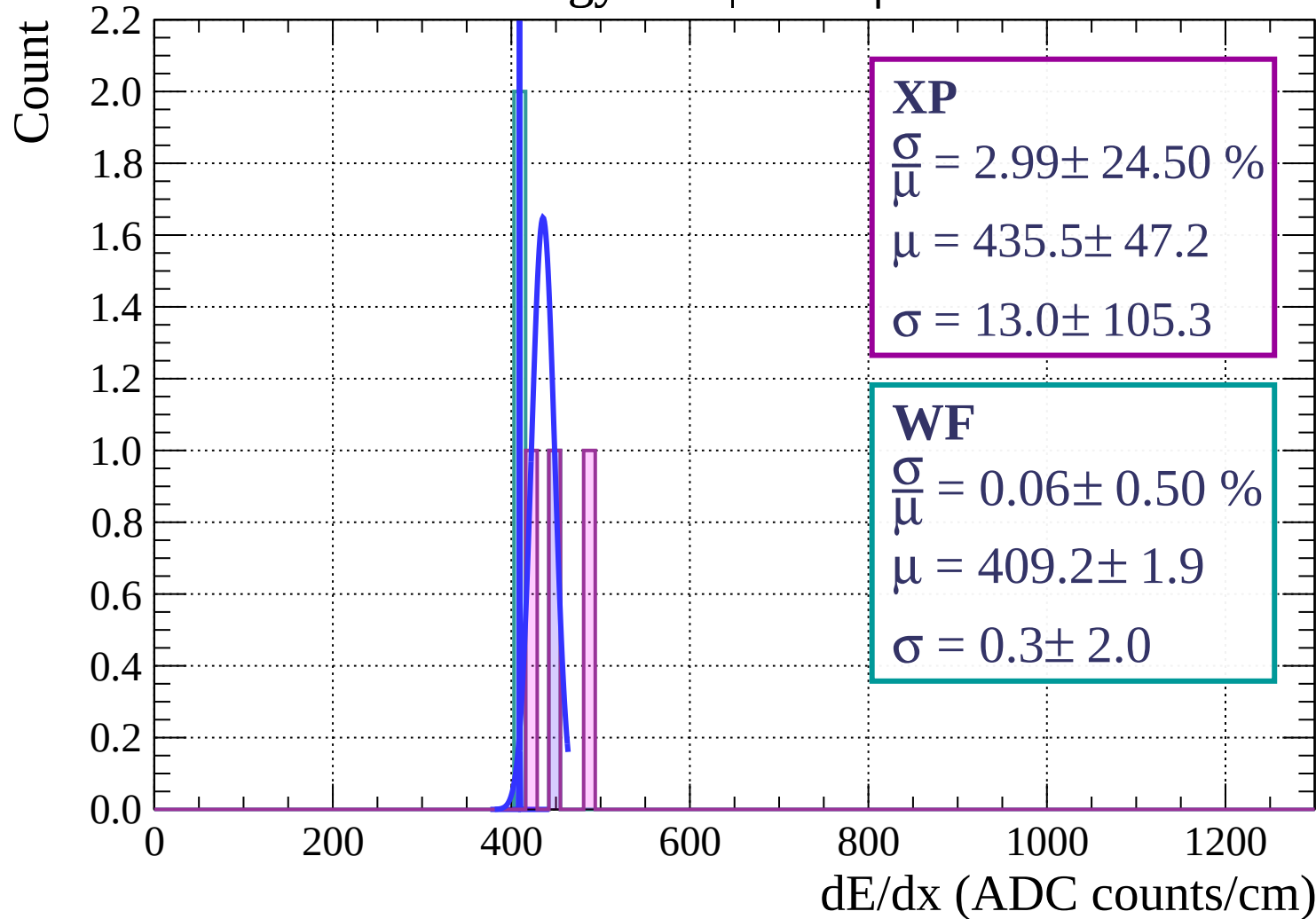
Energy loss | $21 < \phi < 24$

Count



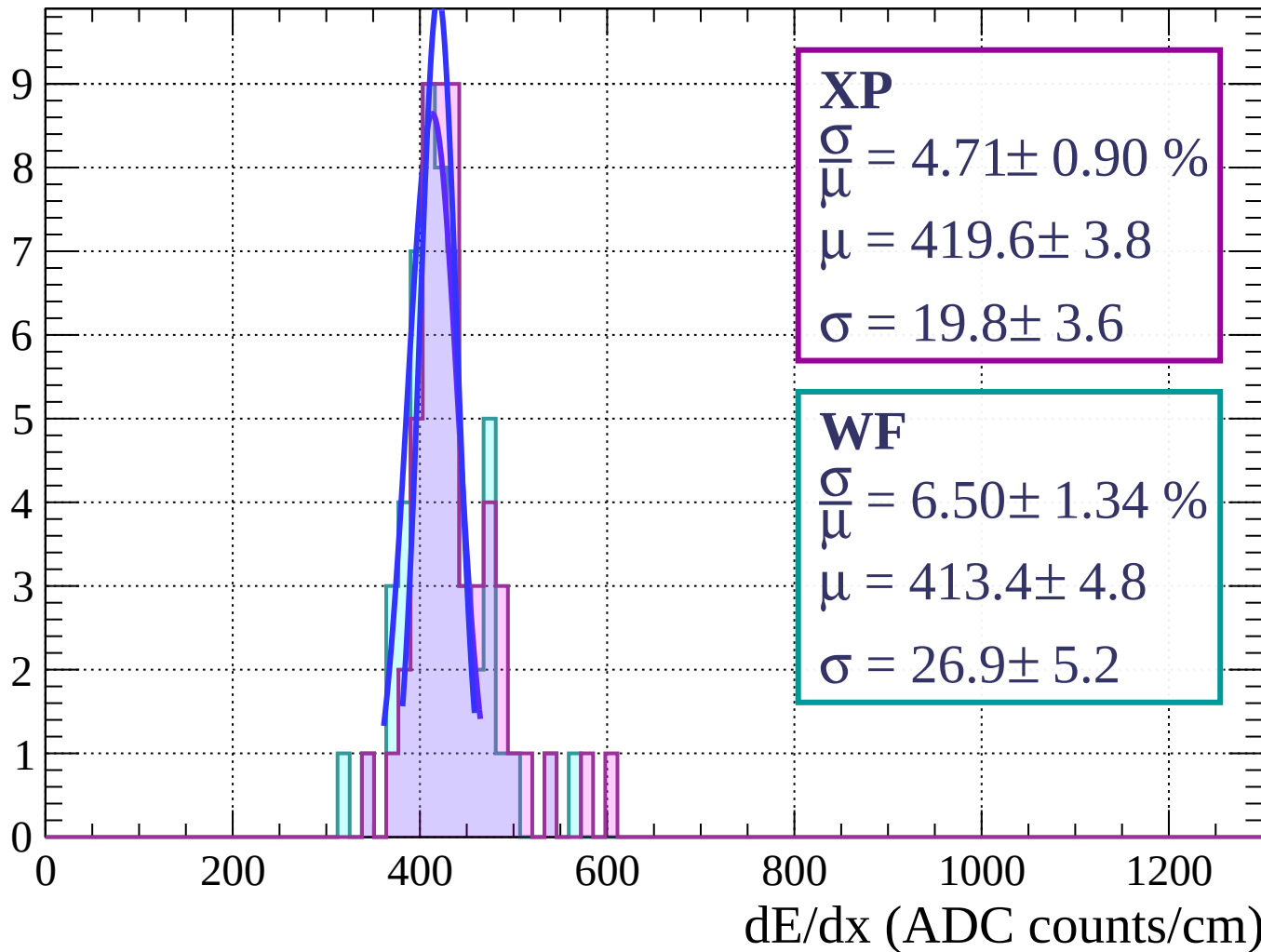
dE/dx (ADC counts/cm)

Energy loss | $24 < \phi < 27$



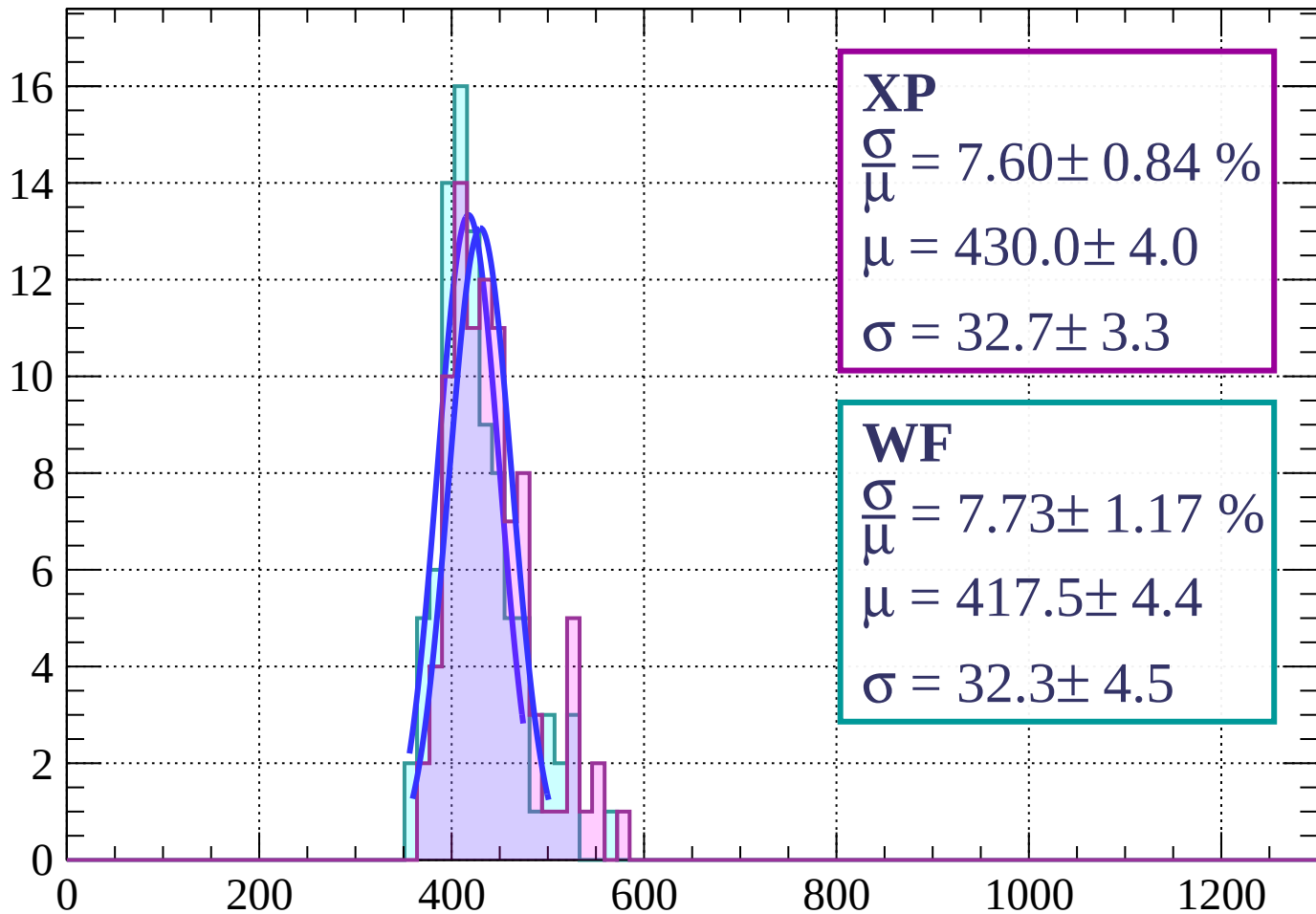
Energy loss | $27 < \phi < 30$

Count



Energy loss | $30 < \phi < 33$

Count



XP

$$\frac{\sigma}{\mu} = 7.60 \pm 0.84 \%$$

$$\mu = 430.0 \pm 4.0$$

$$\sigma = 32.7 \pm 3.3$$

WF

$$\frac{\sigma}{\mu} = 7.73 \pm 1.17 \%$$

$$\mu = 417.5 \pm 4.4$$

$$\sigma = 32.3 \pm 4.5$$

dE/dx (ADC counts/cm)

Energy loss | $33 < \phi < 36$

Count

XP

$$\frac{\sigma}{\mu} = 5.82 \pm 0.72 \%$$

$$\mu = 440.6 \pm 2.7$$

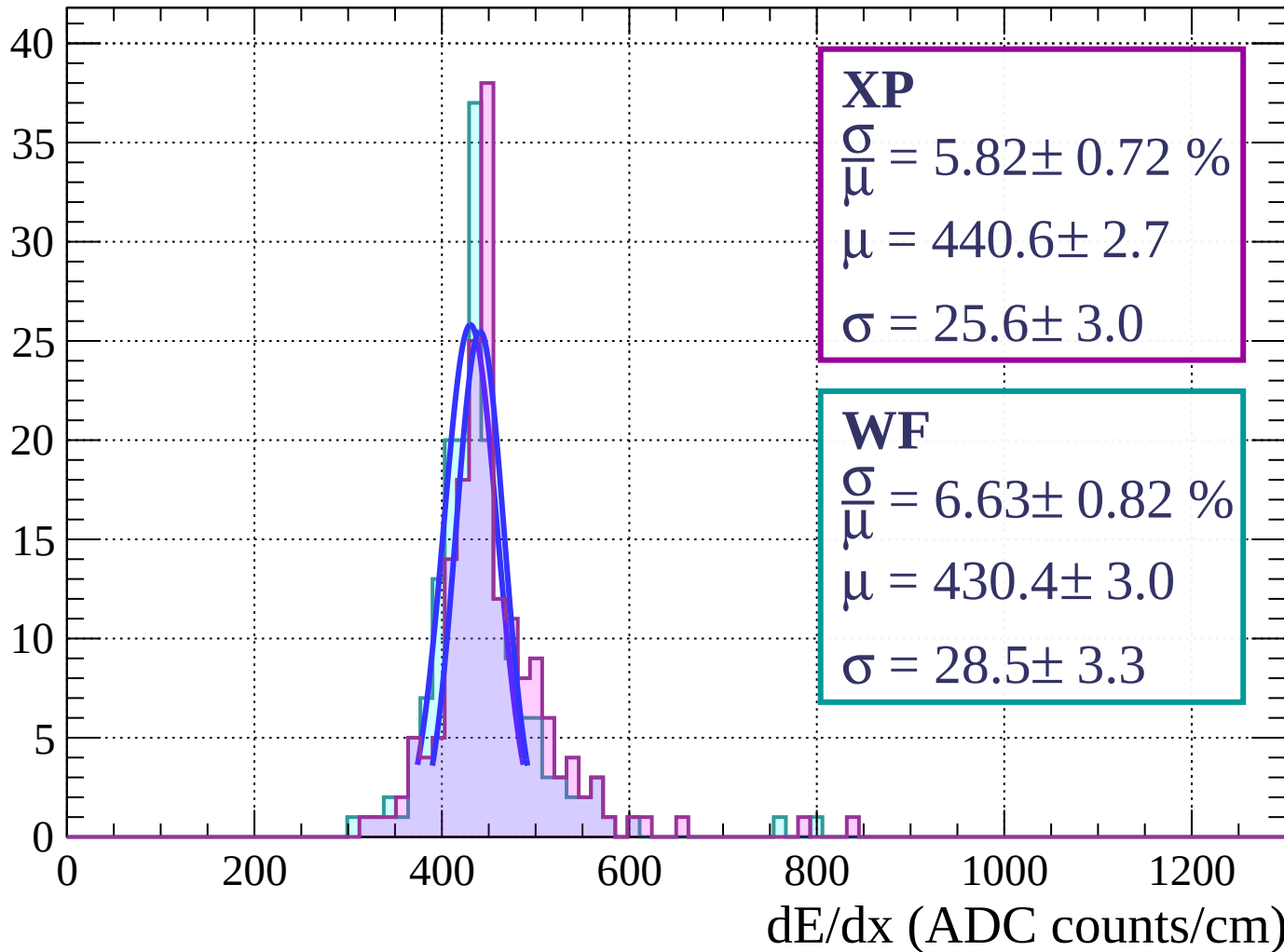
$$\sigma = 25.6 \pm 3.0$$

WF

$$\frac{\sigma}{\mu} = 6.63 \pm 0.82 \%$$

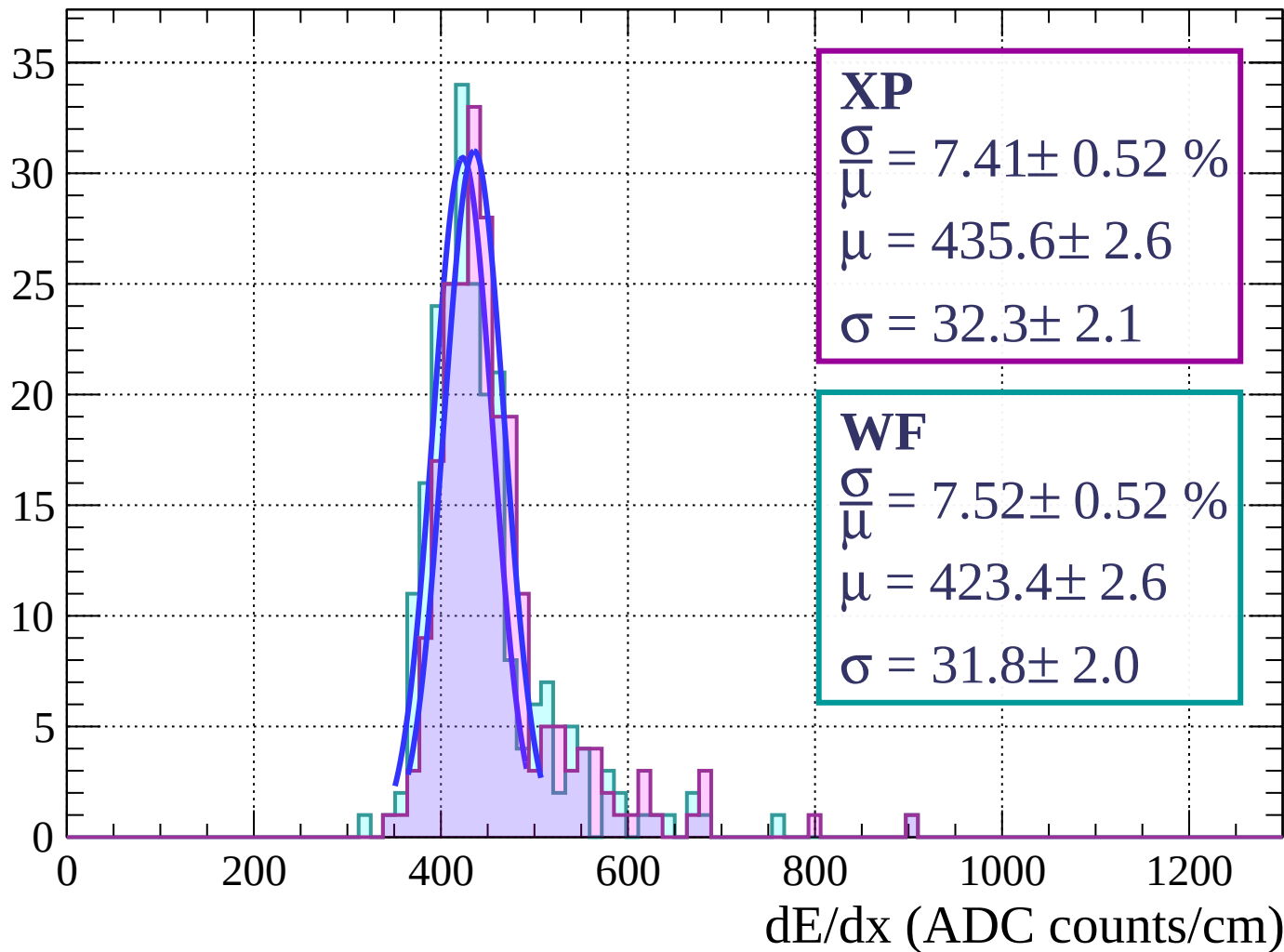
$$\mu = 430.4 \pm 3.0$$

$$\sigma = 28.5 \pm 3.3$$



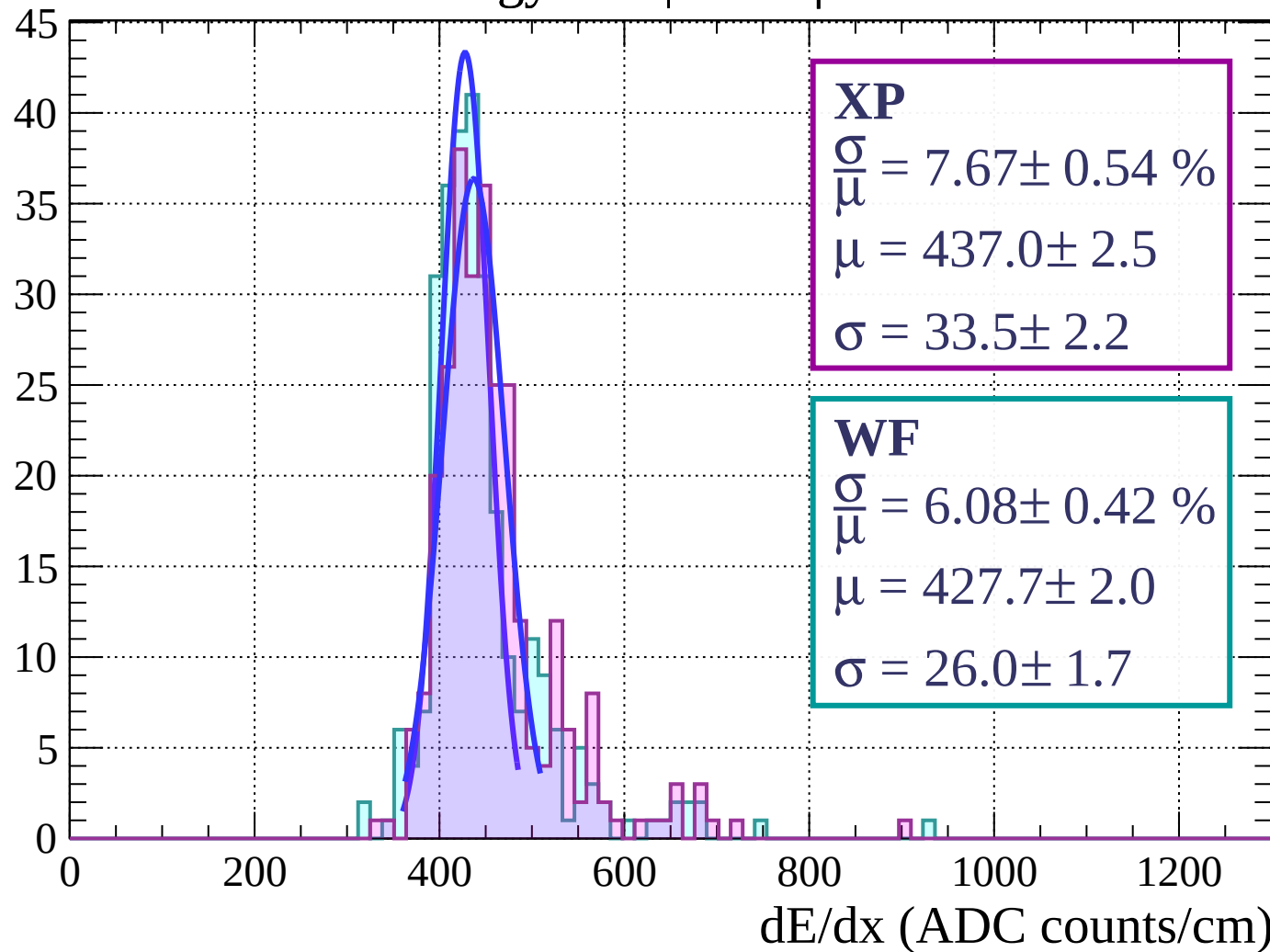
Energy loss | $36 < \phi < 39$

Count



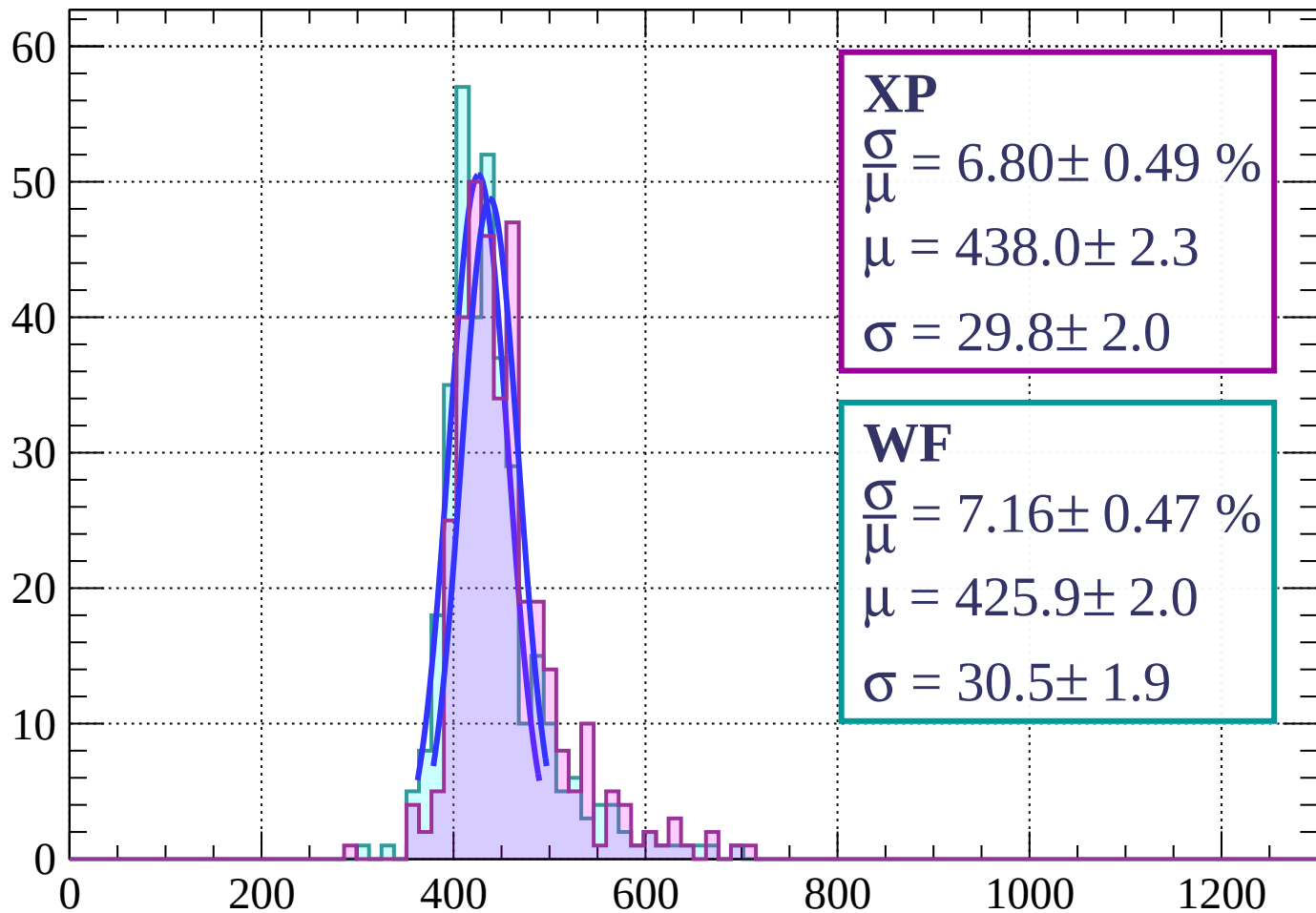
Energy loss | $39 < \phi < 42$

Count



Energy loss | $42 < \phi < 45$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 6.80 \pm 0.49 \%$$

$$\mu = 438.0 \pm 2.3$$

$$\sigma = 29.8 \pm 2.0$$

WF

$$\frac{\sigma}{\bar{\mu}} = 7.16 \pm 0.47 \%$$

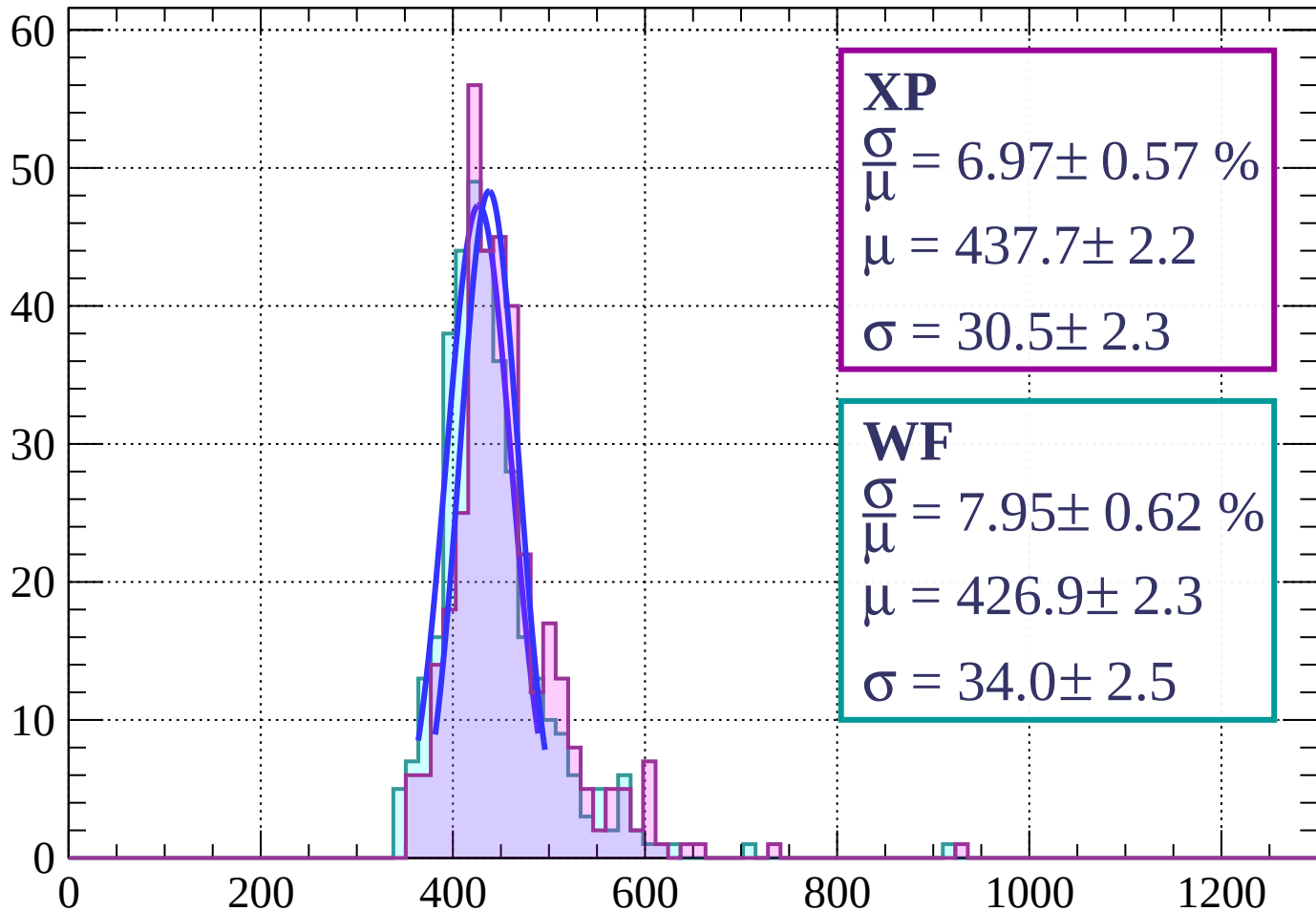
$$\mu = 425.9 \pm 2.0$$

$$\sigma = 30.5 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $45 < \phi < 48$

Count



XP

$$\frac{\sigma}{\mu} = 6.97 \pm 0.57 \%$$

$$\mu = 437.7 \pm 2.2$$

$$\sigma = 30.5 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 7.95 \pm 0.62 \%$$

$$\mu = 426.9 \pm 2.3$$

$$\sigma = 34.0 \pm 2.5$$

dE/dx (ADC counts/cm)

Energy loss | $48 < \phi < 51$

Count

XP

$$\frac{\sigma}{\mu} = 6.78 \pm 0.67 \%$$

$$\mu = 426.1 \pm 2.1$$

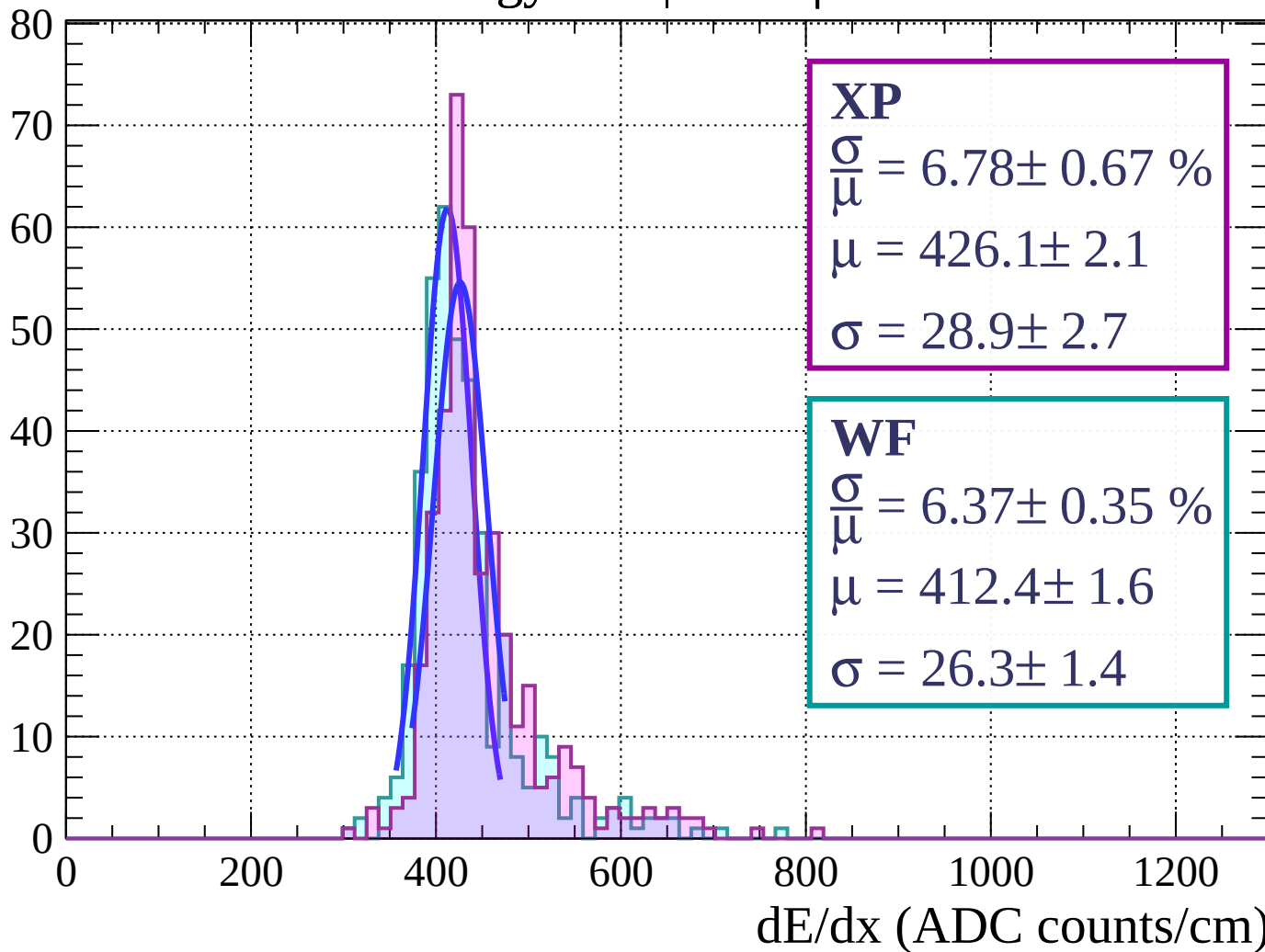
$$\sigma = 28.9 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 6.37 \pm 0.35 \%$$

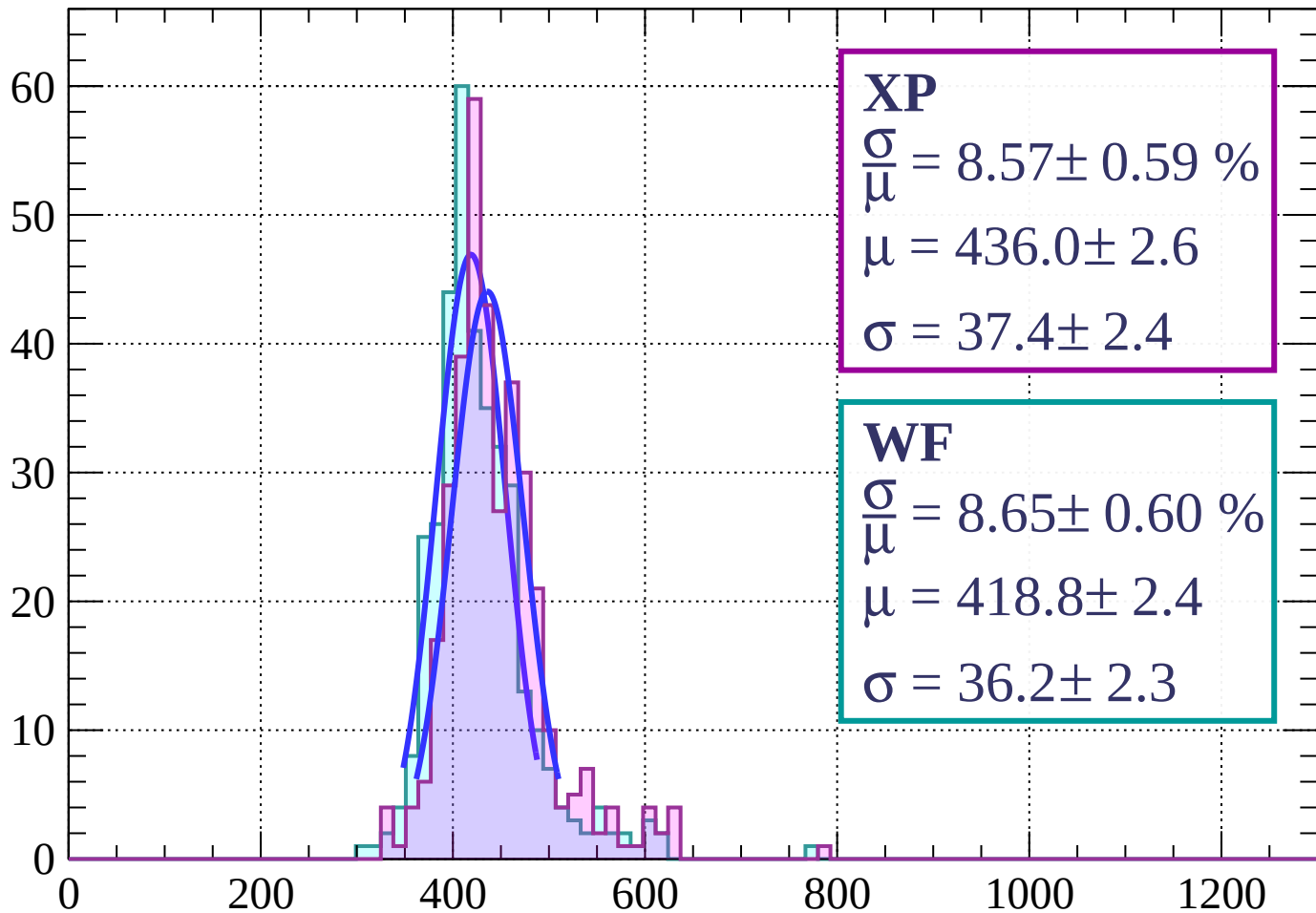
$$\mu = 412.4 \pm 1.6$$

$$\sigma = 26.3 \pm 1.4$$



Energy loss | $51 < \phi < 54$

Count



XP

$$\frac{\sigma}{\mu} = 8.57 \pm 0.59 \%$$

$$\mu = 436.0 \pm 2.6$$

$$\sigma = 37.4 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 8.65 \pm 0.60 \%$$

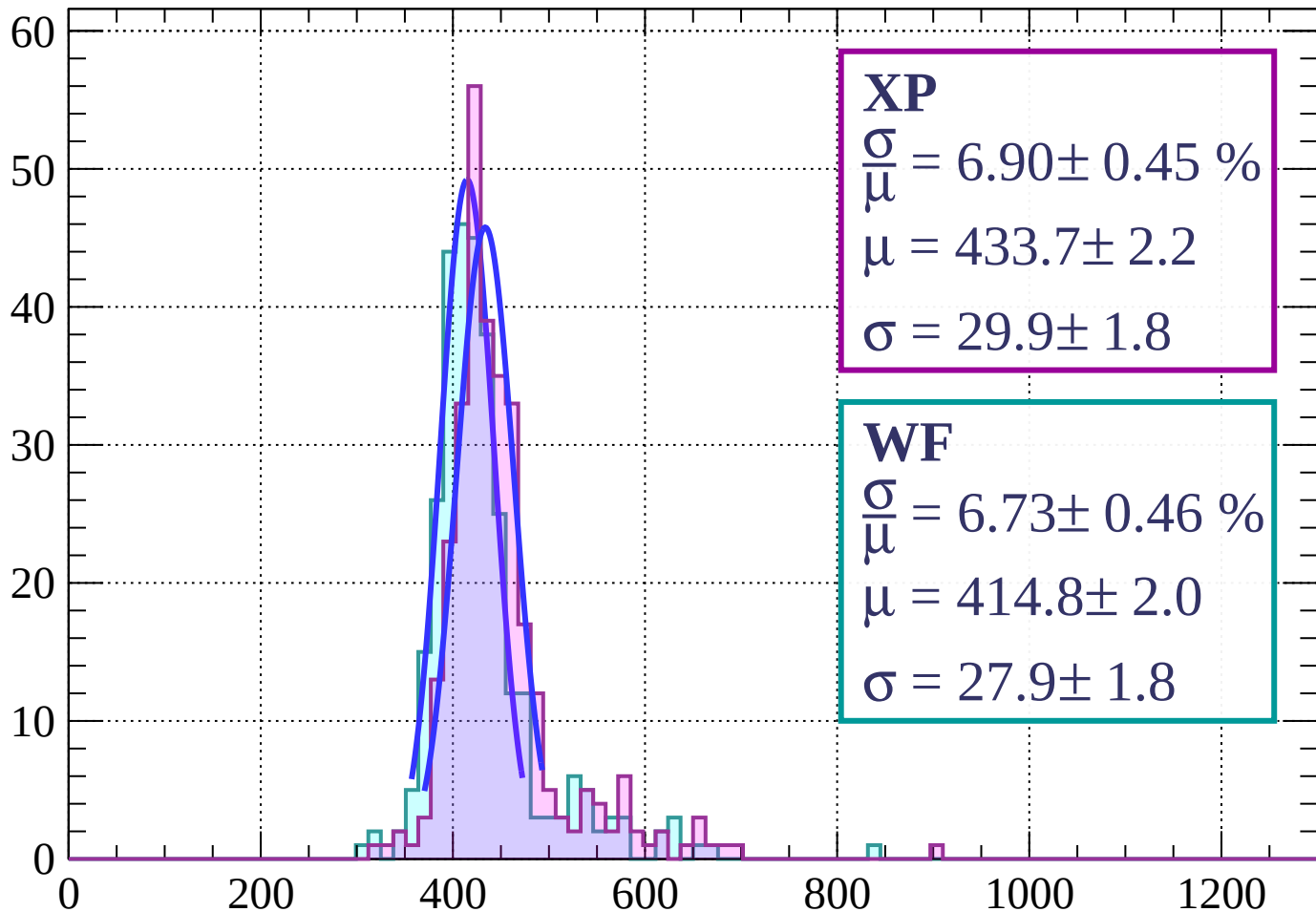
$$\mu = 418.8 \pm 2.4$$

$$\sigma = 36.2 \pm 2.3$$

dE/dx (ADC counts/cm)

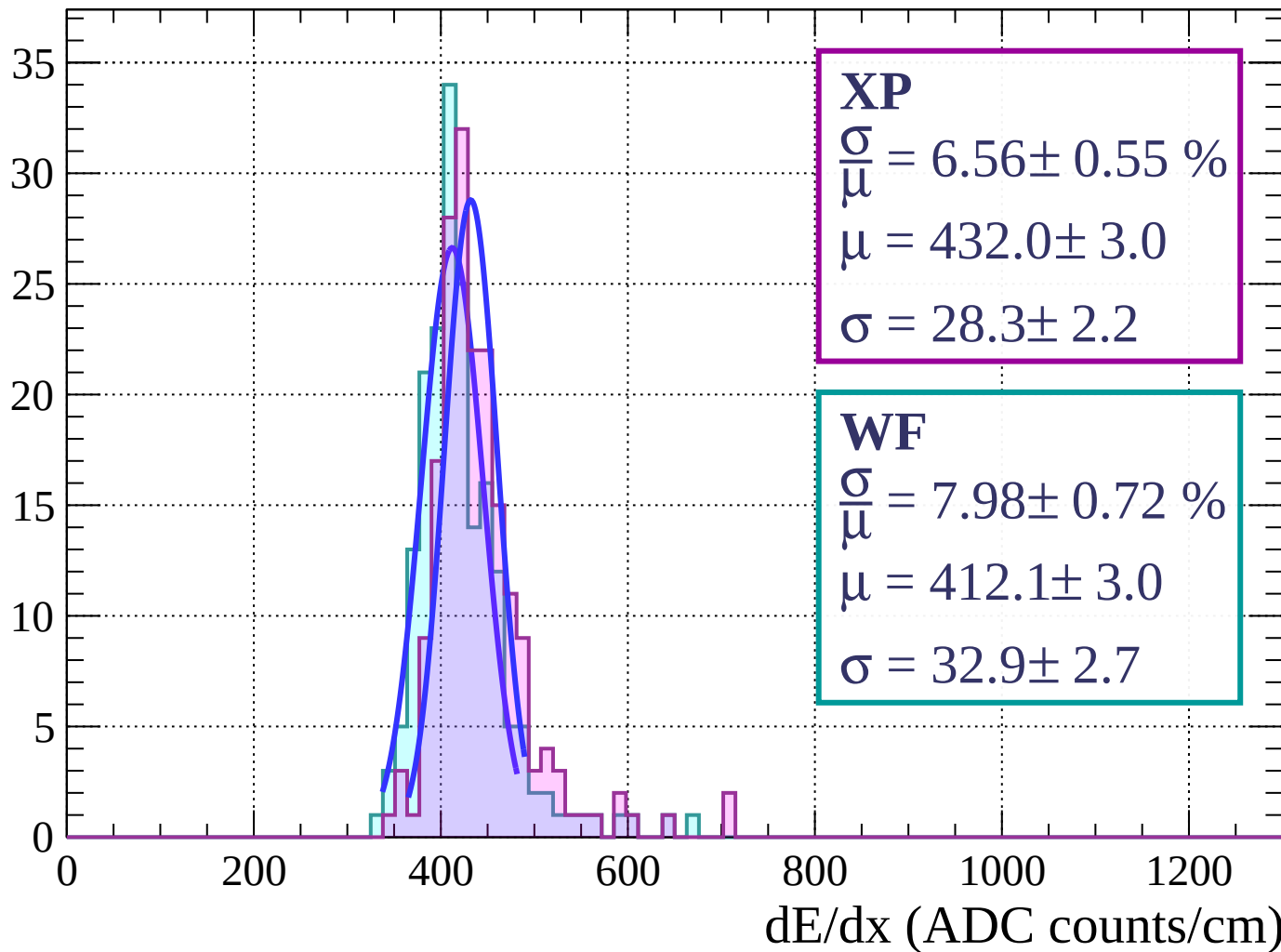
Energy loss | $54 < \phi < 57$

Count



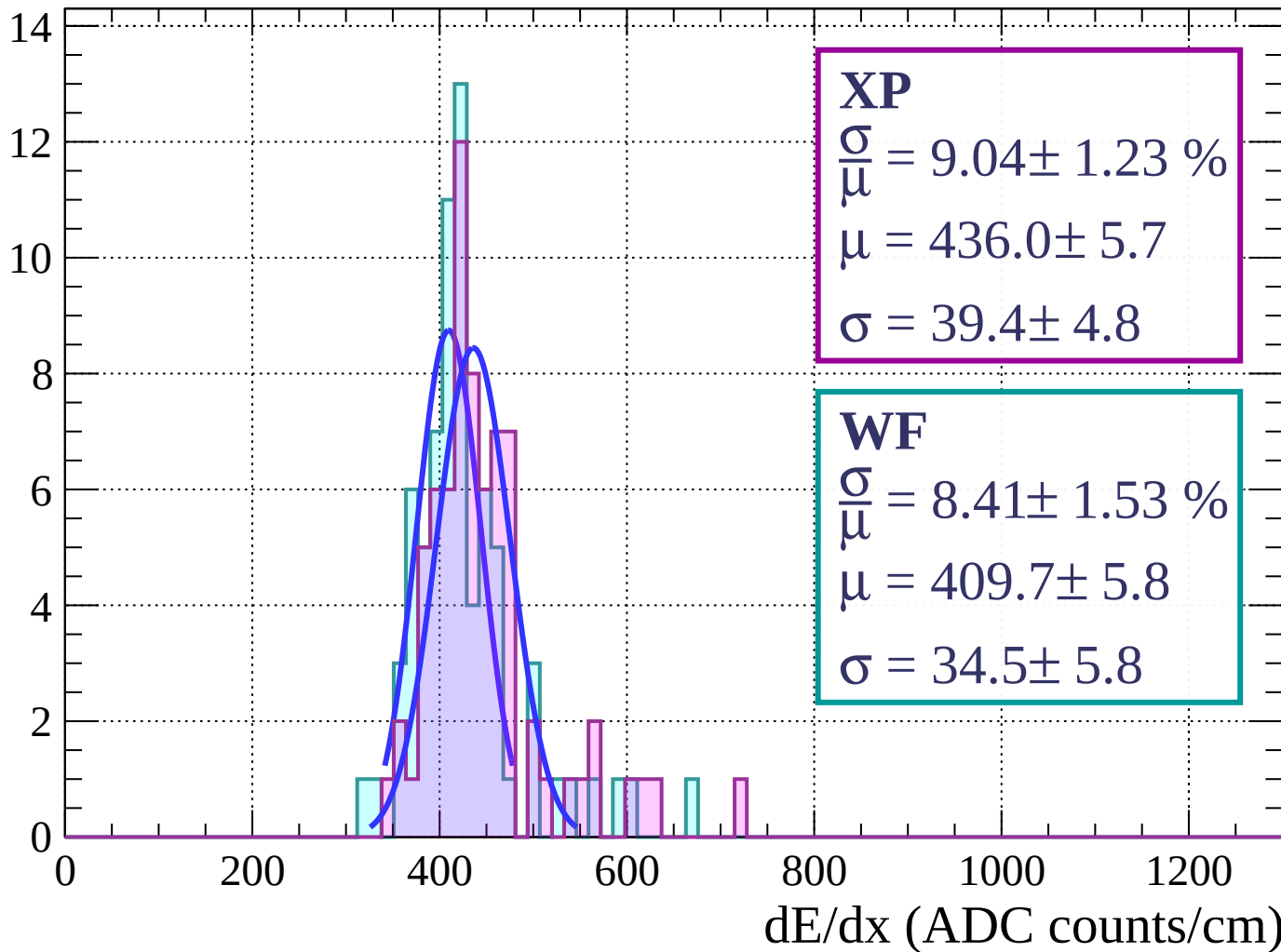
Energy loss | $57 < \phi < 60$

Count



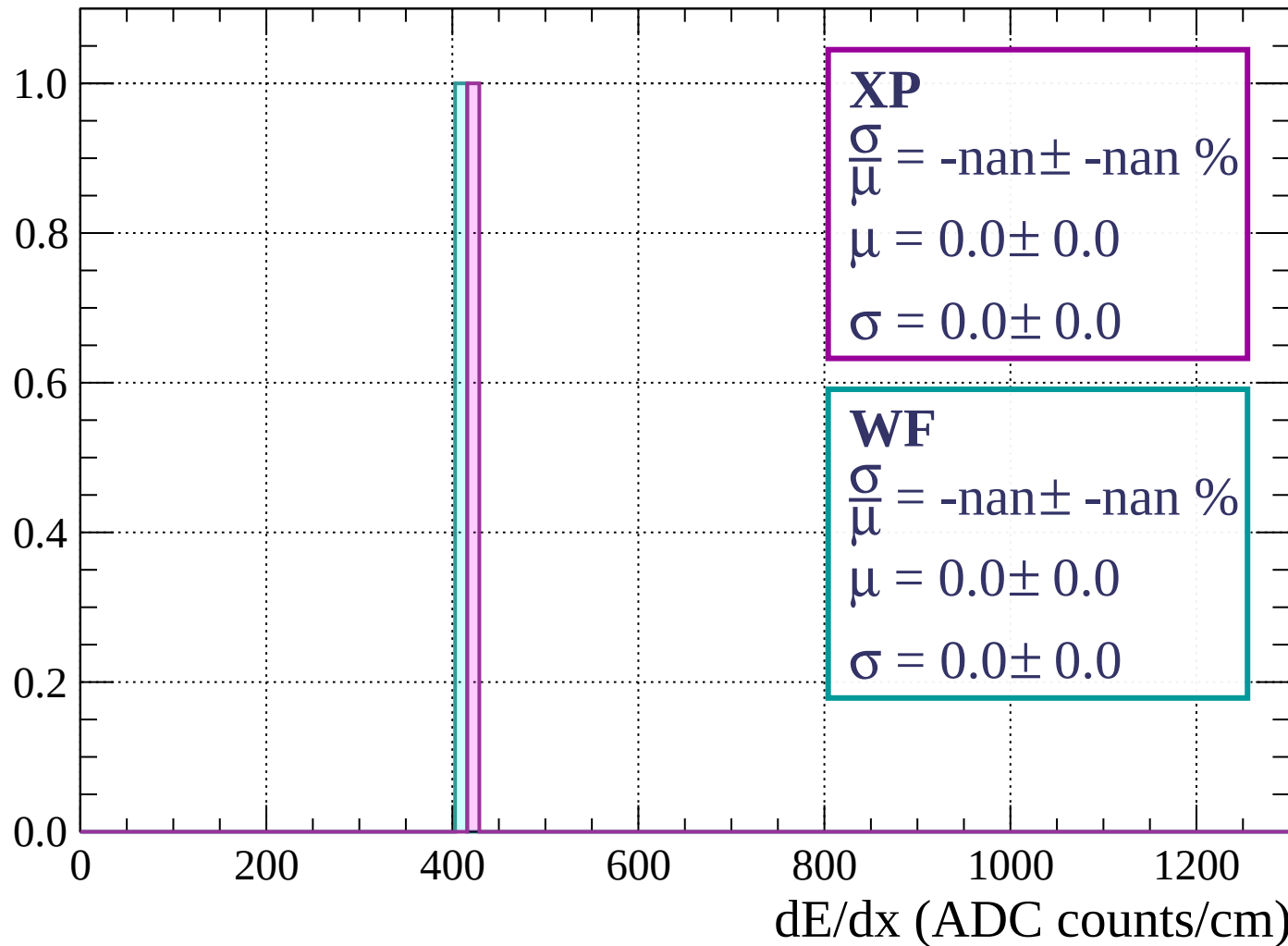
Energy loss | $60 < \phi < 63$

Count



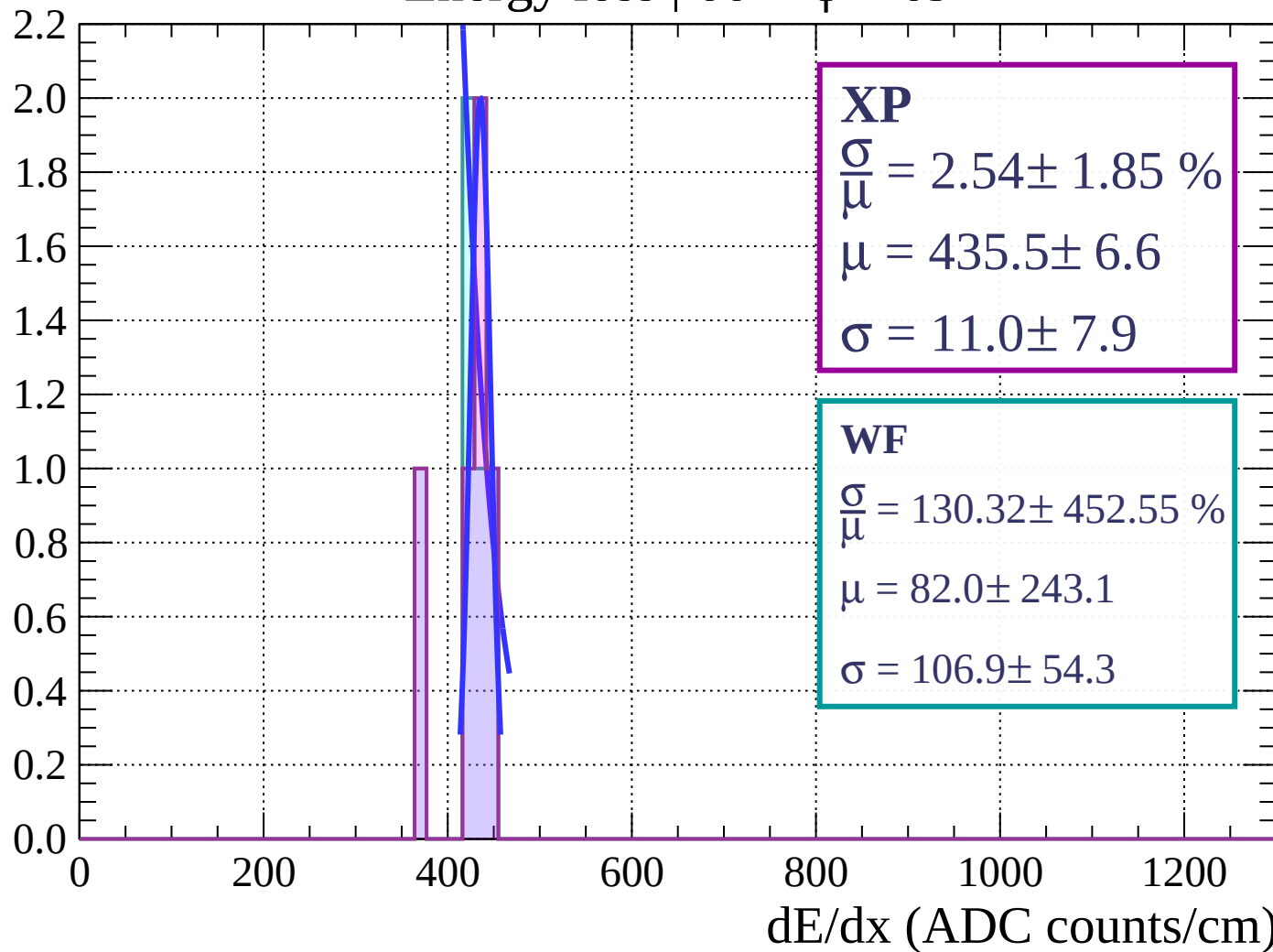
Energy loss | $63 < \phi < 66$

Count



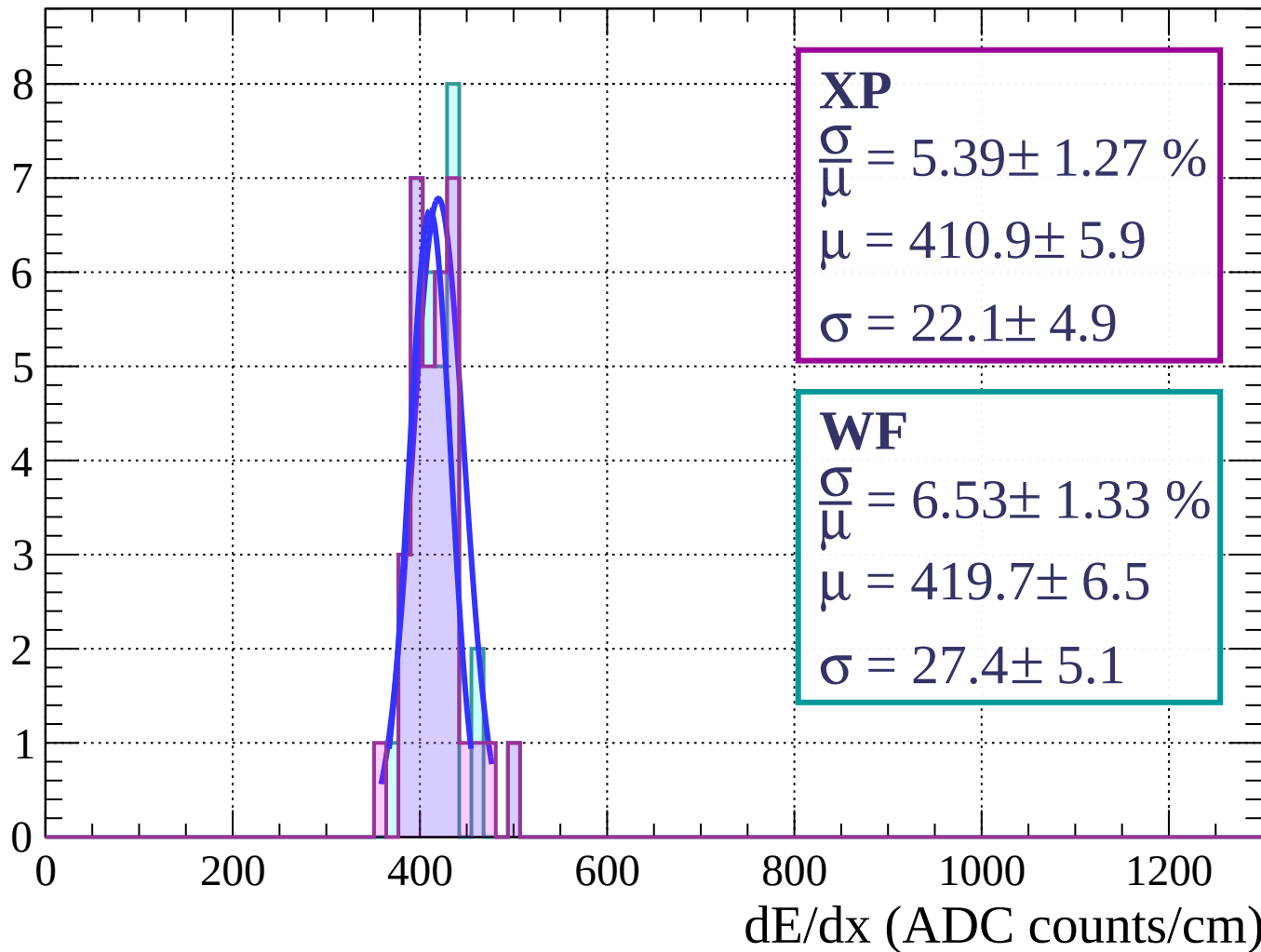
Energy loss | $66 < \phi < 69$

Count



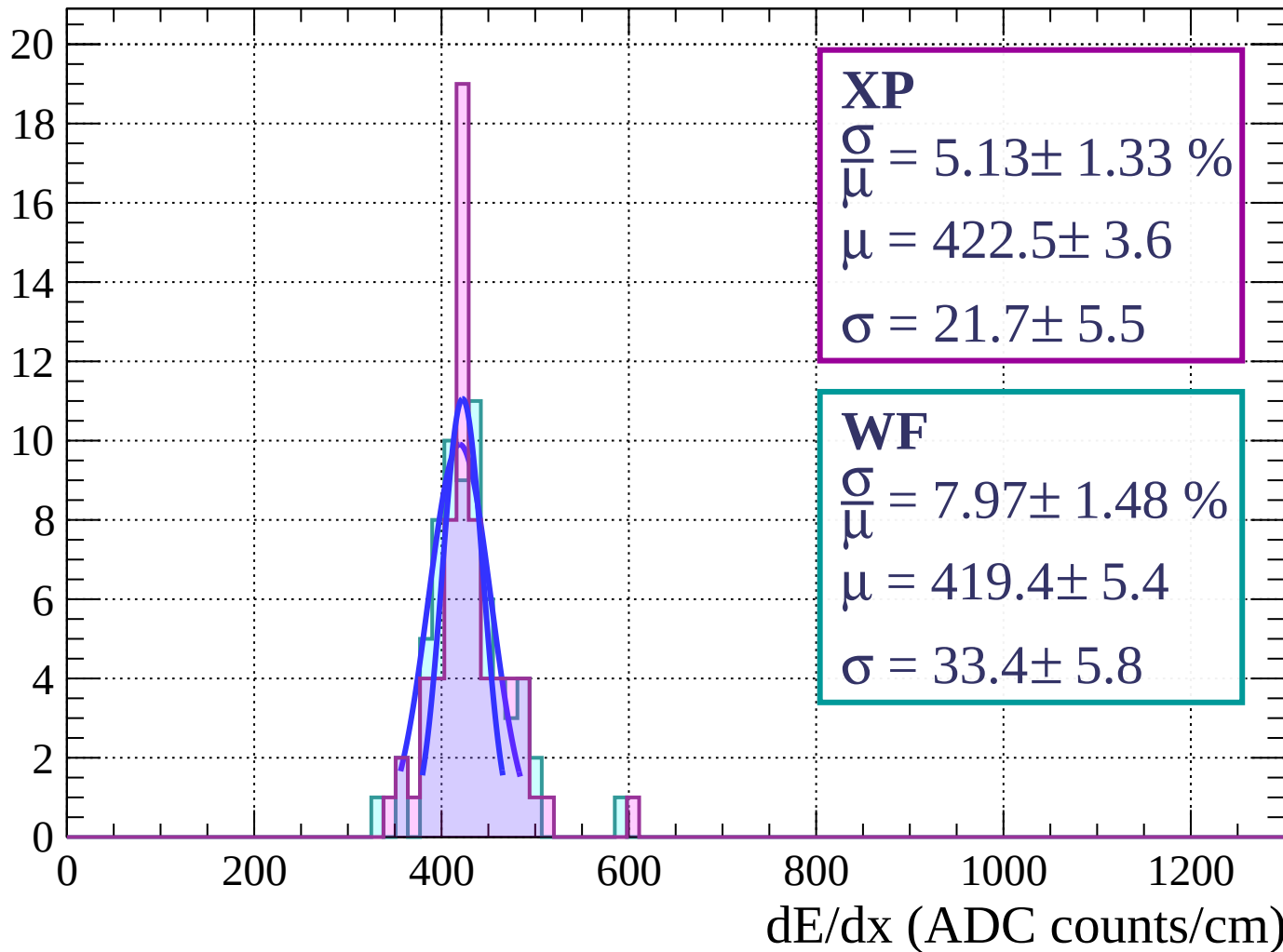
Energy loss | $69 < \phi < 72$

Count



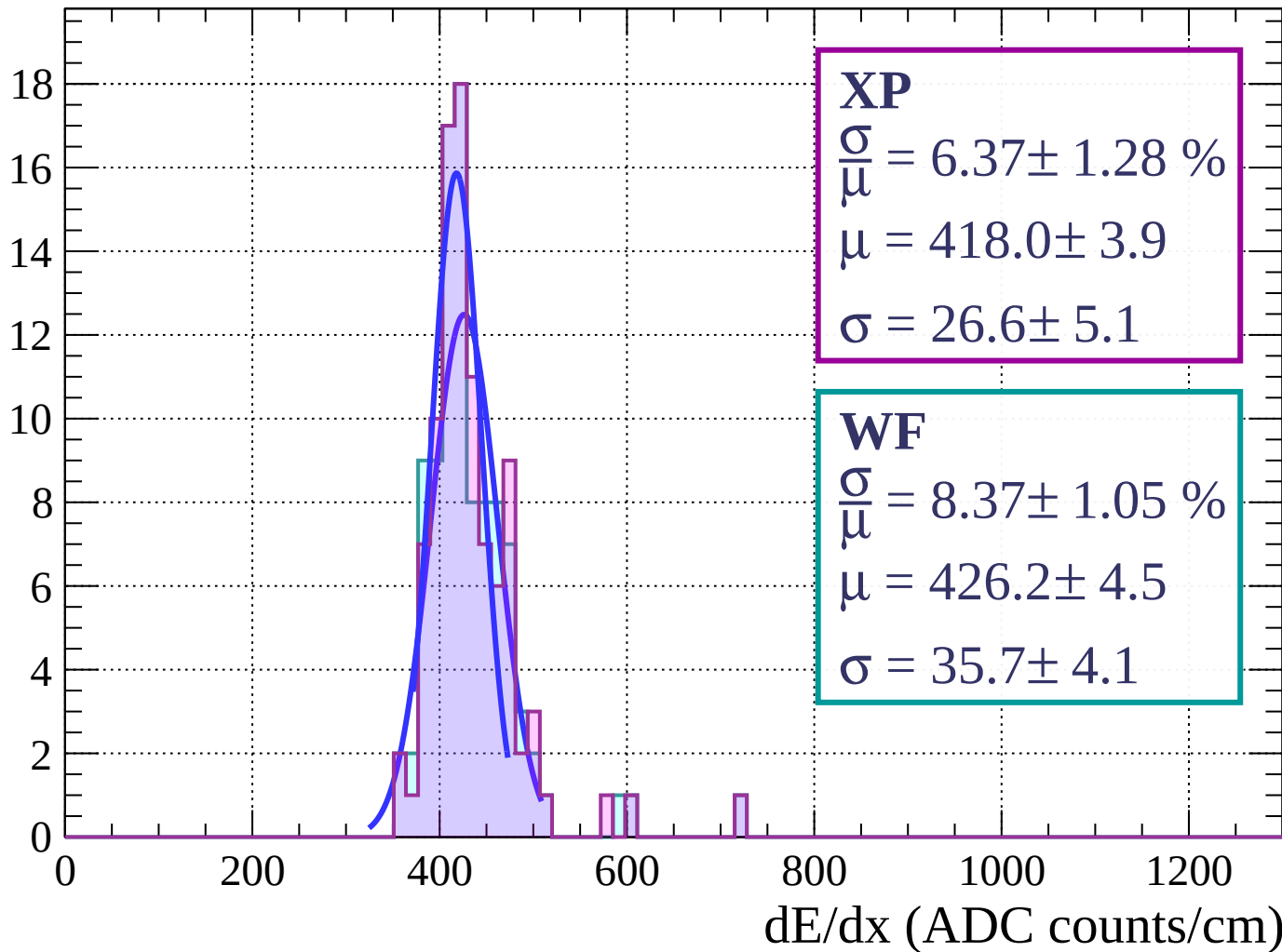
Energy loss | $72 < \phi < 75$

Count



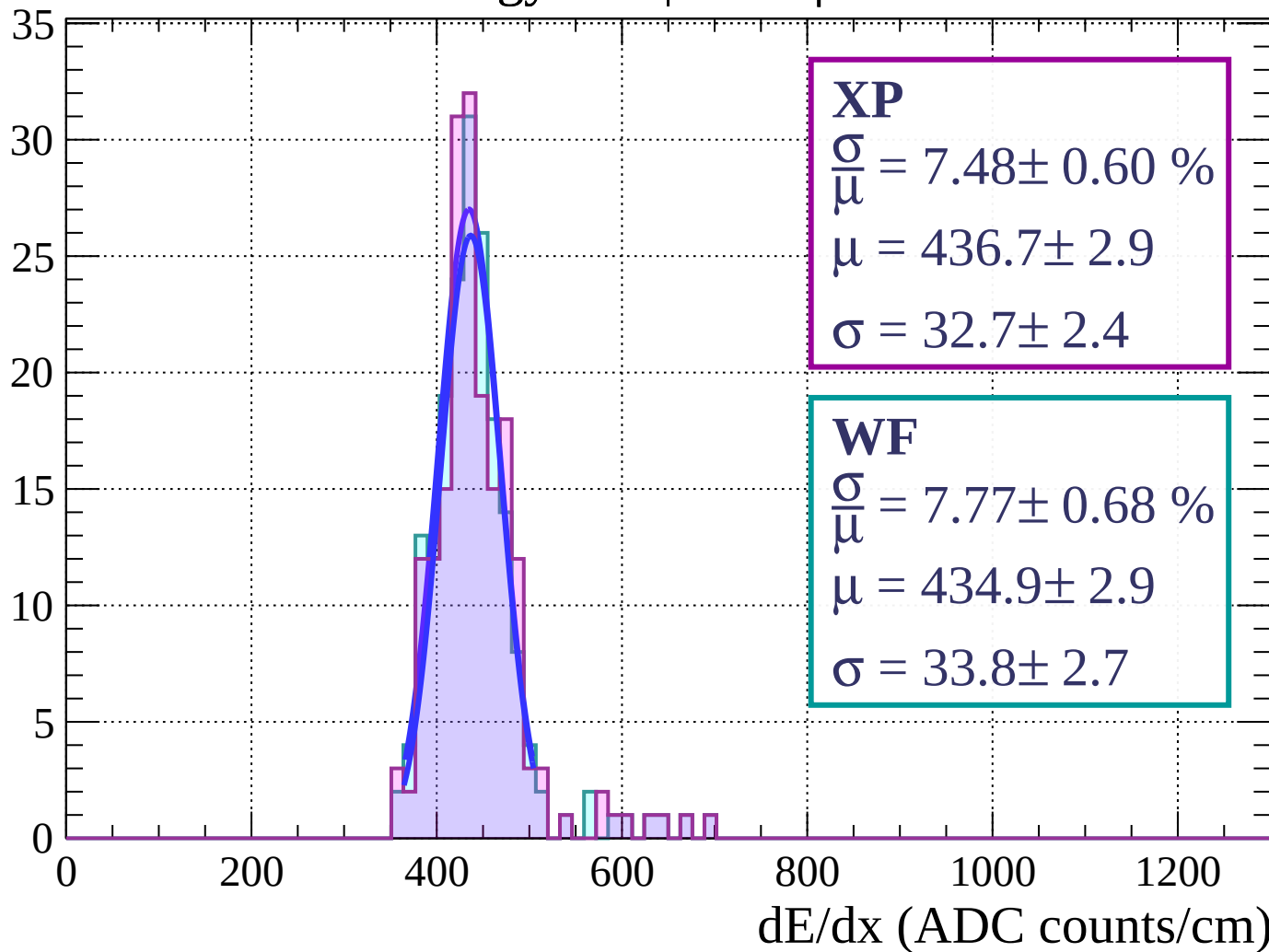
Energy loss | $75 < \phi < 78$

Count



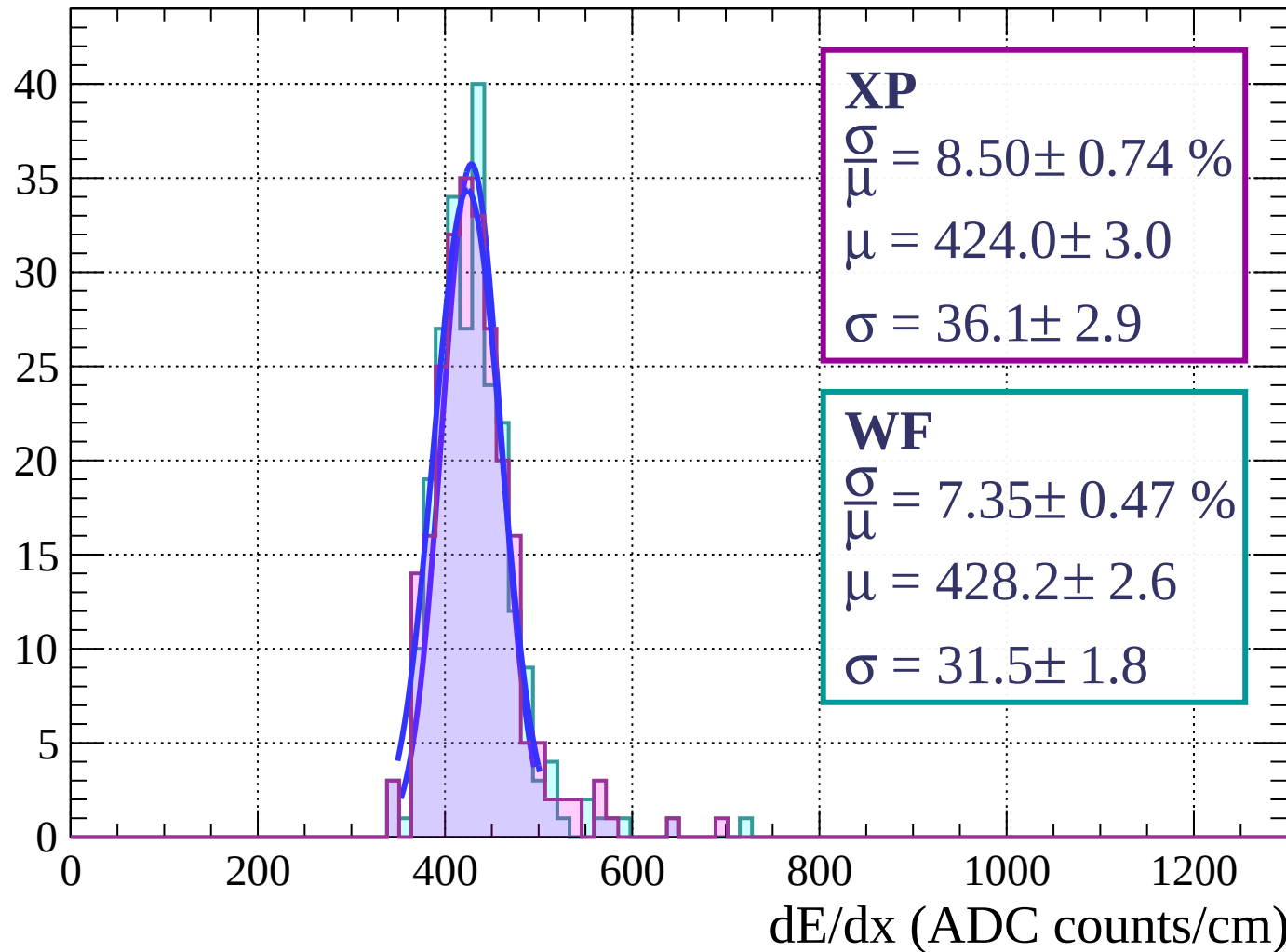
Energy loss | $78 < \phi < 81$

Count



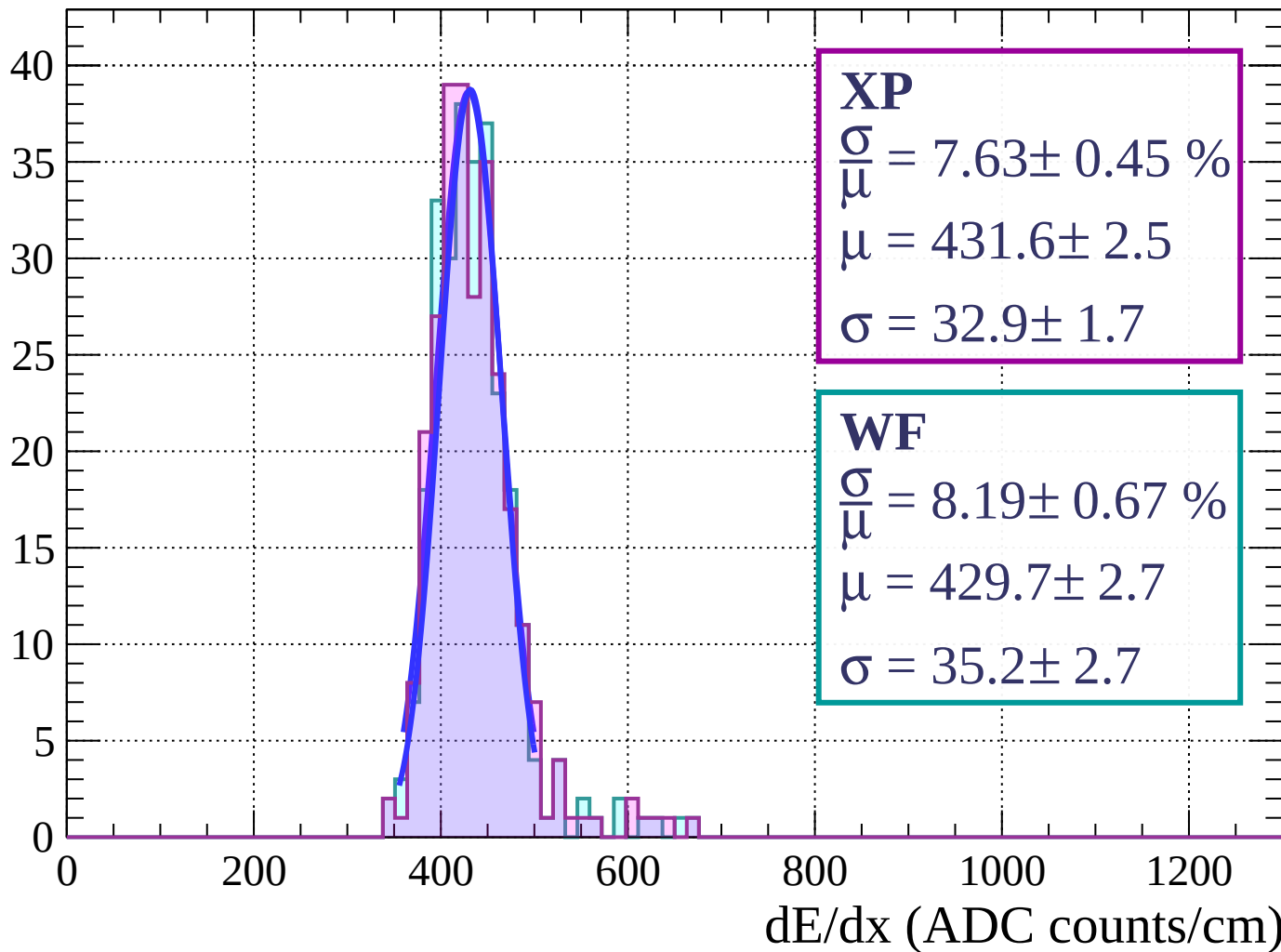
Energy loss | $81 < \phi < 84$

Count



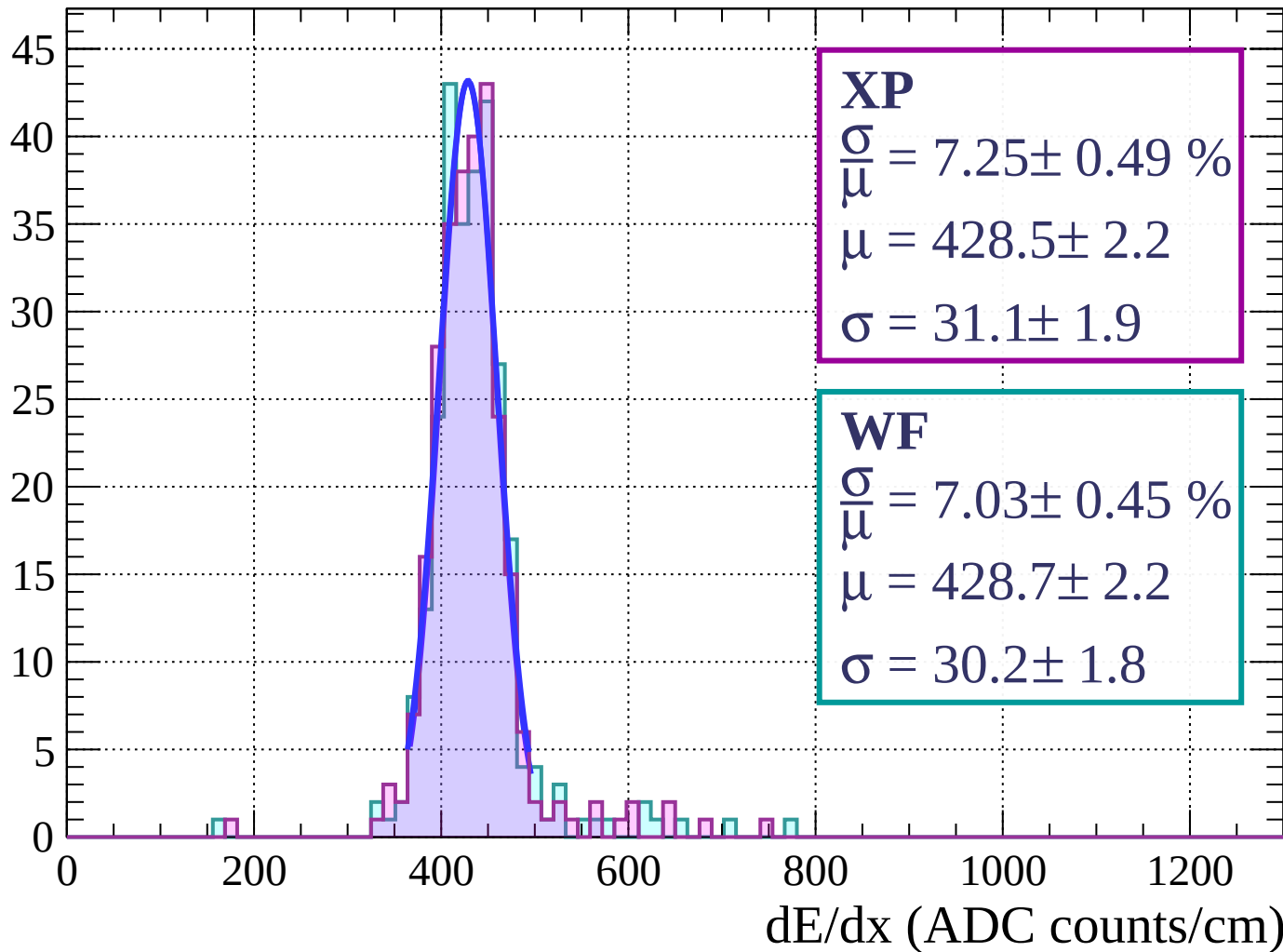
Energy loss | $84 < \phi < 87$

Count



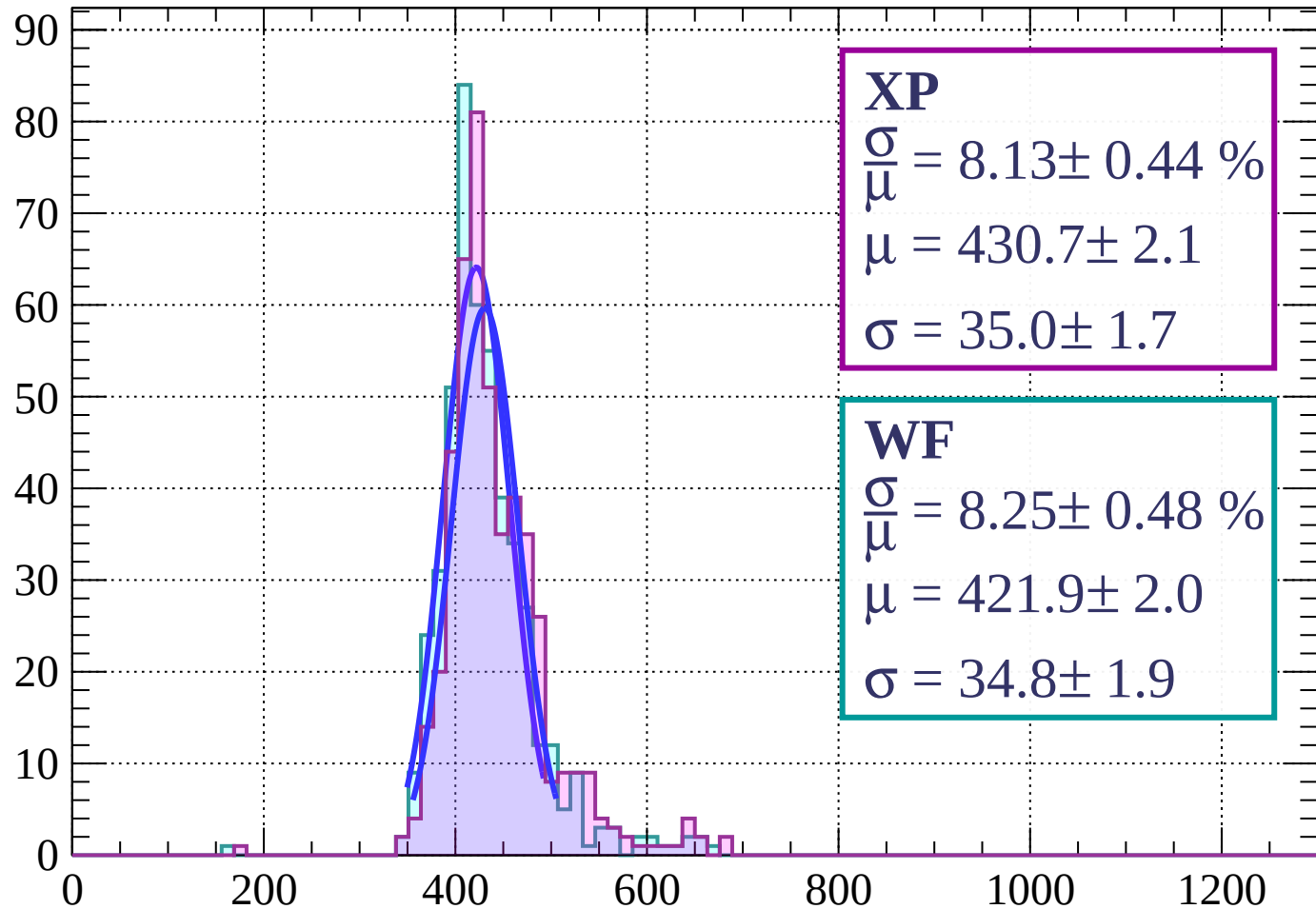
Energy loss | $87 < \phi < 90$

Count



Energy loss | $0 < \theta < 3$

Count



XP

$$\frac{\sigma}{\mu} = 8.13 \pm 0.44 \%$$

$$\mu = 430.7 \pm 2.1$$

$$\sigma = 35.0 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 8.25 \pm 0.48 \%$$

$$\mu = 421.9 \pm 2.0$$

$$\sigma = 34.8 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $3 < \theta < 6$

Count

XP

$$\frac{\sigma}{\mu} = 7.06 \pm 0.39 \%$$

$$\mu = 429.9 \pm 1.8$$

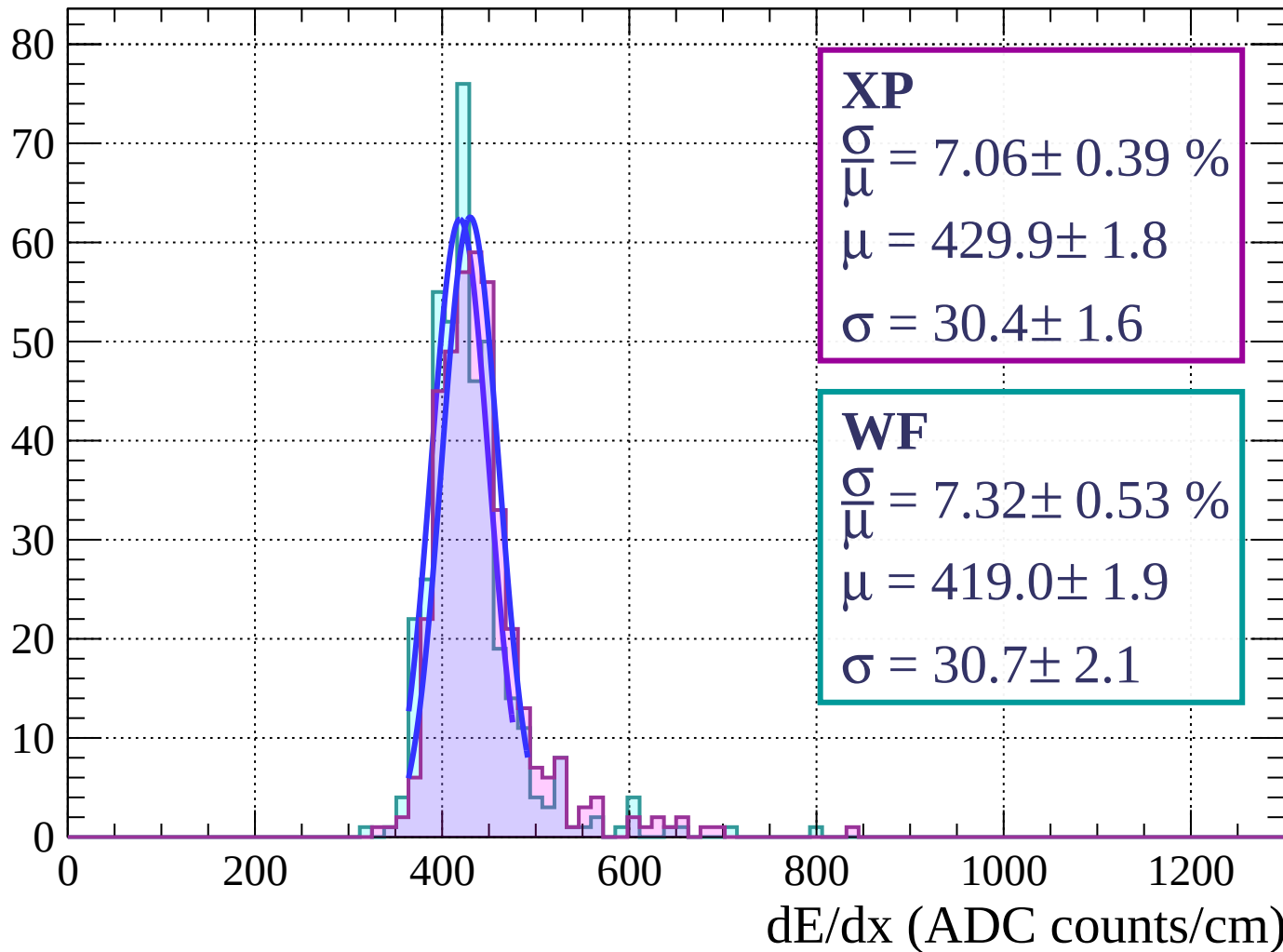
$$\sigma = 30.4 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 7.32 \pm 0.53 \%$$

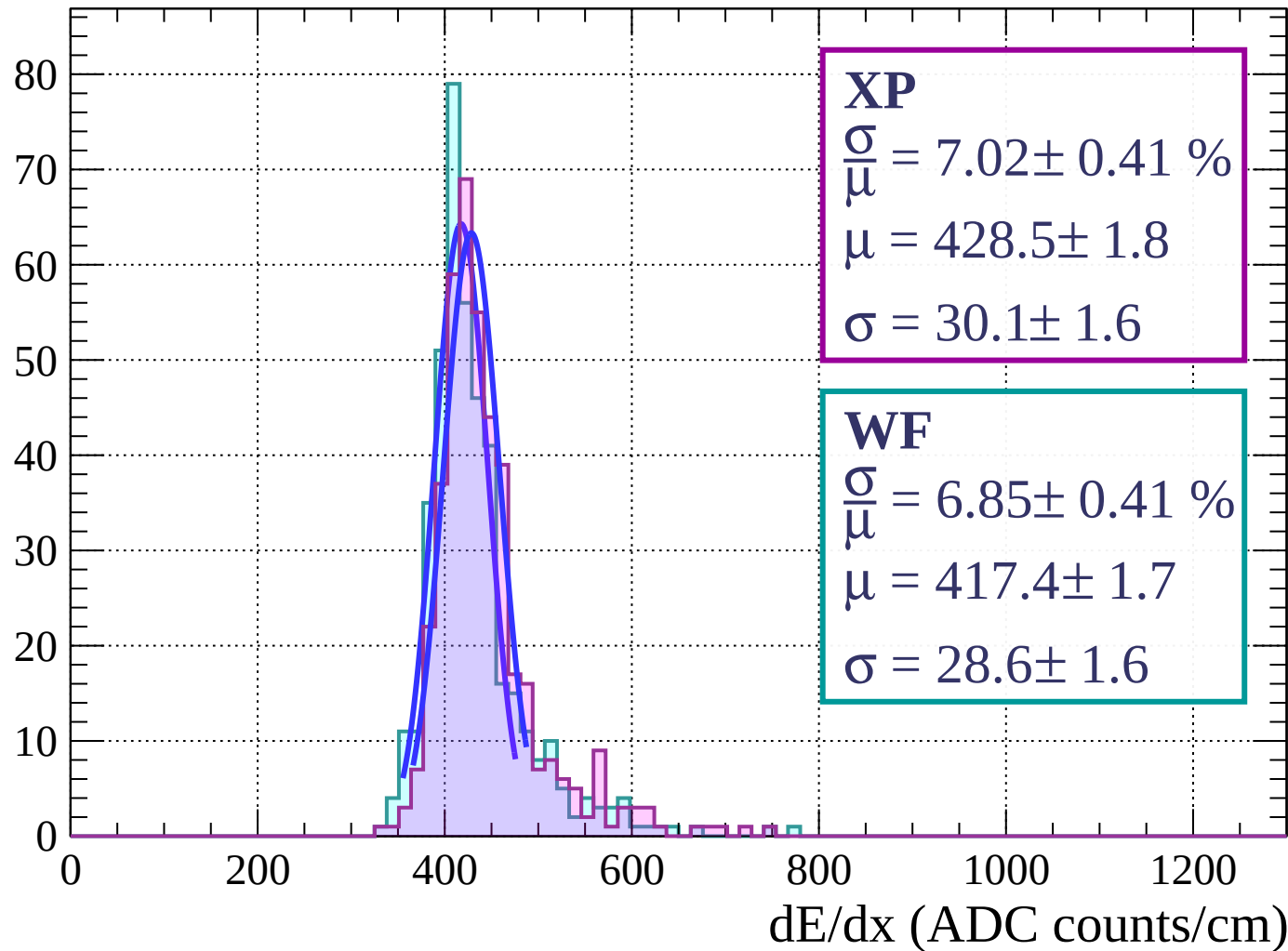
$$\mu = 419.0 \pm 1.9$$

$$\sigma = 30.7 \pm 2.1$$



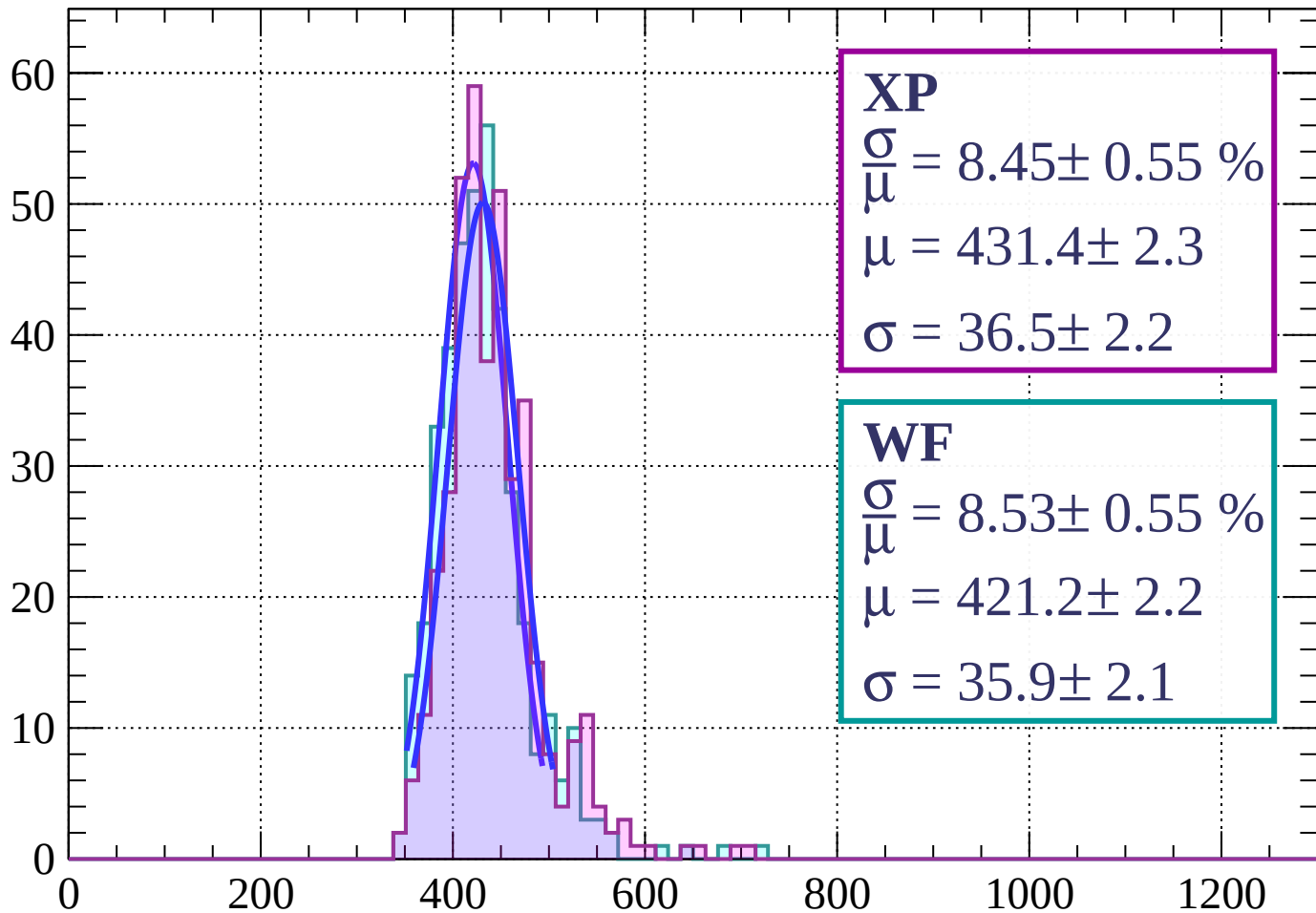
Energy loss | $6 < \theta < 9$

Count



Energy loss | $9 < \theta < 12$

Count



XP

$$\frac{\sigma}{\mu} = 8.45 \pm 0.55 \%$$

$$\mu = 431.4 \pm 2.3$$

$$\sigma = 36.5 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 8.53 \pm 0.55 \%$$

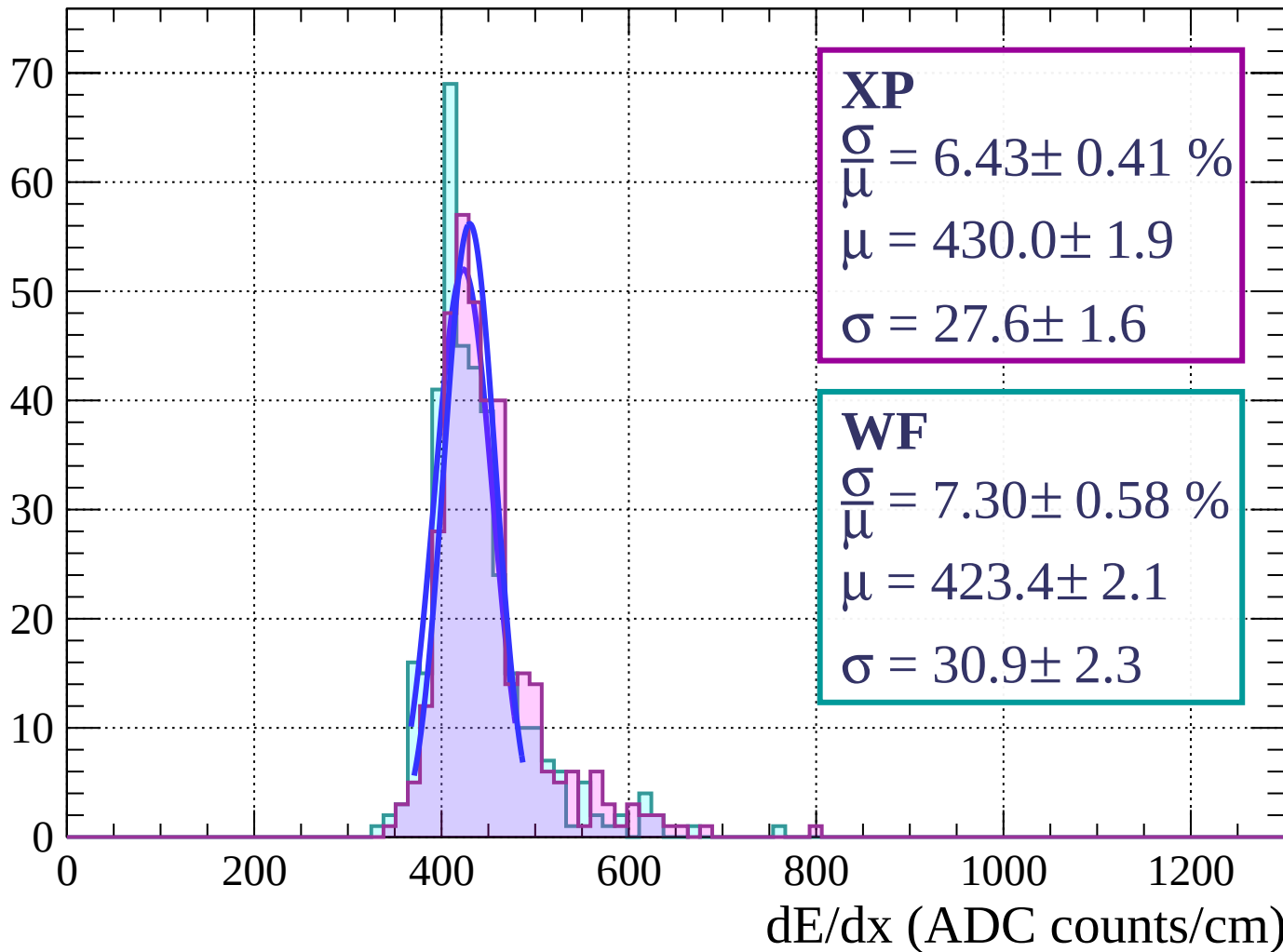
$$\mu = 421.2 \pm 2.2$$

$$\sigma = 35.9 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $12 < \theta < 15$

Count



Energy loss | $15 < \theta < 18$

Count

XP

$$\frac{\sigma}{\mu} = 7.16 \pm 0.60 \%$$

$$\mu = 433.6 \pm 2.1$$

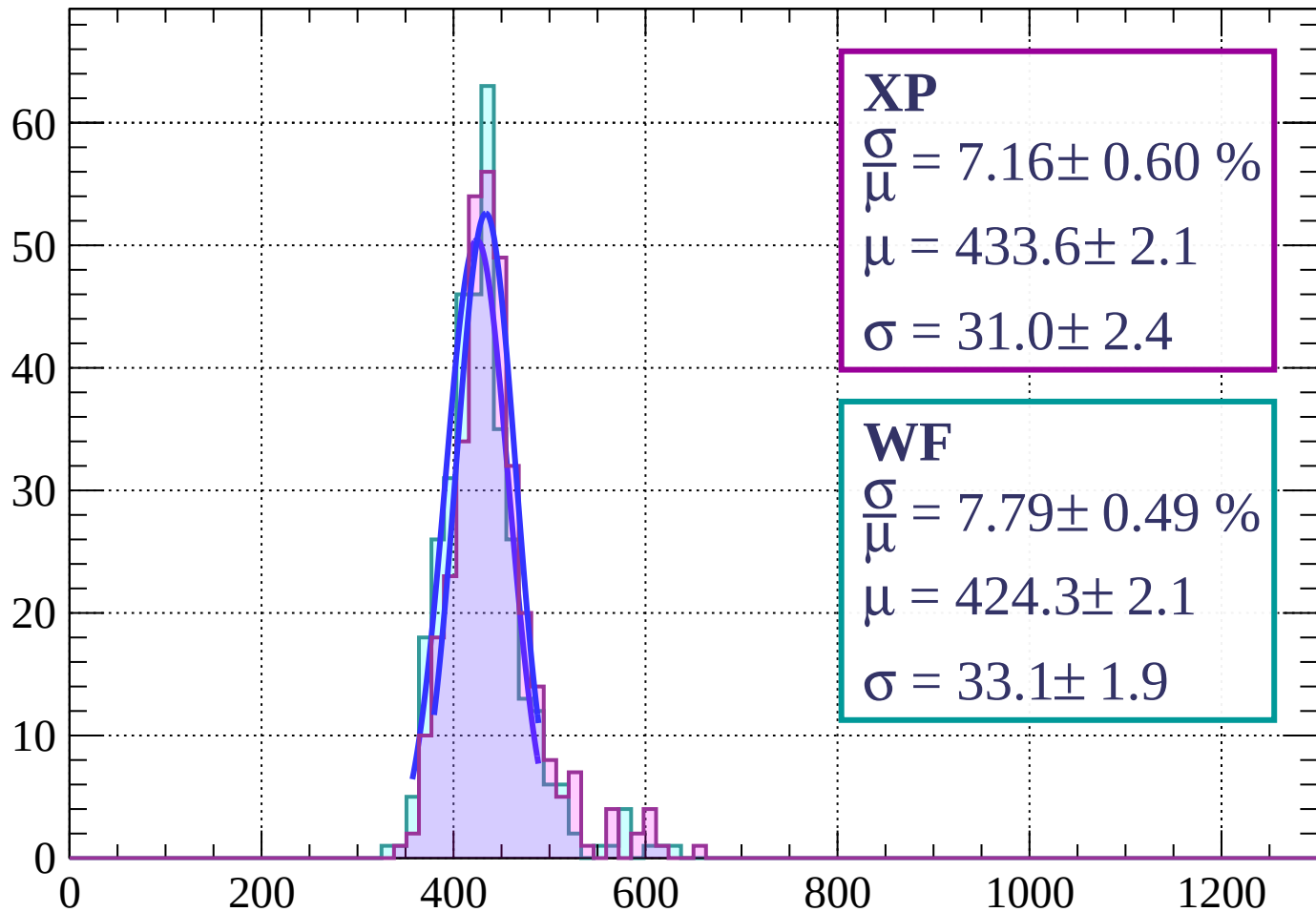
$$\sigma = 31.0 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 7.79 \pm 0.49 \%$$

$$\mu = 424.3 \pm 2.1$$

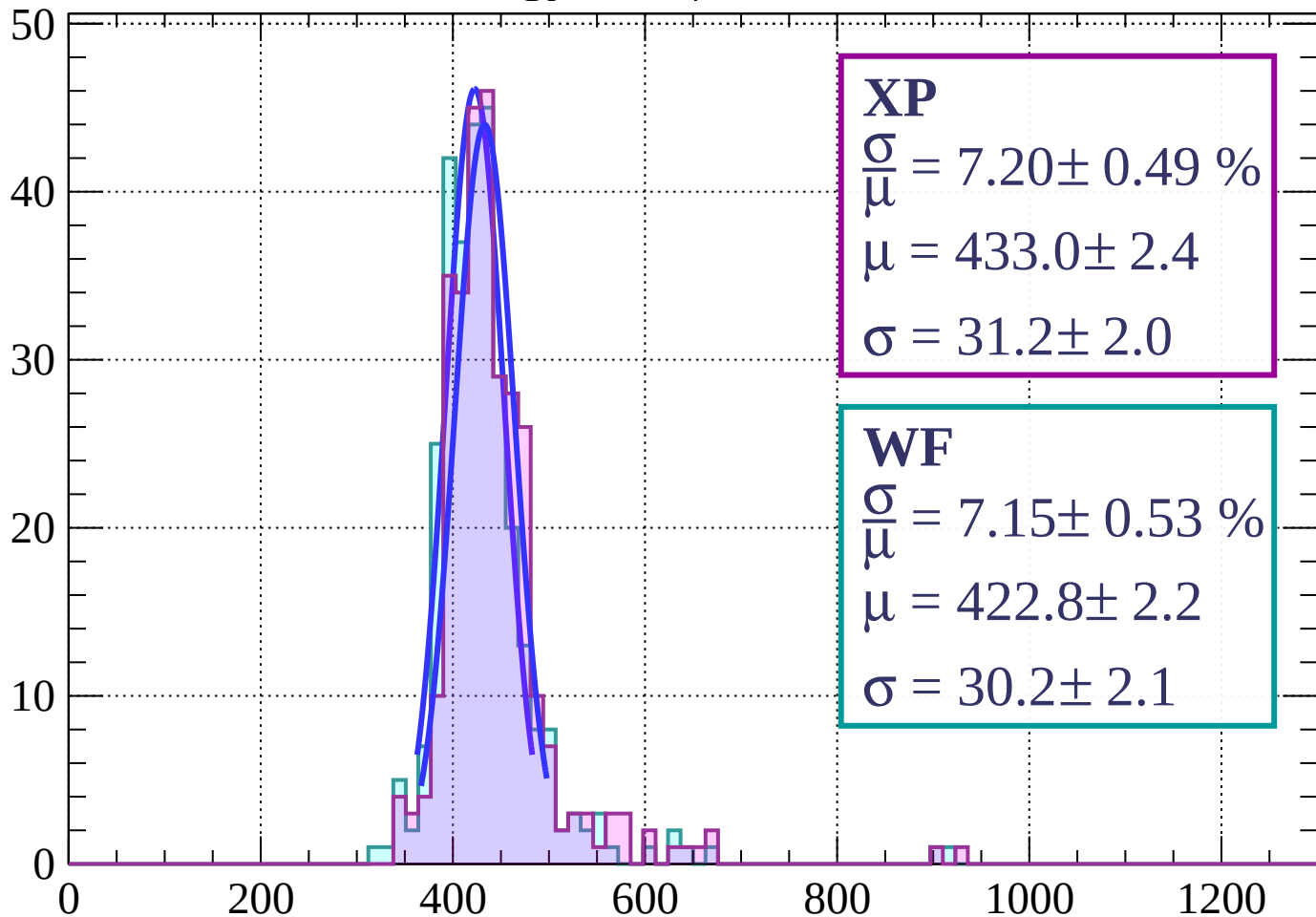
$$\sigma = 33.1 \pm 1.9$$



dE/dx (ADC counts/cm)

Energy loss | $18 < \theta < 21$

Count



XP

$$\frac{\sigma}{\mu} = 7.20 \pm 0.49 \%$$

$$\mu = 433.0 \pm 2.4$$

$$\sigma = 31.2 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 7.15 \pm 0.53 \%$$

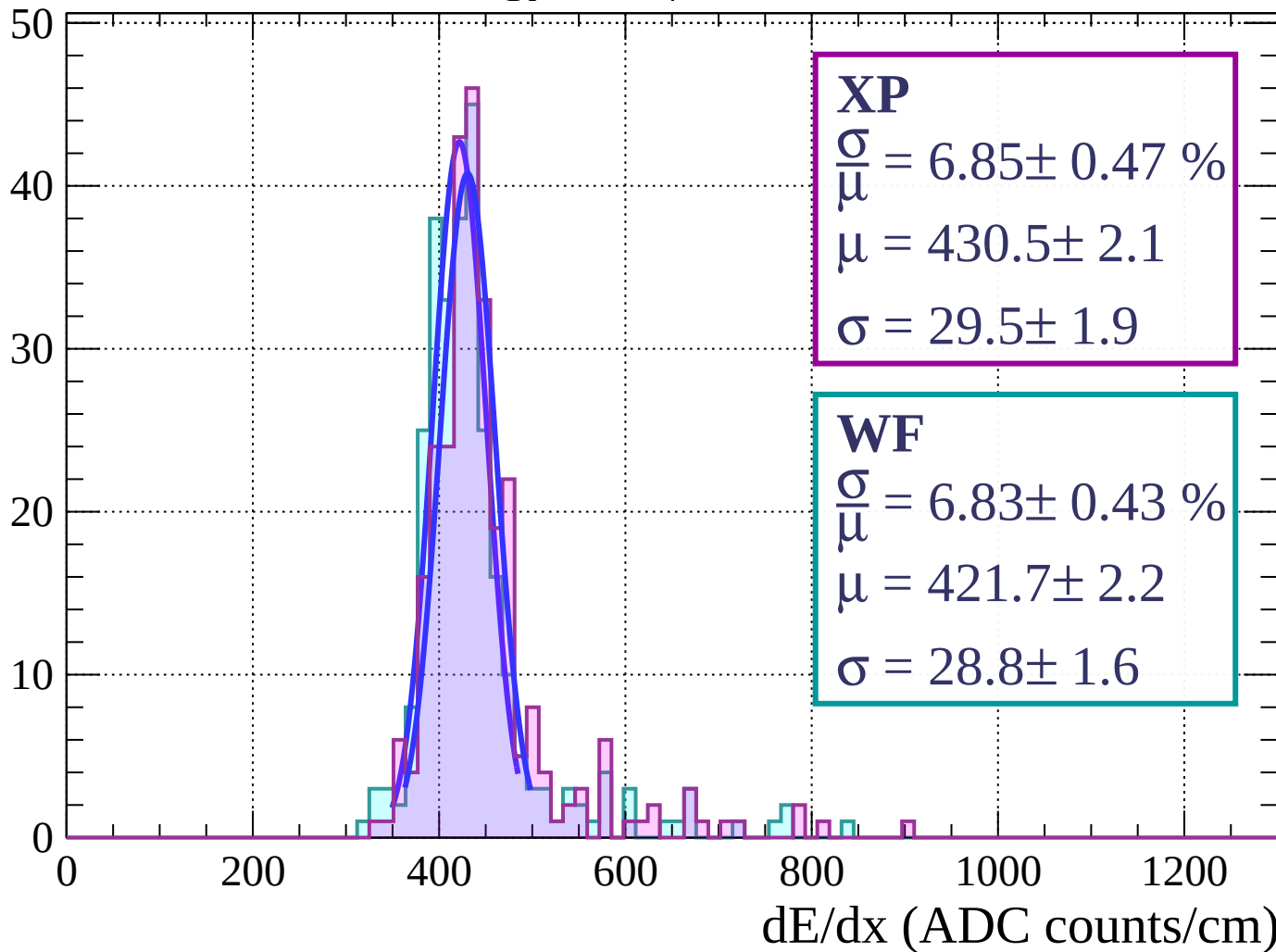
$$\mu = 422.8 \pm 2.2$$

$$\sigma = 30.2 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $21 < \theta < 24$

Count



Energy loss | $24 < \theta < 27$

Count

XP

$$\frac{\sigma}{\mu} = 7.17 \pm 0.62 \%$$

$$\mu = 433.8 \pm 2.7$$

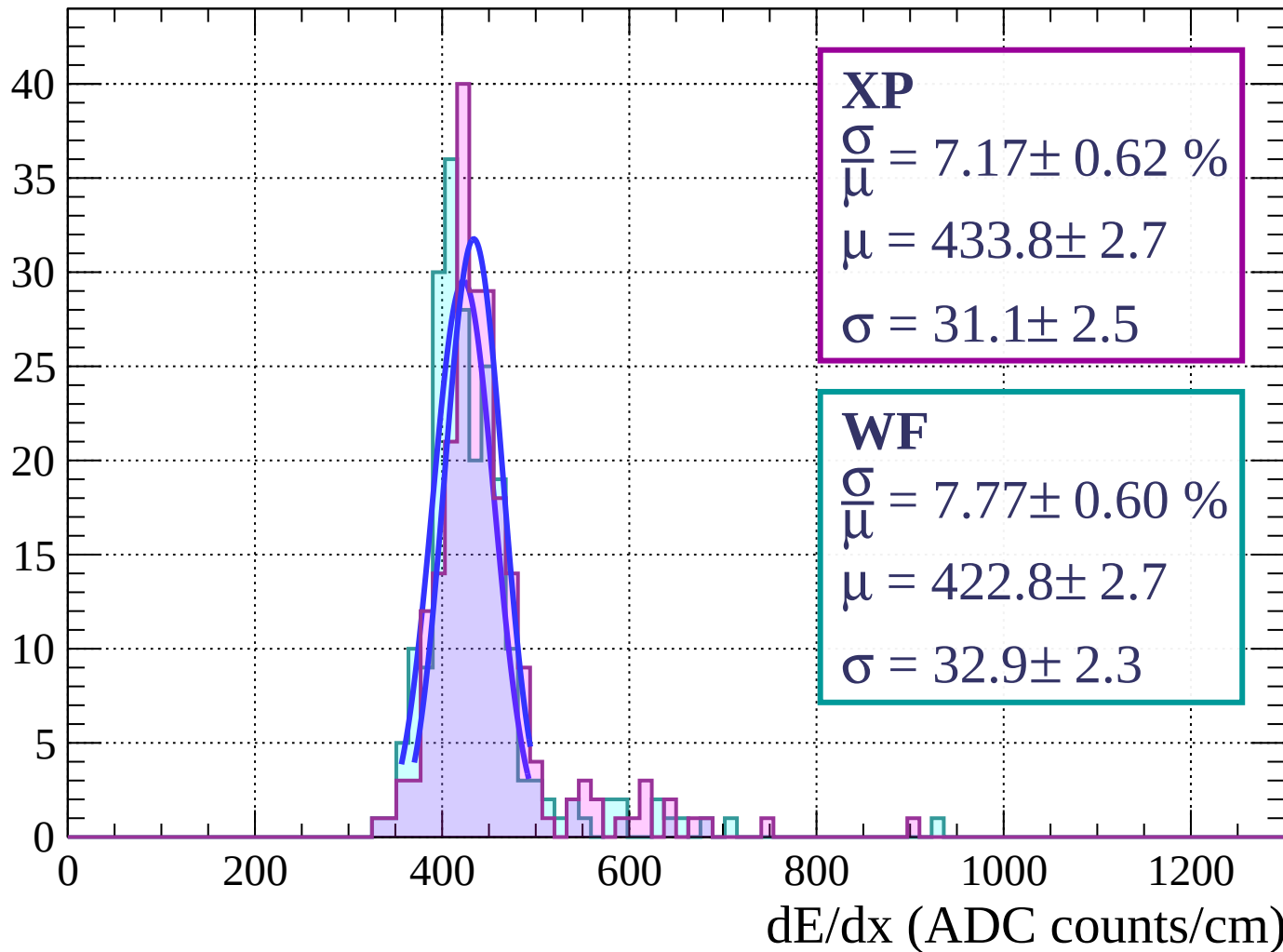
$$\sigma = 31.1 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 7.77 \pm 0.60 \%$$

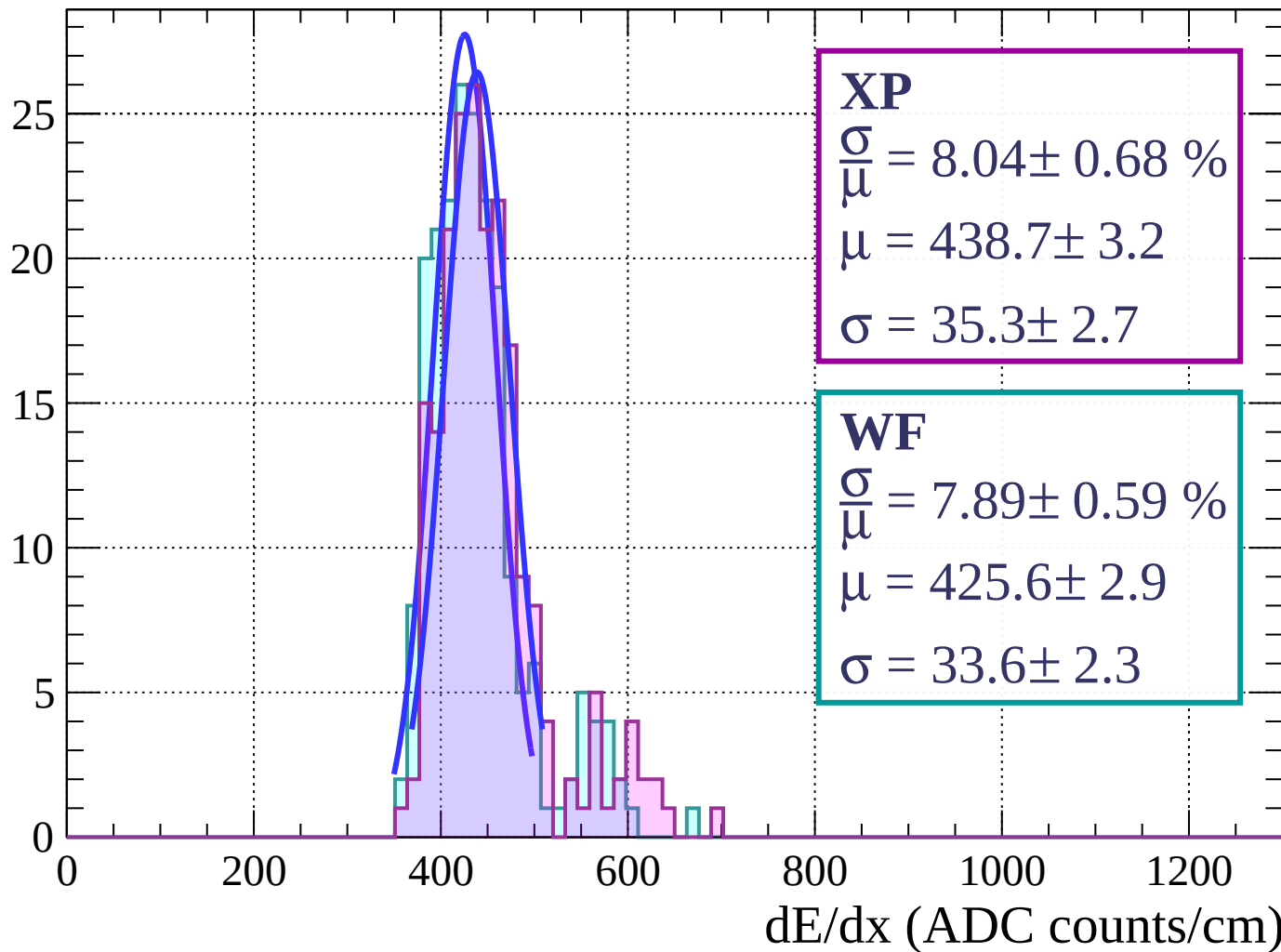
$$\mu = 422.8 \pm 2.7$$

$$\sigma = 32.9 \pm 2.3$$



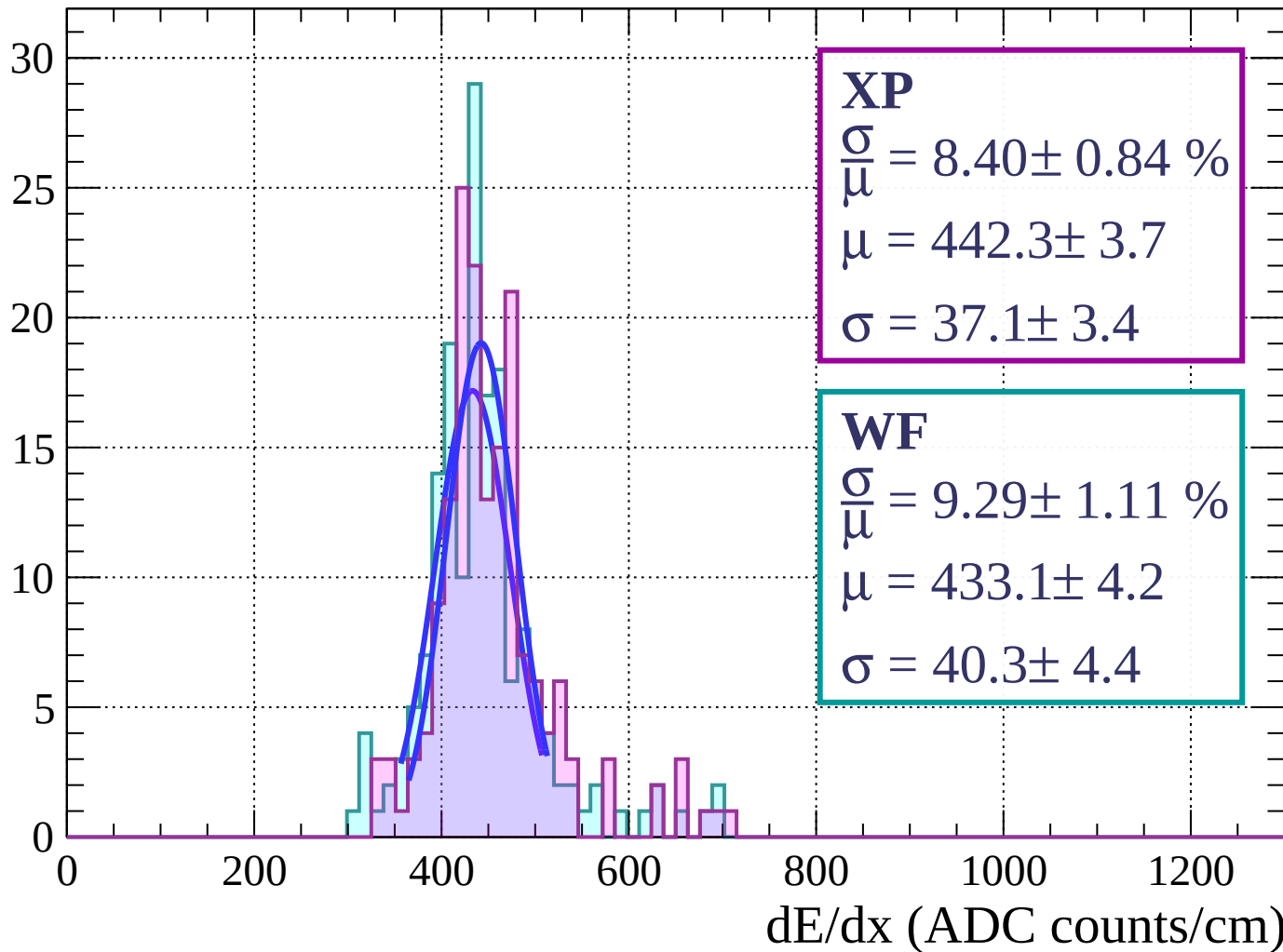
Energy loss | $27 < \theta < 30$

Count



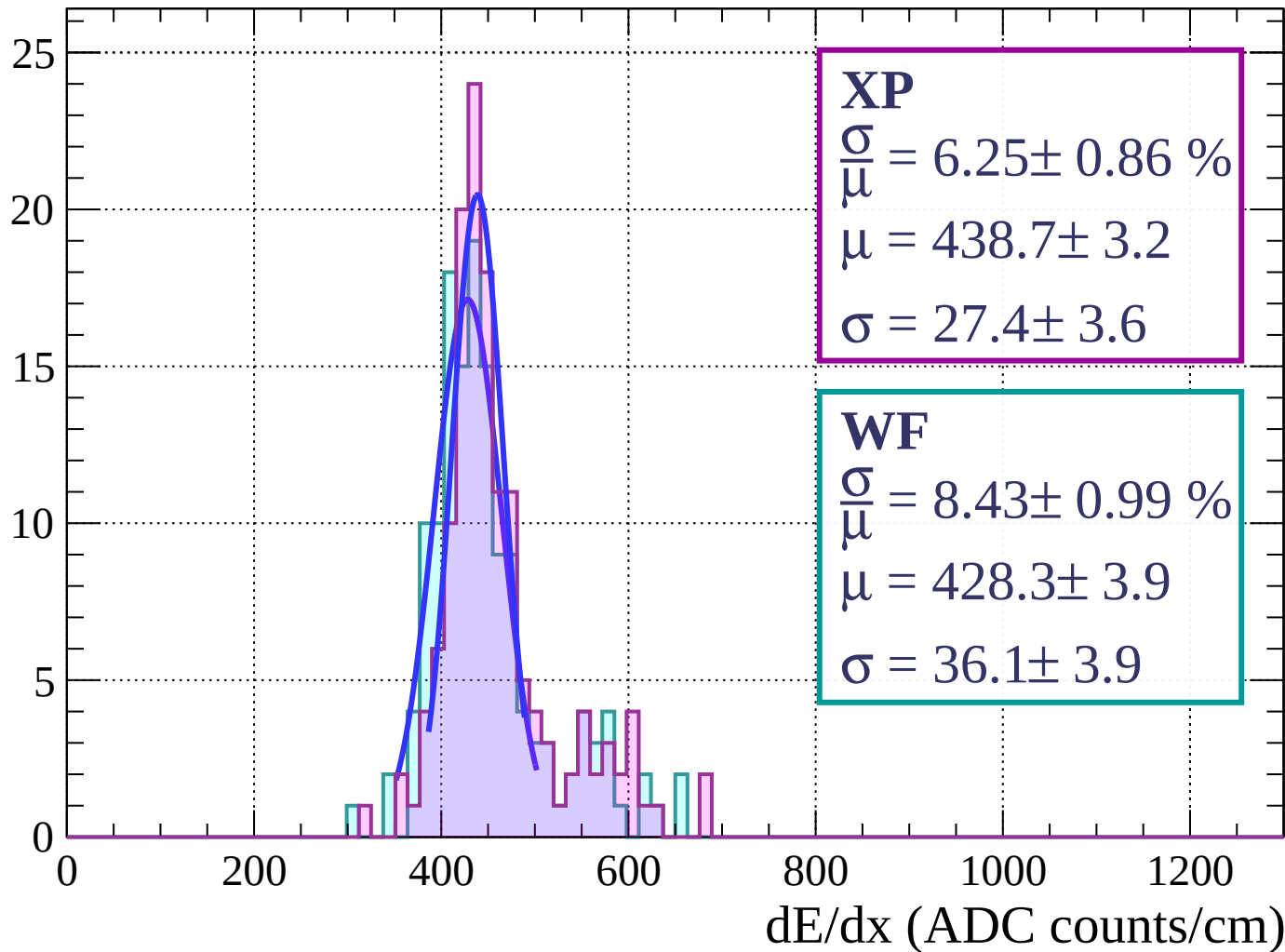
Energy loss | $30 < \theta < 33$

Count



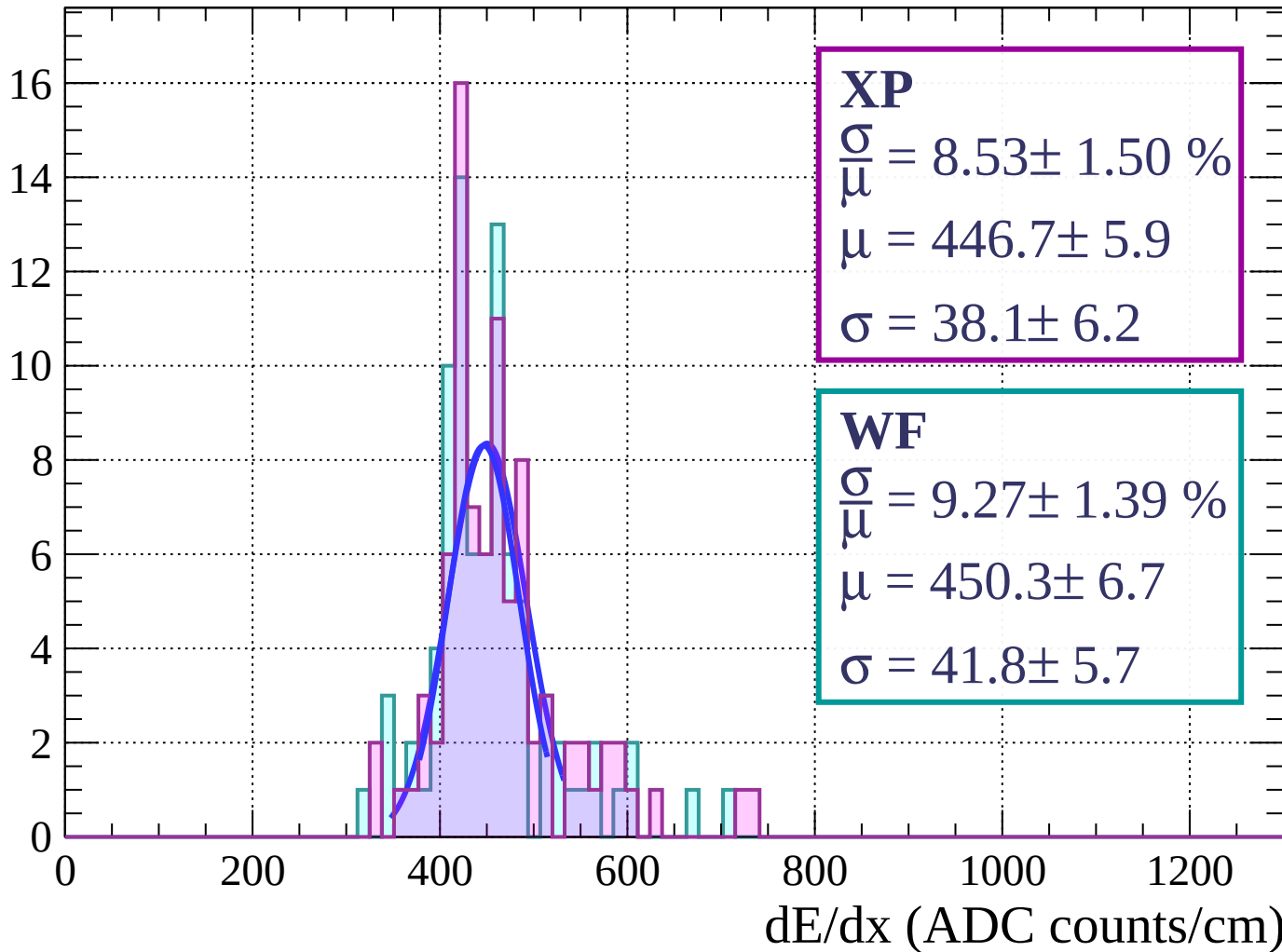
Energy loss | $33 < \theta < 36$

Count



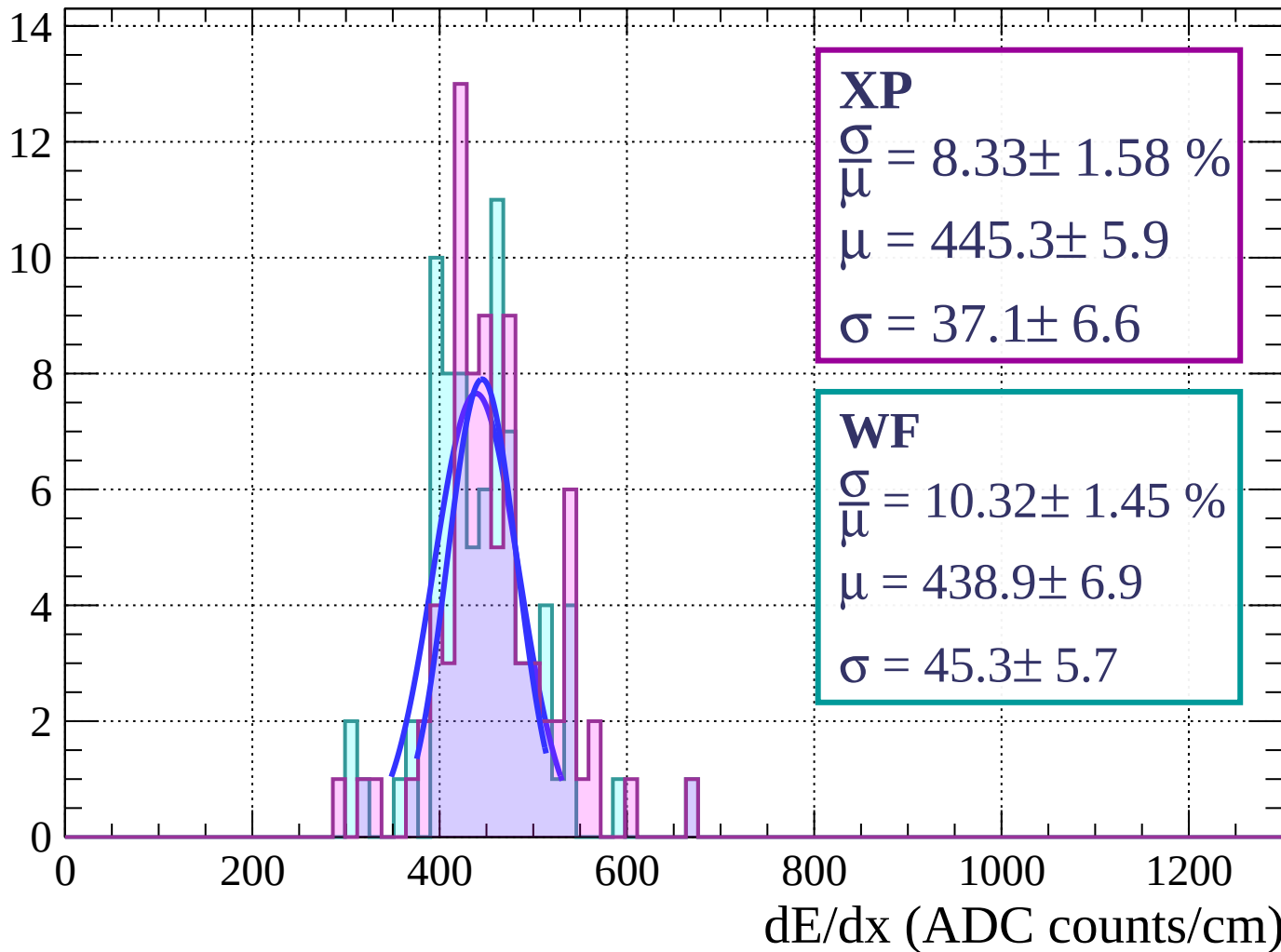
Energy loss | $36 < \theta < 39$

Count



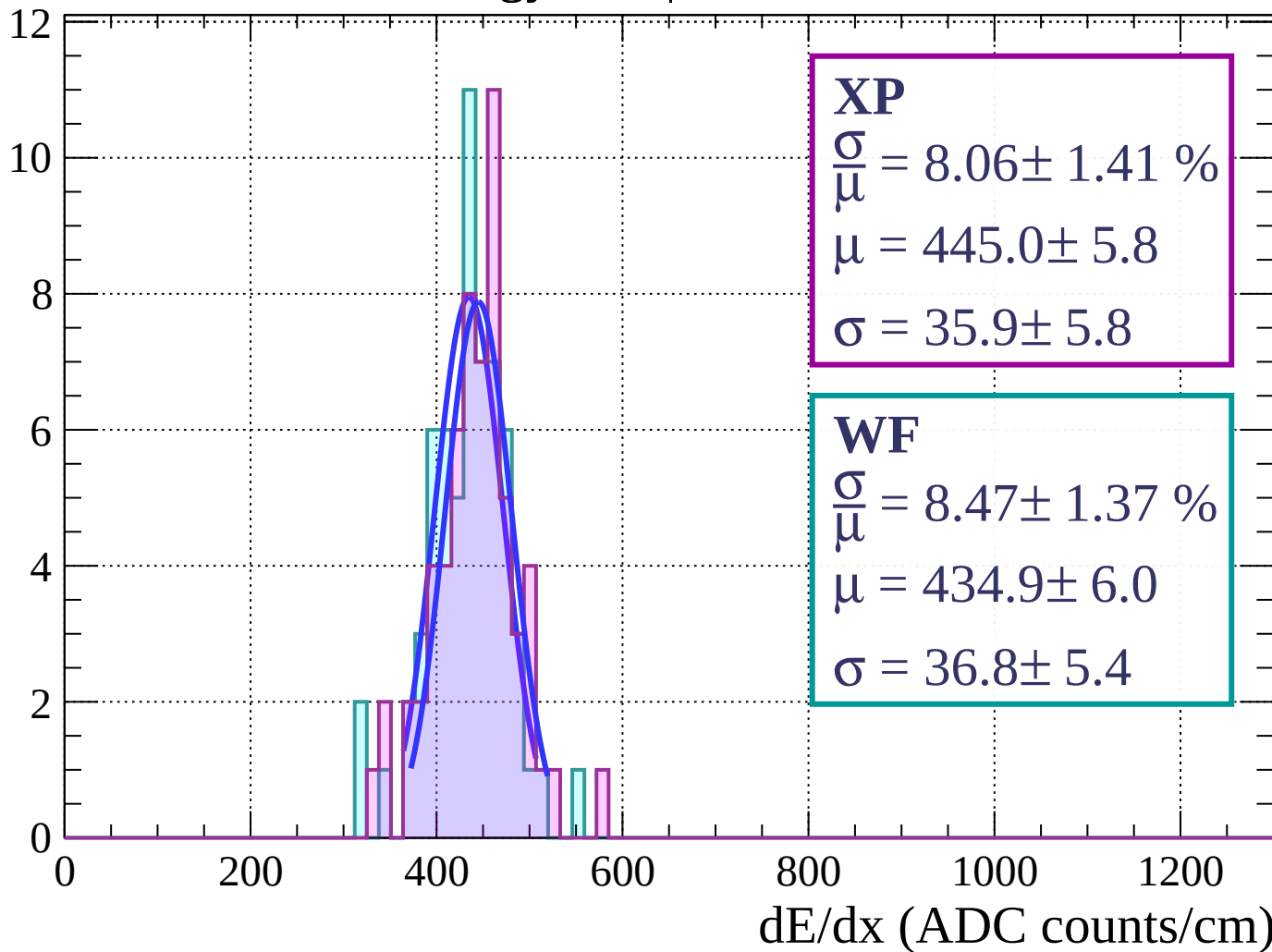
Energy loss | $39 < \theta < 42$

Count



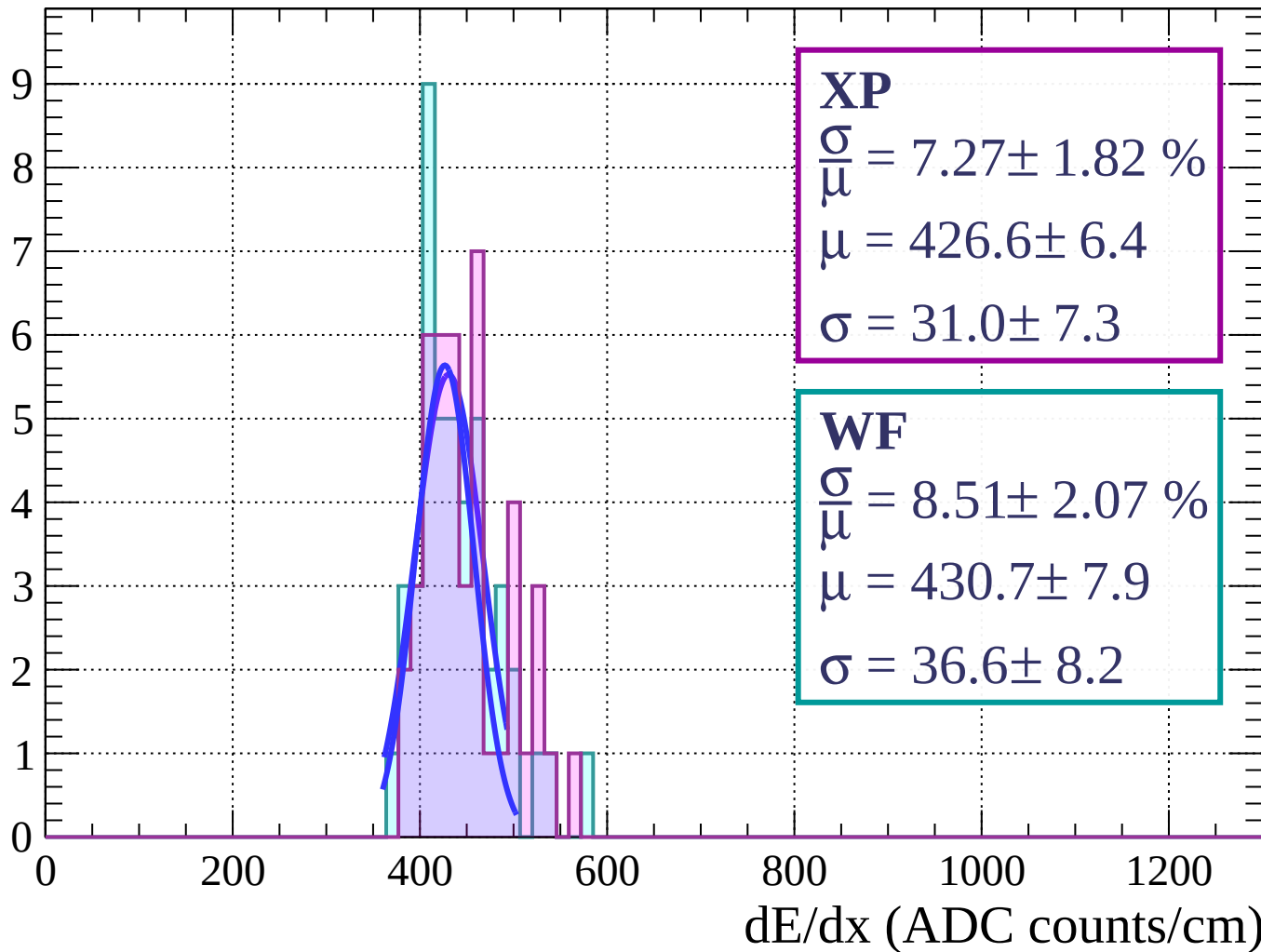
Energy loss | $42 < \theta < 45$

Count



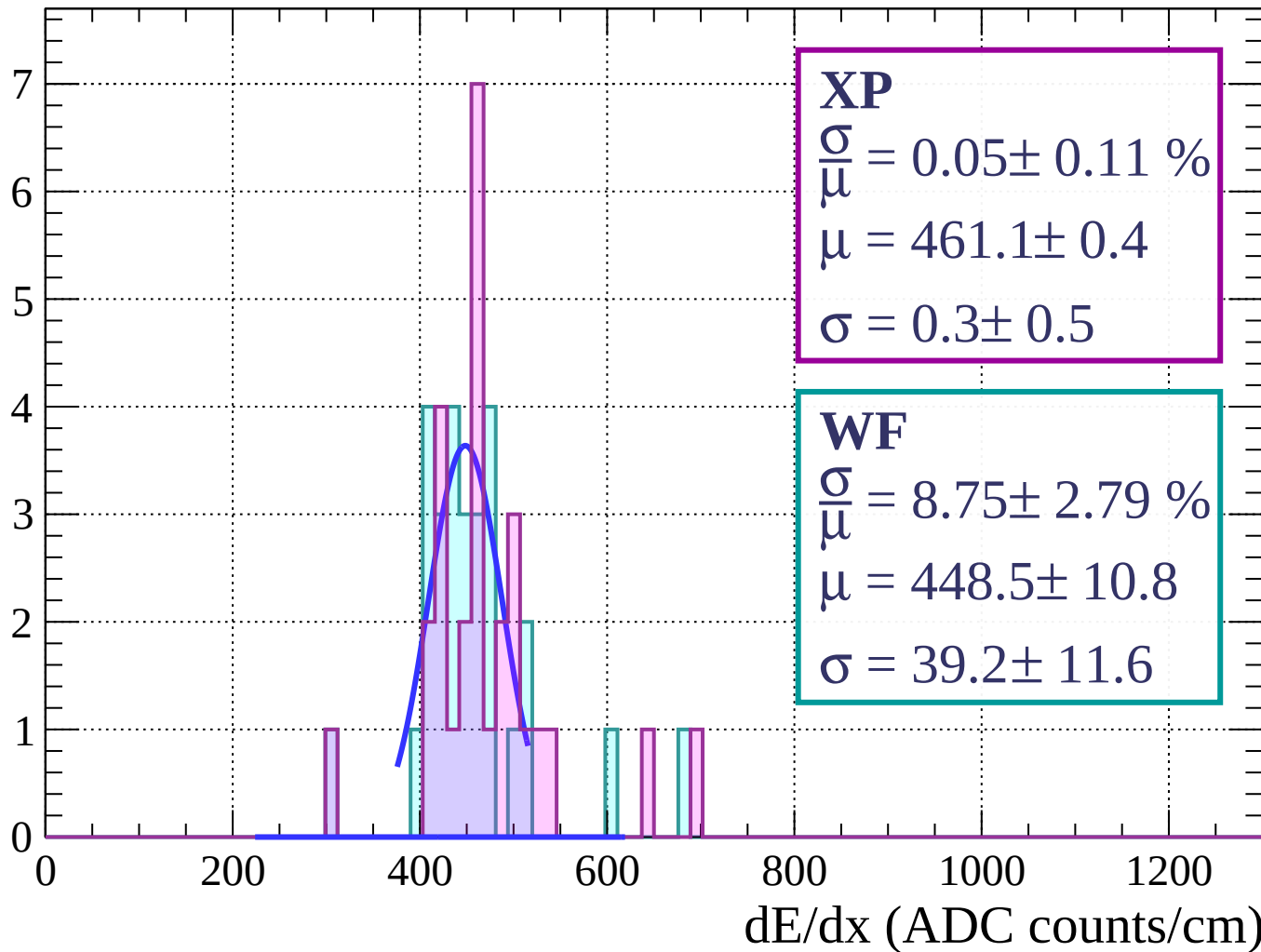
Energy loss | $45 < \theta < 48$

Count



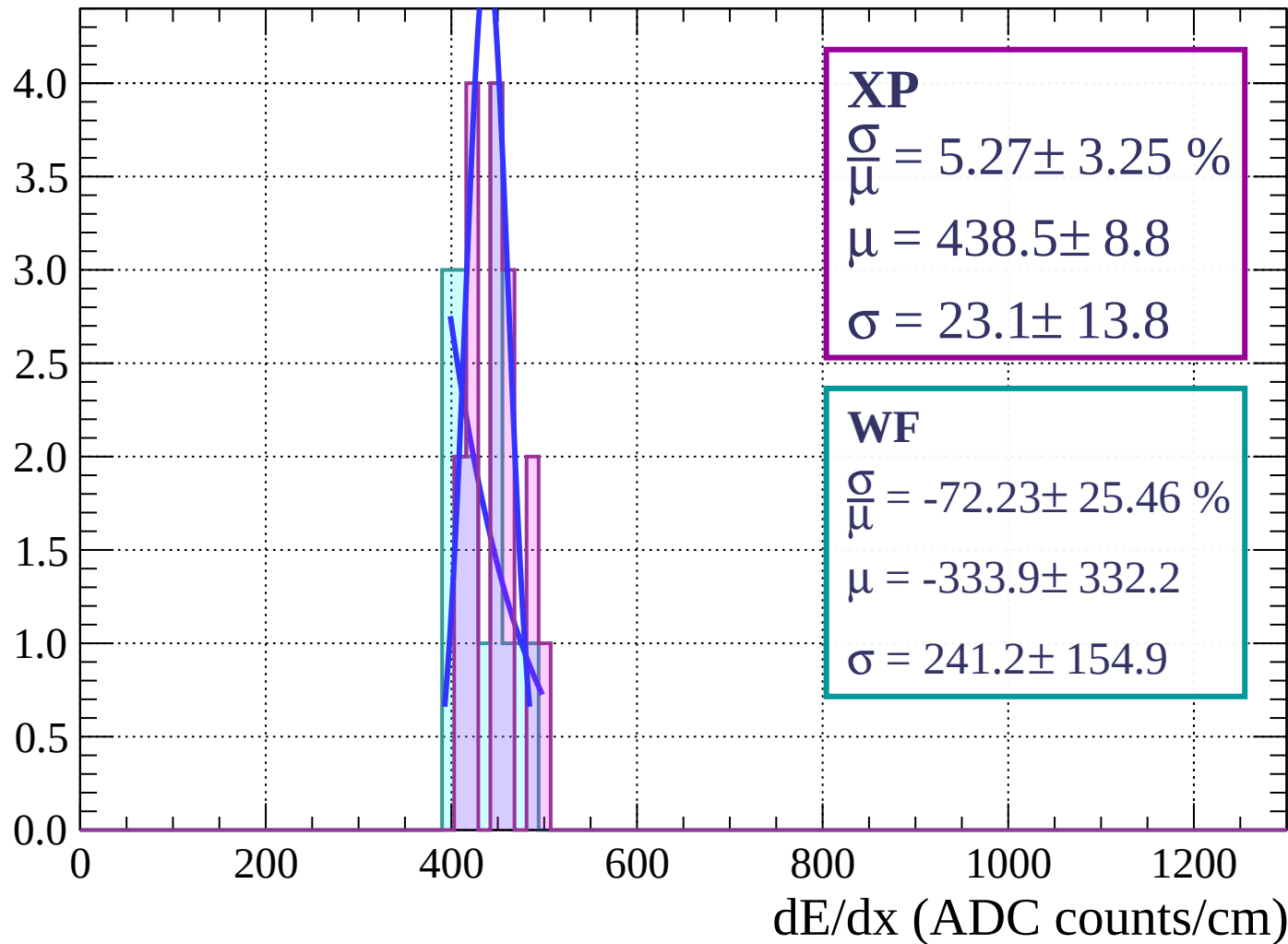
Energy loss | $48 < \theta < 51$

Count



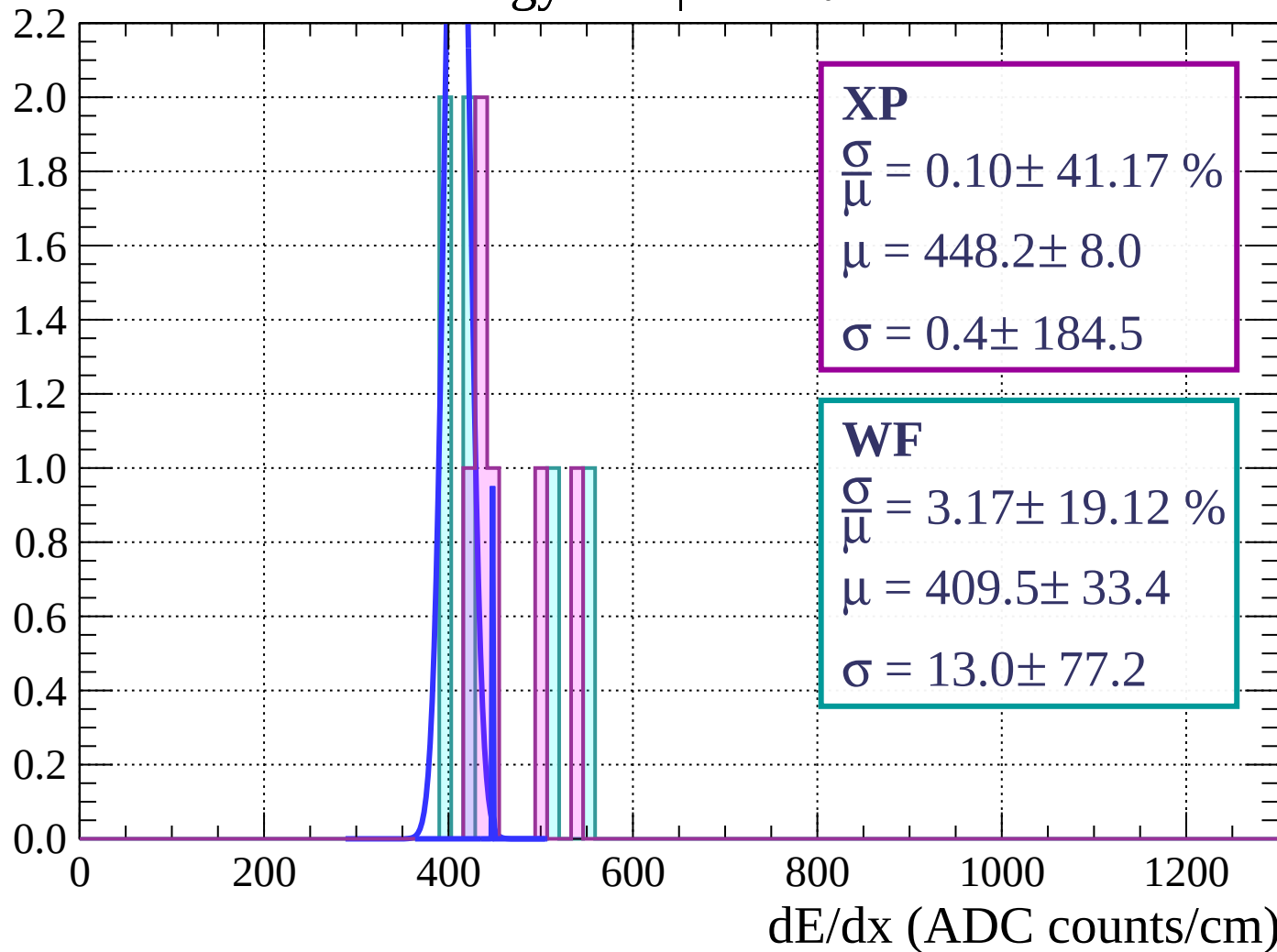
Energy loss | $51 < \theta < 54$

Count



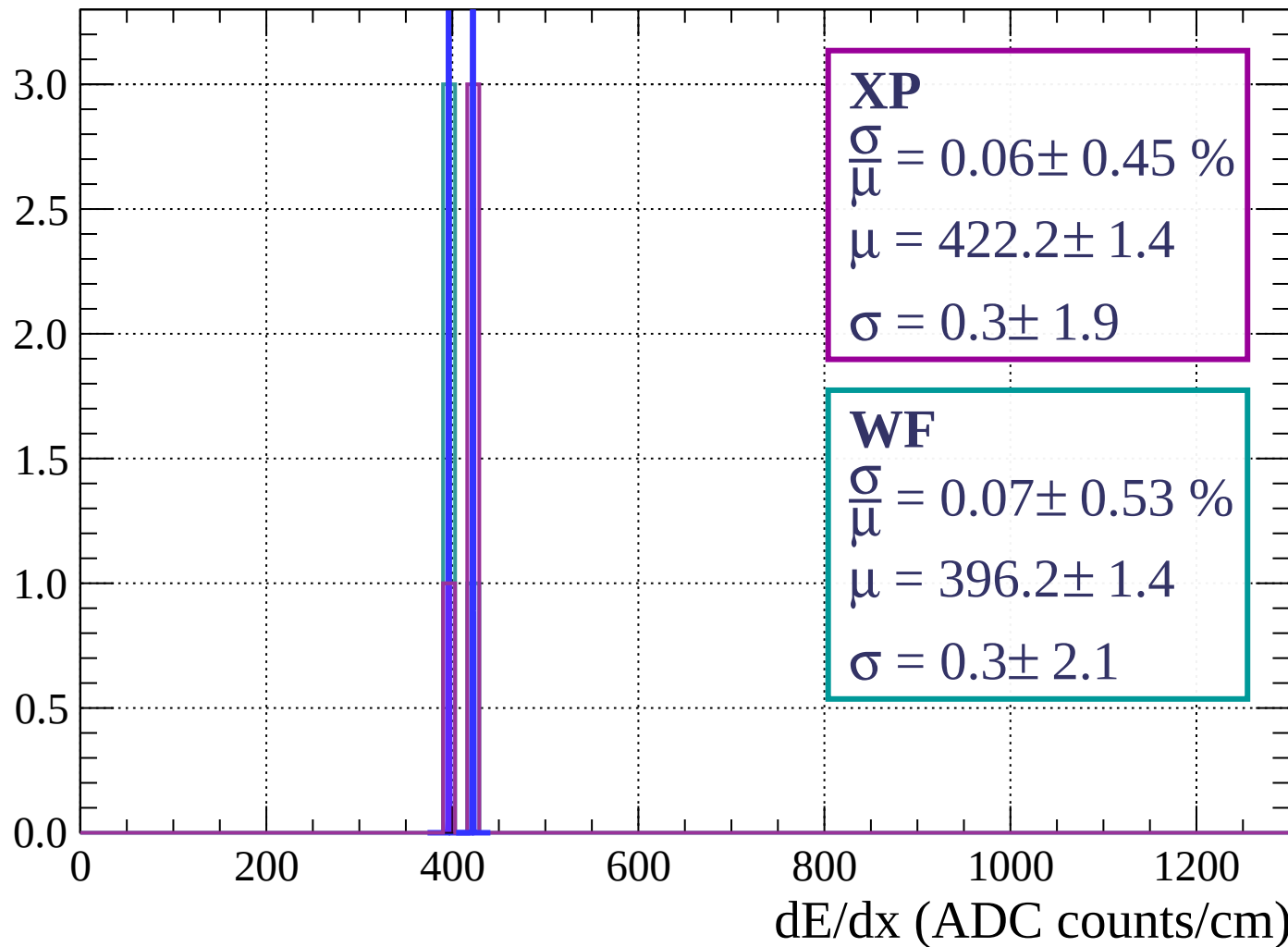
Energy loss | $54 < \theta < 57$

Count



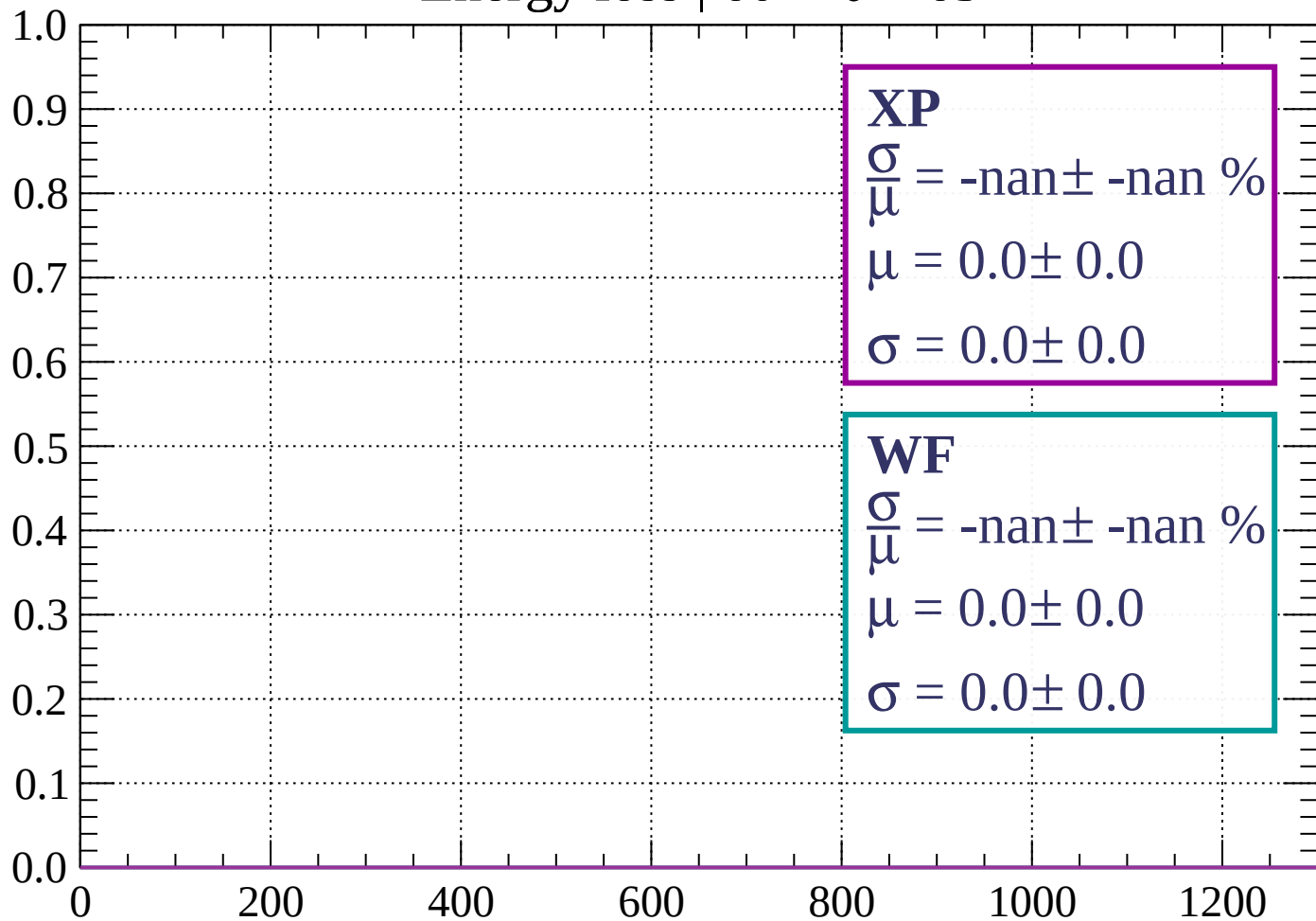
Energy loss | $57 < \theta < 60$

Count



Energy loss | $60 < \theta < 63$

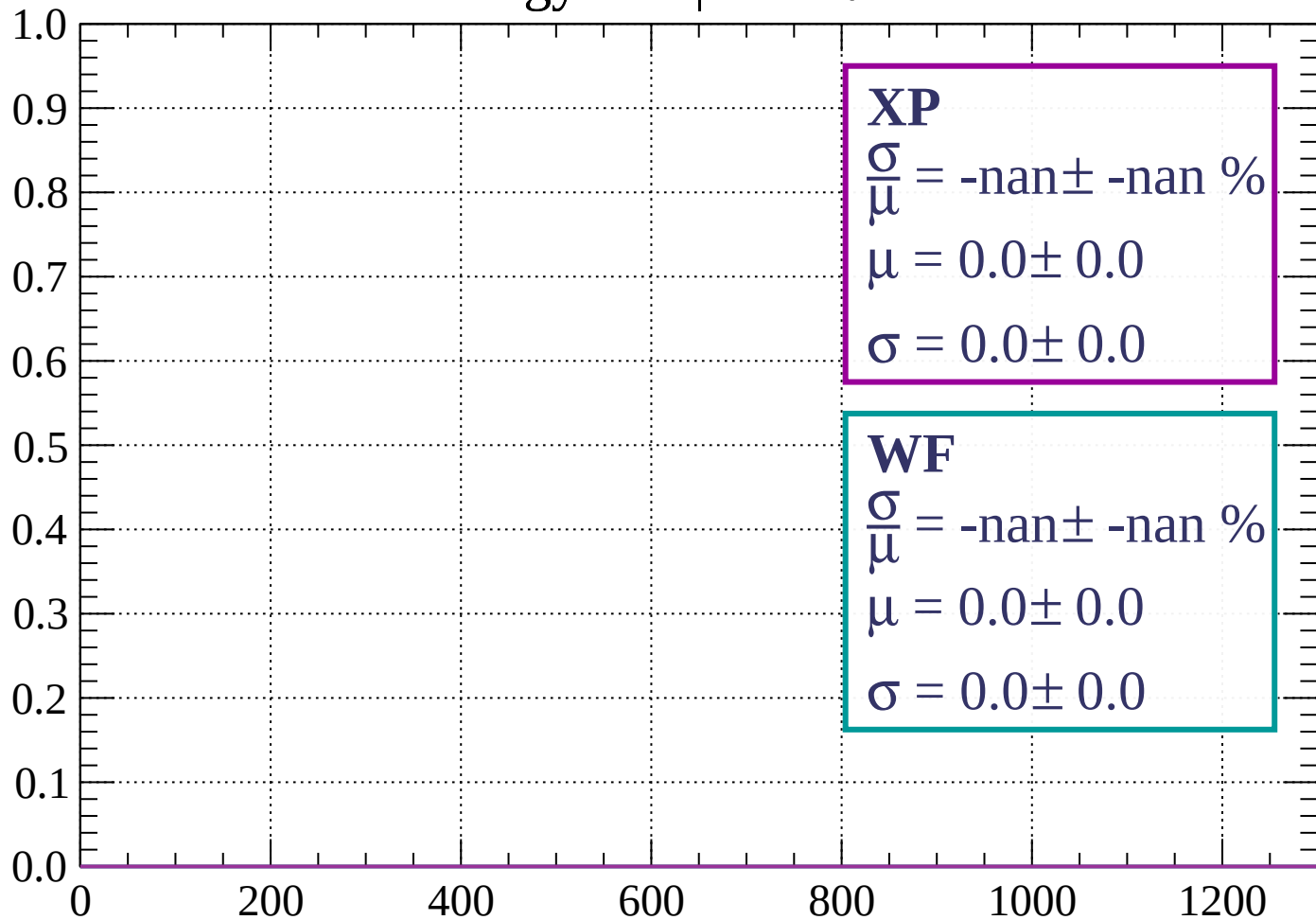
Count



dE/dx (ADC counts/cm)

Energy loss | $63 < \theta < 66$

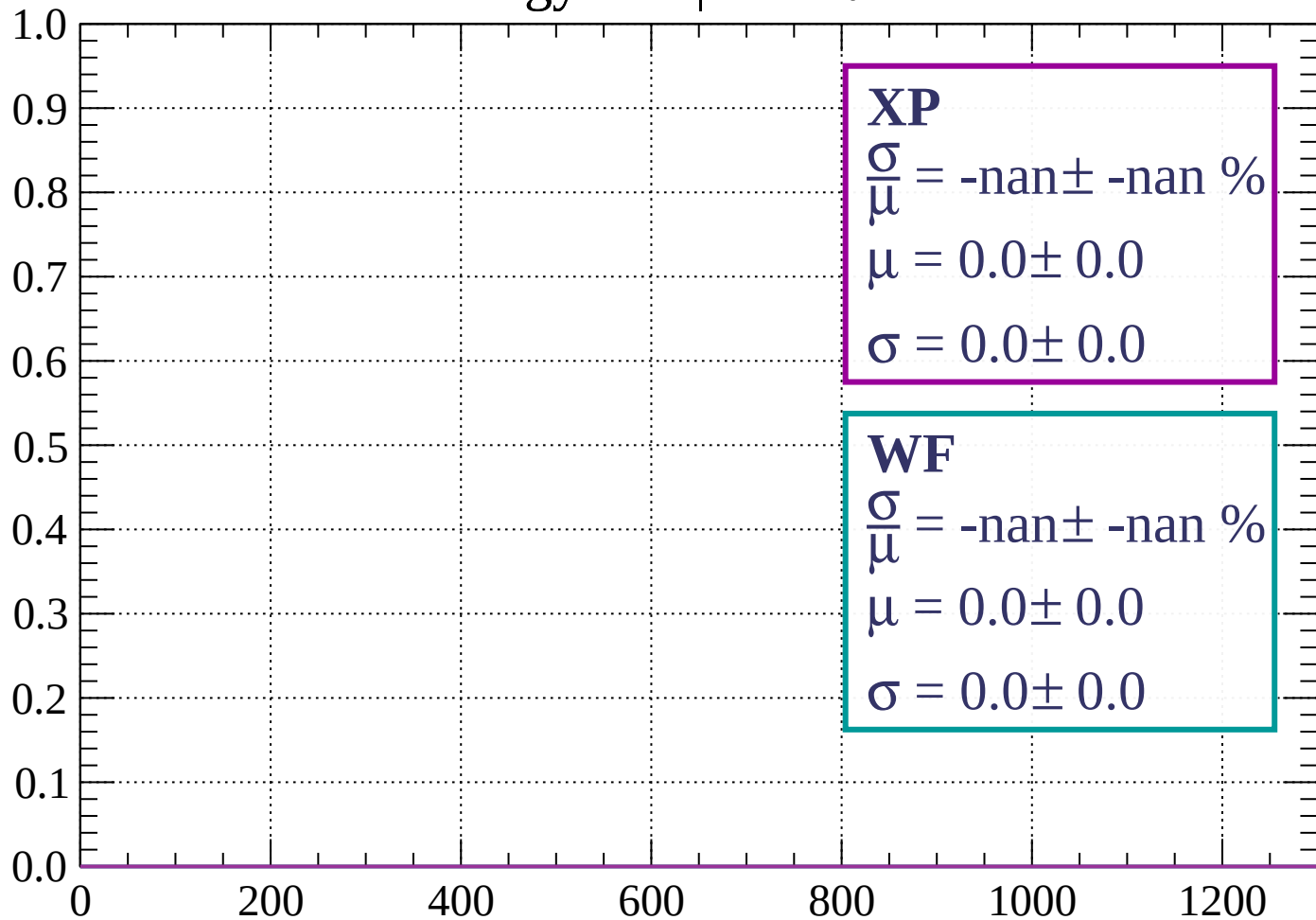
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 69$

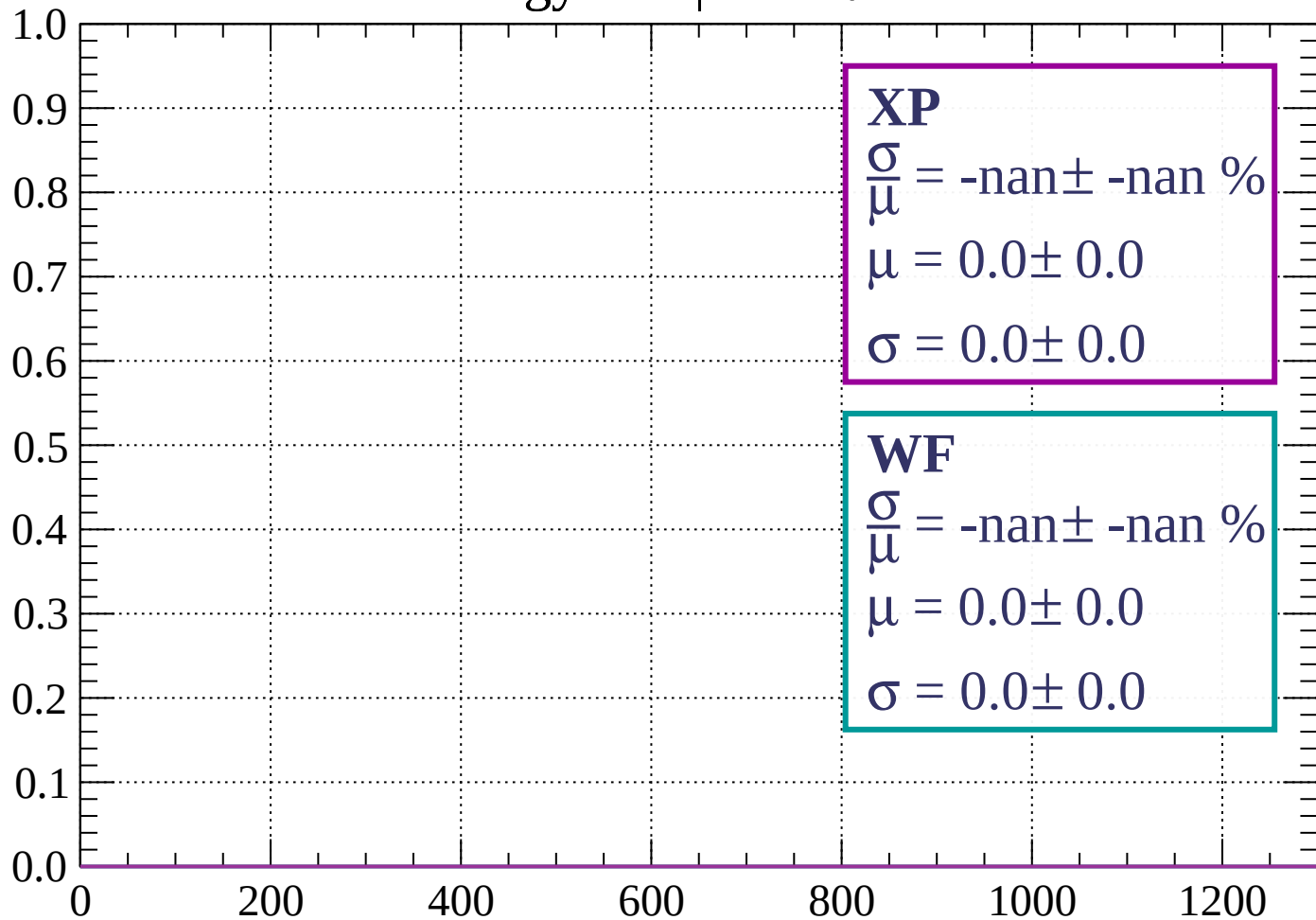
Count



dE/dx (ADC counts/cm)

Energy loss | $69 < \theta < 72$

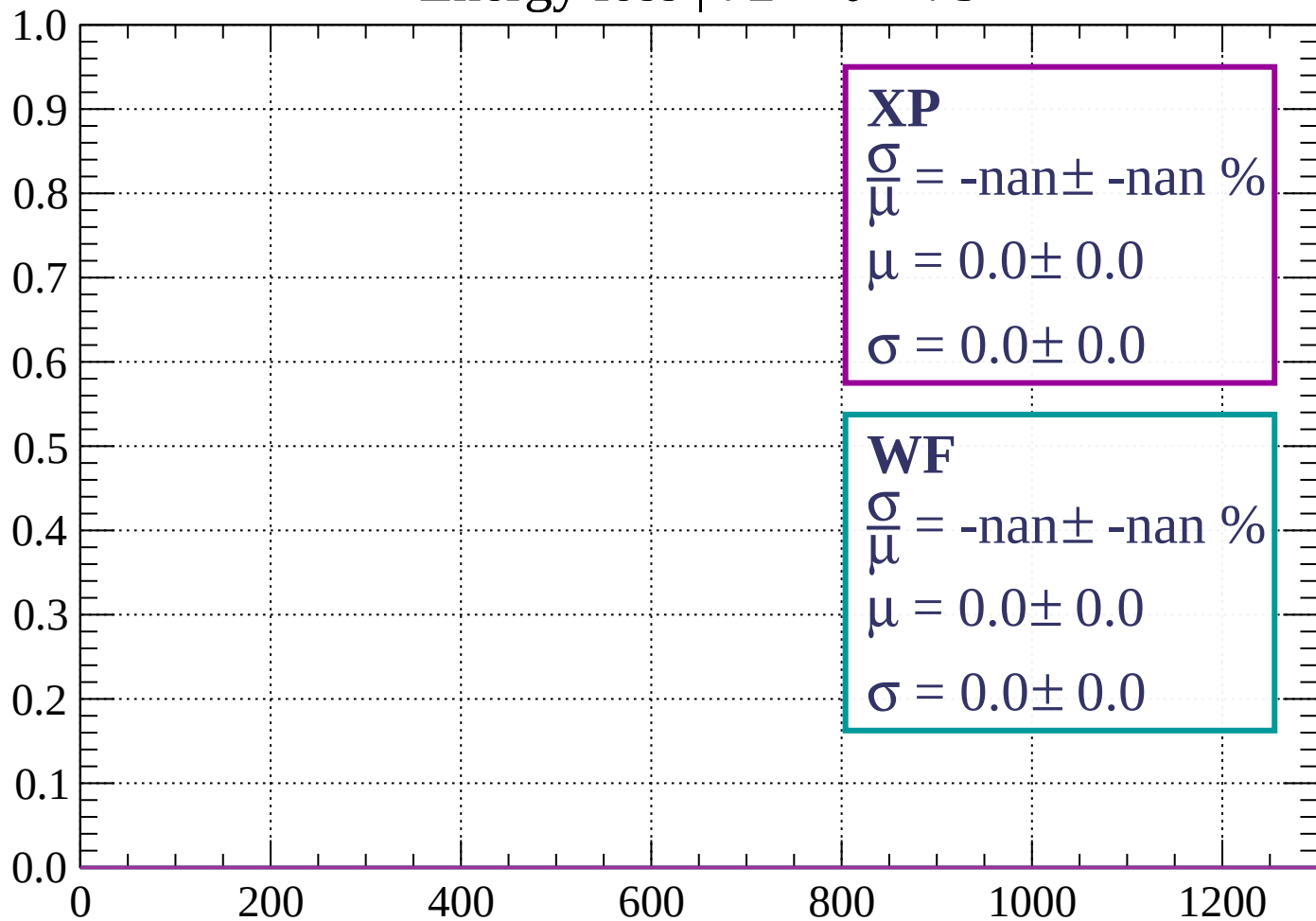
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 75$

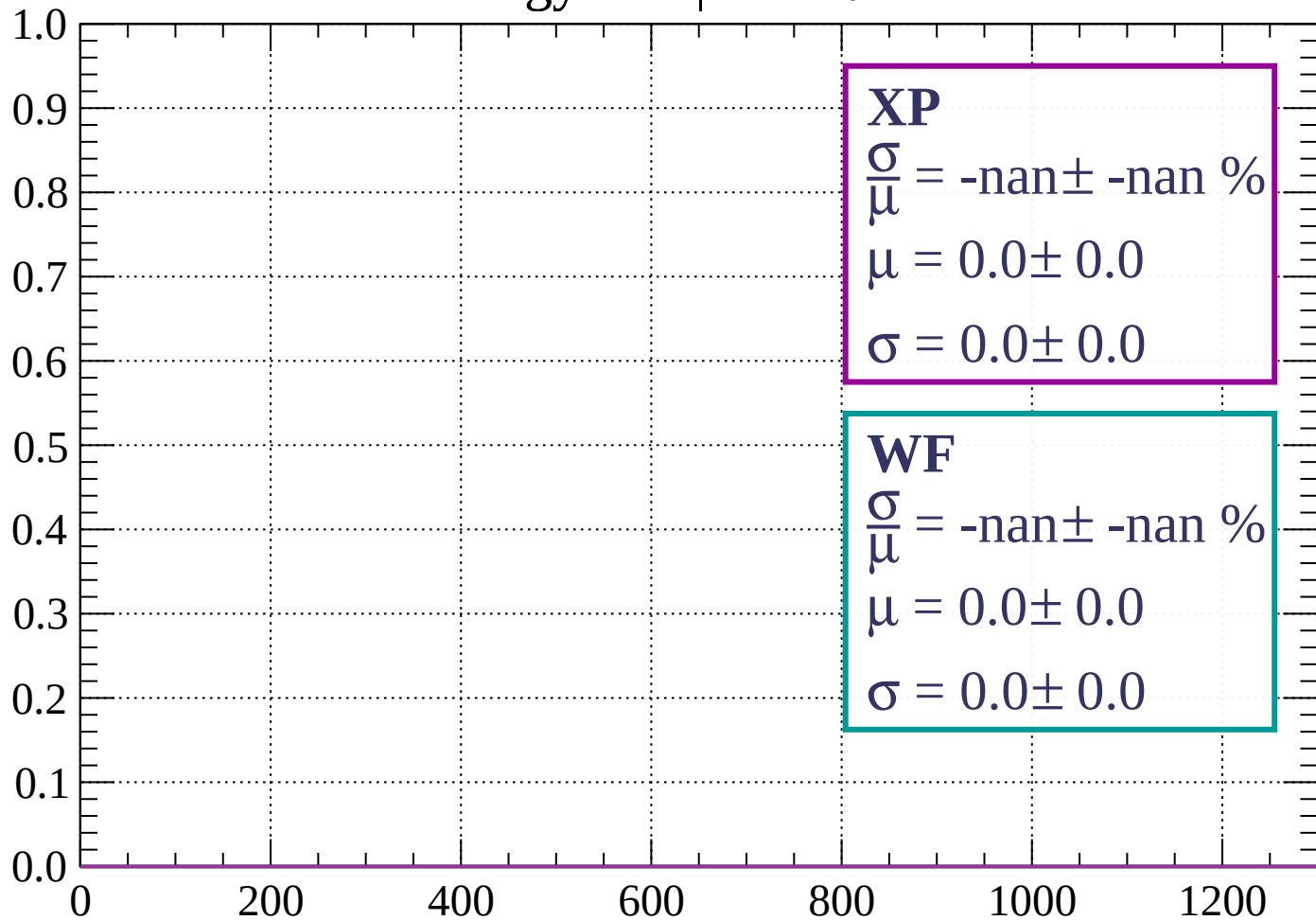
Count



dE/dx (ADC counts/cm)

Energy loss | $75 < \theta < 78$

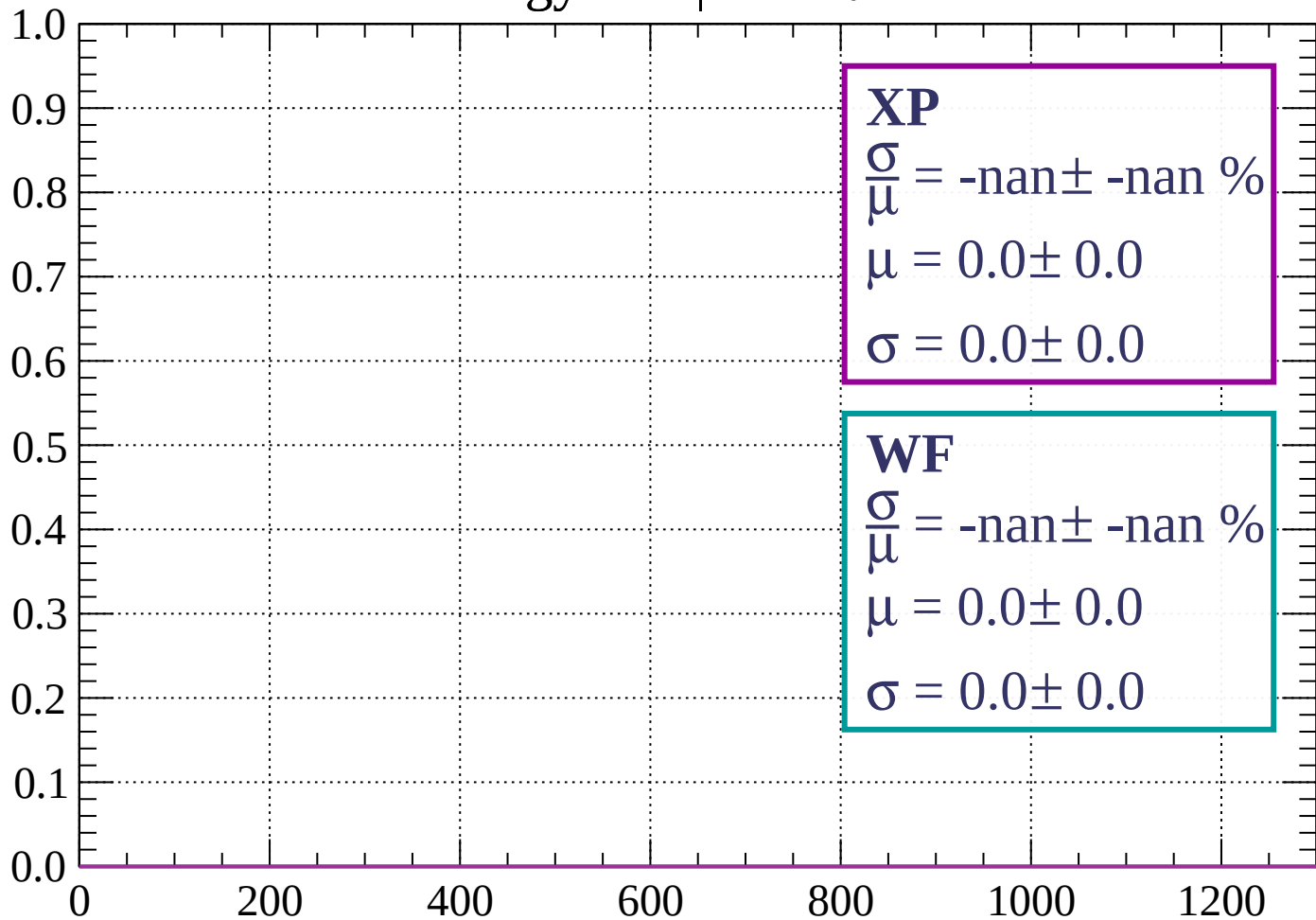
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 81$

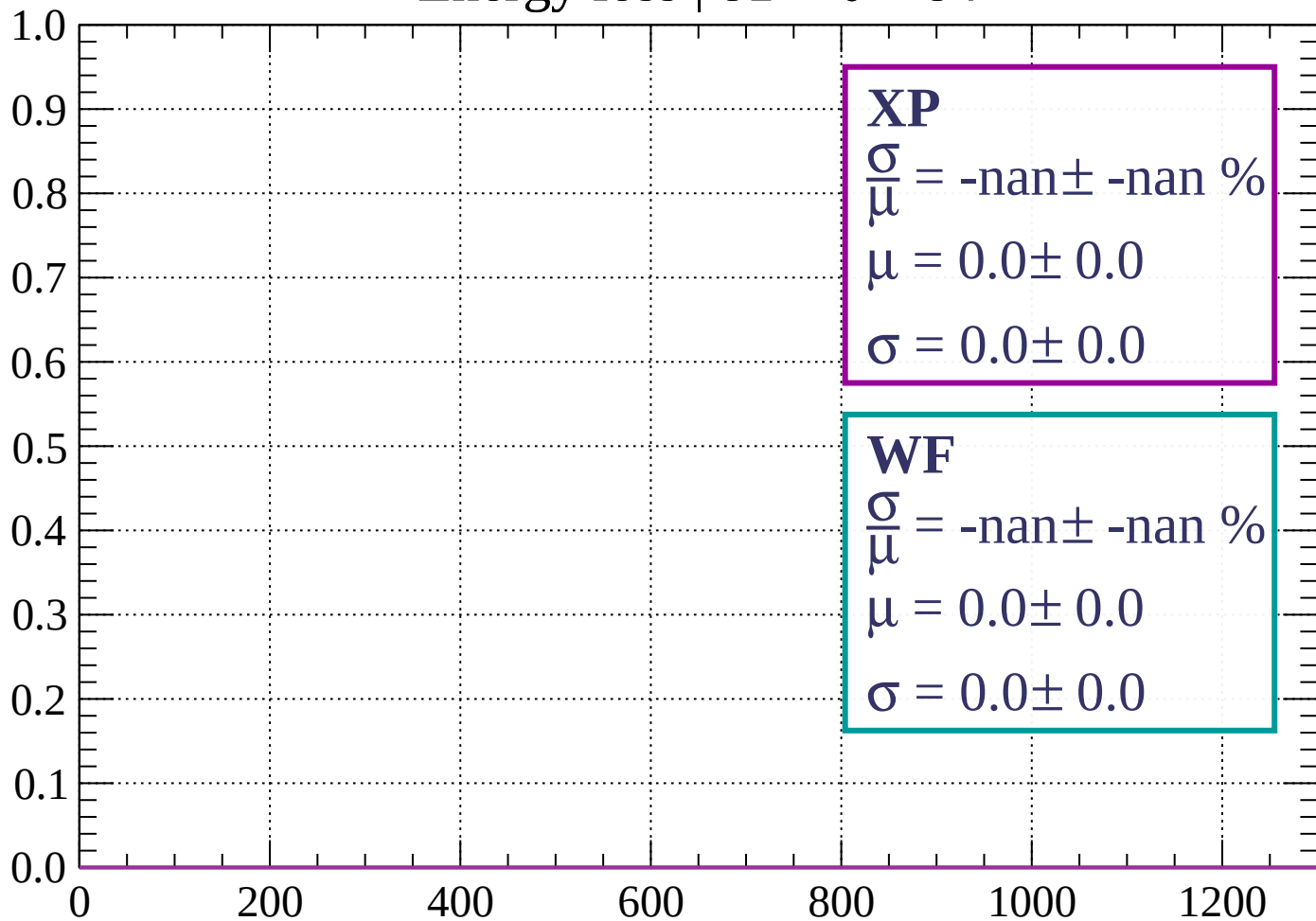
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \theta < 84$

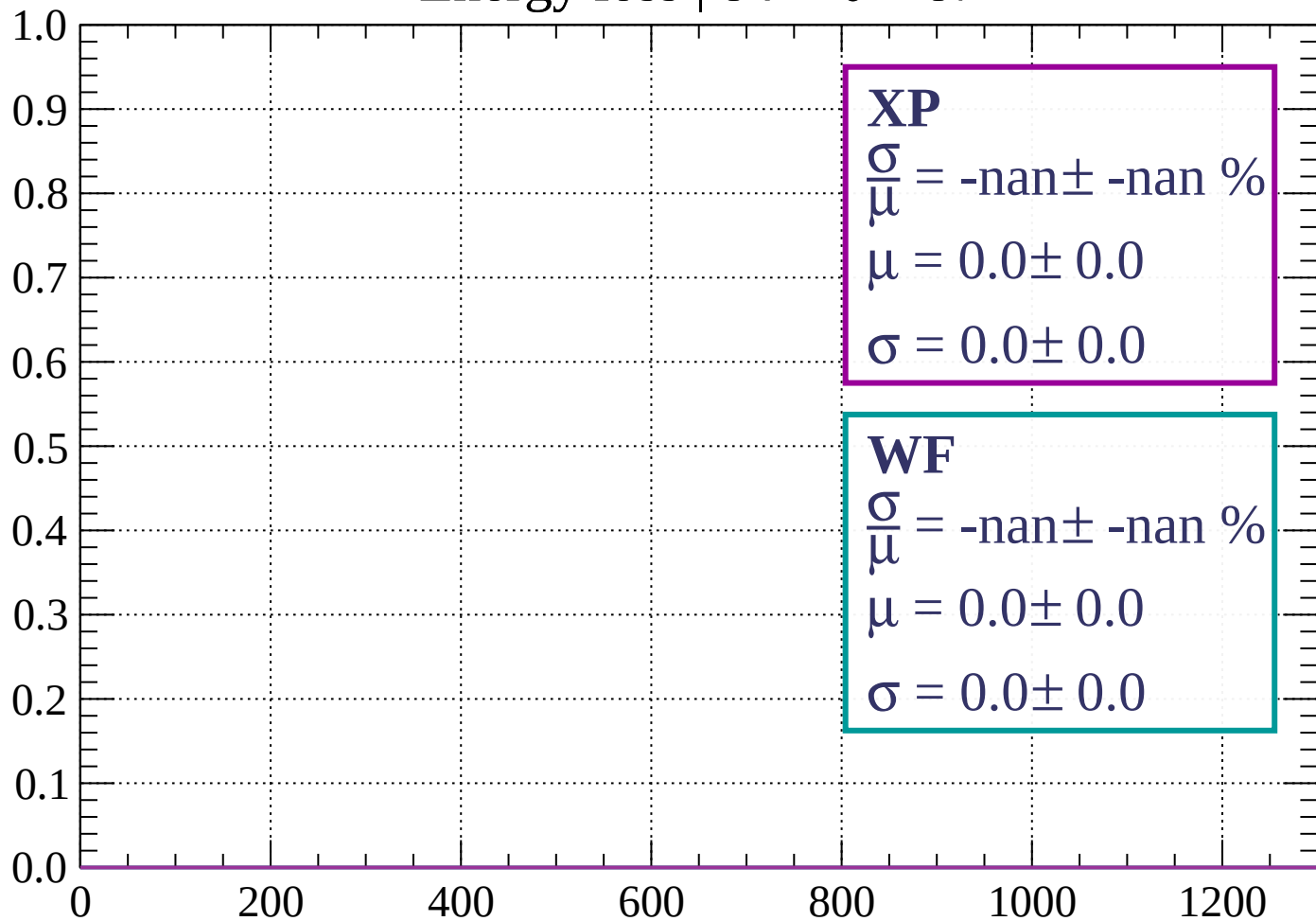
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

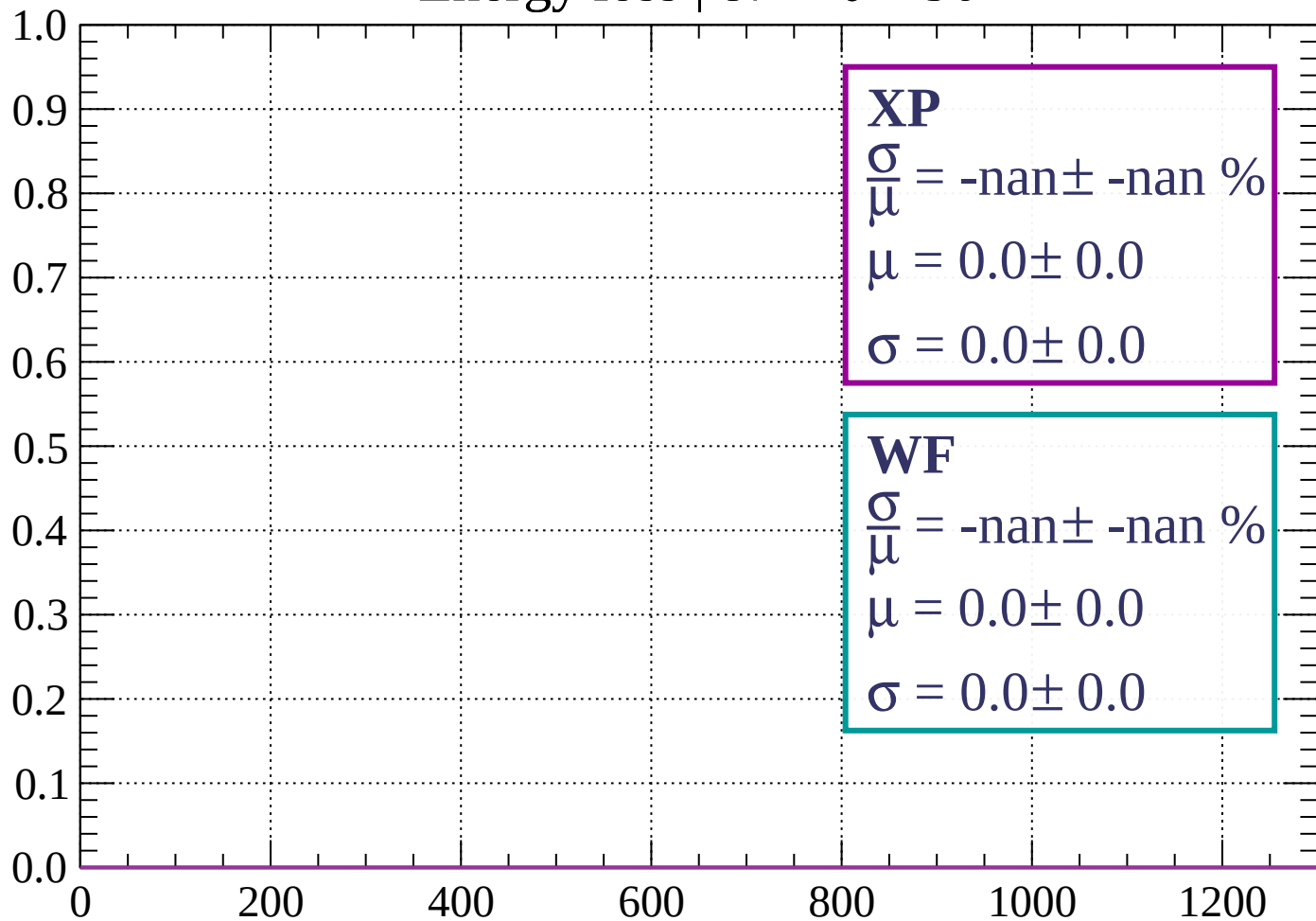
Count



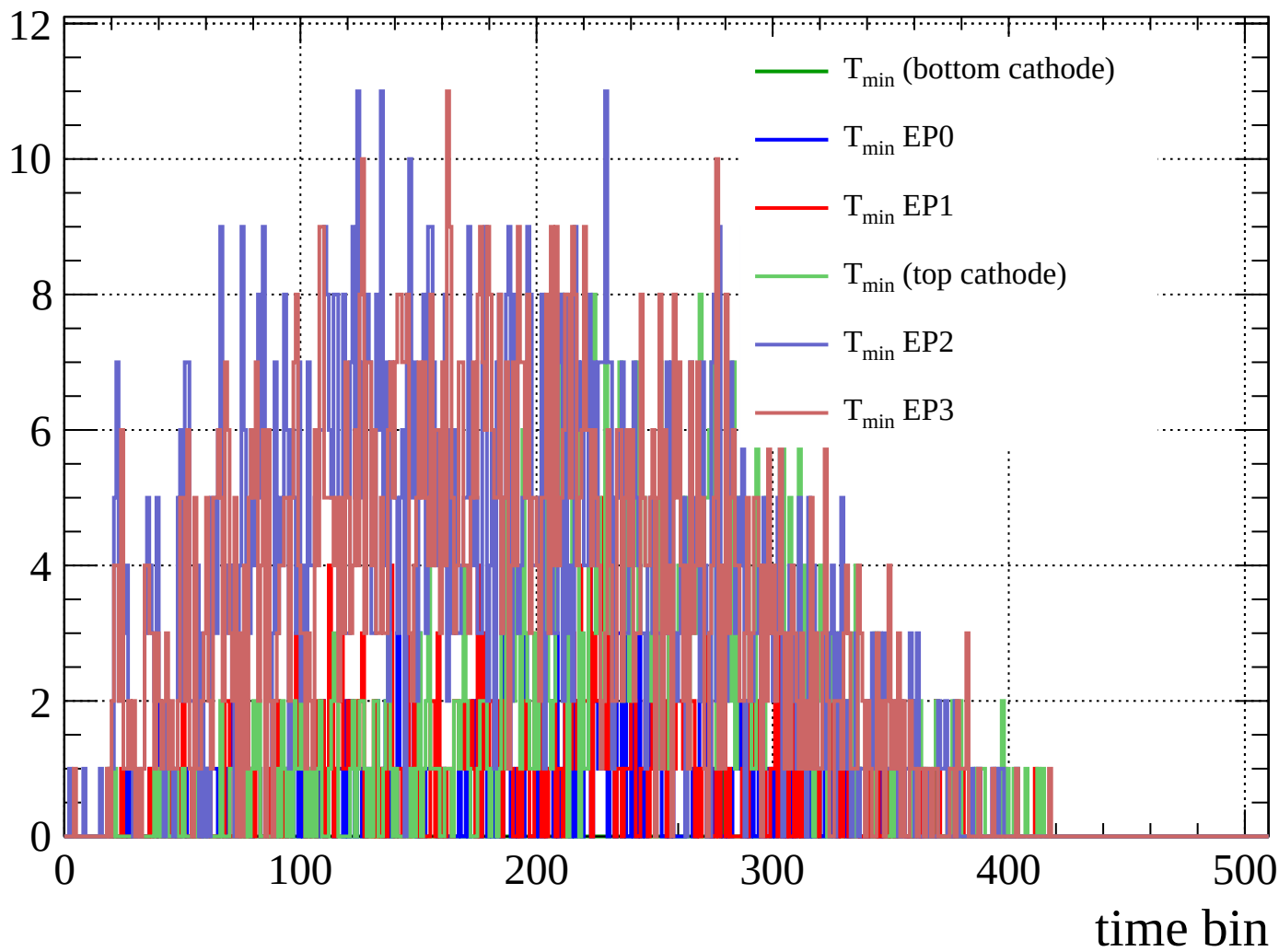
dE/dx (ADC counts/cm)

Energy loss | $87 < \theta < 90$

Count



Count



Count

